TFPT — The Prime Front

Research documentation of the prime / zeta line: seven verified modules, one localized residual, and the complete kill list

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The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and five companions**, best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*

1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.

2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.

3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.

4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.

5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: **R.** `tfpt_research_contracts` — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team); **P.** `tfpt_prime_front` — research documentation of the prime / zeta line: the seven load-bearing modules on the E_8 glue census (Hecke from geometry through the transport ledger), the sandbox localization program that follows them, and the kill list. Documentation, not a result: it makes no claim about the Riemann Hypothesis.

*You are reading the **Prime Front** companion (P): the complete internal record of the prime / zeta line — how a discrete compiler's bookkeeping kept producing classical number-theoretic objects, which of those links became load-bearing modules, and how the remaining question was compressed, over a hundred exploratory probes, down to one strict-positivity margin input — then, once that margin was measured to be an artifact of the requirement, down to a two-object core — then down to a three-point list of which only one point is an inequality — and finally, once the boundary reformulation ran, down to that single inequality in a textbook shape (a Galerkin error with Aubin–Nitsche duality), with the prime comb named as the one structure that refuses compression — and, once the missing lemmas of that inequality were measured against their classical addresses, down to exactly one genuinely new analytic statement — and finally, with the whole chain assembled end to end on one ladder, down to a load-bearing spine that is 96.2% identity or completed Cholesky certificate and whose only hypothesis is a declared accounting convention. It is a **research documentation**, not a results paper. Its purpose is that a reader (or a reviewer) can see the whole route, including everything that died on the way.*

Abstract

A two-axiom discrete compiler ($c_3 = 1/(8\pi)$, $g_{\mathrm{car}} = 5$, $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$) has no business meeting the prime numbers. It does anyway: the μ_4 glue completion carries a quartic character, the signed glue census is a theta series, and the shell counts of that census turn out to be *Hecke* data. It goes further than an analogy: the E_8 shell count is literally $240\,\zeta(s)\zeta(s-3)$ as a Dirichlet series, and the lattice's self-duality gives its theta sum an exact functional equation whose fixed axis is the critical line — a framework picture, stated as such, which the document opens with and then keeps at arm's length. This companion documents the line end to end. Twenty-one modules are load-bearing and ledger-carried (v535–v555): Hecke from geometry on the E_8 census, the Eichler trace layer, the half-integral bridge, one finite relative-trace identity of which the first three are projections, the Weil structure of the resulting family with two explicitly isolated obstructions, the amplitude route with a positive linear carrier, the matching-lemma / transport ledger package, and — out of the second phase — the margin-chain identities, the lumped M-matrix pair, the long-lag support structure, the Hardy-core identities, the capacity-chain identities, the level-lemma identities, the Green/Szegő identities, the gauge/parity identities, the odd-sector identities, the fixed-size Ritz ceiling certificate, the four fixed-size angle instruments, the exact-form identities, the sampling/harmonics identities and the Pareto/total-variation identities. Everything beyond those twenty-one is *sandbox*: an exploratory program, run as 163 preregistered probes with 4267 machine checks, which localized the residual object. That localization is the actual content of this document. The transport ledger closes exactly, leaving one coupled inequality (called **I5**) which is — *by identity, as a typing and not as progress* — equivalent to Weil positivity and hence to the Riemann Hypothesis. The subsequent arc maps I5 geographically (Connes–Consani dictionary, tightness curves, the Sonin crossing, the one-atom band), turns the positivity question into a per-zone handover induction, and then certifies its ingredients one by one until the entire remaining hardness sits in a single Loewner statement on the J-odd half of a window, reducible by Sherman–Morrison/Woodbury to one scalar ratio $r = \kappa/\varepsilon \leq 1$ (measured 0.0053–0.1814 on 16 zones), then, once ε is identified as an exact energy, to two scalars — and finally, with both scalars certified (the wing scalar unconditionally by a graded matrix cap, the boundary value by a residual certificate that carries a cancellation), to a chain that closes 16/16 conditional on exactly one strict-positivity margin input, itself two to six orders of magnitude weaker than the conclusion it buys. The penultimate part then closes the induction circle on the measured zones end to end — a certified base case, fifteen certified handover steps through a graded Loewner minorant, and the structural finding that the atom entry costs nothing — while leaving three sharp, named gaps: no reserve, no scalar step law, and no uniformity in k, with an extrapolated crossing warning at $n \approx 170$. The next part stops extrapolating and measures: on a deep ladder of 199 prime-power zones with 117 certified handovers, the crossing is located between $n = 461$ and $n = 463$ — the $n \approx 170$ figure was a fit artefact, but the wall is real, and $n^* \approx 462$ is an upper bound — and the single warning splits into three separate walls (a measured margin wall, an arithmetic ladder wall set by the twin-prime pair $521 \to 523$, and a vacuous-requirement wall at $n = 727$), while the handover mechanism itself never fails: all 117 steps certify at retention 1.000000, and the chain tears at the ratio, not at a step. The operating variable turns out to be the window depth, not the zone index. The next part acts on that reinterpretation and rebuilds the construction in a gap-coupled scaled frame (D tied to the local prime gap): two of the three walls fall *structurally* — the ladder death becomes a payable cell cost (the killer pair $461 \to 463$ now certifies spectrally with the reserve opening by a factor $\sim$1000) and the requirement wall loses its depth dependence — while the margin wall proves *frame-invariant* at exponent -0.974: it is the substance of the requirement, not geometry. The hardness is thereby traded — three arithmetic walls against two analytic demands, the positivity of one limit operator and a convergence rate below the step reserve — with the prime-gap dependence disclosed rather than hidden. The next part settles the currency question, with a fundamental reinterpretation: the wall carries the same exponent in *every* admissible currency — it is substance — but it measures the *discretisation*, not the spectrum. Under refinement at a fixed window both leading eigenvalues carry the same cell-width power (the continuum window form has no gap), the positive floor survives only as a cancellation of relative size 10^{-7}–10^{-4}, and the T109 requirement chain turns out to divide by an artifact margin; the hardness balance is restated as four parts, led by a margin-free step certificate. The next part builds that certificate and the wall dissolves: the exact Schur complement (Albert 1969, with the Moore–Penrose inverse) certifies every ladder step margin-free — eleven of them beyond the old wall, to $n = 1331$, and all seven zones where the requirement chain tore — the wall was an $O(1)$ numerator divided by an artifact floor, chains now stop only at the compute cap and never at a step, and the (R) demand itself proves grid-bound and is deleted rather than transported, leaving a two-object core: ε relative to κ via the exact Szegő identity, and transport of an $O(0.1)$ Schur complement between grids. The next part splits that core: the transport across a real regrid certifies only mild refinement (a clean split at ratio $\rho^* \approx 1.8$, and for a principled reason — on nested ladders, where the transport error is exactly zero, the Schur floor itself falls like $\rho^{-1.7}$, so no bound can undo the drop), while a two-scale compression breaks the compute cap: a certified margin-free step at $n = 155\,921$ (117× deeper), chains of ten steps, the stopper always cost and never a step — leaving the shortest list the program has produced: three points, of which only one is an inequality (the classical Szegő–Levinson prediction-error bound for one symbol). The next part executes the reformulation point of that list, and the induction step turns out to *be* a boundary process: partial minimisation over the interior (Haynsworth) compresses the window into a state (Σ, r, σ) of $12 \times 12 + 12 + 1$ numbers that carries the global rank-one pole *exactly* (bordered elimination, no truncation) and updates by a Riccati-type recursion whose cost is independent of the window — one unbroken chain of 169 236 prepends ran to a window of 1.35 million cells (903× the old cap) at a flat 76 μs per step, declared honestly as a cost-geometry demonstration and not a Weil certificate. What refuses compression is the *prime comb* itself: the archimedean tail decays cleanly, but the reflection comb couples every incoming cell to an interior cell at $\alpha - \log n_j$ for every prime power in the window, so the full symbol never decays in the width and no sparse faithful state exists. In exchange, the one remaining inequality acquires a textbook shape: ε is the error of a piecewise-constant Galerkin method for the D-independent function $2\sinh(x/2)$, Aubin–Nitsche duality predicts the measured exponent ($\theta = 1.79 \Rightarrow s \approx 0.90$), and the target inequality is $\varepsilon(\alpha, D) \geq c_0 D^{\theta} \alpha^{\varphi}$ across the prime-power jumps. The last part then makes that inequality *theorem-shaped*: the Galerkin reading becomes four exact identities, the lower-bound direction acquires a certified two-level chain that loses no power of D ($\theta' = 1.74$ against $\theta = 1.76$), the jumps acquire exact closed forms — with the correction that prime-power entries *raise* ε, so the factor-120 falls were a sweep artefact of the earlier run — and what remains is three named analytic lemmas about one symbol, two of them constants rather than rates. The next part measures those three lemmas against their classical addresses, and two of the three stand: saturation is an *identity* here; the Schur-floor lemma is rescued by the strengthened Cauchy–Schwarz shift onto the oscillation Gram, whose arithmetic aliasing symbol suppresses the prime comb quadratically — certified positive on half the measured windows, and on a 15 680-point FFT lever the certified floor rises logarithmically and crosses zero, so the failing windows were under-resolved rather than obstructed; and the corner lemma corrects itself — a log singularity is the boundary of all powers, so no power-law corner exponent exists to cite, and the certified chain is repaired at its last step, the rate now living on the oscillation mass. The missing analytic content is thereby exactly one statement: a $D^{1.75}$ lower bound for that mass. The closing part then closes the *arithmetic* half of the remaining list as a theorem — the aliasing symbol is positive for every cell width below an explicit $D_0(\alpha) = \exp(-(\Xi(\alpha) + B))$, with Ξ the prime-power atom count and $B = -1.0474$ universal — proves that the mass bound is *not* derivable from the identities already in hand, and reduces it, through the exact identity $\kappa_{\text{end}} = 1/(1 + R)$, to a single discrete Harnack inequality with a classical address; the chain contains no zeta input anywhere. The final part then *explains* that inequality: the two increment families are the odd and the even half of one sequence, offset by a single fine cell, so given one sign on the corner cells $|R - 1| \leq 0.047$ holds *unconditionally* by a two-line pairing argument — while the per-cell version of the inequality is provably false, and the honest defect count rises from three to four, because the frame's window grows with the zone (the unconditional D_0 route covers not one handover of the real ladder) and the strengthened Cauchy–Schwarz constant decays with the width. The series then closes by *assembling* what it built: on one common resolution, thirty consecutive prime-power atoms form a run in which the output of each certified handover *is* the input of the next (the incoming atom block is bit-exactly zero, the composition residual 2.9×10^{-14}), all five stages complete on 52 of 52 ladder zones with 430 completed Cholesky certificates, and the load-bearing spine turns out to be 96.2% identity or Cholesky certificate with exactly one hypothesis — an accounting convention — so that the Harnack pair leaves the spine and survives only as a second, independent route. What remains is stated as such: not size but *uniformity* in the zone index, quantified in a six-point distance statement (no zeta input, RH unused in either direction, a finite run bounded by the smallest log-gap it contains, a measured ladder that is not all zones, uniformity as the open item, and the resulting object typed as what it is — a lower bound in numerical analysis, not a statement about zeros). A second phase opens past the finale:

its first part dissolves the segment-seam question into a monotone free-resolution construction (re-gridding only at record-small gaps, all twelve affordable real seams certified), turns the continuum step into an exact fractional-Dirichlet identity, and leaves a sixteen-point full-proof map containing exactly *two* genuinely new inequalities — a direction lemma and a zone-uniform seam floor. Its later parts dissect both inequalities (both proof-shaped; one refuted as stated and replaced by a band plus an enumeration), work the resulting four-point list cheapest first (three of the four stand at their preregistered bars; the fourth — a concentration bar — is missed honestly, by 3.6%, systematically in the re-gridding ratio), and then test the law that miss preregistered: the fitted law falls once on 331 fresh transports with the bar frozen, while the structure underneath it is solved with identities — the flat border vector sits at 1 exactly, a linear density profile at 2 exactly, everything above is curvature, and the curvature chain is a per-transport theorem holding on all 436 transports. The two pieces that break leaves are then attacked head on, and exactly one of the two stands: the graded-to-uniform *bridge* stands as an identity — the Céa/Strang defect in matrix form reproduces the directly computed uniform floor on 84 pairs with zero overshoot, explains the measured 8% false positives completely, and carries both deep seams to positive fine floors at up to 3.8× the factorisation cap — while the curvature bound honestly breaks its own frozen shape band on 13 of 545 disjoint test transports and is reduced to a zone-uniform bound on one exponent. **No claim of progress on the Riemann Hypothesis is made anywhere in this document**, and none of the sandbox findings moves any ledger marker. The document is deliberately symmetric about failure: a dedicated section lists the readings and routes that were refuted — an essential-singularity reading, a C/g proxy law, a density chain, an accumulated invariant with self-propagation, the Landau–Widom anchor, the symbol route for the odd channel, three certificate candidates for the avoidance norm plus the Combes–Thomas certificate at the boundary, and the naive margin route — because in a program of this shape the kill list is the more reliable output.

[E] exact / machine-proven	**[C]** conditional	**[O]** open / axiom	**[X]** kill test

1 How to read this document

Scope and status — read this before anything else

1. **No RH claim.** Nothing in this document claims progress towards, or evidence for, the Riemann Hypothesis. Where an equivalence to RH appears (Section 5) it is an *equivalence typing*: a statement about which object carries the difficulty, produced by an exact identity, and deliberately *not* a statement that the difficulty has been reduced.
2. **Two layers, never mixed.** Claims marked [ledger] are load-bearing: they have a row in `verification/status_ledger.csv`, a module in `verification/`, and a script citation here (the blue `[verification/...]` tag). Everything marked [sandbox] is exploratory work in `experiments/tfpt-discovery/`; it carries no ledger row, no marker, and no authority. Section 4 is the only section that is entirely [ledger].
3. **Status markers are quoted, never upgraded.** The markers [E] [C] [O] [X] shown in Section 4 are read verbatim from the ledger rows of v535–v555. This document introduces no new markers and moves none.
4. **Fits are declared as fits.** Every scaling law, slope, exponent and extrapolation quoted from the sandbox is a fit to finitely many measured windows and is labelled as such at the point of use. Sixteen resolved zones are sixteen zones, not all zones.
5. **Classical results are named as classical.** Kneser neighbours, Eichler/Witt, Shimura, Waldspurger, Cohen, Weil/Guinand, Bombieri, Yoshida, Connes–Consani, Gronwall/Robin, Douglas, Slepian–Landau–Pollak, Sherman–Morrison/Woodbury, Grenander–Szegő and the rest are classical and are used as such; the compiler-native part is the wiring, not the theorems.
6. **Numbers are literal.** Every numeric value in this document is transcribed from the research diary `experiments/next.txt` or from a ledger row. Figures that could not be drawn from literal data are marked *schematic* in their caption, and the caption states exactly which quantities in the figure are literal.

The diary parts are cited as **T**n (part n of the research diary, chronological, $n = 11 \dots 163$). The probe file, the contract name, the verdict and the check count of every part are listed in Appendix A. Section 2 explains why a physics compiler meets the primes at all and Section 3 states the framework interpretation that motivated the whole line — a self-dual lattice, its functional equation, and a mirror that returns at the far end of the program; Section 4 is the verified layer; Sections 5, 6 and 7 are the sandbox program in three acts; Section 8 is the kill list; Section 9 is the series balance (cascade, certification degree, the honest RH distance); Section 10 is the opening arc of the second phase (T126–T163: the seam architecture, the two inequalities dissected, the four-point list worked cheapest first, then the κ law tested and broken productively, then the graded-to-uniform bridge established as an identity while the curvature bound is reduced to one exponent, then the self-supply loop built one number short of closed — the secular sandwich and the Perron sign theorem — then the reverse flow: the identity block promoted to v542, a spectrum-only boundary-Dirac coupling and a certificate-floor audit that hardened v379 — then the pole-free floor closed as existence with a sign anatomy (T134) and a bounded seam state where the Weil window provably has none (T135), and finally the M-matrix pair, which closes the a-priori radius item by Varga's identity while the exact D-bookkeeping puts the whole loss in the margin (T136, promoted as v543), and the long lags, whose support turns out to be an arithmetic stripe set and whose absolute-value envelope family is certified *dead* (T137, promoted as v544) — and finally the compensation itself, whose sign law turns out to be *interval geometry* while the m-paired Neumann certificate removes the arithmetic wall on all 77 dead blocks (T138), and the classical decay lemma, refuted at its hypothesis for an arithmetic reason while the sign law is *derived* from one exact telescoping identity and the measured core shrinks to one signed inequality at stripe distance $b \leq 16$ (T139), and finally that one inequality itself, which acquires an *exact finite core per zone* — the telescope identity lifted to the form level, the spectral radius an exact eigenvalue of a closed-geometry covering kernel times a mass-plus-Dirichlet form, the checkerboard split replacing the $O(nb)$ Weyl steps by three D-independent ones, and all the D-dependence located in the geometry (T140), and finally the Hardy ingredient itself, which *resists* but acquires an address: four exact identities put it in classical two-weight shape, the certified constant is not zone-uniform while the object it bounds is, the additive shape is dead as a shape by its own exact Weyl floor and the joint shape fails at the normalisation alone, so the rest list collapses to one closed conductance profile (T141, promoted together with T140 as v545), and at last that profile itself, *constructed* rather than guessed: the capacity decomposition exhibits the optimal Hardy weight exactly — $\Omega = 1$ by a projection identity where T141 had guessed 20.7–2724 — the failure shrinks to a constant factor $\sim$2.3 flat in D, and the rank ladder closes the whole comparison path, forcing the sharp capacity–Rayleigh route (T142), and at last that sharp route itself, which *carries*: the exact capacity–Rayleigh form is an identity on all 26 windows, Maz'ya's capacity criterion applied to the gap form lands inside its window $[1/4, 1]$ with a zone-uniform loss factor ($D^{-0.048\pm0.010}$), and D-uniformity is reduced to *one* named inequality — a non-Markovian Maz'ya capacity bound whose interval structure points at Muckenhoupt's 1972 two-weight calculus (T143), and at last that interval route itself, which *carries*: the interval class is exhausted exactly (11.4 million intervals via a Cholesky prefix-sum identity), the closed two-weight sup lands inside the Maz'ya window flat in D, the family restriction falls entirely by a max-density-subgraph bound over all 2^m sets, the Markov perturbation route is certified dead, and the certified chain has exactly *one* unproven input — the absolute Maz'ya constant c_0, its sharpest shape a Muckenhoupt-type hypothesis (T144), and at last the proof attempt itself, whose verdict is *one step missing*: the classical Maz'ya proof transcribes step by step onto the non-Markovian form, the Markov property sits in exactly *one* line (M4), that line splits with its mass half a theorem that dominates ($\sigma_{\mathrm{tot}} = 0.215$–$0.443 < 1$ everywhere), c_0 becomes explicit (2.248–4.227, flat in D) and S1$'$ is *certified* per window on 64/64 windows, while an explicit no-go proves that an a-priori bound on the minimiser's level profile — the level lemma L1 — is necessary and cannot be replaced by any weaker hypothesis (T145; the identity spine of the capacity cycle is promoted as v546), and at last the level lemma itself, which *stands*: L1 closes on the measurement surface as a chain of theorems and certified window inequalities with no step reading the minimiser — the proof lever is the resolvent identity $\psi = \lambda R\psi$ itself (the $\Theta(D^3)$ smallness turns from obstacle into tool), Davis–Kahan is instrumented and discarded, Maz'ya's classical dyadic 8 falls to 2 because the cake base is free, and the a-priori constant $c_0^{\mathrm{ap}} = 3.90$–4.85 lands at the size of Maz'ya's classical Dirichlet value 4 — with one genuine remainder, D-uniformity for *all* D (T146; the a-priori core is promoted as v547), and at last that asymptotics itself, which acquires an *exact shape*: the last open object becomes the identity $\Gamma = \sqrt{Q^\star}\, S_w$ — all-D uniformity split into a purely spectral and a purely geometric factor, both certified on the surface — the classical decay route is shown by computation to be asymptotically vacuous, delocalisation itself *is* the bound, the mechanism is named (a Toeplitz-minus-Hankel section whose bottom eigenvectors are Fourier modes, by Szegő/Widom theory) with the sharp prediction $Q_B \le 2|B|$ holding, and one statement — a Szegő theorem for the diagonally reweighted section — lifts the chain to all D (T147), and at last that lifting statement itself, dissected: the second factor S_w *closes* on the surface by an LDL^T layer-cake certificate, at the price of an honest negative — the Toeplitz symbol is refuted at 0, so positive definiteness is a finite-section effect — while the lifting statement resists with its hypothesis isolated to a single named scalar, the total variation of the log-whitening weight, proved to be the roughness and not the conditioning by a controlled weight-class experiment; the first end-to-end number arrives — the certified chain delivers 8.4–15.9% of the true gap on all 48 windows (T148; the identity/certificate core of T147+T148 is promoted as v548), and at last that scalar itself, attacked through the gauge freedom: the whitening diagonal is a free choice, the constant gauge *eliminates* the total variation exactly (0.0000) at a certified sandwich price — and thereby refutes the hypothesis, since the gauges that kill the flutter move the smoothness functional by at most 0.9%: the roughness sits in the *form*, not in the multiplier, the missing input relocates to the deep modes of the pure Toeplitz-minus-Hankel section, and the Dirichlet control corrects the question to a ladder bound $\nu_k \le Ck^2$ with m-free C (T149), and at last that ladder itself, which *names the mechanism — parity*: the form is exactly the compression of the full symmetric Toeplitz section onto its antisymmetric parity sector, the sign-changing symbol's entire negative inertia sits in the *even* sector (certified 72/72) — T148's honest negative explained rather than worked around — the flutter amplitude becomes a certified form functional with a closed atom budget, and the arithmetic atoms turn out to be co-responsible for the positivity (the archimedean section alone would be indefinite), while one gate remains: the ladder constant still grows (13.81π, $x^{0.258}$ against the bar 0.25) — the gap from π exactly the arithmetic leakage of the bottom mode (T150; the identity/certificate core of T149+T150 is promoted as v549), and at last that odd-sector ladder itself, which *closes by rerouting off the ladder*: the odd grid steps over the symbol's negative window ($\theta_c/\theta_1 = 0.328$–$0.407$ on 72/72 — the positivity of the odd sector is a grid fact), the bottom spectrum is certified against the parity Laplacian ($\lambda_k \le S\mu_k^P$, $S = 1.10$–2.39), and one discrete Sobolev step yields a per-mode bound *linear* in k with a *non-growing* constant ($C_S = 11.51$–19.57, trend $x^{0.020}$) where the quadratic ladder grew — two of T150's three rest items settled (the archimedean minimum closed and attained exactly; the symbol route a computed dead end: the local model is a matrix statement, not a symbol one), one scalar remaining a fit — the bottom pencil ratio $R = K_{\mathrm{bot}}/\kappa = 3.36$–$9.71$, flat, exactly where the no-go breaks (T151; the identity/certificate core of T151 is promoted as v550), and finally that pencil ratio itself, attacked head-on: both hoped-for gifts from the smooth kernel are *refuted* — the archimedean section is itself negative in the odd sector and the atom part too, so odd-sector positivity is a *cancellation* between geometry and arithmetic, not a property of either half — while the floor nevertheless closes structurally, a Schur two-block criterion with a fixed 16-mode block certifying $\kappa \ge 0.225$–0.250 flat across all windows, leaving the ratio certified on both ends ($R \le K_{\mathrm{bot}}/t = 4.41$–$8.47$) with one term missing — an m-free ceiling on K_{bot} — and 89–92% of the dominant loss Ψ mapped onto the eight modes the certified ladder already controls (T152), and at last that mapped rebuild itself, carried out and *refuted by exactly its own factor*: Ψ is pinned between a_1/λ_1 and $1/\lambda_1$ with a_1 exactly the $8/\pi^2$ of the first parity sine — the reserve was never there — while the theorem-grade collapse closes Ψ as a term, the archimedean part turns out *positive* on the bulk parity block after an m-free peeling, and the two remaining terms lose to one and the same missing object, a Green/alignment estimate of two inverse-iteration columns (T153), and at last that estimate itself, attacked head-on — and the ceiling *closes, exactly and at fixed* *size*: the sixteen-column certificate $\mathrm{span}\{t_1..t_8\} + A^{-1}L_P\,\mathrm{span}\{t_1..t_8\}$ needs no residual argument at all (Ritz values are upper bounds for the eigenvalues of the same index, Courant–Fischer) and agrees with the inertia-certified true bottom ladder constant to 5.2×10^{-7} on every window, retiring the size-m factorisations from the ceiling step; the obstruction acquires a geometry — seven of eight bottom directions agree with the parity Laplacian to a fraction of a degree, *one* sits at 82.93–89.79°, and that single misalignment *is* the collapse price, recovered in full per window by one Cholesky, moving per-window end-to-end to 4.45×10^{-2}–3.13×10^{-1}, inside the target band — with both remaining terms now uniformity terms carrying certified per-window numbers, the most accessible arithmetic-free (T154; the certificate core is promoted as v551), and at last those two floors themselves, attacked head-on — the complement floor becomes an *exact* fixed-size certificate: a 12×12 problem in the exact Kac–Murdock–Szegő numbers plus one 12×8 overlap matrix reproduces the size-m eigenproblem to 0.999999783–0.999999958 on every window, the 83–90° direction is demystified (it lives on modes 9–12, which is exactly why $K = 12$), and the collapse price is recovered 78.8–100% at fixed size — while on the block side the full symbol infimum turns out *negative* on every window against a positive section floor, so no symbol argument can ever produce that floor: the mechanism must be a finite-section Fejér cancellation (T155), and at last those two rest objects themselves, attacked head-on — and both become *single scalars*: the t_1-loss of the bottom Ritz mixing is an exact closed function $F(P, r)$ of two dimensionless numbers (verified to 2×10^{-16}), its two-line ceiling $r \leq 1/(Ls) \leq 1/p_1$ is tight to a factor 1.03–1.16, and the whole first term is *one* m-free lower bound on the angle $p_1 = \cos^2 \angle(t_1, e_1(A)) = 0.2010$–$0.3282$, while the block positivity is re-identified as an *alignment* fact rather than Fejér mass damping — on the minimiser an arch part and an atom part of nearly equal size cancel, the expected arch inequality ≥ 1 is refuted and the additive split is divergent (T156), and at last those two angles themselves, attacked head-on — neither falls, but both change shape: the p_1 floor becomes a *theorem* times one measured fixed-size number (the sine-block confinement, from the certified floor and ladder alone, proves $e_1(A)$ lives 97.1–98.4% inside the first sixteen parity sines, and the resolvent route then pins 97.9% of the measured angle), the whole first term is re-stated as a bound on *one* diagonal entry of a 16×16 Schur complement the chain already forms — which Cauchy–Schwarz misses by two to five orders of magnitude, because the cancellation is nearly complete — and the arch half acquires an *executed* adaptive Lipschitz certificate, while the alignment term stays a per-window domination with a shrinking margin (T157; the four instrument candidates of T155/T157 are promoted as v552), and at last the cancellation-seeing bound itself, which *exists and is a theorem*: the Thomson dual form turns the Schur entry into a Dirichlet maximum, so every trial vector bounds it from the right side — which is exactly why Cauchy–Schwarz missed by three to five orders of magnitude, a maximum evaluated at one direction — a Cholesky ladder of strictly positive terms pins the entry to within 1.13–1.27, flat, from one 16×16 factorisation the chain already builds, the entry is exactly two-dimensional once one Green column is granted, and T157's growth pointer is honestly refuted: the atom mass grows faster than the arch floor in every band, so what carries the domination is precisely the off-band coupling a dyadic argument discards, and the measured-step count falls from three to one (T158), and at last the cancellation itself, executed algebraically — and the algebra is delivered while the bound is not: seven machine-checked identities land at once, among them the gauge identity ($\sum_d w_d = 0$ exactly, because Toeplitz-minus-Hankel annihilates constant lag vectors — the form is exactly *blind* to the lag mass of the arithmetic, which is precisely where the h^2-sized halves live) and a closed 256-term Dirichlet-kernel form of the geometric weights; the honest answer to the kernel question is negative with an exponent — the h^2 does *not* telescope away, both halves grow like $h^{2.1}$ because the cancellation is the common weight, and four hoped-for routes are closed with exponents, including T158's own fixed sixteen-vector — while the genuinely new structural win is a sign law: the archimedean bulk block is exactly a symmetric Z-matrix, raw, with a closed sign-based floor comfortably above target, and the atom block obeys no sign law; the theorem count jumps from six to thirteen (T159; the theorem cores of T158/T159 are promoted as v553), and finally the pairing itself, attacked through its correlation structure — and the watershed of the phase arrives: the tail sum W^1 does not oscillate at all (three sign blocks, with a closed bound from the trigonometric zero count, so the Leibniz block device is refuted by being *worse* than the pricing it was meant to beat), three closed sign-definite moment laws evaluate the smooth archimedean half of the pairing down to the double-precision floor (a second numerical horizon, declared), and a machine-checked *sampling identity* shows that the atom half *is*, identically, a finite combination of Λ-weighted prime sums at thirty-two explicit frequencies, needed to relative depth h^{-2} and measured to cancel only to 0.00–0.37 of the trivial bound — the h^2 cancellation is the intrinsic arithmetic hardness of the problem, not an assembly artefact (T160), and finally the circularity triage itself, whose answer is the most consequential of the phase: the thirty-two frequencies are exactly the Fourier harmonics of the log-window (a theorem), the measured cancellation is fully the prime-number-theorem main term, and the required depth exceeds RH strength while sitting below the boundary term of every partial summation — the chain is *not* circular, it needs *more* than RH-strength input would supply, so the h^2 sits in the *split* and not in the primes; the analytic half of the arch bound closes m-free with a scale-free Bernstein rate and a closed head split, the sign inequality is refuted in all four readings and replaced by a certified fraction bound, and the closed cores of T160+T161 are promoted as v554 (T161); then the third split itself, which *exists* while the exhaustion saturates — the archimedean Mellin ladder lowers the demand in closed cell moments and turns at its second rung, one Abel step makes the demand prime-free with all arithmetic in the single Chebyshev constant, and the Fejér split goes below the yardstick on every window at a price growing like $h^{2.86}$: relocated, not closed (T162); and finally the Pareto front itself, which resists *as a theorem* rather than as a failed search — the exchange law makes price and demand two coordinates of one exact identity, a four-line total-variation floor $\mathrm{TV}(x) \geq 1/\mu_1^P \sim h^2$ closes the flat-price route for every admissible trial vector in the parity sector, and the hardness is thereby identified as the spectral gap of the parity Laplacian meeting the entry normalisation — never the primes — with the theorem cores of T162+T163 promoted as v555 (T163)); Section 11 is the status statement.

2 Motivation: why a compiler meets the primes

TFPT is a discrete compiler with two inputs: the seam constant $c_3 = \frac{1}{8\pi}$ (P1) and the carrier rank $g_{\text{car}} = 5$ (P2). They force the lattice completion $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, and the readouts of the Standard-Model skeleton are taken off that completion. None of this is about number theory. The prime line begins with an accident of the *glue*.

The completion is not $D_5 \oplus A_3$ but $D_5 \oplus A_3$ *plus* a glue group μ_4 — a cyclic group of order four. A cyclic group of order four carries a quartic character, and the quartic character on $\mathbb{Z}$ is χ_{-4}, the non-trivial character mod 4. Once the census of glue vectors is counted *with that sign*, three things happen at once, and all three are elementary consequences rather than choices:

1. The signed glue count is a **theta series**. Signed counting of a quadratic form is a theta series by definition; here the compiler's own count θ_3^2 is exactly the Gaussian-integer theta, $r_2(n)$, so the census counts $\mathbb{Z}[i]$-ideals (T85).
2. The character makes the count a **Dirichlet object**. χ_{-4} is the character of $\mathbb{Q}(i)$; the associated L-function is the first Dirichlet L-function, and the split law "$p = a^2 + b^2$" becomes a compiler-visible fibre property of each prime (T21, channel B: χ_4 fibre $\iff$ $p = a^2 + b^2$, verified 81/0/87/0 on $p < 1000$; χ_3 fibre $\iff$ $p = a^2 + 3b^2$, 80/0/86/0).
3. The shell counts of the census carry **Hecke eigenvalues**. That is the surprise, and it is the content of the first load-bearing module.

The first question of the series was therefore not "does this prove anything about primes" but "which of these links are *mechanism* and which are pretty coincidences". Parts T11–T13 established the signed glue theta, its L-series and the glue-character atlas (19/19, 16/16, 11/11); T14–T19 tested six candidate contracts in parallel and produced two self-corrections against the project's own earlier readings (T14 DEFLATED: the measured 2π is the universal Bisognano–Wichmann/Unruh

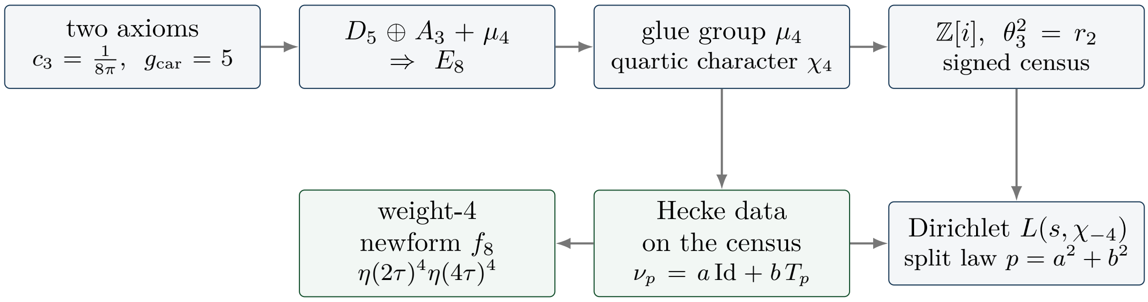

Figure 1: The chain by which the compiler acquires arithmetic. Nothing here is chosen for number theory: the glue group is forced by the completion, the quartic character is the character of a cyclic group of order four, and the signed count of a quadratic form is a theta series. The green nodes are where the line becomes load-bearing (Section 4).

conversion, not "the self-dual width"). The cross-finding of that batch is worth stating because it survived the entire series: the structure is exact at the census/lattice level (weight 4), and the gap is always the *same* gap — the canonical transport to the GL(1) / $\xi(s)$ world (multiplicity one, weight 1/2, the archimedean place, the derivation of the Hecke action). One meta-contract, `ZETA.CENSUS.TO.GL1` [O].

Parts T20–T25 then killed, in four stages, the slogan "the archimedean term comes from the seam": the raw seam density of states is flat/slightly decreasing (loglog slope ≈ -0.40) while the classical digamma kernel grows logarithmically (measured 0.15917 against $1/(2\pi) = 0.159155$, rms 3×10^{-6} for $t \geq 20$) — KILLED-AS-NAIVE (T20, 25/25); the non-DOS successor located the missing logarithm in the interval cut but failed the one-constant dictionary at $2/\pi$ (T22, PARTIAL); and both dictionary closures plus the scattering-phase observable died (T23–T25, $23/23 + 25/25 + 29/29$). This matters for the rest of the document: the archimedean place was never going to come for free, and when it finally arrived (T82) it arrived from *inside* the same family, not from the seam.

3 The big picture: a self-dual lattice and its own mirror

What this section is

A **framework interpretation**, not a result. It says how the pieces fit together and why the line was pursued for a hundred parts. Every factual statement it uses is either classical (and named as such), one of the twenty-one modules of Section 4, or a sandbox probe with its part number. Nothing here is an argument about the Riemann Hypothesis, nothing here is used later as a premise, and the picture itself carries no status marker.

3.1 (i) The zeta function is literally in the geometry

Count the E_8 lattice points shell by shell. The even shells satisfy $N(2n) = 240\,\sigma_3(n)$ — direct lattice counting gives 240, 2160, 6720, 17 520, 30 240, 60 480 for $n = 1 \ldots 6$ (T6, one of the parts that predate the indexed range of Appendix A) — and σ_3 is multiplicative, so the counting function of the lattice is a Dirichlet series with an Euler product:

$$\sum_{n\geq 1} \frac{N(2n)}{n^s} = 240 \sum_{n\geq 1} \frac{\sigma_3(n)}{n^s} = 240\,\zeta(s)\,\zeta(s-3) \qquad (\text{at } s = 5:\ \ 1.705672 \text{ vs } 1.705678).$$

The Riemann zeta function stands in E_8's own counting function, with a simple pole at $s = 4$ of residue $240\,\zeta(4) = 8\pi^4/3$ (T12). This is classical lattice theory — Eisenstein series of weight 4 — and it is stated here only to fix where the arithmetic enters: not through a modelling choice, but through the object the compiler already had. The contrast is instructive: the same multiplicativity *fails* for the $D_5 \oplus A_3$ sublattice alone (52, 576, 1584, 16 128 with $576 \cdot 1584 \neq 16\,128 \cdot 52$, T6). Without the glue there is no Euler structure in the count.

3.2 (ii) The glue carries the character of the Gaussian integers

The completion is $D_5 \oplus A_3 + \mu_4$, and a cyclic group of order four carries a quartic character, which on $\mathbb{Z}$ is χ_{-4} — the character of $\mathbb{Q}(i)$. Two consequences make the primes a *fibre property* of the compiler rather than an application of it. First, the split $p \equiv 1$ versus $p \equiv 3 \pmod 4$ is visible to the census directly: the χ_4 fibre holds exactly for $p = a^2 + b^2$ (81/0/87/0 on $p < 1000$, T21). Second, the character kills a pole. Of the three character channels of the E_8 counting function, the trivial one is $240\zeta(s)\zeta(s-3)$ with its pole at $s = 4$, the spinor one is $64(1-2^{-s})\zeta\zeta$ and keeps it, while the μ_4-signed channel $16 \cdot 2^{-s}\eta(s)\eta(s-3) - 8L(f_8, s)$ is *entire* ($L_c(4.001) = -6.976$, finite; T12). The signed channel is the only pole-free one — exactly the ζ versus $L(\chi)$ pattern.

3.3 (iii) Hecke structure is an output of geometry, not an input

The step that made the line load-bearing is `v535`: the Hecke action on the census is produced by *pure lattice adjacency* — Kneser p-neighbours — with no modular input anywhere in the derivation. The marked neighbour sum is exactly an affine Hecke element, $\nu_p = a\,\mathrm{Id} + b\,T_p$, and the cusp eigenvalues come out of the glue census as $a_p = -c(p)/8$. Prime structure is what the geometry *emits*; nothing about primes was put in. The first four modules of Section 4 make this precise and end in one finite relative-trace identity of which the first three are projections.

3.4 (iv) The core point: self-duality, and a mirror that keeps coming back

E_8 is unimodular — self-dual — and self-duality is exactly the hypothesis of Poisson summation. The lattice theta sum therefore satisfies an exact functional equation, $\sum_x e^{-t|x|^2} = (\pi/t)^4 \sum_x e^{-(\pi^2/t)|x|^2}$ (verified to 25 digits, T12), whose unique self-dual width is $t = \pi$. A functional equation is a *reflection*, and the critical line is the fixed axis of that reflection. That is the geometric content of the picture: on the axis sits the symmetry, and beside it sits everything the lattice actually carries.

Two findings, at opposite ends of the program, put the same mirror on the table twice.

- **The reflection product is the transport operator** (T83, `fe_symmetrization_probe.py`, 27/27, INVARIANT-NULL). The involution algebra was computed exactly: ζ's functional equation is the reflection $J_{1/2}$ and is absorbed in evenness (on autocorrelations it acts as the identity), while the family's functional equation is J_1. Their *product* $J_1 \circ J_{1/2} = e^{\pm u}$ is the unit-line shift — the difference of the two centres *is* the transport operator, algebraically. The two reflections generate an infinite dihedral ladder; the value side is invariant under the whole ladder, the spectral cone only under its own reflection, which is what anchors it to the critical line; and $\mathrm{Fix}(J_1) \cap \{\text{Weil cone}\} = \{0\}$ exactly, so no symmetrisation can relieve I5. The verdict was a null with a structural finding: the wall is *transversal* to the functional equation, not positional.
- **The same mirror sits in the last hard residue** (T106, Section 7). The mirror parity J with $JQJ = Q$ and $JB = -BR$ splits the Weil pole exactly into $ss^\mathsf{T} - tt^\mathsf{T}$: a positive rank-one lift in the J-even channel and a negative rank-one pressure in the J-odd one. The even channel closes; everything that is still hard lives in the odd half — the anti-symmetric part of the same reflection.

Stated as a picture and nothing more: the recurring question of this line is whether the self-dual object is entirely at peace with its own mirror. That sentence is an interpretation of where the difficulty sits, in the same spirit as the classical reading of the critical line as a symmetry axis. It is *not* a reformulation of the Riemann Hypothesis, it is not used as a premise anywhere below, and it produces no claim.

3.5 (v) What the program did with the picture

It converted it, step by step, into smaller and more explicit objects. The transport ledger closes exactly (`v541`), which leaves the single coupled inequality I5; I5 is by identity equivalent to Weil positivity, an equivalence *typing* and not a reduction (Section 5); the positivity question is then reorganised as a per-zone handover induction (Section 6); the induction is certified ingredient by ingredient until only one Loewner statement (R) on the J-odd half of a window remains; and (R) reduces first to one scalar ratio, then to two scalars, and finally — with both scalars certified — to one strict-positivity margin input (Section 7). The second phase continued the same conversion, and by T160 the chain localizes its residual hardness in a single classical object: by a machine-checked identity, the last open term of the pairing *is* the cancellation of $\sum_{n \le X} \Lambda(n)\, n^{-1/2} \cos(t \log n)$ at thirty-two explicit frequencies (Section 10) — a statement about a finite sum over prime powers, reported as a localization and not as progress on it. T161 then priced that object against the classical ladder, and the result relocates the hardness once more, soberly: the required cancellation depth exceeds what RH-strength input would supply and sits below the boundary term that every partial-summation bound carries, while the measured cancellation is fully accounted for by the prime-number-theorem main term — so the residual hardness is a *split-design* problem sitting above the classical prime sum, not an arithmetic statement hiding inside it (Section 10); that comparison of strengths is a triage, not a claim. The full cascade, with the type of every arrow marked, is Figure 8; the reason it is a route and not a proof sketch is stated there and in Section 11.

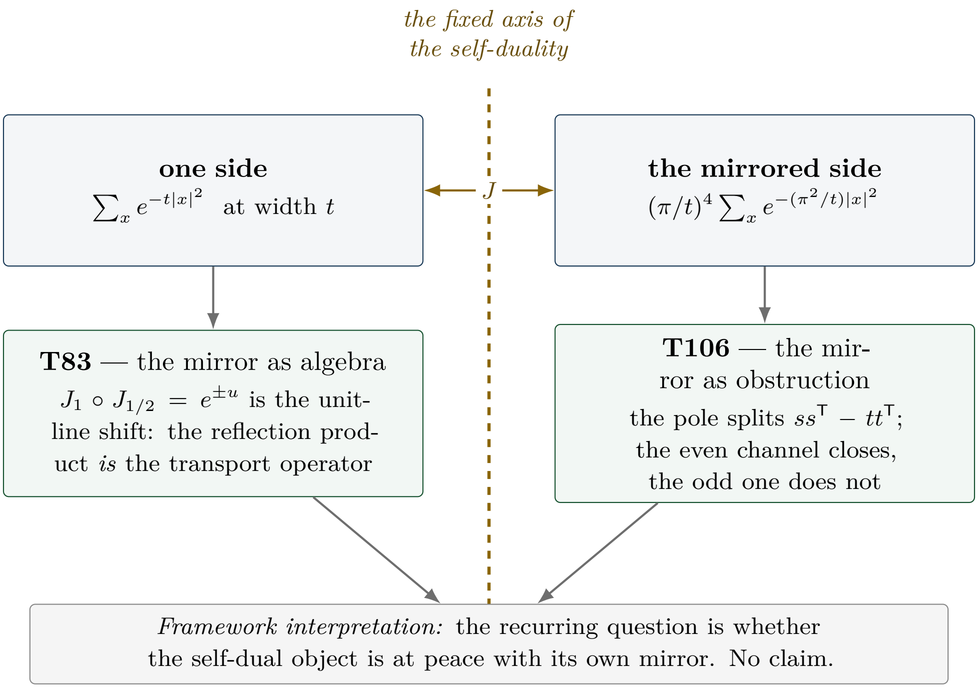

Figure 2: *Schematic.* The self-duality picture. *Literal:* the Poisson identity and its unique self-dual width $t = \pi$ (25 digits, T12); the algebraic identity $J_1 \circ J_{1/2} = e^{\pm u}$ and $\mathrm{Fix}(J_1) \cap \{\text{Weil cone}\} = \{0\}$ (T83); the exact pole splitting $ss^\top - tt^\top$ at 10^{-18} (T106). *Schematic:* the geometric arrangement, and the reading of the axis as "order" with the lattice's arithmetic content beside it. The bottom line is an interpretation and carries no status.

4 The verified layer: twenty-one modules

This section is entirely [ledger]. Each subsection states the claim as the ledger states it, cites the module, and quotes the ledger's own status markers. Nothing here depends on Sections 5–7.

The twenty-one modules and their ledger markers

Module	Claim id	Content	Markers
v535	`HECKE.GEOM.01`	Hecke from geometry on the E_8 census channels	[E] + [C] + [O]
v536	`HECKE.GEOM.EICHLER.01`	Eichler trace layer at the E_8 lattice	[E] + [C] + [O]
v537	`HECKE.GEOM.HALFINT.01`	half-integral bridge	[E] + [C]
v538	`HECKE.GEOM.RTF.01`	compiler relative trace identity	[E] + [C]
v539	`RTF.GNS.WEIL.01`	Weil structure of the compiler family	[E] + [C]
v540	`RTF.GNS.AMP.01`	amplitude route and positive linear carrier	[E] + [C]
v541	`RTF.GNS.LEDGER.01`	matching lemma and transport ledger package	[E] + [C]
v542	`PRIME.MARGIN.IDENT.01`	margin-chain identities of phase 2 (per instance)	[E]
v543	`PRIME.MMATRIX.IDENT.01`	lumped M-matrix pair of phase 2 (per instance)	[E]
v544	`PRIME.LONGLAG.SUPP.01`	long-lag support structure of phase 2 (per instance)	[E]
v545	`PRIME.HARDY.IDENT.01`	Hardy-core identities of phase 2 (per instance)	[E]
v546	`PRIME.CAPCHAIN.IDENT.01`	capacity-chain identities of phase 2 (per instance)	[E]
v547	`PRIME.LEVEL.LEMMA.01`	level-lemma identities of phase 2 (per instance)	[E]
v548	`PRIME.GREEN.SZEGO.IDENT.01`	Green/Szegő identities of phase 2 (per instance)	[E]
v549	`PRIME.GAUGE.PARITY.IDENT.01`	gauge/parity identities of phase 2 (per instance)	[E]
v550	`PRIME.ODD.SECTOR.IDENT.01`	odd-sector identities of phase 2 (per instance)	[E]
v551	`PRIME.RITZ.CEIL.01`	fixed-size Ritz ceiling certificate of phase 2 (per instance)	[E]
v552	`PRIME.ANGLE.INSTR.01`	four fixed-size angle instruments of phase 2 (per instance)	[E]
v553	`PRIME.EXACT.FORM.IDENT.01`	exact-form identities of phase 2 (per instance)	[E]
v554	`PRIME.SAMPLING.HARM.01`	sampling/harmonics identities of phase 2 (per instance)	[E]
v555	`PRIME.PARETO.TV.01`	Pareto/total-variation identities of phase 2 (per instance)	[E]

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

Every row carries `NO marker moves` in the ledger. The [O] on v535 and v536 is the same open gate throughout: *weight-4* $\to$ GL(1) *transport*.

4.1 Hecke from geometry (v535)

The Hecke structure of the μ_4-glue census is lattice-native: it comes out of Kneser p-neighbour geometry, not out of modular input [`verification/v535_hecke_from_geometry.py`] (25 checks, $\sim$11 s; consolidated promotion of T27/T31/T32).

- **Kneser carries Hecke** [E]: #iso_pts $= p^7 + p^4 - p^3$ and #iso_lines $= \sigma_3(p) \cdot \#\mathbb{P}^3(\mathbb{F}_p)$ identically and by full enumeration at $p = 2, 3, 5, 7$ (135/1120/19656/137600 lines). The Eisenstein eigenvalue $\sigma_3(p) = 1 + p^3$ is an exact factor of a lattice-native correspondence count.
- **The marked neighbour sum is an affine Hecke element** [E]: $\nu_p = a\,\mathrm{Id} + b\,T_p$ exactly (overdetermined, residual 0), with $b = \sigma_3 + a_p$; the cusp rule $a_p = b - \sigma_3$ recovers $a_3 = -4$ and

$a_5 = -2$.

- **Census redundancy is 2-adic oldform structure [E]**: $\dim V = 7 = 5{+}2$, $\pi_{\text{cusp}} = (28{-}T_3)/32$, only 2-adic levels occur.
- Fences carried *inside* the claim: the $p = 5$ geometric $S[1]$ block is a T31-validated *freeze* [C]; the weight-4 $\to$ GL(1) transport is [O].

4.2 The Eichler trace layer (v536)

The smooth (Eisenstein) background and the coherent (cusp) interference separate exactly [`verification/v536_eichler_trace_layer.py`] (23 checks, ∼30 s; promotion of T33/T36/T37/T42). The Witt layer gives $\lambda_{\text{Eis}} = \sigma_3(\#\mathbb{P}^3 - \sigma_3) = L - \sigma_3^2$ identically (anchors 336, 3780, 19264 at $p = 3, 5, 7$) [E], and the Eichler identity $\lambda_{\text{geom}} = \lambda_{\text{Eis}} + a_p^2$ holds two-sidedly (352/3784/19840). The signed cusp reading is $a_p = -c(p)/8 \Rightarrow (-4, -2, +24)$, and the assembler $R(p) = a_p^2$ is exact for $p \le 31$. The $p = 5$ type-(ii) block is again a validated freeze [C]; the transport gate stays [O].

4.3 The half-integral bridge (v537)

In the 70-element compiler weight-5/2 theta monoid there is *exactly one* object g with $\mathrm{Sh}_{t=2}(g) = -8f_8$ coefficientwise [E], it is an exact $T(p^2)$-eigenform with the weight-4 eigenvalues $(-4, -2, 24, -44, 22)$ [E], and $U_4(g) = 0$ places it outside the Kohnen plus space — a scope fence, not a defect [E] [`verification/v537_halfintegral_bridge.py`] (20 checks, ∼90 s; promotion of T38/T41/T44/T45). The Waldspurger/Baruch–Mao constant $R = 23.1873585645\ldots$ is constant on ten discriminants $d \equiv 1 \bmod 8$ (spread $\le 10^{-12}$), typed [C] because it is an AFE-numeric measurement.

4.4 One finite relative trace identity (v538)

The strongest structural statement of the verified layer: v535, v536 and v537 are three *projections* of one finite relative-trace identity of Jacquet type [`verification/v538_relative_trace_identity.py`] (18 checks, ∼14 s; promotion of T53). $\mathrm{Tr}_V(\nu_p \circ \pi)$ is computed twice, independently — geometrically from the glue census with no modular input, and spectrally from $\eta(2\tau)^4\eta(4\tau)^4$ with no geometry — and the two agree exactly for all $(p, \pi) \in \{3, 5, 7\} \times \{\mathrm{Id}, \pi_{\text{Eis}}, \pi_{\text{cusp}}, \pi_{E_4}, \pi_{f_8}\}$ (anchors, e.g. $p = 7$: full trace 727680, f_8 channel 19840) [E]. The Eichler split *is* the orbit decomposition: principal orbit $a\,\mathrm{Id} \to \lambda_{\text{Eis}}$, elliptic orbit $b\,T_p \to a_p^2$, cross terms cancel exactly. And the period side is literally compiler geometry: g is the quaternary lattice form $n = (x^2 + y^2)/2 + 2z^2 + u^2 + 2w^2$ with sign $(-1)^{|u|+|w|}$, giving [signed lattice count]$^2 = R\,|d|^{3/2}L(f_8 \times \chi_d, 2)$ at relative 10^{-12} [C]. The identity is *finite*; the infinite RTF is explicitly open.

4.5 The Weil structure of the family (v539)

The family carries a fully identified Weil structure *up to two explicitly isolated obstructions*, and the obstructions are promoted content, not a footnote [`verification/v539_weil_structure_family.py`] (25 checks, ∼6 s; promotion of T55/T61/T62/T63). The canonical GNS pairing is a direct integral over the Gelfand spectrum with fibre mixing $\le 5.04 \times 10^{-16}$; the trivial Sato–Tate isotype is algebraically a GL(1) core, $G_0 = (1 + Y)/(1 - Y)$; the prime side is an exact linear relation to shifted ζ-Weil forms at relative $\le 9 \times 10^{-16}$. The two obstructions are (i) a sign/doubling residue (metaplectic, sym^2) that survives every preregistered canonical projection, and (ii) a Plancherel correction which is irreducibly non-automorphic. Finite-class positivity of Q_{fam} is measured [C] ([4.369, 11.486]); dense-class positivity is open. The ledger row says it in one line: *not almost-RH*.

4.6 The amplitude route and the positive linear carrier (v540)

The route out of the square plane [verification/v540_amplitude_linear_carrier.py] (34 checks, $\sim$3 s; promotion of T67–T72). An amplitude Dirac operator $D = \begin{pmatrix} 0 & V \\ V^\top & 0 \end{pmatrix}$ with $V[d,m] = \sqrt{w_d}\, b(d)\chi_d(m)$ exists and is Hecke-equivariant [E]; the geometric polarisation $b = N_+ - N_-$, $\Theta = N_+ + N_-$ is lattice-exact with $N_\pm \geq 0$ for all $n \leq 50\,000$ [E]; the Cohen seed $\Theta(d) = -48L(-1, \chi_d)$ is exact-rational on 159 live d [E]; the prime side is a *pure plus* combination $Q = Q_\zeta(g_-) + Q_\zeta(g_+)$ at relative 2×10^{-15} [E]; the functional equation of the linear family holds at relative 1.4×10^{-40} [E]. The open boundary is *inside* the claim: the gap functional λ^* on $n \equiv 6 \bmod 8$, with an exact Farkas witness (0.0511), is a named limit, and positivity is asserted only in the Euler region.

4.7 The matching lemma and the transport ledger (v541)

The largest module of the line [verification/v541_matching_lemma_ledger.py] (33 checks, $\sim$10 s; promotion of T78–T82 and T85; every core check recomputed, not cited).

- **Window proof** [E]: the matching lemma is proved exact-integer on $[4, 10^6]$ — $7S(j) < 40A(j)$ at every atom, 0 violations over 939 870 clash atoms, no-overflow proven in exact integers, 268 big-integer rechecks; exact margin $X = 22242575693/270725400000 = 0.082159$ at $j^* = 554400$; $\rho_{\text{crit}} = 1.143989 > 21/20$. A maximal-greedy identity makes the certificate uniform in the target set and the greedy weights.[1]
- **Transport ledger** [E]: $Q_{\text{Weil}} = Q_{\text{cert}} + \Delta_{\text{arch}} + \Delta_2$ closes with residual 4.4×10^{-16} on a 24-row battery, and $\Delta_{\text{pole}} \equiv \Delta_{\text{conv}} \equiv 0$ are *proven*, not measured (kernel collapse in `sympy`, exact scale invariance).
- **Character-exact envelope** [E]: $-\psi = (\chi_{-4} + \frac{1}{4}\chi_8 + \frac{1}{4}\chi_{-8})\Theta$ coefficient-exact for all odd $n \leq 50\,000$; confinement holds as a *set equality* on 10^6.
- **Archimedean term internal** [E]: the Legendre duplication bridge $(2\pi)^{-s}\Gamma(s) = \frac{1}{2}\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s{+}1)$ and the kernel identity $k_\zeta = K_{\text{fam}} - K_{\text{shift}}$ at 7.9×10^{-31} pointwise.
- **Frontier facts** [C]: unlifted coherent chain crossing at $k^* = 14$ ($\log_{10} N^* = 23.1$); Robin tail factor 6.16 at $J = 10^6$ and divergent — the constant route stays open.
- **Named limits as content.** Two limits are part of the claim rather than hidden by it: one classically-shaped open lemma (correlated cancellation on credit-rich non-coherent tail atoms), and the I5 equivalence typing discussed next. The row states explicitly: *NOT "almost RH"*; `ZETA.HP.CARRIER` untouched.

4.8 The margin-chain identities of phase 2 (v542)

The first promotion *out of* the second phase — and the narrowest module of the line [verification/v542_margin_chain_identities.py] (44 checks, $\sim$0.2 s; promotion of the identity/theorem block of T128–T131, recomputed on small frame-A windows, nothing cited from the sandbox). What is promoted is exactly the algebra: nine identities and per-instance theorems, each a numerical residual against a *preregistered* tolerance plus at least one mutation control that must fail by $\geq 10^{-3}$ relative. What is *not* promoted is everything the sandbox still owes: no fit, no graded floor, no Lanczos value, and no statement uniform in the zone index. The battery is 24 seam instances ($n = 3 \ldots 97$, $\rho = 1.25 \ldots 4.00$, $h_c = 24 \ldots 214$, $h_f = 36 \ldots 299$; every inverted or diagonalised matrix ≤ 300), 43 graded/fine grid pairs ($m \leq 172$) and 400 randomised profiles.

[1] Prior-art note (literature pass, 2026-07-27): the coefficient laws L1–L4 that T77/T78 use to shape the lemma are specialisations of Cohen 1975 (Math. Ann. 217: the Hurwitz class-number series $H(r,n)$ and the Cohen–Eisenstein series of weight $r + \frac{1}{2}$) and of Pei–Wang 2003 — corollaries of known theory, not discoveries. The restricted divisor inequality itself remains the independent part.

- **Border geometry and the (T) identity [E]**: $\sum_i (1-g_i) = t_l + t_r$ exactly (5.5×10^{-13}), so the κ weights are a probability vector and the flat border vector gives $\kappa_{\text{flat}} = 1$; and $\tau_{\text{dn}} = \|y\|^2 - m_{\text{prot}} + m_{\text{fill}} - V_{\text{bord}}$ at 2.2×10^{-16}, with τ recomputed independently from the *full* odd overlap matrix (1.1×10^{-13}).
- **The profile identity and the curvature chain [E]**: $p_{N-1} = 2 - p_0 + 2E$ at 2.3×10^{-16} (instances) and 1.4×10^{-15} (400 random profiles), hence the equivalence $2E > p_0 \iff p_{N-1} > 2$ with 0 exceptions and both branches realised (60/400); the Abel/Hölder chain holds per instance with 0 violations, and the *same* chain without the curvature term is violated on 88/400 profiles — which is the honest statement of why the curvature term is in it.
- **Assembly and the compression defect [E]**: the matrix-free two-scale assembly equals $J^{\mathsf{T}}QJ$ (5.2×10^{-16}), the graded overlap operator equals J (5.2×10^{-14}), and $S_{\text{graded}} - S_{\text{uniform}} = R^{\mathsf{T}}X^{-1}R$ holds as a *matrix* identity on all 43 grids (2.1×10^{-14}) with a PSD defect — the compression error is one-sided *by* the identity, which is what T130 needed and all it gives.
- **Sandwich, sign, counting [E]**: the secular sandwich holds on all 48 pole-free sections at which $A>0$ *and* $\varepsilon>0$ are *checked* (relative width ≤ 1.4448), with Albert's sign equivalence verified in both directions (6 rescaled controls with $\varepsilon<0$ all give $\lambda_{\min}(Q)<0$); $S^{-1}>0$ entrywise (measured on 48/48 blocks) implies a sign-constant and simple ground vector, with a control matrix where hypothesis and conclusion fail together; and $n_{\text{run}} \le n_{\text{cross}} + 1 \le s_2 + 2$ together with $\sum_j |\Delta p_j - \overline{\Delta p}| = \mathrm{TV}(P)$ hold without exception on a battery that is not vacuous (s_2 up to 31, n_{run} up to 33).
- **Named limits as content.** Every statement is per instance on small windows; nothing is uniform in the zone index. The κ *law* is *not* claimed — T129 found it false in general (Section 10) — only the chain underneath it. The sandwich carries no floor value: positivity of the pole-free section at depth (M25d) stays open. The Perron step is an implication with a *measured* hypothesis. And the defect identity says nothing about the *size* of the defect, which is the open exponent of T130.

4.9 The lumped M-matrix pair (v543)

The second promotion out of the second phase: the identity block of the M-matrix question that T134 posed and T136 answered [`verification/v543_lumped_pair_identities.py`] (35 checks, $\sim$1 s; promotion of T136, recomputed on small frame-A windows). The battery is 20 seam windows ($n = 4\ldots59$, $\rho = 1.25\ldots4.00$, $h = 25\ldots299$; every factorised or diagonalised matrix ≤ 300) giving 40 symmetric matrices — pole-free odd sections and border Schur blocks — of which 39 carry positive off-diagonals at all, plus 312 shifted ladder rungs and 8 whitened two-grid rungs. Every statement is a residual against a *preregistered* tolerance with at least one mutation control that must fail by $\ge 10^{-3}$ relative.

- **The lumped pair and the congruence [E]**: with Δ the positive off-diagonal part and $L_\Delta = \operatorname{diag}(\Delta\mathbf{1}) - \Delta$, the pair $A_B = A + L_\Delta$ is Stieltjes, L_Δ is a graph Laplacian, so $A_B\mathbf{1} = A\mathbf{1}$ (1.6×10^{-15}) and $A_B \succeq A$ — an *upper* bound on $\lambda_{\min}(A)$ and a floor never. The congruence $A = A_B^{1/2}(I-W)A_B^{1/2}$ with $W = A_B^{-1/2}L_\Delta A_B^{-1/2} \succeq 0$ holds as a *matrix* identity (4.4×10^{-15}), $\lambda_{\max}(W) = \rho(L_\Delta A_B^{-1})$ (4.4×10^{-15}), and the equivalence $\lambda_{\max}(W) < 1 \iff A \succ 0$ is checked in *both* directions (39 shifted controls in the indefinite range all give $\lambda_{\max}(W) \ge 1$) — the Loewner sandwich is circular by itself, and that declared circularity is the content.
- **The anchor and the empty shift ladder [E]**: with $x = A_B^{-1}\mathbf{1} \ge 0$ and $A_B^{-1} \ge 0$ entrywise (Ostrowski 1937, *checked*), $\lambda_{\min}(A_B) \ge 1/\max_r x_r$ on all 39 rows (min slack 3.0×10^{-3}, sharpest ratio 0.9070) — one solve, one sign check, no eigenvalue. Lumping commutes with diagonal shifts, $(A-sI)_B = A_B - sI$ $(2.5\times10^{-16}$ on 312 rungs), and $(A_B - sI)^{-1} \ge A_B^{-1} \ge 0$ entrywise (Berman–Plemmons 1979), so $\max_r x_r(s)$ and the a-priori margin are non-decreasing along the

whole ladder: the shifted ladder is *structurally* empty — a theorem, not the measurement T136 first made.

- **Varga's identity and the a-priori Jacobi radius** [E]: for the regular splitting $A_B = D_0 - N_0$ (all three hypotheses checked), $\rho(J) = \tau/(1+\tau)$ with $\tau = \rho(A_B^{-1}N_0)$ at 4.8×10^{-15} ($\rho(J) = 0.7347$–0.9860), and the Collatz–Wielandt bracket at the anchor vector gives an *a-priori* $\rho(J) \le 0.9861 < 1$ with no eigenvalue anywhere, sharp on the measured gap by 1.0062–1.0531. Controls: a non-regular splitting breaks the identity (1.1×10^{-1}); at a sign-changing test vector the quotient bound is false on 39/39.
- **The k-scanned M18 majorant — a negative result** [X]: the whitened mass identity $\|L^{-1}\hat{s}\|^2 = \hat{s}^\mathsf{T}U^{-1}\hat{s}$ holds at 7.2×10^{-16} and the triangle chain is a *valid* majorant at every k separately (39 (rung, k) pairs over $k = 2\dots64$, 0 violations), so minimising over the ladder is legitimate — and it does not reach the preregistered bar 1/2 on any rung (best 3.080) while the *measured* bad mass is ≤ 0.0120. The localisation term alone already exceeds the bar (0.542): the loss is the bound, and the bar is not moved.
- **Named limits as content.** Per instance on small windows; nothing uniform in the zone index, in D or in the gap. The T136 D-exponents (anchor $D^{-0.56}$, gap $D^{2.72}$, slack $D^{0.12}$) are fits and stay in the sandbox. The congruence carries no floor; the anchor and Varga rows are implications with hypotheses *checked at the instance*; the ladder row kills a route and says nothing about the true $\lambda_{\max}(W)$.

4.10 The long-lag support structure (v544)

The companion module: what the positive off-diagonal part *is*, and which bounding families are dead [`verification/v544_long_lag_support.py`] (24 checks, ∼0.2 s; promotion of T137 on the same 20 windows, 3–25 prime-power atoms, 20555 positive off-diagonal edges, 455 loaded stripes, 39 bracketed lumped comparisons).

- **The support is a comb stripe set** [E]: the kernel splits exactly as $c = c_{\mathrm{arch}} - c_{\mathrm{comb}}$ with $c_{\mathrm{comb}} \ge 0$, and the archimedean section alone has *all* off-diagonals ≤ 0 on 20/20 windows, so every positive off-diagonal of A is comb-generated (control: the full section carries $+2.0 \times 10^{-1}$). All 20555 positive edges have their Hankel index $M - 1 - r - t$ inside the atom-hat index set — per-window fraction exactly 1.000000 — so $\operatorname{supp}(\Delta)$ is a union of anti-diagonal stripes at prime-power atom positions (the converse is reported, not claimed), and each stripe is a perfect matching (0 endpoint collisions), so its edge vectors are *exactly* orthogonal and L_Δ restricted to a stripe is a direct sum of 2×2 blocks.
- **The amplitude certificate** [E]: $A_{rs} = A^{\mathrm{arch}}_{rs} - c_{\mathrm{comb}}[|r-s|] + c_{\mathrm{comb}}[\mathrm{anti}]$ gives $\Delta_{rt} \le c_{\mathrm{comb}}[M - 1 - r - t]$ on all 20555 edges (excess -1.9×10^{-6}, sharpest ratio 0.99998532) — no eigenvalue, no norm; the Toeplitz lag in place of the Hankel index breaks it on 20415/20555. Honestly undecided and *not* promoted: at $h \le 300$ the atom hats never overlap, so the tighter one-atom refinement (T137: worst ratio 2.268 at depth) coincides with the certificate here.
- **The Green function, bracketed two-sidedly** [E]: for the lumped comparison $B = D_0 - N_0$, $J = D_0^{-1}N_0 \ge 0$, the anchor identity $q_J = \max_r (Jx)_r/x_r = 1 - \min_r 1/(d_r x_r) < 1$ (8.2×10^{-16}) needs no measurement; the Neumann partial sum to $K = 8$ is then an *entrywise* lower bound for B^{-1} (-2.1×10^{-4}) *because the discarded terms are nonnegative*, and the anchor-weighted ceiling is an entrywise upper bound (-1.4×10^{-1}) on 39 rows. This delivers item (2) of the T136 rest list; dropping the tail makes the upper bound false.
- **The absolute-value envelope family is certified dead** [X]: with $E = B^{-1}L_\Delta$, Gershgorin, every row-sum norm and every Collatz–Wielandt quotient at every *positive* weight are upper bounds for $\rho(|E|)$, so one rigorous *lower* bound decides the family at once — $\rho(|E|) \ge 1.3141 \ge 1$ on 20/20 rows (the cheapest four members all ≥ 1.5937). No member can ever certify $\lambda_{\max}(W) <$

1. The *signed* radius on the same rows is $\rho(E) = \lambda_{\max}(W) \le 0.99995781$: the absolute value alone destroys the cancellation, and what survives is a sign-preserving bound that does not exist yet.

- **Named limits as content.** Per instance on small windows, no decay profile and no amplitude law (T137's decay measurements are fits). The support row is *one* inclusion; the amplitude row bounds the amplitude and *not* $\lambda_{\max}(W)$, and the structure bound assembled from it does not beat the margin (T137: 0 of 35 windows) and is not promoted; the bracket is for the *lumped comparison*, not for the target, whose entrywise inverse positivity at depth stays open; the envelope row closes a route, not the question, and moves no marker. Demko–Moss–Smith 1984 is cited as a model and *not* applied — the comparison is dense.

4.11 The Hardy-core identities (v545)

The identity block of the finite core: what the signed inequality *is*, in closed form, and which two bound shapes are *valid* on it [`verification/v545_hardy_core_identities.py`] (36 checks, $\sim$1 s; promotion of T140 and T141 on 11 frame-A windows, $n = 8 \dots 139$, $h = 58 \dots 285 \le 300$, core $m = 57 \dots 284$, 193–5102 positive off-diagonal edges, 10–87 loaded stripes, $D = 7.578 \times 10^{-3} \dots 2.388 \times 10^{-2}$ — a factor 3.15, so the last row is a comparison and not a single point — exact $\rho(W) = 0.997759$–0.999981; the explicit $n_e \times n_e$ signed Gram is formed only where $n_e \le 900$).

- **The form lifting** [E]: with M the edge–interval incidence matrix and $C = \mathrm{diag}(\sqrt{\Delta})M$, the T139 telescope lifts from the *entries* of the signed Gram to its *form*, $\mathrm{Gram} = CHC^{\mathsf{T}}$ at 3.0×10^{-15}, hence $\mathrm{rank}(\mathrm{Gram}) \le h-1$ (measured 57–183 against 193–882 edges); the unweighted incidence breaks it (1.6).

- **The exact finite-core reduction** [E]: the covering kernel in closed geometric form, $K_{rs} = W([r \wedge s,\ r \vee s])$ — entrywise nonnegative and monotone by inclusion on 11/11 — equals $M^{\mathsf{T}}\Delta M$ (1.3×10^{-15}), and $\rho(W) = \lambda_{\max}(K^{1/2}HK^{1/2}) = 1 - \lambda_{\min}(A, A_B)$ (5.4×10^{-14}) $= \lambda_{\max}(\mathrm{Gram})$ (7.8×10^{-16}): three independent routes to the same number, no truncation anywhere. The factorwise product $\lambda_{\max}(K)\lambda_{\max}(H)$ misses by 7.8×10^2, which is why the joint shape below is needed. Energy reordering $H = \mathrm{diag}(s) + L_N$ at 4.7×10^{-16}; the Dirichlet weights are positive on 0.676–0.872 of the off-diagonal area (a measurement, no law).

- **The checkerboard split** [E]: at group length exactly b, $\mathrm{Band}_b = D + A_{\mathrm{even}} + A_{\mathrm{odd}}$ *entrywise* (0.0) on all 9 (window, b) pairs over $b = 4, 8, 16$, and the three-step Weyl sequence dominates $\lambda_{\max}(\mathrm{Band}_b)$ with 0 exceptions (worst ratio 1.6940) at a step count independent of the up to 9 groups; group length $b/2$ loses entries (3.5×10^{-1}).

- **The four Hardy identities** [E]: (i) $\rho(W) = u^{\mathsf{T}}Hu/u^{\mathsf{T}}K^{+}u$ at $u = K^{1/2}v$ (6.6×10^{-13}); (ii) $L_N = D^{\mathsf{T}}BD$ with $B_{kl} = \sum_{r \le k \wedge l,\ k \vee l < s} N_{rs}$, the same closed covering formula one grid in (4.9×10^{-16}); (iii) $Q = B_{+}\mathbf{1}$ (1.4×10^{-15}) — the Cauchy–Schwarz weight of T140 *is* a Gershgorin row sum; (iv) $DKD^{\mathsf{T}} = L_{\Delta}$ on the interior nodes (2.7×10^{-14}) with the endpoint edge mass 0.4886–15.15 as diagonal, a *local* quantity. The crossing kernel of $|N|$ does not reproduce L_N (2.6×10^{-1}): the signs are load-bearing.

- **The closed Muckenhoupt quantity** [E]: $A_{M0} = \max_k Q_k(\Delta\mathbf{1})_{k+1}$ by two closed routes agreeing to 8.3×10^{-15}; its value (65.5–4244) is *reported* and neither its size nor its zone behaviour is promoted. Replacing the local endpoint mass by $\lambda_{\max}(K)$ inflates it by ≥ 358.

- **Two certified shapes, as *valid* bounds** [E]: every Loewner step is certified by a completed Cholesky (-1.5×10^{-14}, -2.3×10^{-10}). The additive Hardy chain dominates $\rho(W)$ on 11/11 (factor 3.074–10.758); its own exact two-term Weyl split dominates too (1.691–2.503) and lies *below* the chain everywhere, so it is the floor of the additive family; the joint shape $\Lambda(H, Y) \cdot \Omega(Y)$ with a positive definite Jacobi Hardy form Y dominates on 11/11 with the closed-form geometric profile (8.815–74.872 at $\Omega = 8.33$–36.2) and on 11/11 with the per-zone tridiagonal read-off of K^+. Promoted is the *dominance*; the factors are not. Controls: the same Y without its mass term is singular — the constant vector lies in its kernel (5.7×10^{-17}) while H does not vanish there — so the certificate *refuses* on 11/11; and dropping the *positive* part instead of the negative one falls below $\rho(W)$ on 11/11.

- **The sign-versus-absolute separation, as a comparison** [X]: over the measured D range the exactly bounded object spreads by 1.36× while every absolute variant spreads by at least 54.94× (Loewner drop 62.03, Cauchy–Schwarz 54.94, product 56.92), and dropping the positive part of H costs 9.96–464.63×. The power laws are *fits*, labelled as fits: $D^{-0.138\pm0.094}$ (signed) against $D^{-0.776\pm0.352}$ (certified Muckenhoupt) and $D^{-2.748}$, $D^{-2.820}$, $D^{-2.918}$ (absolute).
- **Named limits as content.** Per instance on small windows; nothing uniform in the zone index or in D, and every exponent above is a fit that stays in the sandbox. The bound rows promote *validity* and not size: neither shape beats the target, the additive shape is dead as a *shape* by its own Weyl floor, and the joint shape is decided at the normalisation Ω. The finite-core row is an *identity* and not a bound — the zone-uniform discrete Hardy inequality it reduces the question to stays open, and no marker moves toward it.

4.12 The capacity-chain identities (v546)

The identity spine of the capacity cycle T142–T145 of Section 10: what the certified chain *is*, per instance and machine-checked, with the one open lemma fenced out [`verification/v546_capacity_chain_identities.py`] (26 checks, ∼0.5 s; promotion of the identity blocks of T142–T145 on 12 frame-A windows, $n = 4\ldots139$, $h = 26\ldots285 \le 300$, $D = 7.578 \times 10^{-3}\ldots2.789 \times 10^{-2}$, 32–5102 positive off-diagonal edges; the covering kernel is positive definite on 11 of them, where the first two items run; each statement carries a mutation control that must fail by $\ge 10^{-3}$).

- **The capacity decomposition (T142)** [E]: $K^{-1} = D^\top J^{-1} D + xx^\top/\mathrm{cap}$ with $J = DKD^\top$ (assembled entrywise, 3.8×10^{-15}), $x = K^{-1}\mathbf{1}$ the equilibrium charge and $\mathrm{cap} = \mathbf{1}^\top K^{-1}\mathbf{1}$, as a matrix identity ($1.4\times10^{-13}$); under the congruence $u = K^{1/2}v$ the Dirichlet half is an *orthogonal projection* ($P^2 = P$ at 1.9×10^{-12}, $\Omega = \lambda_{\max}(P) = 1$ to 5.1×10^{-12} — the T142 normalisation, where T141 had guessed 20.7–2724), its complement *is* the capacity rank-one, and it annihilates the equilibrium direction $K^{-1/2}\mathbf{1}$. Controls: dropping the rank-one (3.4×10^{-1}), a 1% cap perturbation (3.4×10^{-3}).
- **The exact capacity–Rayleigh identity (T143)** [E]: with $H = \mathrm{diag}(s) + L_N$ (4.7×10^{-16}), $L_N = D^\top BD$ for the signed crossing kernel (4.9×10^{-16}) and $\rho(W) = 1 - \lambda_{\min}(A, A_B)$ (5.4×10^{-14}), the assembled quotient $[\,d^\top(J^{-1} - B)d + (x^\top v)^2/\mathrm{cap} - \sum_k s_k v_k^2\,]/[\,d^\top J^{-1}d + (x^\top v)^2/\mathrm{cap}\,]$ equals $v^\top(K^{-1} - H)v/v^\top K^{-1}v$ for *every* v (9.5×10^{-14} on 24 random vectors per window), reproduces $1-\rho(W)$ at the exact top eigenvector (1.7×10^{-9} under the *declared* 1/gap conditioning bar 5.4×10^{-8}), and every random quotient stays $\ge 1-\rho(W)$ (the inf property); on the constant vector the whole denominator *is* the capacity term. Controls: dropping the cap term (rel = 1), the $|N|$ crossing kernel (2.6×10^{-1}).
- **Cholesky nestedness (T144)** [E]: $\mathrm{cap}_E([a, a+j))$ equals the prefix sum of squared forward-substitution entries of *one* triangular solve — 3404 interval capacities from 36 solves at 2.0×10^{-14} against independent Schur solves, exhaustive on the median window (every prefix, 2.8×10^{-14}), monotone in b by construction. Control: the reversed ordering computes *suffix* capacities and deviates by $\ge 4.6 \times 10^{-1}$.
- **The whitening certificate (T144)** [E]: $\lambda_{\max}(W) \le \kappa_{\mathrm{up}}$ (completed Cholesky) $\le$ the Gershgorin ceiling on every window ($\kappa_{\mathrm{up}} = 1.2245$–$1.4236$ against ≤ 2.1213); the licensed chain direction $\lambda_{\min}(E)/\kappa_{\mathrm{up}} \le \lambda_{\min}(A, A_B)$ holds on every window — a *denominator* comparison, untouched by T142's numerator obstruction — and the Cholesky refuses at the shrunken shift $0.995\,\lambda_{\max}(W)$.

- **The M4 split and the certified chain (T145)** [E]: the mass half of Maz'ya's step M4 — $\sum_k f_k^2 \le \psi^2$ *pointwise* (a theorem for nonnegative truncations, exact slack ≥ 0), the row-wise implication on rows with $(E\mathbf{1})_i \ge 0$, and the full step $\sigma_{\text{tot}} = 0.2197$–$0.4425 < 1$ on every window although the Markov hypothesis is absent. Licence 4 is stated with $|R|$ and *not* R^+, with the witness $R = \left[\begin{smallmatrix} 1 & -1/2 \\ -1/2 & 1 \end{smallmatrix}\right]$, $x = (1, -1)$ $(3.0 > 2.0)$ built in as a counterexample check. The sign-free chain is ordered on every window, and S1′ is *certified per window on both routes*: cert_lam_max$(R) \le 12\,\sigma_{\text{tot}}\Psi_{\text{pos}}$ (energy, $c_0 = 2.6369$–5.3099) and $\le G_{\text{dy}}\Psi_{\text{abs}}$ (sign-free, $c_0 = 2.5154$–6.1449), better route $c_0 = 2.5154$–4.2265 against the classical Dirichlet value 4; the all-sets density bound carries Charikar's cited constant, bracketed by exhaustive enumeration of all 4095 subsets of a small block. Control: $c_0 \times 10^{-3}$ breaks the certificate on every window.
- **Named limits as content.** Per instance on small windows; nothing uniform in the zone index or in D, and every T142–T145 exponent is a fit that stays in the sandbox. The c_0 of the last row is *measured* at the minimiser's own layer cake — the a-priori *level lemma* (T145's L1) stays open, unclaimed and unapproached, and no marker moves toward it. Maz'ya's strong-type half is *not* used as a theorem anywhere: it is the thing the cycle transcribes.

4.13 The level-lemma identities (v547)

The a-priori core of T146 (Section 10): the same level constant that v546 certified with a *measured* input, now supplied as a functional of the form alone [`verification/v547_level_lemma_identities.py`] (20 checks, $\sim$0.4 s; promotion of the four a-priori pieces of T146 on 12 frame-A windows, $n = 4 \ldots 139$, $m = 26 \ldots 285 \le 300$, $D = 7.578 \times 10^{-3} \ldots 2.789 \times 10^{-2}$, plus 9 random positive definite forms for the form-independent items and a stress ladder $m = 64 \ldots 288$; each statement carries a mutation control that must fail by $\ge 10^{-3}$).

- **The union identity for nested cakes (T146)** [E]: for a nested decreasing chain with positive coefficients, $\sum_{j,l} c_j c_l\,|S_j \cup S_l| = 2T\|\psi_t\|_1 - \|\psi_t\|^2$ with $T = \sum_j c_j$ and $\psi_t = \sum_j c_j \mathbf{1}_{S_j}$ — form-independent and $O(m)$, the reason the cake base is movable at all; checked against T145's $K \times K$ double sum on all 12 window minimisers and 9 random vectors (max rel 0.0) *and* against the union sizes computed from the sets themselves with no nestedness used. Controls: shuffling one proper level set breaks it (3.2×10^{-2}), a 1% perturbation of the heaviest coefficient breaks the double sum (4.8×10^{-3}).
- **The layer-cake counting lemma at a general base (T146)** [E]: $G(b) \le 2b^2\,\|v\|_\infty \|v\|_1/\|v\|^2 + \varepsilon(b)$ for *any* vector v at all 6 preregistered bases (margin ≥ 1.08 on the battery and on every window minimiser); at $b = 2$ the leading constant is the classical dyadic 8 *exactly*, and the base is *free* because the chain consumes only the lower domination $|v| \le \psi_t$: the a-priori constant gains $4/b_*^2 = 3.7612$ over base 2 on every window. Controls: the bound $\times 10^{-3}$ fails everywhere, and the one-level-down cake (ψ_t/b) breaks the domination on every draw — the $+1$ in the cake exponent is load-bearing.
- **The resolvent delocalisation bound (T146)** [E]: $\psi = \lambda R\psi$ to 9.0×10^{-14} — an *identity*, so no Davis–Kahan transfer step appears anywhere — and per coordinate (Cauchy–Schwarz), with $\lambda_{\text{up}} = \min_j \text{Rayleigh}(Re_j) \ge \lambda_{\min}$ (Rayleigh–Ritz; window ratio 1.0592–1.2883) and the column norms certified from above with the linear-solve residual paid for,

$$\|\psi\|_\infty \;\le\; \lambda_{\text{up}} \max_j \|Re_j\|\,\|\psi\|$$

  on every window, within a factor 1.061–1.337 of the truth. *The lever*: the trivial ceiling $\max_j \|Re_j\| \le 1/\lambda_{\min}$ would give $\Gamma \le \sqrt{m} = 5.1$–$16.9$ (useless); the certified columns sit a factor 2.85–10.17 *below* it (preregistered bar 2.0), giving $\Gamma = 1.8356$–2.2797 — read off the form, never off the minimiser. Controls: a 1% λ perturbation breaks the identity (1.0×10^{-2}), the sup bound $\times 10^{-3}$ fails on every window.

- **The composite level lemma (T146)** [E]: $c_0^{\mathrm{ap}} = 2b^2\,\Gamma\min(1,\Gamma_1) + \varepsilon = 3.9042$–$4.8488$ dominates the measured level constant on all 12 windows (slack 1.986–2.418) and stays under the preregistered ceiling 16 = 4× Maz'ya's Dirichlet value 4; the certified window inequality $\mathrm{cert\_lam\_max}(R) \le c_0^{\mathrm{ap}}\Psi_{\mathrm{abs}}$ holds on 12/12 (both sides by completed Cholesky, the density the $|R|$ one with Charikar's cited constant), closing the chain at loss factor 0.0423–0.1272 of the exact gap — v546's certified chain with its one measured input replaced by a functional of the form. *The built-in discriminator*: on T145's no-go form $R = aa^{\mathsf{T}} + \varepsilon I$, $a_i = i^{-1/2}$, Γ grows $3.676 \to 6.798$ over $m = 64\ldots288$ (ratio $1.85 \ge$ the preregistered 1.5) and $c_0^{\mathrm{ap}} = 11.10$ *exceeds* every window value — the bound must be worse there, since a candidate that did not fail on the no-go would prove a false statement — while the theorem half survives on every no-go size and the Dirichlet path-Laplacian control stays flat (max/min 1.0063). Control: $c_0^{\mathrm{ap}} \times 10^{-3}$ breaks the certificate on every window.
- **Named limits as content.** Per instance on small windows — a *finite list* of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D: the asymptotic delocalisation of the Green columns (T146's remainder) stays open and typed open. Only the sign-free / Green constant is bounded a priori — the energy-route σ_{tot} stays *measured* — and every T146 exponent is a fit that stays in the sandbox. Davis–Kahan 1970 is cited as the transfer step this route deliberately avoids; Maz'ya's strong-type half is *not* used as a theorem anywhere.

4.14 The Green/Szegő identities (v548)

The identity/certificate core of T147 and T148 (Section 10): the two scalars the level constant of v547 is made of, certified on the same class of windows [`verification/v548_green_szego_identities.py`] (21 checks, ∼0.4 s; promotion of the identity/certificate pieces of T147+T148 on 12 frame-A windows, $n = 4\ldots139$, $m = 26\ldots285 \le 300$, $D = 7.578\times10^{-3}\ldots2.789\times10^{-2}$, plus 7 random positive definite forms for the form-independent items and a weight-class model ladder $m = 64\ldots288$; each statement carries a mutation control that must fail by $\ge 10^{-3}$).

- **The factorisation identity (T147)** [E]: for any symmetric positive definite form with Green matrix R,

  $$\Gamma = \sqrt{m}\,\lambda_{\mathrm{up}}\max_j \|Re_j\| \;=\; \sqrt{Q^\star}\cdot S_w\,, \qquad S_w = \lambda_{\mathrm{up}}\|R\|_F\,, \qquad Q^\star = m\,\max_j (R^2)_{jj}/\operatorname{tr}(R^2)\,,$$

  exact at 2.2×10^{-16} on all 12 windows and 7 random forms — the all-D question splits into a purely spectral and a purely geometric scalar and into nothing else; the certified factorisation (residual-paid column norms, Frobenius bracket) dominates the true Γ everywhere, with $S_w = 1.1186$–1.5777 and $Q^\star = 2.0879$–2.8634 on this list. Controls: a 1% $Q^\star$ perturbation breaks the identity (5.0×10^{-3}), perturbing the largest Green column breaks the factorisation.
- **The bottom-block equidistribution constant (T147)** [E]: with Π_B the spectral projector onto $B = \{k : \lambda_k \le 10\hat{\lambda}\}$ (preregistered rule, $|B| = 3\ldots5$), $\sum_j (\Pi_B)_{jj} = |B|$ *exactly* (3.3×10^{-16}), hence $Q_B = m\max_j(\Pi_B)_{jj} \ge |B|$ is a theorem (max ≥ mean) — and the *measured* $Q_B/|B| = 1.5091$–1.7980 lies in the preregistered band $[1, 2.2]$ against the Szegő/Widom sharp reference $Q_B \le 2|B|$ (a measurement on a finite list, typed as such; T148's honest negative stands beside it: $f(0) < 0$ on every surface window, so the mechanism enters nowhere as a theorem). Controls: the rank-$|B|$ coordinate projector gives $Q_B = m$ and overshoots the band by ≥ 3.9; a 1% basis scaling breaks the trace identity.
- **The second-difference ℓ^1 ceiling (T148, the lever)** [E]: from the exact Dirichlet eigenpair $L_0 s_k = \mu_k s_k$ (KMS 1953) and $\mu_k \ge 4k^2/(m+1)^2$ ($\sin t \ge 2t/\pi$), the coefficient bound $|a_k| \le \min(1, \|s_k\|_\infty \|L_0^p\psi\|_1/\mu_k^p)$ for $p = 1, 2$ holds against the *true* coefficients on the bottom mode of every window and on random unit vectors (worst excess ≤ 0); the chain $\|a\|_1 \le \mathrm{ceil}_\mu$, $\mathrm{ceil}_\nu \le 2\sqrt{\nu}+1$ with $\nu = (m+1)^{3/2}\|L_0\psi\|_1/(2\sqrt{2})$ — every power of m cancelled, ν alone carries

the m-dependence — and $m\|\psi\|_\infty^2 \le 2\|a\|_1^2$ holds everywhere. *Calibration*: for the Dirichlet bottom mode $\|a\|_1 = 1$ exactly and $\nu = 3.1415 \to \pi$ at every size. Controls: the ceiling $\times 10^{-3}$ fails on every window, a one-index μ shift breaks the coefficient identity (1.5×10^{-1}).

- **The layer-cake S_w certificate (T148)** [E]: the completed LDL^T inertia count at every rung of the preregistered ladder $\tau_i = \rho_i \lambda_{\mathrm{lo}}$ ($\rho = 1.5 \ldots 1024$, λ_{lo} a certified positive floor) equals the direct spectral count *exactly* on all 96 (window, rung) pairs (Sylvester 1852; Bunch–Kaufman 1977 — a certificate, never a sorted list), and the layer cake $\sum_k \lambda_k^{-2} \le N_1/\lambda_{\mathrm{lo}}^2 + \sum_i (N_{i+1} - N_i)/\tau_i^2 + (m - N_L)/\tau_L^2$ certifies $S_w \le 1.3288$–1.8220 per window (true 1.1186–1.5777, slack ≤ 1.216); nothing here says m-free — that reading lives in the sandbox as a labelled trend. Control: *dropping the bottom layer* breaks the bound on every window — the bottom eigenvalue is the weight.
- **The Liouville transform and the weight-class discriminator (T148)** [E]: the chain

$$m\|\psi\|_\infty^2 \;\le\; \kappa_\Lambda\, m\|\hat\varphi\|_\infty^2 \;\le\; 2\kappa_\Lambda\, \|a(\hat\varphi)\|_1^2 \;\le\; 2\kappa_\Lambda\, \mathrm{ceil}(\hat\varphi)^2\,, \qquad \varphi = \Lambda^{-1/2}\psi\,,$$

  holds stepwise on every bottom-block mode of every window — the exact pencil equivalence $E\psi = \lambda\psi \Leftrightarrow A\varphi = \lambda\Lambda\varphi$ is a change of variables, not a perturbation (Davis–Kahan 1970 cited as the route T148 computed to be vacuous and *not* used) — with $\kappa_\Lambda = \max\Lambda/\min\Lambda = 1.3868$–$2.5805$ *a priori* (a ratio of two diagonal entries of the form). *The discriminator*: on a model section with a genuine order-2 zero ($f(0) = 0$, $f''(0)/2 = 1$, verified exactly), over the ladder $m = 64 \ldots 288$ at matched κ_Λ (mismatch 0.0013), the bounded-variation multiplier keeps ν flat (max/min 1.003) while the rough multiplier ($\mathrm{TV}(\log\Lambda)$ larger by 162.6) grows by 19.9 — the rough class *must* be worse, and it is: the operative hypothesis is the roughness of the multiplier, not its condition number. Control: the first Liouville step $\times 10^{-3}$ fails on every mode, provably (the step slack is bounded by κ_Λ^2).
- **Named limits as content.** Per instance on small windows — a *finite list* of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D: the one open lifting input ($\nu_L \le F(\text{form})$ with F m-free; equivalently bounded $\mathrm{TV}(\log\Lambda)$ or any weaker smoothness of the multiplier) stays open and typed open. The symbol-level order-2 hypothesis is *refuted* (T148: $f(0) < 0$ on every surface window) and enters nowhere as a theorem; σ_{tot} stays retired as a route; every T147/T148 exponent is a fit that stays in the sandbox.

4.15 The gauge/parity identities (v549)

The identity/certificate core of T149 and T150 (Section 10): the gauge *freedom* around the machinery v548 certified in the Jacobi gauge, and the parity *structure* underneath it [`verification/v549_gauge_parity_identities.py`] (21 checks, $\sim$0.3 s; promotion of the identity/certificate pieces of T149+T150 on 12 frame-A windows, $n = 4 \ldots 139$, $m = 26 \ldots 285 \le 300$, $D = 7.578 \times 10^{-3} \ldots 2.789 \times 10^{-2}$, with 48 gauge assemblies, the full-section parity items on the 6 windows with $M \le 300$, and a parity control model at $m = 2001$; each statement carries a mutation control that must fail by $\ge 10^{-3}$).

- **The gauge freedom (T149)** [E]: for *any* positive diagonal $\tilde\Lambda$ the whitening $\tilde E = \tilde\Lambda^{-1/2} A \tilde\Lambda^{-1/2}$ is an exact congruence, so $\lambda_{\min}(A) \ge \mathrm{cert\_lam\_min}(\tilde E) \cdot \min_j \tilde\Lambda_j$ is a *valid* certified lower bound for every gauge — an identity plus one Rayleigh step — and the family maximum is valid too (all 48 assemblies: Jacobi, const, moving-geometric-mean, and the arithmetic-free archimedean diagonal); the factorisation identity $\Gamma = \sqrt{Q^\star} S_w$ holds *gauge-invariantly* (max rel 2.4×10^{-16}); the const gauge (the geometric mean of the Jacobi diagonal) has $\mathrm{TV}(\log\tilde\Lambda) = 0$ *exactly* on every window — T149's elimination, not reduction, of T148's blocking scalar — with the certified entrywise sandwich $\sigma = c_2/c_1 = 1.3868$–$2.5805$. Controls: a 1% $Q^\star$ perturbation breaks the identity (4.99×10^{-3}); on the shifted indefinite form $A - 1.05\lambda_{\min} I$ the completed-Cholesky floor *refuses* on 12/12.

- **The two-regime discriminator (T149, the refutation as a check)** [E]: the flutter-smoothing gauge (moving geometric mean, preregistered half-width $m/16$) removes $\mathrm{TV}(\log\Lambda)$ by $\geq 2.88\times$ on every window while moving the Liouville-*transported* bottom-block smoothness functional $\nu_{\tilde{L}}$ by at most 0.0088 — and even the matched-amplitude *rough* re-gauge (lattice-scale oscillation) moves it by at most 0.0046 (bar 0.01 for both): the roughness $\nu_{\tilde{L}}$ responds to is *not* in the multiplier — T149's withdrawal of the TV hypothesis, wired as a per-instance check. Control: the same functional read *without* the transport (on the whitened modes ψ, which carry the multiplier) jumps by 0.41–9.76 under the same rough re-gauge — the invariance is carried by the transport, not by numbness of the functional.

- **The explicit zone diagonal (T150)** [E]:

$$\Lambda_r \;=\; c_0 - c_{M-1-2r} + \sum_{s\neq r} \max\bigl(c_{|r-s|} - c_{M-1-r-s},\, 0\bigr)\,,$$

  assembled lagwise with no matrix anywhere, equals the lumped-pair diagonal on every window (max rel 5.9×10^{-16}) — the whitening diagonal is an *explicit* functional of the zone's lag arithmetic and of nothing else; the lag vector splits *exactly* as $c = c^{\mathrm{arch}} + c^{\mathrm{atom}}$ with $c^{\mathrm{atom}} \leq 0$ entrywise, and the arithmetic mass carries the closed budget $\|c^{\mathrm{atom}}\|_1 \leq \sum_j \mu_j \leq 4B\sqrt{N}$ with $B = 1.038821$ the *verified* maximum of $\psi(n)/n$ over the whole atom table (Chebyshev 1852, computed and never assumed; Abel summation carries the closed form). *The honest negative as content*: the archimedean section alone is *indefinite* on 12/12 windows (completed $\mathrm{LDL}^{\mathsf{T}}$, up to 5 negative eigenvalues) while the full section has zero — the atoms are *co-responsible* for the positivity, so the additive perturbation reading (Weyl 1912 / Bauer–Fike 1960 / Davis–Kahan 1970, cited, computed dead by T150 at 2.4–5.8 orders, *not* used) is structurally empty and only the multiplicative / Loewner gauge step transfers. Control: dropping the positive part in the zone formula breaks the identity (4.25×10^{-1}).

- **The parity compression (T150, the named mechanism)** [E]: with U_-, U_+ the antisymmetric / symmetric parity isometries of the full window,

$$U_-^{\mathsf{T}}\, T_M(c)\, U_- \;=\; T - H\,, \qquad U_+^{\mathsf{T}}\, T_M(c)\, U_+ \;=\; T + H\,, \qquad U_+^{\mathsf{T}}\, T_M(c)\, U_- \;=\; 0\,,$$

  exactly (max rel 3.9×10^{-16}, cross block 1.4×10^{-16}) on all 6 windows with $M \leq 300$: the reflection-odd section used since T106 *is* the compression of the full symmetric Toeplitz section onto its antisymmetric parity sector (Basor–Ehrhardt 2009: the parity sectors are the natural objects of the Toeplitz+Hankel algebra, cited as the address and never as an authority); the completed $\mathrm{LDL}^{\mathsf{T}}$ counts *localise* the negative inertia — the full section's negative eigenvalues (forced by the sign-changing symbol, $f(0) = -29.2\ldots - 6.3 < 0$; T148's honest negative *re-read*, never re-derived) sit entirely in the *even* sector while the odd sector counts zero: $f(0) < 0$ is explained rather than worked around. Control: the sign-flipped basis does not reproduce $T - H$ (3.99×10^{-1}).

- **The parity-sine calibration and the ladder certificate (T150)** [E]: the parity sines $t_k(r) = \frac{2}{\sqrt{N}}\sin(2\pi k(r{+}1)/N)$, $N = 2m{+}1$, are the exact eigenpairs of the parity Laplacian (reflecting corner 3, forced by antisymmetry, not chosen) at $\mu_k^P = 4\sin^2(\pi k/N)$ — eigen-residual 2.3×10^{-13} — and the smoothness ladder *calibrates* on the exact model: $\nu_k^P = \pi k^2$ for $k = 1\ldots 12$ at $m = 2001$ (max deviation 3.0×10^{-5}), so the ladder form $\nu_k \leq Ck^2$ is the right question and π its exact floor; on every window, in the const gauge ($\varphi = \psi$ exactly — the ladder of the *pure* Toeplitz-minus-Hankel section, no multiplier, no arithmetic weight), the parity coefficient licence holds, $\nu_k^P \leq Ck^2$ with the per-window certified constant $C = 11.70$–17.28, and the derived per-mode ceiling $m\|\psi_k\|_\infty^2 \leq 2(2\sqrt{C}k + 1)^2$ is a certified instance inequality — an m-free C is exactly the one open input and is *not* claimed. Controls: the ladder $\times 10^{-3}$ fails everywhere, a one-index calibration shift misses by 0.75.

- **Named limits as content.** Per instance on small windows — a *finite list* of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D: the one open input after T150 (an m-free constant C in the odd-parity-sector ladder $\nu_k \le Ck^2$ for a sign-changing-symbol Toeplitz section) stays open and typed open; $\mathrm{TV}(\log\Lambda)$ stays withdrawn as a hypothesis (T149), σ_{tot} stays retired as a route (T148); every T149/T150 exponent is a fit that stays in the sandbox.

4.16 The odd-sector identities (v550)

The identity/certificate core of T151 (Section 10): the *reroute* off the quadratic ν ladder that v549 certified — through the pencil against the parity Laplacian and one elementary Sobolev step, to a per-mode bound *linear* in k [`verification/v550_odd_sector_identities.py`] (19 checks, $\sim$0.2 s; promotion of the identity/certificate pieces of T151 on 12 frame-A windows, $n = 4 \dots 139$, $m = 26 \dots 285 \le 300$, $D = 7.578 \times 10^{-3} \dots 2.789 \times 10^{-2}$, with parity control models at $m = 241$ and $m = 2001$ and a no-go size ladder $m = 64 \dots 256$; each statement carries a mutation control that must fail by $\ge 10^{-3}$).

- **The grid step-over and the honest negative (T151)** [E]: the parity sines live on the *shifted* grid $\theta_k = 2\pi k/N$, $N = 2m{+}1$, which never contains $\theta = 0$, with $\mu_k^P = 2 - 2\cos\theta_k$ *exactly* (2.2×10^{-16}; eigen-residual 2.3×10^{-13}); the symbol has $f(0) = -47.6 \dots -6.3 < 0$ (T148's honest negative, re-read and never re-derived) with a nonempty negative window $[0, \theta_c)$, and $\theta_c/\theta_1 = 0.371$–$0.485 \le 0.9$ on *every* window: the odd grid *steps over* the negative window — odd-sector positivity is a grid fact, stepped over rather than cancelled. *The honest negative beside it, promoted as content* (T150's rest item 3 answered negatively): the symbol-only pencil floor $\kappa_{\mathrm{sym}}(1-\rho)$ is vacuous ($\rho \ge 1$) on 12/12 windows and the symbol moves by 0.45–3.27 relative across *one* mode spacing — no pointwise symbol statement is admissible at the bottom mode's resolution: the local model is a *matrix* statement, not a symbol statement. Controls: the Dirichlet corner breaks the eigenpairs (1.0), a one-index shift misses by 6.5×10^{-3}.
- **The certified bottom ladder (T151)** [E]: $\lambda_k(A) \le S\mu_k^P$ for $k \le 8$, certified per window by 8 completed $\mathrm{LDL}^{\mathsf{T}}$ inertia counts (Sylvester 1852; Bunch–Kaufman 1977 — a certificate, never a sorted list), $S = 1.1019$–6.4725: the odd sector's bottom spectrum *is* the parity-Laplacian ladder up to a printed $O(1)$ factor — order 2 in k, with no trace of $f(0) < 0$ in it (the Rayleigh floor $L_P \ge \mu_1^P I$ absorbs it); the count equals the direct spectral count on all 96 (window, k) pairs. Control: $S \times 10^{-3}$ certifies no eigenvalue below it anywhere.
- **The discrete Sobolev step and the π price (T151)** [E]: $\|\nabla\psi\|_2^2 = \langle\psi, L_0\psi\rangle - \psi_{\mathrm{last}}^2$ exactly (7.6×10^{-15}), $\|\nabla\psi\|_2^2 \le \langle\psi, L_P\psi\rangle$, and $\|\psi\|_\infty^2 \le 2\|\psi\|_2\|\nabla\psi\|_2$ on the true window modes, the parity sines and random unit vectors — elementary, licensed by the odd sector's *virtual node* $\psi_{-1} = 0$ (Hardy–Littlewood–Pólya 1934 / Agmon 1965 the continuum address); calibration: on the exact parity model the pencil is $(1,1)$ (certified $|\kappa - 1| = 1.1 \times 10^{-12}$ at $m = 241$) and the Sobolev per-mode price is π to 5.1×10^{-8} at $m = 2001$ — the honest, m-free cost of leaving the ℓ^2 world. Controls: dropping the boundary term breaks the identity (4.8×10^{-2}), the Neumann energy undercounts a first-node vector by 0.50.
- **The closed archimedean minimum (T151; T150's rest item 2 closed and attained)** [E]: the archimedean lag kernel has two shape properties, checked per window and never assumed — $c_i^{\mathrm{arch}} < 0$ for $i \ge 1$ and c_i^{arch} non-decreasing on $i \ge 1$ — and given those, $\min_r \Lambda_r^{\mathrm{arch}} = c_0^{\mathrm{arch}} - c_{M-1}^{\mathrm{arch}}$ *exactly* (max rel 0.0; value 2.1719–3.4674), attained at $r = 0$: a closed two-term functional of the smooth kernel, no matrix, no eigenvector, no loss. Control: the full atom-including kernel violates the monotone shape on every window and the closed formula misses its minimum by $\ge 8.7 \times 10^{-2}$.
- **The linear per-mode bound and the no-go discriminator (T151)** [E]: with $\kappa = 0.1939$–0.3599 the certified pencil floor (*one* completed Cholesky of $A - \kappa L_P$ per window, the

floating-point floor carried as fl/μ_1^P; the direction re-read on the actual bottom eigenvalue: $\kappa\mu_1^P \le \lambda_1(A) \le S\mu_1^P$) and $K_{\rm bot} = S$, the chain

$$\langle \psi_k, L_P\psi_k\rangle \;\le\; \frac{\lambda_k(A)}{\kappa} \;\le\; \frac{K_{\rm bot}\,\mu_k^P}{\kappa} \qquad\Longrightarrow\qquad b_k := m\|\psi_k\|_\infty^2 \;\le\; 2m\sqrt{\frac{K_{\rm bot}\,\mu_k^P}{\kappa}} \;\le\; C_S\,k$$

holds with $C_S = 12.081$–$26.953 \le 2\pi\sqrt{K_{\rm bot}/\kappa}$ on the bottom 8 modes of every window — *linear* in k where the quadratic ν ladder grew (v549), every m-power cancelled, nothing additive anywhere (the $\Theta(D^3)$ bottom that killed the Weyl / Bauer–Fike / Davis–Kahan route never enters) — and the bound dominates the truth $m\|\psi_k\|_\infty^2$ and the parity Dirichlet energy on every read mode; the bottom pencil ratio $R = K_{\rm bot}/\kappa = 3.7362$–$18.6607$ is the *one* scalar the chain still owes, printed per window and typed *open* in m. *The discriminator*: on the T145 no-go form the ratio *explodes* on the size ladder ($1.686\times10^3 \to 2.667\times10^4$, factor 15.8 over $m = 64\ldots256$) while the exact parity model holds $R = 1$ to 1.1×10^{-12} — a route that did not fail there would prove a false statement. Controls: the ladder $\times10^{-3}$ fails against the true sups, the inflated floor 1.5κ makes the Cholesky *refuse* on 12/12.

- **Named limits as content.** Per instance on small windows — a *finite list* of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D: the one open input after T151 (the m-freeness of the bottom pencil ratio $R = K_{\rm bot}/\kappa$ for the odd parity sector of a sign-changing-symbol Toeplitz section) stays open and typed open — its flat trend ($x^{0.037\pm0.015}$) is a fit that stays in the sandbox; the symbol-only route stays certified vacuous; $\mathrm{TV}(\log\Lambda)$ stays withdrawn as a hypothesis (T149), $\sigma_{\rm tot}$ stays retired as a route (T148).

4.17 The fixed-size Ritz ceiling certificate (v551)

The certificate core of T154 (Section 10): the *fixed-size* replacement for the ceiling step that v550 certified by per-window LDL^T counts of size m — the sixteen-column subspace $\mathrm{span}\{t_1..t_8\} + A^{-1}L_P\,\mathrm{span}\{t_1..t_8\}$ carries a Courant–Fischer ceiling that *attains* the inertia-certified true bottom ladder constant, so the ceiling step needs no factorisation of size m and, by direction, no residual argument at all [`verification/v551_ritz_ceiling_certificate.py`] (16 checks, $\sim$0.4 s; promotion of the certificate core of T154 on the 12 *deepest* prime-power zones admitting a frame-A window inside the cap, $n = 29\ldots139$, $m = 96\ldots291 \le 300$, $D = 7.248\times10^{-3}\ldots1.815\times10^{-2}$ — the deep end of the accessible surface, mirroring T154's own surface selection and declared rather than tuned: on the shallowest zones $n \le 11$ more than one bottom direction leaves the sine block, and the fixed-depth certificate is *not* claimed there; a no-go size ladder $m = 48\ldots288$, parity controls at $m = 64, 128, 256$, and 3 Dirichlet control windows; each statement carries a mutation control that must fail by $\ge 10^{-3}$).

- **The direction lemma (T154)** [E]: for orthonormal Q and $W = Q^\mathsf{T}AQ$, $\lambda_k(A) \le \theta_k$ for *every* $k \le \dim$ — Ritz values are *upper* bounds for the eigenvalues of the *same index* (Courant–Fischer 1920 / Cauchy 1829), verified against the exact spectrum on the certificate subspace of every window and on 24 random subspaces (one-sided excess 0.0) — so the fixed-size ceiling needs *no* residual argument; Temple 1928 / Kato 1949 is a floor device and is used nowhere in the module. Control: the *reversed* reading (Ritz values as lower bounds) is recognised as false on 12/12 windows, θ_k overshooting λ_k by a factor ≥ 18.6 somewhere on every window.
- **The sixteen-column certificate (T154)** [E]: the subspace has *fixed* dimension 16, independent of m (one Green step $A^{-1}L_Pt_k$ by Cholesky back-solves, no inverse formed), and its ceiling $K^F = \max_k \theta_k/\mu_k^P$ (8 completed Choleskys of matrices $\le 8\times8$) equals $K^F = 1.1019$–1.5845 with $K^F \ge \max_k \lambda_k/\mu_k^P$ on every window *and* $|K^F/K_{\rm bot} - 1| \le 9.9\times10^{-7}$ against the declared cap 10^{-4} on 12/12, where $K_{\rm bot}$ is the inertia-certified true ladder constant (8 completed LDL^T counts of size m per window; count = direct spectral count on all 96 pairs): the fixed-size ceiling *attains* the truth, and the size-m counts are retired from the ceiling step. Controls: the parity

sines alone overshoot by 41.6–739 (failing the cap everywhere — T152/T153's $O(m^2)$ overshoot reproduced at fixed size), and corrupting the Green columns ($L_P t_k$ instead of $A^{-1} L_P t_k$) breaks the attainment on 12/12 (Kaniel 1966 / Paige 1971 quoted as the mechanism, never used as a bound).

- **The one-direction anatomy (T154)** [E] (angles measured): the median of the seven aligned principal angles between the bottom eight of A and the bottom eight of L_P is 0.03–0.14° while the largest is 83.79–89.70° on every window (Björck–Golub 1973) — the misalignment is carried by *one* nearly orthogonal direction, and that single direction *is* the collapse price $\lambda_1(A)/(t\,\mu_1^P) = 4.408$–$7.042$ ($t = 0.2250$–0.2500 the certified Schur two-block floor, $k_b = 16$), recovered *in full* per window by one completed Cholesky of $A - \gamma I$ (size m, certified, not fixed size, labelled as such): the price is a uniformity gap, not a size gap. Control: $A = L_P$ through the same instrument returns $K_{\mathrm{bot}} = K^F = 1$ and zero misalignment (deviations $< 10^{-5}$, the inertia ladder's backoff step), with the price exactly $1/t$.
- **The no-go discriminator (T154)** [E]: on the T145 no-go reconstruction $c(l) = 1/(1 + l)$ the odd section is positive definite and the Schur floor survives ($t = 0.05$ on all 4 sizes), but K_{bot} explodes ($\times 35.4$) and the sixteen-column ceiling explodes *with* it ($\times 57.5$) over $m = 48 \dots 288$ — the instrument does not report flatness where flatness is false and manufactures no closure; the Courant–Fischer direction is violated nowhere (16/16 positive-definite cases); and the *Dirichlet* control (the Hankel half removed, arithmetic kept bit for bit) stops being positive definite — $\lambda_1 = -5.60 \dots - 3.55$ certified on 3/3 — so no ceiling exists there: the reflection half is load-bearing for positivity (T154's second refutation of the boundary-reflection reading).
- **Named limits as content.** Per instance on small windows — a *finite list* of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D: the attainment $K^F = K_{\mathrm{bot}}$ is exhibited per window and *not* promoted to a theorem (T154's no-go stress shows the certificate overshoots there by 1.12–2.19 — exact attainment is a property of the real section, not of the instrument); the two open inputs after T154 (R1′, an m-free floor for $\lambda_{\min}(B_{HH})$ on the bulk parity block; R2′, an m-free lower bound for the L_P floor on the complement of the eight bottom Ritz directions — arithmetic-free, worth 91–100% of the collapse price) stay open and typed open; every floor consumed is certified per window at size m; v550's ladder stays as stated, and nothing is re-opened.

4.18 The four fixed-size angle instruments (v552)

The instrument cores of T155 and T157 (Section 10): the four fixed-size instruments the angle campaign produced — none of them closes an angle, and none is promoted as anything but a per-instance instrument [`verification/v552_angle_instruments.py`] (18 checks, ∼0.4 s; everything recomputed on v551's declared surface, the 12 deepest prime-power zones admitting a frame-A window inside the cap, $n = 29 \dots 139$, $m = 96 \dots 291 \le 300$; a no-go size ladder $m = 48 \dots 288$; parity and Dirichlet controls at $m = 64, 128, 256$; each statement carries at least one mutation control that must fail loudly).

- **The complement-floor certificate (T155)** [E]: $\min_{v \perp W} v^{\mathsf{T}} L_P v \ge \mu_{13}^P - \lambda_{\max}(M^{1/2}(I - GG^{\mathsf{T}})M^{1/2})$ with $M = \mathrm{diag}(\mu_{13}^P - \mu_k^P)$ and $G = T_{12} Q_W$ — an identity plus *one* certified Rayleigh bound on the high block, a theorem for every m and every subspace, direction checked: the certificate *never* exceeds the exact size-m complement bottom (12/12) and agrees with it to $1 - \mathrm{ratio} \le 1.01 \times 10^{-3}$ against the declared cap 2×10^{-3} on this small surface (the agreement tightens with depth: T155 measured $\le 2.2 \times 10^{-7}$ at $h = 142 \dots 1293$, sandbox, not consumed here). Exactness controls in *both* known configurations: $W = \mathrm{span}\{t_1..t_8\}$ returns μ_9^P and the *worthless* configuration $W = \mathrm{span}\{t_2..t_{13}\}$ returns μ_1^P, both to 2.0×10^{-11} — the instrument reports its own worthlessness. Mutation: one corrupted overlap row moves the certificate by ≥ 0.549, only downward.

- **The sine-block confinement (T157)** [E]: from the certified pencil floor $A \succeq tL_P$ (Schur two-block, $k_b = 16$) and the certified ladder ceiling $\lambda_1 \le S\mu_1^P$ (v551's sixteen-column certificate) *alone*, with $\gamma = Te_1(A)$ the parity coordinates of the bottom eigenvector, $\|\gamma_H\|^2 \le \lambda_1/(t\,\mu_{17}^P) \le (S/t)/\rho_{17} = 0.0153$–$0.0245 < 1$ on 12/12: $e_1(A)$ lives 97.5–98.5% inside the first 16 parity sines by a *per-instance theorem* (measured tail 6.7×10^{-6}–3.0×10^{-5}; T146's measured "98 per cent" replaced by one line). Negative control: substituting the pencil ceiling κ for the ladder S makes the bound *vacuous* (17.6–477 > 1 on 12/12, at least 997× above the S form) — κ carries the whole atom mass, and the choice of ceiling is the whole content. Mutation: the inflated floor κL_P is refuted at $e_1(A)$ itself by 3.3×10^3–1.0×10^5.
- **The Schur-entry identity (T157)** [E]: $s = \mu_1^P\, t_1^\top A^{-1} t_1 = (B^{-1})_{11}$ exactly (worst relative 1.1×10^{-11}) and $(B^{-1})_{LL} S_L = I$ to 5.8×10^{-11} with $S_L = B_{LL} - B_{LH} B_{HH}^{-1} B_{HL}$ the *same* fixed-size 16×16 Schur complement the T152..T157 chain already forms, with the literal-entry direction $1/s \le (S_L)_{11}$ on 12/12: $1/s = 4.0460$–5.8962 is an $O(1)$ number carried by *one* fixed-size object, so T156's R2″ loss ceiling $r \le 1/(Ls)$ is a statement about one entry. Negative control (T157's refutation wired as a check): the inversion-free Cauchy–Schwarz ceiling $\hat a - \|b\|^4/(b^\top B_{HH} b)$ is valid and *useless*, off by 250–7.27×10^3 ($\hat a = 1.13 \times 10^3$–$3.11 \times 10^4$ is the same entry *without* the Schur subtraction) — the cancellation in $b^\top B_{HH}^{-1} b$ is nearly complete and Cauchy–Schwarz cannot see it. Mutation: a one-index shift breaks the identity by 1.55–2.15.
- **The adaptive Lipschitz ceiling (T157)** [E]: interval bisection with the *local* Lipschitz constant $\mathrm{lip}(a) = d/g(a) + 2f/g(a)^2$ certifies $\inf f^{\mathrm{arch}}/(4\sin^2(\theta/2)) \ge t = 0.25$ on 12/12 windows in 519–1201 symbol evaluations (certified floor 0.2505–0.3062, never above the fine-grid minimum — direction checked): the arch one-variable statement carries an *executed* certificate per window, not a grid observation. The *global* θ_{lo} constant is documented insufficient (a uniform grid would need 2.14×10^3–7.33×10^4 points, up to 61.4× the adaptive cost). Mutation: fed a target above the true grid minimum the certificate *refuses* on 12/12 — it cannot be talked into certifying a false floor.
- **The no-go discriminator and the controls** [E]: on the T145 family $c(l) = 1/(1+l)$ the certified Schur floor *survives* ($t = 0.05$ on all 4 sizes) while both new instruments break — the confinement bound goes vacuous (7.93–413 > 1, ×52.0 over $m = 48 \ldots 288$) and $(S_L)_{11}$ explodes ($1185 \to 7.11 \times 10^4$, ×60.0) against the flat 4.05–5.90 real band: the instruments do not report confinement or $O(1)$ entries where both are false. Parity control $A = L_P$ exact to 4.9×10^{-14} ($B = I$, $s = 1$, zero tail); the Dirichlet residual hits the predicted $2/\sqrt{N}$ to 7.4×10^{-5}.
- **Named limits as content.** Per instance on small windows — a *finite list* with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D: *nothing about the angles is closed* — the two open terms after T157 (an m-free lower bound on $\hat g_1^2$, equivalently an m-free upper bound on the one Schur diagonal entry $1/s$, since the resolvent route is a theorem times that one measured fixed-size number; and an m-free R1″ domination with a non-shrinking margin, which T157 certifies per window with a 7.3×10^{-4} margin at the largest window and a *shrinking* trend) stay open and typed open; every floor t and ceiling S consumed is certified per window and its m-freeness is exactly what remains open; v551's ceiling and v550's ladder stay as stated, and nothing is re-opened.

4.19 The exact-form identities (v553)

The theorem cores of T158 and T159 (Section 10): the algebra of the sixteen-form itself — the dual variational structure, the exact lag sum with its gauge identity, the two closed scalars, and the one exact sign law the campaign found — and it closes *neither* open term [`verification/v553_exact_form_identities.py`] (25 checks, ∼0.2 s; everything recomputed on v551/v552's declared surface, the 12 deepest prime-power zones admitting a frame-A window inside the cap, $n = 29 \ldots 139$, $m = 96 \ldots 291 \le 300$; no-go ladders $m = 48 \ldots 288$ on *both* T145 forms;

parity and Dirichlet controls at $m = 64, 128, 256$; each statement carries at least one mutation control that must fail loudly).

- **The Thomson dual form and the positive Cholesky ladder (T158, P1)** [E]: $s = (B^{-1})_{11} = \max_x(2x_1 - x^\mathsf{T}Bx)$ (Maz'ya 1985 / Miclo 1999) with the direction *executed*, not asserted — 8 random trial vectors per window never exceed s, so every trial is an upper bound on the entry $1/s$: exactly the variational structure Cauchy–Schwarz lacks. The ladder $g_K = \sum_{j\le K} y_j^2$, $y = L^{-1}e_1$ from *one* Cholesky of the leading 16×16 block (nested Schur complements, Schur 1917 / Haynsworth 1968): all 16 terms strictly positive, partial sums strictly monotone, every $g_K \le s$, and $1/s \le 1/g_{16} \le 1/g_1 = B_{11}$ literally on 12/12; tight to $(1/g_{16})/(1/s) = 1.0642$–$1.1522$ against the declared cap 1.5 (T158 measured ≤ 1.2738 at $h = 254 \ldots 1393$, sandbox, not consumed here). Mutation: a 5% corruption of one off-diagonal entry destroys positive definiteness — the Cholesky *refuses* on 12/12.
- **The two-dimensionality (T158, P2)** [E]: $\mathrm{span}\{t_1, A^{-1}L_P t_1\}$ attains the Dirichlet maximum s exactly ($g/s = 1$ to 9.0×10^{-13} on 12/12) for the theorem reason $L_P t_1 = \mu_1^P t_1$ (KMS 1953; eigenpair residual 8.1×10^{-13}) — the entry is a *two-dimensional* quantity the moment one Green column is granted. Negative controls: $\mathrm{span}\{t_1, t_2\}$ (no Green column) misses s by a factor 1.42–4.8, and the corrupted Green column $L_P t_1$ (no solve) misses by 255–7.7×10^3.
- **The gauge identity and the TmH signature (T159, P4/P7)** [E]: the section map annihilates constant lag vectors *entrywise with zero tolerance* ($\mathrm{odd\_toeplitz}(\mathbf{1}) = 0$ exactly — the mechanism in one line), hence the gauge identity $\sum_d w_d = 0$ holds to 1.1×10^{-14} of $\sup |w|$ for the per-window optimiser and 3 random sixteen-vectors: the form is *blind* to the lag mass of c, and the lag mass is exactly where the arch and atom halves of the sixteen-form are of size h^2 and cancel. The premise — the Toeplitz-minus-Hankel signature $A_{r,s} - A_{r+1,s+1} = f(r + s)$ — is constant along every anti-diagonal to 1.4×10^{-16}. Mutations: a one-index Hankel shift breaks the gauge identity by 2.7×10^{-7}–4.2×10^{-5}; the T145 form $R = aa^\mathsf{T} + \varepsilon I$ has signature defect 7.4×10^{-2} — not a parity section, the gauge identity is not even *defined* for it.
- **The exact lag sum and the two closed scalars (T159, P2/P5)** [E]: $x^\mathsf{T} B_{LL} x = \sum_d c_d w_d$ ($w_d = T_d - H_d$, T the autocorrelation and H the convolution of $y = \sum_k (x_k/\sqrt{\mu_k^P})\, t_k$) against the direct assembly to 2.7×10^{-16} of the absolute scale $\sum_d |c_d w_d|$ — and the honest price *measured*: relative to the cancelled total the same identity holds only to 1.4×10^{-11} on this small surface (the sandbox at $h \le 1293$ loses 4.0–8.1 digits; that depth *is* the h^2 obstruction). The two closed scalars: $w_0 = \sum_k x_k^2/\mu_k^P$ (the $N^2/4\pi^2$ factor itself) to 4.2×10^{-16} and $2w_0 - w_1 = \|x\|^2$ to 7.4×10^{-16}, because $L_P = \mathrm{odd\_toeplitz}(2, -1, 0, \ldots)$ and the t_k are its exact eigenvectors (KMS 1953) — the huge scale and the $O(1)$ scale sit in the *same* two weights. Mutations: a 1% perturbation of the largest lag breaks the lag sum by 2.2×10^{-3}; a one-index μ shift breaks w_0 by 0.60.
- **The Z-matrix law and the Collatz–Wielandt floor (T159, U1/U2)** [E]: B_{HH}^{arch} *is* a symmetric Z-matrix, *raw* — every off-diagonal entry ≤ 0 with no tolerance on 12/12. The closed sign-based floor $\min_k(B_{HH}^{\mathrm{arch}}\mathbf{1})_k = 0.9093$–$1.1284 \ge t = 0.25$ with the theorem direction verified per instance: floor $\le \lambda_{\min}(B_{HH}^{\mathrm{arch}}) = 0.9138$–$1.1432$ exactly (the symmetric M-matrix criterion, Perron 1907 / Frobenius 1912 / Collatz 1942 / Wielandt 1950), beating the sign-free norm route by 0.69–8.8 on every window. The honest limit as content: the atom block obeys *no* sign law (0.44–0.53, a coin toss), the full block is not a Z-matrix and its row-sum floor is *negative* ($-48.3 \ldots -0.79$) against a positive exact bottom — admissible and vacuous, R1$''$ stays open. Mutation: the checkerboard similarity $S = \mathrm{diag}((-1)^k)$ — an isometry, $\lambda_{\min}$ moves by 0 exactly — destroys the raw law (fraction $1.0000 \to 0.494$–0.498): the Z law is an entrywise fact, not a spectral one.
- **The no-go discriminator and the controls** [E]: the T145 no-go breaks on three axes of this module's structure (Z law the exact opposite at fraction 0.000, signature broken, diagonal

varies by 0.94) while on the $c(l) = 1/(1+l)$ reconstruction the ladder *theorem* survives and the size explodes ($1/g_{16} = 1187 \to 7.11 \times 10^4$, $\times 59.9$, tracking the true $1/s$ — the instrument reports the collapse instead of hiding it). Parity control: $A = L_P$ *is* the section of the lag vector $(2, -1, 0, \ldots)$ entrywise, and the same machinery returns $B = I$, $s = 1$, every ladder partial sum $g_K = 1$ exactly constant in K, a zero gauge sum and lag sum $= \|x\|^2 = 1$ (worst 4.9×10^{-14}); the Dirichlet residual hits the predicted $2/\sqrt{N}$ to 7.4×10^{-5}.

- **Named limits as content.** Per instance on small windows — a *finite list* with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D: *neither open term is closed* — R2″ (one explicit pairing inequality: a bound on $\sum_d (\Delta c)_d W^1_d$ that uses the *correlation* between Δc and W^1 rather than their sizes, since $\ell^1 \times \sup$ pricing is refuted at every Abel rung) and R1″ (a device for the atom off-band block that is neither a norm nor a sign) stay open and typed open; T159's refutations (the fixed sixteen-vector, the one-sine rung, the power-law ansatz family, the Abel ℓ^1-pricing, the checkerboard law) are sandbox negatives and *not* promoted; the numerical horizon T159 declared ($\mathrm{cond}(B_{LL}) > 10^{12}$ past $h \approx 1292$) does not reach this surface and is not consumed; v552's instruments and v551's ceiling stay as stated, and nothing is re-opened.

4.20 The sampling/harmonics identities (v554)

The closed theorem cores of T160 and T161 (Section 10): what the sixteen-form *pairs against* — the atom half as a Λ-weighted sampling at the Fourier harmonics of the log-window, the closed moment laws and the total-variation theorem of the smooth half, the closed head split of the archimedean kernel with its scale-free Bernstein rate, and the certified off-diagonal fraction bound that replaces the refuted sign law — and it closes *no* open term [`verification/v554_sampling_harmonics_identities.py`] (21 checks, ~0.5 s; everything recomputed on v551–v553's declared surface, the 12 deepest prime-power zones admitting a frame-A window inside the cap, $n = 29 \ldots 139$, $m = 96 \ldots 291 \le 300$, plus a declared Λ-battery of six log-spaced cut-offs $X = 10^3 \ldots 3.2 \times 10^5$ for the PNT item — finite von-Mangoldt sums only, no matrix anywhere; each statement carries at least one mutation control that must fail loudly).

- **The sampling identity (T160, P2)** [E]: $\sum_d c^{\mathrm{atom}}_d w_d = -\frac{1}{2} \sum_{n \le X} (2\Lambda(n)/\sqrt{n})\, \widehat{w}(\log n)$ (plus one reflected term below the grid), $\widehat{w}$ the linear interpolant of the lag weights — exact to 5.6×10^{-15} of the absolute atom scale on 12/12 windows: the atom half of the pairing *is* a Λ-weighted sampling at the prime-power positions, hence identically a finite combination of sums $\sum_n \Lambda(n)\, n^{-1/2} \cos(t \log n)$ at 32 explicit frequencies. Mutation: a one-index shift of the sampling grid breaks by 5.1×10^{-5}–3.1×10^{-3}.

- **The moment laws and the TV theorem (T160, P1/P4)** [E]: $m_0 = \sum_d w_d = 0$ (1.0×10^{-16} of $\|w\|_1$), $m_1 = \sum_d d\, w_d = -[S_0^2 + 2\sum_j P_j^2]$ (9.2×10^{-16}) and $m_2 = \sum_d d^2 w_d = -[2S_1 - (M-1)S_0]^2$ (3.4×10^{-14}, a perfect square), both sign-definite forms strictly negative on every window; and U3 as a *theorem with checked hypotheses* ($c^{\mathrm{arch}} \le 0$ and monotone on $d \ge 1$, checked on 12/12, never assumed): $\mathrm{TV}(c^{\mathrm{arch}}) = 3.9817$–$4.8991 \le |c^{\mathrm{arch}}_0| + 2 \sup_{d \ge 1} |c^{\mathrm{arch}}_d| < 6$, a closed window-independent constant. Mutations: a one-index prefix shift breaks m_1 by $\ge 1.25 \times 10^{-2}$, an $(M-2)$ mis-normalisation breaks m_2 by $\ge 6.1 \times 10^{-2}$, and the full atom-including kernel violates the monotone hypothesis on 12/12.

- **The harmonic-frequency theorem and the PNT main term (T161, P3)** [E]: $t_j\,(2\alpha) = 2\pi j$ exactly (2.8×10^{-14}) on every window and every battery cut-off — the 32 frequencies are precisely the Fourier harmonics of the log-window $[0, 2\alpha]$, so the atom half is the vector of the first 32 Fourier coefficients of the measure $\sum_n (2\Lambda(n)/\sqrt{n})\, \delta_{\log n}$; at those frequencies $\theta = 2\pi j$ collapses the partial-summation main term to the closed, parameter-free prediction $(\sqrt{X} - 1)/(\frac{1}{4} + t^2)$ (de la Vallée Poussin 1896; the Mellin factor of $x^{-1/2}$), and on the battery the finite sums agree with it to 0.0007–0.097 of the trivial mass against the declared cap 0.12, the subtraction reducing the aggregate residual to 0.008–$0.466 \le 0.5$ of $\sum |S|$ on every cut-off. Mutations: a wrong Mellin pole worsens the aggregate residual by $\times 1.8$–$\times 95.8$, and a mass-matched *uniform* fake measure misses by 0.23–0.51 $\geq 1.5\times$ the cap — the agreement is a property of the von-Mangoldt measure, not of any measure with the same mass.

- **The head split and the scale-free Bernstein rate (T161, P1)** [E]: $c_d^{\mathrm{arch}} = \Psi_d + D\,\widehat{G}(dD, D)$ with $\Psi_d = -\frac{1}{2}[(d+1)\log(d+1) - 2d\log d + (d-1)\log(d-1)]$ closed, D-free and m-free (the triangle-smeared simple pole; the $\log D$ cancels identically) and $\widehat{G}$ the smear of the symbolically regularised $g_{\mathrm{reg}} = g - 1/(2w)$ — exact to 3.2×10^{-16} of $\sup|c^{\mathrm{arch}}|$ at every lag of every window, the two Ψ representations agreeing to 2.9×10^{-14}; and $\rho^* = (3+\sqrt{5})/2 = 2.618034$: the root of $x^2 - 3x + 1$ (0.0), returned exactly by the $D \to 0$ limit of the peeled interval $[0.4\alpha, 2\alpha]$, invariant under $\alpha \times 10$ (it depends only on the interval ratio 5), every window's $\rho(D) = 2.588$–2.608 below it (Bernstein 1912). Mutations: dropping Ψ misses c^{arch} by 0.99–1.00; the wrong interval ratio moves the limit to 3.0.
- **The off-diagonal fraction bound and the sign refutation (T161, P2)** [E]: the gauge identity licenses one boundary-free Abel step and $Q^{\mathrm{arch}} = \sum_{k,l} a_k a_l \sum_{d\geq1} (\Delta c^{\mathrm{arch}})_d\, R^1_{kl}(d)$ exactly (2.0×10^{-13}); the certified per-window inequality is $|\text{off-diagonal block}| \leq \frac{1}{4}\,|Q^{\mathrm{arch}}|$ (measured 0.171–0.212 on 12/12; the sandbox measures 0.1035–0.1713 over $h = 50$–1218, flat — a fit, not consumed) — and the *refuted* sign law is wired as the negative control: the arch half is negative while its off-diagonal block is positive on 12/12 (the harmful direction for the Thomson dual) and 176 of 240 off-diagonal pairs violate the per-pair form on every window, so a one-sign statement was never the right object and nothing single-signed is promoted. Mutation: a one-index R-kernel shift breaks the decomposition by $\geq 6.8\times10^{-5}$.
- **The no-go discriminator and the controls** [E]: the head split is a statement about *the* T115 kernel — on the T145 no-go reconstruction $c(l) = 1/(1+l)$ the same closed formula misses by 2.39 of $\sup|c|$; the closed Dirichlet cosine-sum identity agrees with the brute-force sum to 1.8×10^{-14} on 64 random triples including the degenerate branch; $L_P t_k = \mu_k^P t_k$ to 1.4×10^{-14} with the basis orthonormal to 1.0×10^{-14} (KMS 1953); odd_toeplitz$(c^L) = L_P$ at zero tolerance.
- **Named limits as content.** Per instance on small windows — a *finite list* with an explicit maximum; nothing uniform in the zone index or in D; finite Λ-sums are allowed and used, *no* RH statement is made, and the δ triage of T161 (required depth against RH strength) is a documentation item, *not* a claim of this module; the open terms after T161 — R-A′ (the prime-free log-moment $\sum_d \Psi_d w_d$, outside the polynomial ladder), R-B′ (the m-free 16×16 Gram fraction bound, Fejér 1915 / Schur 1917), R-C′ (*one* split of the pairing with $\delta_{\mathrm{eff}} < \frac{1}{2}$; T161 refutes the arch/atom and arch+smooth/fluctuation splits with numbers), R-D (a fifth R1″ device) — stay open and typed open; v553's algebra stays as stated, and nothing is re-opened.

4.21 The Pareto/total-variation identities (v555)

The closed theorem cores of T162 and T163 (Section 10): the *price structure* of the pairing — the exchange law that makes demand and price two coordinates of one object, the total-variation floor that closes the flat-price route in the parity sector, the chain-derived price axis of the Fejér knob, the Mellin cell moments and the boundary-free Abel step with its closed loss reason, and the Lerch/Frullani log-moment on three independent routes — and it closes no open term beyond the negative closure of R-C‴ that the floor inequality itself is [`verification/v555_pareto_tv_identities.py`] (23 checks, ∼0.4 s; everything recomputed on v551–v554's declared surface, the 12 deepest prime-power zones admitting a frame-A window inside the cap, $n = 29\ldots139$, $m = 96\ldots291 \leq 300$; the one arithmetic input $\kappa = 0.038821$ *verified* at every jump point of $\psi(x)/x$ on the von-Mangoldt table in $[100, 3.2\times10^5]$, reproduced to 4.2×10^{-7}; each statement carries at least one mutation control that must fail loudly).

- **The exchange law and the crossing criterion (T163, R1.2)** [E]: $\delta_{\mathrm{bnd}}(x) =$

$\frac{1}{2} + \log(2\kappa\, g_{16}\,\mathrm{TV}(x)/P(x))/\log X$ is an identity at all 612 grid points of the declared Fejér knob (50 geometric σ on $[1, 4096]$ plus ∞, 12 windows; 5.1×10^{-16} relative, admissibility $Q(x) > 0$ everywhere), and the crossing criterion $\delta_{\mathrm{bnd}}(x) < \frac{1}{2} \iff P(x) > 2\kappa\, g_{16}\,\mathrm{TV}(x)$ holds as an exact equivalence at every point: price and demand are two coordinates on *one* object, and the crossing price is a closed formula. Mutation: the wrong constant (κ for 2κ) moves the law by 6.5×10^{-2}–3.4 at every grid point.

- **The total-variation floor (T163, R2.2 — the theorem that carries the verdict)** [**E**]: (T1) $w_0(x) = \|a(x)\|^2$ to 4.9×10^{-16} (the parity rows are orthonormal); (T2) $\mathrm{TV}(x) \geq |w_0(x)|$ by telescoping with $w_M = 0$; (T3) the normalisation $x_1 = 1$ forces $\|a\|^2 \geq 1/\mu_1^P$, hence $\mu_1^P\,\mathrm{TV}(x) = 5.00$–$28.84 \geq 1$ on all 708 trial vectors built in the module (51 knob settings plus 8 random admissible sixteen-vectors per window), against the closed floor $1/\mu_1^P = 1/(4\sin^2(\pi/N)) = 943.6$–$8609.6 \sim h^2$; (T4) with the crossing criterion, every sub-$\frac{1}{2}$ grid point pays $P > 2\kappa g_{16}/\mu_1^P = 15.4$–$142.8$ (all 54 sub-$\frac{1}{2}$ points respect it) — a flat-price sub-$\frac{1}{2}$ demand is impossible in the parity sector on this surface, the inequality that closes R-C$'''$ negatively. Negative control: the *inadmissible* vector $0.01\,x^*$ ($x_1 \neq 1$) does undercut the floor ($\mu_1^P\mathrm{TV} = 2.5$–2.9×10^{-3}) and is recognised as inadmissible — the floor is a property of the normalisation meeting the smallest parity eigenvalue (Kac–Murdock–Szegő 1953), not of the machinery.
- **The chain-derived price axis (T163, R1.0)** [**E**]: $\sigma = 1$ gives $x = e_1$ exactly with $P = g_{16}/g_1$ (the $K = 1$ rung of the T158 Cholesky ladder, the T157 route-(0) certificate; 4.1×10^{-16}) and $\sigma = \infty$ gives $g_{16}\,Q(x^*) = 1$ (the $K = 16$ top rung; 1.2×10^{-12}) — the Fejér sweep is a path between two certificates the chain already owns; the ladder carries all 16 terms strictly positive with strictly monotone partial sums, and the knob is monotone in both coordinates on every window (the knob curve *is* the Pareto front). Mutation: a corrupted off-diagonal sixteen-block entry makes the Cholesky *refuse* on 12/12.
- **The Mellin cell moments and the Abel step (T162, C2/C5)** [**E**]: the k-th rung of the archimedean Mellin ladder is the closed sum $\sum_d w_d\, C_d(-(2k + \frac{1}{2}))$ with $C_d(s)$ the exact tent integral (Mellin 1896), reproduced by an independent per-cell Gauss–Legendre quadrature to 2.8×10^{-13} on every rung $k = 1 \ldots 4$ (shifted tent centres break by $\geq 1.3 \times 10^{-2}$); the boundary-free Abel identities hold at levels $k = 1 \ldots 3$ to 8.3×10^{-17} (Abel 1826), the dropped-boundary level-1 form — the one the chain uses — agrees because the gauge identity kills the boundary term exactly, the optimal level is *one* for T162's closed reason (net factor $32\pi/\alpha = 40.6$–$59.4 > 1$, confirmed by the measured level-2/level-1 bound ratio 30.8–$69.1 > 1$), and a gauge violation breaks the dropped form by its closed prediction $v_0 c_0 M$ to 1.2×10^{-15} — T162's positive control.
- **The Lerch/Frullani log-moment (T162, R-A$'$)** [**E**]: $L = \sum_d \Psi_d w_d$ agrees on three independent routes on every window — the direct sum with Ψ_d closed; the double-Abel form *with* its Wronskian boundary term $-\frac{1}{2}M\log M\, w_{M-1}$ (2.8×10^{-14}; dropping the term breaks by 2.2×10^{-4}–1.3×10^{-2}); and the Lerch/Frullani integral with the $d = 1$ term peeled in closed form (1.8×10^{-15}; 44 declared panels) — the peel constant $\int_0^\infty ((1 - e^{-y})/y)^2 dy = 2\log 2$ reproduced to 2.2×10^{-16} and $\Psi_1 = -\log 2$ exactly (Frullani 1828 / Lerch 1887 / Clausen 1832).
- **The no-go staircase and the controls** [**E**]: coarsening the lag grid over $\nu = 4/2/1/0.5$ degrades the arch/atom cancellation *monotonically* in the median (damage $1.00 \to 3.49 \to 10.81 \to 24.98$, the $\nu = 4$ column an exact null control) — the T145 resolution no-go breaks monotonically in the collision rate; the closed Dirichlet cosine-sum identity to 2.4×10^{-14} including the degenerate branch; KMS eigenpairs to 1.4×10^{-14}; odd_toeplitz$(c^L) = L_P$ at zero tolerance.
- **Named limits as content.** Per instance on small windows — a *finite list* with an explicit maximum; nothing uniform in the zone index or in D; the T163 h-exponents ($P_{\mathrm{aff}} \sim h^{3.05}$, $P_{\mathrm{cross}} \sim h^{1.91}$, the TV exponent $\sim h^{2.00}$) are sandbox fits and are *not* consumed — what is consumed is the closed per-window floor; finite Λ-sums are allowed and used, *no* RH statement is made ($\frac{1}{2}$ appears in exactly one role: as the strength of a classical statement a required depth is compared against); the open terms after T163 — R-E (the successor, both arms prime-free: a

growing $1/s$ ceiling for the downstream chain, or a sector whose gap does not vanish like h^{-2}), R-B‴ (an h-uniform positivity margin), R-D (a fifth R1″ device) — stay open and typed open; v554's identities stay as stated, and nothing is re-opened.

5 The localization program (sandbox)

Provenance

Everything from here to Section 11 is [sandbox]: exploratory probes in `experiments/tfpt-discovery/`, no ledger row, no marker, no authority. The findings are reported because the *shape* of the residual is the useful output, not because any of them is established.

5.1 I5: the ledger closes and one inequality is left (T79)

The transport wall — the thing that had blocked the line since T20 — was fully decomposed in T79 (`transport_ledger_probe.py`, 22/22). The key observation is that the odd-prime side of the Weil functional agrees *exactly* with the certified plus combination (relative 2.6×10^{-16}): the prime side was never the wall. What remains after the ledger closes is a minimal transport set of two exact identities (proved), two classically-provable bounds, and exactly one coupled inequality:

$$\textbf{(I5)} \qquad Q_{\text{cert}}(h) + \Delta_2(h) + A_{\text{fam}}(h) - A_{\text{shift}}(h) \;\geq\; 0 \quad \text{for all autocorrelations } h.$$

I5 is the prime ↔ archimedean coupling. By the closed ledger it is, as an identity, equivalent to Weil positivity and hence to RH. The diary is explicit about what that means: *the built-in safeguard fired*. RH does not follow from finite bookkeeping; the hidden infinite step is exactly I5 — and the value of T79 is that the step was *found and named* rather than concealed. This is an equivalence typing, not progress.

T82 then changed I5's *type* (`heat_arch_probe.py`, 22/22, ARCH-INTERNAL): Δ_{arch} is exactly the internal Γ difference $A_{\text{fam}} - A_{\text{shift}}$ via Legendre duplication, verified on 18/18 battery rows at max relative 6.6×10^{-15}. I5 stopped being a coupling between two worlds and became a *self-consistency statement about one heat family*. Again: a type change is not a proof.

T83 (`fe_symmetrization_probe.py`, 27/27, INVARIANT-NULL) then asked the obvious question of the self-duality picture (Section 3): can the functional equation itself do the work? It cannot, and the way it fails is informative. The ζ reflection $J_{1/2}$ acts as the identity on autocorrelations — the symmetry is already absorbed in evenness, and $Q_{\text{Weil}}(Jh) = Q_{\text{Weil}}(h)$ holds floating-point-exactly on 15 test functions — and the family reflection J_1 moves nothing in the value geography (5/24 overlap invariant, λ^* covariant-exact) while pushing m_+h out of the Weil cone in closed form, with $\text{Fix}(J_1) \cap \{\text{Weil cone}\} = \{0\}$ exactly. The structural finding underneath the null is the one quoted in the big picture: the product of the two reflections is the unit-line shift, i.e. the transport operator, and the wall is transversal to the functional equation rather than positioned by it. Symmetrisation is therefore not a route; the same mirror returns, as an obstruction, in T106.

5.2 The Connes dictionary (T87) and where positivity is anchored

T87 (`i5_connes_dictionary_probe.py`, 22/22, DICTIONARY-EXACT-CORE) identified the I5 one-family form as a compiler instance of the semi-local Connes structure. Four exact matches: the two-shell orbit route equals Connes' expression at relative 6.3×10^{-16}; the orbit kernel of Connes–Consani (2021) is literally the compiler's Poisson ladder at $\rho = e^u$ (`sympy` plus 1.7×10^{-50}); h_+ is by definition the compiler kernel $k_\zeta = K_{\text{fam}} - K_{\text{shift}}$; and $k_\zeta = 2\theta'_{\text{RS}}$ — the derivative of the *Riemann–Siegel phase*, at relative 0.0. The dihedral shift is the scaling group $\mathbb{R}_+^*$.

The most useful part of T87 is not a match but a *boundary*. The two positivity programs are complementary: Connes–Consani anchor positivity *before* the primes (the prime-free Sonin window, support $(\frac{1}{2}, 2)$, i.e. $u \in (-\log 2, \log 2)$), the compiler anchors it *after* the primes (atom certificates, all supports). The window boundary is exactly the first prime atom $u = \log 2$ (`sympy`-exact; on narrow rows the prime side, Q_{cert} and Δ_2 all vanish identically). The sectors *touch* and do not overlap; the I5 coupling lives in the crossing region between them, outside both proved sectors.

5.3 The tightness map (T88) and the crossing (T89–T91)

T88 (`i5_tightness_probe.py`, 27/27) mapped where I5 is actually tight: a band-limited zero plateau (classical), safe zones, and *eight* near-vertical tight curves whose spacing follows the Γ-side density law (ratio 1.01 ± 0.08, computed zero-free — no spectral identification is claimed or made). There are no negatives on genuine autocorrelations; the earlier T76 negatives were the shadow of missing archimedean/$p = 2$ bookkeeping, exactly as the ledger predicted.

T89 (`sonin_crossing_probe.py`, 19/19) established that the window compression of the Weil form *is* Bombieri's classical object, and settled the sharpness question: the boundary at $\log 2$ is *not* sharp. The margin falls smoothly from $a \approx 0.5$ towards the floor with no structure at $\log 2$; the prime-free sector still has margin 0.557 at its own boundary and keeps ≥ 0.25 of the reference comfort to $a \approx 0.75$ (+0.057 beyond the proved boundary). The residual subspace grows softly $0 \to 1 \to 2 \to 3 \to 4$ (out of 32) and carries the global minimum. T90 (`residual_dissection_probe.py`, 17/17) made those residual vectors explicit — n-stable to $\leq 2.1°$, closed Gaussian$\times\cos$/Gaussian$\times\sin$ fits with 99+% capture — and decided the core question: the residual is *not* just pole coupling (3/10 vectors are controlled by no structure at all; beyond $a \approx 0.85$ genuinely pole-free atom$\leftrightarrow$arch content remains).

T91 (`thin_band_analytic_probe.py`, 19/19) identified the band as the *one-atom zone* and found the mechanism that dominates the rest of the document: beyond an inner edge the prime atom $u = \log 2$ becomes *load-bearing* — the prime *rescues* the form where the archimedean margin is exhausted (rescue term $+0.22 \ldots + 0.41$, purely odd sector). Two uncertainty constants were decided in closed form: $a \cdot t_{\mathrm{rms}} \to \pi$ (Wirtinger, 0.3%) and $a \cdot t_{\mathrm{cent}} \to 2\,\mathrm{Si}(\pi) - 4/\pi = 2.4306$ (two independent routes, 10^{-25}).

The band that these three parts map out is narrow and its landmarks are explicit. Reading outwards from the classical side, the map is: classical edge 0.34657, prime-free crossing 0.3723–0.3763, band top $\alpha^* = 0.46265$, second atom $\log 3/2 = 0.54931$. Everything in Sections 6 and 7 happens inside that interval, between the first and the second prime atom, and the four numbers are the coordinates used throughout. The map was re-derived by a third independent implementation (T93, `band_reconciliation_probe.py`, 41/41) before it was used further.

An attempt to convert the band into a proof was made and did not succeed. T92 (`zone_extension_proof_probe.py`, 36/36, T-SKELETON) certified something small and real — $Q \geq 6.7 \times 10^{-12} > 0$ on an 8-dimensional sine-window subspace across the whole band, with an error certificate and a 1057-point covering — but not the lattice zone extension it was aiming at, and it named two independent walls: $\lambda_{\min}$ collapses geometrically (factor 11.4 per mode), and the high-mode complement would need $N^* \gtrsim 5069$ modes. The finding, and the reason it is quoted here rather than buried, is a statement about instruments: *the finite-block route is structurally the wrong instrument for this target, not an under-resourced one.* That is what motivated the change of approach in Section 6.

5.4 The blind prime demonstration (T94)

A separate, deliberately modest exercise. T94 (`prime_blind_demo_probe.py`, 16/16, BLIND-100) used the exact geometric criterion "odd n is prime $\iff r_4(n) = 8(n+1)$" (Jacobi 1834, on the $\mathbb{Z}^4 = $ rank-$|\mu_4|$ theta tower) to predict all 753 primes of the never-seen window [1 000 001, 1 010 000] — count and MD5 declared *before* any ground truth, then evaluated: exactly 753, zero false positives, zero false negatives, 100.00% on all 10^4 integers, with an AST-enforced prediction path containing no divisibility test (no `isprime`, no sieve, no `%`, `//` or `/`). The cost is part of the result: a sieve wins by $\sim$820$\times$ (1 ms against 0.83 s). This is structure completeness, not an algorithmic advance — and the *spectral* prediction (zeros $\to$ primes) remains bound to I5/RH and is not claimed.

6 The handover induction (T95–T101)

The band work left a picture: each prime atom rescues its own zone and hands over to the next. T95–T101 turned that picture into an induction, and then measured whether the induction can run.

6.1 The relay is real (T95, T96)

T95 (`continuum_extension_probe.py`, 28/28) proved the C1 chain unconditionally ($|h_f(\log 2)| \leq 1/2$ by disjoint supports with margin 0.2305; the windowed symmetrised shift has $\|S\| = 1/2$ exactly, 1.1×10^{-16}) and found that the binding mechanism is *not* the two-bump extremal (share 0.046) but the atom rescue: above $\alpha_{\text{free}} = 0.371654$ the prime-free minimiser has $h(\log 2) < 0$ down to -0.328, so the atom adds positivity exactly in the losing direction.

T96 (`relay_test_probe.py`, 21/21) is the model of a self-correcting run. The apparent "edge" of T95 at α^* was a map artefact: the value $+7.8 \times 10^{-6}$ reproduced exactly, but it sits on a smooth, strictly positive curve with no feature at all ($\lambda_{\min} > 0$ on the whole of $[0.38, 0.86]$, decay law $\lambda_\infty \sim \exp(-49.2\alpha)$). The *relay*, however, is real — and it lives entirely in the counterfactual: remove the $\log 3$ atom and $\lambda_{\min}$ crashes to -0.445, with the loser being exactly the anti-double-bump at distance $\log 3$ (alignment -0.99) and a rescue identity at 5×10^{-15}. All four handover windows are positive: each atom arrives strictly *before* it is needed ($+0.025/+0.009/+0.011/+0.007$ for $k = 1..4$). Three consequences were recorded: it is a margin problem, not an edge problem; numerics as a witness is exhausted beyond $\alpha \approx 0.55$; and the counterfactual is the proof target, because it works on $O(0.1)$-sized quantities instead of 10^{-6}.

6.2 The induction step and its one missing lemma (T97, T98)

T97 (`relay_induction_probe.py`, 105/105) gave the induction step proof shape. Alignment is true and sharp with a constant law; the $t = 0$ killer loss is proved by three structure identities (old atoms vanish identically on E_-; the pole rescue flips with factor $1 - \cosh(u/2)$; $|\hat{f}|^2 = 2(1 - \cos(tu))|\hat{g}|^2$), giving $k_{\text{eff}} = (1 - \cos(tu))k(t)$ with infimum $-1.11 \ldots - 2.59$ instead of $k(0) = -5.37$.[2] The structural pearl: E_0 is *literally* the same form at a smaller window (7×10^{-14}) — the induction hypothesis, by self-similarity. What remained was a single cross-block lemma about a single vector.

T98 (`cross_lemma_probe.py`, 44/44) attacked it and produced the most instructive negative result of the series: the measured $\sqrt{\lambda_0}$ law is Douglas (1966) range inclusion, i.e. a *tautology* of the positivity one is trying to prove — the identification *dissolves* the lemma rather than proving it. Three T97 premises were refuted in the same run. The replacement target is an exact scalar inequality with no constants and no near-null vector:

$$D_k(\alpha) \quad := \quad -\lambda_{\min}\Big(Q|_{E_-} - Q_{-0}(Q|_{E_0})^{-1}Q_{0-}\Big) \quad \leq \quad \tfrac{\mu_k}{2}.$$

T99 (`dk_recursion_probe.py`, 23/23) then found the a-priori structure: an exact mirror-parity selection rule ($J_- Q_{-0} J_0 = -Q_{-0}$ at 2×10^{-15}) giving $D_k = \max(D_{\text{even}}, D_{\text{odd}})$, with the binding even channel coupling only to the *odd* spectrum of the smaller window — so the fragile near-null mode is identically absent from the binding channel. T100 (`bulk_remainder_probe.py`, 27/27) closed the bulk remainder from 11/24 to 24/24 samples and showed the 10.8% drift was a grid artefact.

6.3 The race (T101)

T101 (`k_asymptotics_probe.py`, 31/31, CROWDING-TRENDS) resolved 16 zones ($n = 2 \ldots 29$) and separated two questions that had been fused. The *mathematics* trends self-supportingly: the

[2] In closed form (a T92 side result) $k(0) = -\gamma - 3\log 2 - \pi/2 - \log \pi$. Prior-art note (literature pass, 2026-07-27): in substance this is the classical digamma value $\psi(1/4) = -\pi/2 - 3\log 2 - \gamma$, shifted by $-\log \pi$, as it appears in Suzuki 2023 (J. London Math. Soc. 107) — a good consistency check, not a find.

primitive quantities are flat over 16 zones (w/g slope $+0.002 \pm 0.178$; the relative D_k margin does not trend; $D_k \le \mu_k/2$ holds 64/64 and never fails). Atom *strengths* do not fall ($\mu \sim n^{-0.087}$), only the *spacings* do ($n^{-0.615}$, exact) — pure arithmetic. What loses the race is the *closure instrument*: the demand ratio r_k decays like $\exp(-0.1622k)$ (fit, ± 0.0562). The collapsed one-parameter law $w_k = 0.0838 \cdot (\text{atom spacing})/\mu_k$ reduced the scatter from 8.85× to 2.66× with zero residual trend.

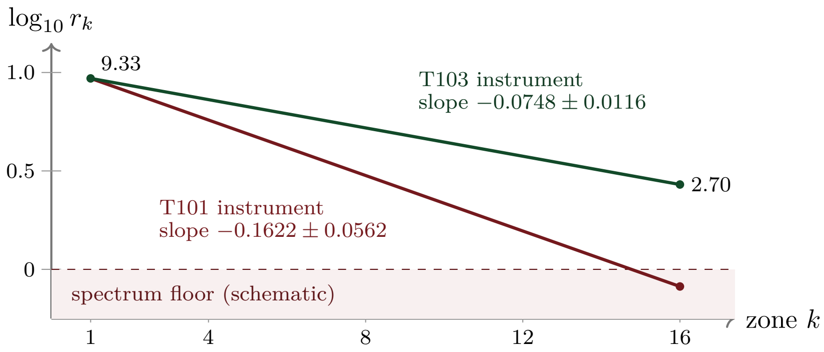

Figure 3: *Schematic.* The instrument race over the 16 resolved zones. *Literal:* the two decay slopes (-0.1622 ± 0.0562 measured in T101; -0.0748 ± 0.0116 measured in T103, a factor 2.2 flatter), and the T103 endpoints $r_1 = 9.33 \to r_{16} = 2.70$ with the statement that the improved instrument never leaves the spectrum. *Schematic:* the straight-line interpolation in $\log_{10}$, the common anchor of the two lines, and the position of the "spectrum floor". Both slopes are declared fits in the diary. What loses the race is the instrument, not the inequality: the true inequality $D_k \le \mu_k/2$ holds 64/64 and its relative margin does not trend.

T101 also stated what an asymptotic theorem would need, and that list has not changed since: (A) the collapsed lower bound $w_k \ge c\, g_k/\mu_k$ with a fixed $c > 0$ — a sharp, non-perturbative, *arithmetic* lower bound at a Weil-positivity edge, with Taylor arguments forbidden by the $\exp(-c/\delta')$ onset; (B) a uniform relative margin; (C) a replacement for the bulk remainder; (D) a finite check of the low zones. If crowding ever wins, the hardness sits entirely in (A).

7 The certification arc (T102–T125)

The last twenty-four parts stopped improving measurements and started converting them into certificates — and, where that failed, into explicit refutations. The arc closes with T125, which assembles the certificates into one chain and states what that chain is and is not (Section 9).

7.1 The mechanism: a crossing, not a singularity (T102)

T102 (`arithmetic_bound_probe.py`, 42/42, MECHANISM-IDENTIFIED) computed the mechanism exactly. The k-th atom acts on $E_-/E_0/E_+$ as $\mathrm{diag}(-\frac{1}{2}, 0, +\frac{1}{2})$, so $Q_{k-1} = Q_{\text{full}} - \frac{\mu_k}{2} P_- + \frac{\mu_k}{2} P_+$: the atom strength μ_k enters exactly once, linearly. That gives a two-sided sandwich through the Schur profile $\sigma_k(\delta)$, with $\mu_k/2 \le \sigma_k$ implying that the handover holds, and $2w_k$ lying between the crossings in 16/16 zones.

Two corrections came out of the same run. First, the onset is *anchored* at $\delta_c = 2w_k$ — a crossing of two finite quantities ($R^2 = 0.968$) — and exponential-singular versus power behaviour is not separable; T96's essential-singularity reading is compatible but no longer distinguished. Second, and more consequentially, the binding constraint *flips*: no concentration condition binds. The bare E_- form is strongly positive ($2.65 \ldots 3.52$, i.e. 4–14× $\mu_k/2$), the pole flip and band mass saturate their classical ceilings (96.9–97.1% of Cauchy–Schwarz and Landau–Pollak/prolate respectively), and the onset is manufactured *entirely* by the Schur dressing against $E_0 \oplus E_+$, which eats 35.7%–97.3% of the bare eigenvalue. The loser does not live in E_-.

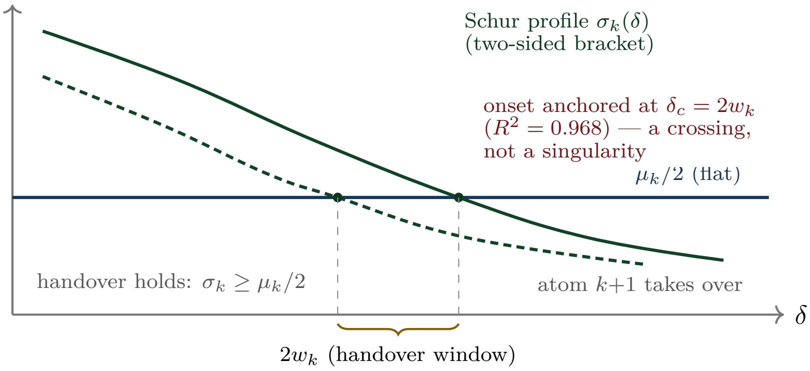

Figure 4: *Schematic.* The crossing mechanism of T102. *Literal:* μ_k enters exactly once and linearly, so the condition is a two-sided sandwich through the Schur profile $\sigma_k(\delta)$ against the flat level $\mu_k/2$; the window width $2w_k$ lies between the crossings in 16/16 zones; the onset is anchored at $\delta_c = 2w_k$ with $R^2 = 0.968$; the candidate bound $w_k \geq A_k\mu_k^{1/q_k}$ holds 16/16 with ratio 0.749–0.940. *Schematic:* the curve shapes and all absolute scales.

The same run performed the most useful kill of the arc: a triple decomposition test showed that the T101 law in C/g was a *proxy*. g_k is causally impossible (the form Q_{k-1} is blind to the next atom), statistically dispensable (a causal law $C \cdot u$ has CV 23.3% against 27.6%), and arithmetically only a ceiling ($C_k \leq g_k\mu_k$ exactly; the 16-zone extrapolation violates the ceiling from $k = 69$; checked over 18 120 prime-power atoms to $n = 200\,000$).

7.2 The instrument, rebuilt (T103), and the two doors

T103 (`instrument_probe.py`, 29/29, INSTRUMENT-IMPROVED) rebuilt the closure instrument with certified weights $w_j \geq 1/\lambda_j$ and a θ-weighted band sum, reproducing both anchors exactly. With *one* fixed, k-uniform instrument ($\Lambda_0 = 3$, $r = 2$, no per-zone tuning) the closure map jumps from 7/64 to 44/64 and the race slope halves (Figure 3). The price is explicit: the number of explicit modes grows $2 \to 232$. And the verdict relocated the remainder: it is no longer the bulk (the bulk is measurably not low-rank — effective rank up to $0.579 \dim E_-$) but the *wing slack* $S = 1 - \rho$, which falls $0.2091 \to 0.0392$.

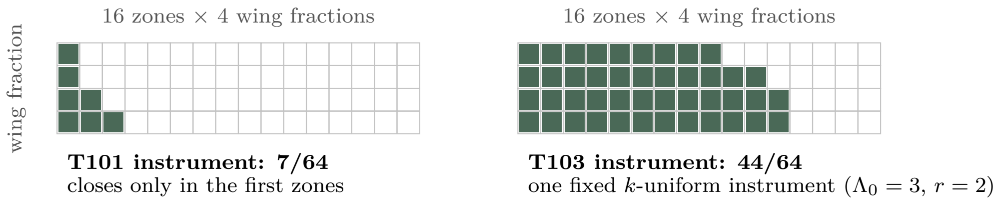

Figure 5: *Schematic.* The closure map. *Literal:* the grid is 16 zones × 4 wing fractions = 64 cells; the old instrument closes 7 of them, the rebuilt one closes 44 with a single fixed k-uniform setting ($\Lambda_0 = 3$, $r = 2$) and no per-zone tuning. *Schematic: which* cells are shaded — the diary records the totals, not the pattern; the shading is drawn as a monotone wedge purely to visualise the counts.

The convergence recorded at the end of T102 is the pivot of the whole arc: T103's wing slack $S = 1 - \rho$ and T102's Schur dressing are *the same object seen from two sides.* Both doors lead to the wing near-null direction.

7.3 Inductive chains, and the death of the margin route (T104)

T104 ran as *two independent arms* after a file collision between two parallel workers was resolved by keeping both implementations of the same contract — `schur_profile_bound_probe.py` (21/21)

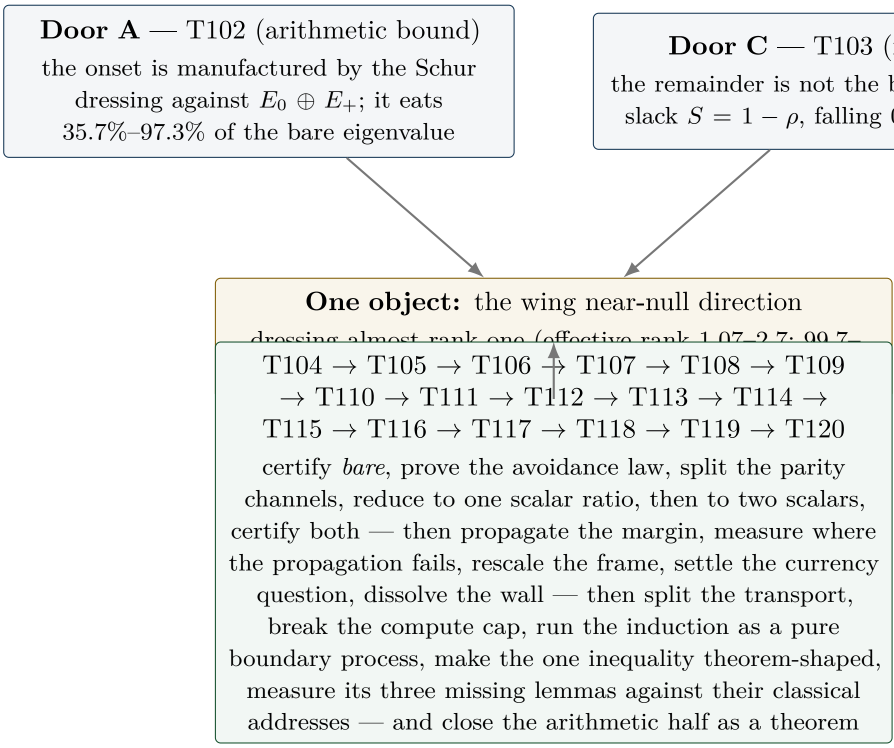

Figure 6: Two doors, one object. Both entries are literal diary findings; the convergence statement ("the same object from two sides") is quoted from the T102 closing note.

and `schur_profile_chain_probe.py` (47/47). All core findings converged, which makes it a de-facto independent replication.

- **The naive margin route is dead**: $M \geq m \cdot \mathrm{Id}$ holds in 0/16 zones; the coupling mass to margin ratio is 10^3–10^6; m vanishes in the continuum like $M^{-1.7}$ (fit).
- **The mechanism is avoidance**: the coupling B_- decouples from the soft bulk modes — the softest mode carries 0.00% of the coupling mass for $n \geq 3$, and the mass sits in mid/high bands (lowest band only 2–12%). Positivity is not saved by a margin but by structural avoidance.
- **Exact spectral-split chains close** 16/16 with finite data: arm A via the exact block-inverse identity with matrix caps in PSD order (headroom 1.15–1.44, reach $\gamma_{\max} = 0.50$–0.90); arm B via a threshold split with an $O(1)$, resolution-stable threshold $w = 2.47$–5.90 — but requiring $r_{\min} = 64 \ldots 1024$ soft modes $\propto M$, which makes the continuum statement a spectral-density condition.
- A recommendation from T96/T103 was tested and *rejected*: the prolate wing basis loses against the raw basis in both arms.

Consequence: the hard core migrated from σ_k as a whole to four named ingredients — a lower bound on $\mathrm{bare}_k = \lambda_{\min}(Q_{\mathrm{full}}|_{E_-})$, the soft dressing scalar L, finite induction data, and a one-sided edge estimate instead of a fit. The chain around them is classical (Schur complement, exact block inverse, Parseval/Bessel, Weyl) and fit-free.

7.4 The certified bare bound and the avoidance theorem (T105)

T105 (`bare_wing_bound_probe.py`, 28/28, ONE-OF-TWO) delivered one of the two missing ingredients in closed form. First it resolved the T104 arm discrepancy: *bare* depends only on (u_k, δ) — the window centre cancels ($\leq 0.74\%$ drift under $8\times$ refinement) — with the exact split bare $= \mu_k/2 + b_0$. Then three classical steps (Bessel; a Legendre/level-set optimisation at $k(\pi/\delta)$; Cauchy–Schwarz at the pole pair) collapse to a closed lower bound:

$$b_0 \quad \geq \quad \frac{\delta}{\pi}\int_0^{\pi/\delta} k(t)\,dt \quad - \quad \delta\,K_{\max} \quad - \quad 4\,(\cosh\tfrac{u}{2}-1)\sinh\tfrac{\delta}{2}.$$

This is the band average of the archimedean kernel over the band conjugate to the wing, of size $\sim\log(1/2\delta)-1$. It is positive in 16/16 zones (1.27–3.54) at 81.1–92.7% of the measured value (median 92%), it sharpens as the wing flattens, and it needs no eigenvalue or induction data.

Second, the *avoidance law* became a theorem rather than an observation: $Q_{\text{full}} \succeq 0 \Rightarrow \|B^{\mathsf{T}}v_i\|^2 \leq w_i\Lambda_-$ (a semi-inner-product argument; coupling proportional to the mode eigenvalue), verified on every mode of every zone with worst ratio 0.377. That explains the measured 0.00% avoidance exactly. Alongside it, an exact parity superselection: $JQJ = Q$ (5.7×10^{-15}) and $JB = -BR$ (3.4×10^{-15}) — the two channels do not mix, and the softest mode is J-even in 16/16 zones. Notably the *Euclidean* angles are not small (cosine up to 1.0): the avoidance lives in the form metric, as a Friedrichs angle.

What T105 left is one Loewner statement: $Q_{\text{full}}(\alpha) \succeq (\mu_k/2)P_-^{(k)} \iff \rho \leq 1-(\mu_k/2)/\text{bare}_k$ — a Friedrichs-angle / spectral-density condition, explicitly *not* a margin condition. The free cap $L \leq \Lambda_- = 5.63$–7.52 diverges like $\log(1/D)$, so the certificate has to come from the spectral density near the soft end.

7.5 Parity superselection: the pole splits (T106)

T106 (`friedrichs_angle_probe.py`, 32/32, DENSITY-MAPPED) killed two routes and found the structural result of the arc. The kills are in Section 8. The finding:

$$ab^{\mathsf{T}} \ + \ ba^{\mathsf{T}} \quad = \quad ss^{\mathsf{T}} \ - \ tt^{\mathsf{T}} \qquad (10^{-18};\ Ja \quad = \quad b)$$

Here J is the mirror parity of Section 3, arriving for the second time and now as the obstruction itself. The rank-2 Weil pole splits *exactly* into a positive rank-one lift in the J-even channel and a negative rank-one pressure in the J-odd channel. The Toeplitz part has exactly one negative eigendirection; it is J-even, and ss^{T} removes it precisely there. The odd channel never needed the pole at all.

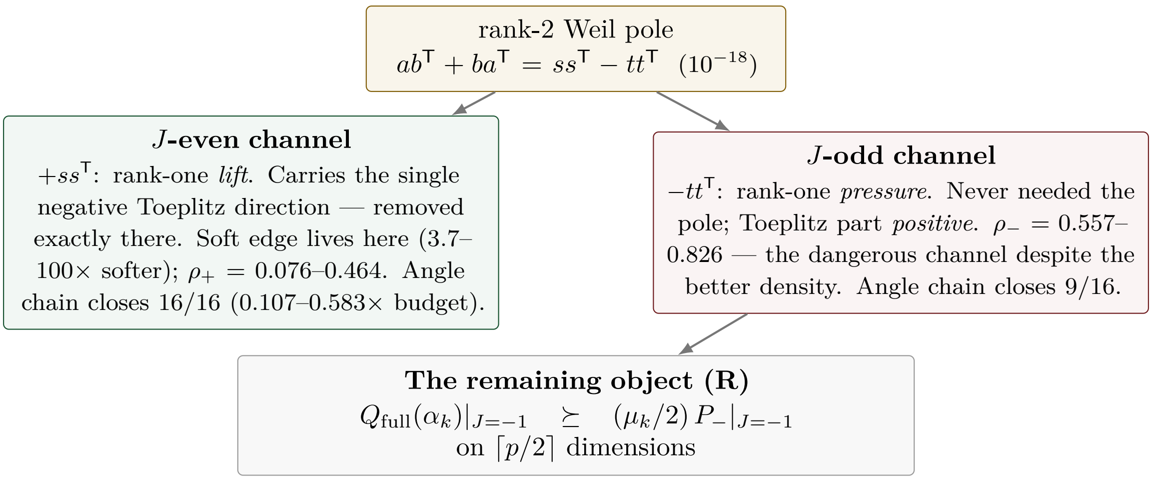

Figure 7: Parity superselection (T106). All numbers are literal; the layout is illustrative. The practical consequence: it is not the density that decides the angle but the *avoidance* — the dangerous channel is the one with the better density. T105's certified bare bound sits 16/16 exactly in the hard channel, and ρ_- is resolution-stable ($\leq 17.6\%$).

The channel-wise angle chain closes 16/16 in the even channel and 9/16 in the odd channel, against only 3/16 unsplit; all 16 handovers close at operating depth in both channels under the exact Schur criterion, with certified b_0 at 89–93%. Certified after T106: bare, the parity split, the pole splitting, the channel inertia, the avoidance law, the growth inequality, the accumulation bounds. Measured: ρ_-.

7.6 One scalar ratio (T107)

T107 (`odd_channel_closure_probe.py`, 30/30, SCALAR-TRACTABLE) reduced the matrix statement (R) to a scalar one. Sherman–Morrison gives exactly (R) $\iff s^* \le 1$ (identity at 3.4×10^{-11}; equivalence confirmed 16/16 against Rayleigh–Ritz; precondition $G \succ 0$ via Cholesky 16/16) — but s^* is *saturated* ($0.99885 \ldots 0.99999$, with only $n = 2$ at 0.2569), so no uniform s^* chain can work. The Woodbury decomposition rescues it: $s^* = \tau + \kappa$ and

$$\text{(R)} \iff r := \frac{\kappa}{\varepsilon} \le 1, \qquad \varepsilon := 1 - \tau, \qquad \text{measured } r = 0.0053 \ldots 0.1814.$$

Two orders of magnitude of headroom instead of five decimal places, and s^* is resolution-stable (spread $\le 2.60\%$ over $M = 600/900/1200$). The synthesis is stated with the usual asymmetry: certified-closed 0/16, measured 16/16 (with $q = 1.31$–4.16 at $M = 1200$, replacing T106's 0.747–1.156 — a factor 1.7–2.5 gain). Exactly one object is missing: a certified positive lower bound for $\varepsilon = 1 - \tilde{t}^{\mathsf{T}} T_{\text{odd}}^{-1} \tilde{t}$ — how far the pole vector is from exhausting the odd channel — or, since ε and κ both fall with M ($\varepsilon \sim M^{-1.99\ldots-1.69}$, fit), a certificate for the *ratio* r itself. The named route is via a wing statement of the T105 type plus an avoidance norm, explicitly not via Szegő.

7.7 The ε identity, and two scalars (T108)

T108 (`ratio_certificate_probe.py`, 44/44, EPSILON-IDENTITY) turned the missing positivity of ε from a bound to be won into an *identity*. Everything hangs on the pole response $x := T_{\text{odd}}^{-1} \tilde{t}$, and three exact identities (verified at 10^{-11}–10^{-13}) pin it down: $\varepsilon = x^{\mathsf{T}} Q|_{\text{odd}} x / \tau$ — so ε *is* a $Q|_{\text{odd}}$-energy, not an accident of subtraction — $1/\varepsilon = 1 + \tilde{t}^{\mathsf{T}} Q|_{\text{odd}}^{-1} \tilde{t}$, and the overlap of ε and κ is exactly one rank-one term $hh^{\mathsf{T}}/\varepsilon$, so that (R) $\iff \hat{r} = (\mu/2)\lambda_{\max}(\Gamma + hh^{\mathsf{T}}/\varepsilon) \le 1$. The third route supplies the classical reading: $\tilde{t}$ is exactly two-term geometry, τ a 2×2 Christoffel–Darboux form, and ε exactly the square of the last Cholesky pivot — the Szegő/Levinson prediction error. *The positivity of ε coincides with the induction positivity in the odd channel*; it is not a second, independent thing to certify.

Two consequences, and one correction of the previous picture. The consequence: (R) reduces from an $M/2$-dimensional Loewner statement to **two scalars** — $\omega < 1$ (measured 0.2366–0.7531, 16/16; the T105 structure survives the parity split verbatim, $V^{\mathsf{T}} S|_{\text{odd}} V = -\frac{1}{2} I$ at 4.9×10^{-14}) and one statement at the vector x. The chain (R) $\Leftarrow [\omega < 1] \wedge [\lambda_{\min}(Q|_{\text{odd}}) \ge (\mu/2)\tau\theta/(1-\omega)]$ has $C1$ closing 16/16 (margin 5.5–178; ≥ 1 at all 48 ladder points, 2.7–343), $C2$ 15/16, $C3$ 0/16 — the first time a margin chain in this arc is *not vacuous*: the induction margin it needs is explicitly 9.3×10^{-8}–3.7×10^{-3}, i.e. two to six orders of magnitude *below* $\mu/2$. The correction: the soft-edge picture of T107 was wrong in its mechanism — τ sits to 99.66% on modes $\lambda \ge 1$, and $\|h\|^2$ is small by sign cancellation ($\sum |k| = 401$ versus $\sum k = 1$), not by softness. T107's Cauchy–Schwarz chain turns out to be exactly the Weyl bound (slack 1.0000).

What is exactly open after T108 is one boundary value. The remaining object is a certified upper bound for the avoidance norm $\|V^{\mathsf{T}} T_{\text{odd}}^{-1} \tilde{t}\|^2$, and on 8 zones that is literally the single edge entry x_0 of an explicit vector (suppressed at $|x_0|/\max|x| = 2.9 \times 10^{-3}$–$5.7 \times 10^{-3}$, with a linear edge profile $r^{1.01}$ quoted as a fit) — a boundary-decay statement about an explicit vector, the same object class as the T105 carrier separation. $\hat{r}$ is flat in M ($6.1\times$ more stable than ε) and $r, \hat{r} < 1$ at all 92 (zone, M) points up to $M = 3000$, tightest at 0.9648: measured, on resolved zones, as always. That boundary value, and the still-uncertified ω, are exactly what the next part attacked.

7.8 Boundary certificates: cancellation, not decay (T109)

T109 (`boundary_decay_probe.py`, 29/29, BOUNDARY-CERTIFIED) closed the arc: both remaining scalars are now certified, and the whole chain for (R) closes 16/16 conditional on exactly one explicit input. Three findings and one honest remainder.

First, the mechanism. The edge value x_0 is not small because anything *decays* — it is small because a sum *cancels*. Carrier separation is excluded geometrically: $\tilde{t}$ has its maximum at $r = 0$;

the source sits on the boundary itself. The Combes–Thomas hypothesis is satisfied (T_{odd} is of Jaffard class, with local rate $\rho(s) = D(\frac{1}{2} + 2/(e^{2s} - 1))$, matched at a factor 0.85–1.00, quoted as a fit) — but its conclusion is empty: the Green sum for x_0 cancels by a factor 44–4.6×10^4 on 15/16 zones (the known outlier $n = 2$ does not cancel), and the edge profile is a nearly linear zero, $r^{0.77\ldots1.28}$ as a fit. A boundary condition, not a decay length — and no absolute-value bound can survive a cancellation.

Second, the x_0 certificate that works. Four candidates were run certified-against-measured: Combes–Thomas, evaluated with the *exact* Green row so that no constant could rescue it, is *refuted* (1/16; an upper bound on its whole class); a T-metric Cauchy–Schwarz at the edge entry closes 1/16; the exact Szegő/Levinson two-point evaluation exhibits the cancellation but bounds nothing. The fourth is the new object: a residual / goal-oriented certificate (Prager–Synge; second order in the two residuals) which *carries* the cancellation by construction instead of bounding it away. Sharpness 1.0000–2.4937 against the measured norm, 16/16 within budget.

Third, ω is cracked — unconditionally. The certificate is a graded matrix cap in PSD order: one trial level per soft direction, validity by a single Cholesky factorisation (Sylvester inertia) plus Loewner antitonicity of the inverse, $n_{\mathrm{top}} = 17$–512. It gives $\omega_{\mathrm{cert}} = 0.2561$–$0.7569$ against the measured 0.2366–0.7531 — a factor 1.004–1.124 — with *no* hypothesis input. The compression Schur distance that blocked T108 shrinks from 0.18–0.52 to 0.0024–0.039, and the ungraded cap (one level for the whole soft block, which is what T108's global route amounted to) is vacuous everywhere: the grading is the whole difference. The trial data are numerically generated; their *validity* is trial-independent (identity + Cholesky + Loewner), only their sharpness is not.

The synthesis feeds both certificates from one object ($\Gamma_{\mathrm{cert}} = S_{\mathrm{cert}}^{-1}$): the chain closes 16/16 with margin 1.3–10.4 and is M-stable (44/48 ladder points to $M = 3000$; the certification price 1.5–28.6 does not grow). What remains open is exactly **one input**: the *strict* positivity margin of the odd channel, $\lambda_{\min}(Q|_{\mathrm{odd}}) \geq 1.7 \times 10^{-6}$–$7.5 \times 10^{-3}$ against a measured 1.4×10^{-5}–5.1×10^{-2} — two to six orders of magnitude *below* $\mu/2$, i.e. strictly weaker than the conclusion it buys; the induction hypothesis as it stands supplies only the non-strict $Q|_{\mathrm{odd}} \succeq 0$. Declared caveats: floating point is not audited, and $\lambda_{\min}$ is read through Rayleigh–Ritz. Whether that one number propagates itself through the induction is exactly what the next part measured.

7.9 The margin propagates: the circle closes on the measured zones (T110)

T110 (`margin_propagation_probe.py`, contract `MARGIN.PROPAGATION`, 28/28, MARGIN-PROPAGATES — with a precision statement) asked the only question T109 left open: does the strict scalar margin $m_k = \lambda_{\min}(Q|_{\mathrm{odd}})$ propagate itself along the zone ladder, against the explicit demand need109(k) of the T109 chain? Three findings close the circle on the measured zones; three gaps state precisely what a theorem would still need.

The margin map. On a common grid (16 zones, $\delta_0 = 0.00284$) the margin falls monotonically, 6.95×10^{-2} ($n = 2$) to 1.66×10^{-5} ($n = 29$), against need109 $= 3.09 \times 10^{-7}$–3.87×10^{-4}: $m_k \geq$ need109 on 16/16 zones, with ratio 2.09–179.6. Dilution during pure window growth is weak — a factor 1.02–2.05 per atom gap, i.e. 0.993–0.997 per cell, a $1 - O(1/M)$ loss; the scaling fits are honestly poor (the power law wins 7/16). The structural finding of the part is the atom entry: it costs *nothing*. The naive expectation was a $-(\mu/2)$ pressure on the odd channel; measured, the entry *raises* $\lambda_{\min}$ on 15/15 handovers (the first-order Kato term has helping sign), and at $n = 29$ the atom-free window is not even positive (-9.2×10^{-3}): the atom makes the window co-positive. The T106 killer argument — transport destroys a demand that *points* somewhere — has no analogue for the scalar floor.

The step law. The growth step splits exactly (embedding error $\leq 2.6 \times 10^{-14}$), and $\max|\mu N| = 0.0$ on 15/15: the new atom lies outside the old lag reach (gap 0.0606 against $\delta_0 = 0.00284$), so its restriction to the old window is the exact zero matrix — everything is decided by the bordering. Scalar routes fail decisively: bordered Weyl closes 0/15 and the Schur/Friedrichs scalar cap is vacuous 15/15 (both recorded in the kill list). What passes certified is the **graded Loewner minorant** — its validity condition is exactly the induction hypothesis, and the level is then found

RH $\iff$ Weil positivity $\iff$
I5 (one-family form; T79/T82, ledger `v541`)

per-zone **handover induction**: $D_k(\alpha) \leq \mu_k/2$ (T98; holds 64/64 on the 16 resolved zones)

Schur profile sandwich: $\sigma_k(\delta) \geq \mu_k/2$ (T102; μ_k linear, onset anchored at $\delta_c = 2w_k$)

bare_k **certified** + soft dressing L (T105; b_0 closed form, positive 16/16 at 81.1–92.7%)

parity split: even channel closes 16/16; hardness $\to$ odd channel (T106)

(R): $Q_{\text{full}}|_{J=-1} \succeq (\mu_k/2)P_-|_{J=-1}$ on $\lceil p/2 \rceil$ dimensions (T106)

$\iff s^* \leq 1$ (saturated) $\iff r = \kappa/\varepsilon \leq 1$ (T107; measured 0.0053–0.1814)

$\iff$ two scalars: $\omega < 1$ and one statement at $x = T_{\text{odd}}^{-1}\tilde{t}$ (T108; ε identity; boundary value x_0)

both scalars certified: $\omega_{\text{cert}} < 1$ unconditional (graded matrix cap) + residual certificate carrying the boundary cancellation (T109; closes 16/16, margin 1.3–10.4)

open **input**: one strict-positivity margin $\lambda_{\min}(Q|_{\text{odd}}) \geq 1.7 \times 10^{-6}$–$7.5 \times 10^{-3}$ (T109; 10^2–10^6 below $\mu/2$ — the cascade ends here)

equivalence *typing*

architecture of a *sufficient* route, on finite windows

exact reduction

exact reduction

two certificates

one input left

The asymptotic wall. Every step below the top line is established on *16 resolved zones* at finite resolution. Sixteen zones are not all zones; no uniform-in-k constant is certified anywhere in this cascade.

Figure 8: The reduction cascade, with its arrow types made explicit. The top line is an *equivalence typing* produced by an exact identity (Section 5); the arrows below are the architecture of a would-be sufficient route, verified on finitely many windows only. Since T109 the cascade ends at one strict-positivity margin input; T110 shows that this input propagates itself along the measured ladder (certified base case, graded Loewner minorant), leaving three gaps — which T111 then drives deep and measures as three separate walls (the two subsections below). The dashed box is the reason this figure is not a proof sketch.

by bisection through a Cholesky/Sylvester factorisation: with one soft direction ($n_{\text{soft}} = 1$) it holds 15/15 with retention 1.0000; the strictly scalar variant ($n_{\text{soft}} = \dim$) closes 0/15.

Base case and circle test. The base case $n = 2$ is *certified* by an explicit Cholesky of $Q|_{\text{odd}} - \lambda I$ at three grid resolutions ($\lambda_{\min} \geq 6.93 \times 10^{-2}$, a factor 88–180 over need). The circle test then runs the whole chain from that certified base value: the strictly scalar induction breaks at $k = 3$ (its Schur factor compounds to 2.7×10^{-36}), but the graded chain runs through completely — 16/16 zones above need109, each step one Cholesky, with the propagated floor as the only input. On the measured zones, the induction circle closes end to end.

The three gaps — the asymptotic mountain, made precise

1. **No reserve.** f_{crit} at the first handover is 1.00; the chain lives on a retention of $1 - 2 \times 10^{-7}$, and a factor-2 loss anywhere would break it within two steps. What a theorem needs is a step law *without* loss.
2. **No scalar step law.** Structurally excluded, not merely missed: the new cells attach at the boundary layer where the pole source sits, so any one-number summary of the bordering is blind to exactly the direction that decides it.
3. **No uniformity in k.** Every certificate above is one finite Cholesky; flat-in-k is a measurement, not a theorem. And the measured trend warns rather than reassures: $m_k \sim n^{-1.93}$ against need109 $\sim n^{-0.98}$, ratio $\sim n^{-0.96}$ (fit, rms 0.735) — extrapolated crossing at $n \approx 170$. Propagation gets *harder* with k, not easier. (T111 subsequently *measured* that crossing: it sits at $n^* \approx 462$, not 170 — the extrapolation was a fit artefact of the 16-zone window, but the wall itself is real. See the next subsection.)

7.10 The three walls (T111)

T111 (`deep_zone_stress_probe.py`, contract `DEEP.ZONE.STRESS`, 23/23, CROSSING-CONFIRMED) stopped extrapolating and measured. On exactly the T110 cell grid — the first 16 zones reproduce T110 bit-exactly — the ladder was extended to *every* prime-power zone $n \leq 1000$ plus a thin tail to 3001: 199 zones in the margin map, with the chain ladder running $n = 2 \dots 521$ through 117 handovers.

The margin wall, measured. The ratio m_k/need109 falls $179.59 \to 0.0174$ across the deep ladder, and the crossing is now a measurement, not a fit: the last zone with ratio ≥ 1 is $n = 461$, the first below is $n = 463$ (ratio 0.935). The $n \approx 170$ figure of T110 was a fit artefact of the 16-zone window — the wall sits much further out, but it is real. The failure follows the *prime branch*: prime-power zones (512, 529, 625, 729), whose atoms are weaker, survive longer than the primes around them. The fit families were again compared honestly (best by AIC: a broken-scale law with $n^* = 501$; the jackknife band of the pure power law is $[298, 352]$), and a depth-preserving resolution study shows that refinement pushes the ratio *down*: $n^* \approx 462$ is an *upper* bound, not an estimate to be rounded up.

The ladder wall, arithmetic. The uniformity object is no longer flat: n^*_{soft} runs $1 \dots 8$ across the 117 handovers. But retention stays 1.000000 at every step (largest single loss 5.96×10^{-8}), and the atom entry stays constructively free on 117/117 — the restriction of the new atom to the old window is the exact zero matrix, deep into the ladder. The new and harder finding is the *ladder wall*: the nested step needs a lag gap exceeding twice the window depth, and the twin-prime pair $521 \to 523$ (gap $0.003831 < 2D = 0.005617$) kills the ladder *arithmetically* — at frozen cell width, $n = 521$ is the deepest reachable zone, and the bound is set by twin primes, not by spectra. An arithmetic scan to 40 000 finds 4052/4284 consecutive prime-power pairs endangered; letting D shrink like $1/n$ instead forces the cell count to grow like $n \log n/4$ (horizon $n \approx 690$ at $h \leq 1500$).

The circle tears at the ratio, not at a step. The no-reserve finding of T110 softens: $f_{\text{crit}} \sim n^{-0.39}$ (fit) — the reserve *opens* with depth, and the bottleneck stays at the first step $2 \to 3$. The circle test runs from the certified base case (factor 179.6) and breaks first at $k = 109$, $n = 463$ — exactly where the margin map measures the crossing. All 117 handovers certify at retention 1.000000, including beyond the break: what fails is the demand ratio, never the handover mechanism.

The three walls — and the reinterpretation

The hardness balance names the driver: the growth of $1/\kappa$ (exponent $+0.86$) against the margin (exponent -0.68). Three separate walls, in order of appearance:

1. **The margin wall**, $n^* \approx 462$: measured, an upper bound (refinement pushes it down); the T109-chain demand ratio crosses 1 between $n = 461$ and $n = 463$.
2. **The ladder wall**, $521 \to 523$: purely arithmetic — at frozen cell width the nested handover dies at the first twin-prime gap below $2D$, independently of any spectral quantity.
3. **The requirement wall**, $n = 727$: $\omega_{\text{cert}} \geq 1$ makes need109 vacuous on 46 deep zones — beyond it the T109 chain no longer states a demand at all.

The reinterpretation that comes with the measurement: the operating variable is the window *depth*, not the zone index n — doubling the cheeks moves the margin wall from 463 to 47. A proof along this route must beat the exponent gap $-0.974 = -0.681 - 0.293$ and couple the depth δ_0 to the local prime gap.

T112 subsequently ran the construction in exactly that coupled frame (next subsection): walls (2) and (3) fall *structurally*, wall (1) proves frame-invariant.

7.11 The scaled frame: two walls fall, the margin wall is frame-invariant (T112)

T112 (`adaptive_scaling_probe.py`, contract `ADAPTIVE.SCALING`, 20/20, SCALING-PARTIAL) stopped freezing the cell width and coupled it to the local scale. Two couplings were built: frame A, gap-coupled, $D_k = g_k/(2\nu)$ with g_k the local prime-power gap and ν the resolution constant of the frame; and frame B, mean-field, $D_k = \log n/(2\nu n)$ — the smooth PNT version of the same idea. The first finding is a structural asymmetry: *frame B has no ladder* (only 38% of consecutive pairs nest; the first failure is already at $n = 3$) — the nested handover lives exclusively in the gap coupling, not in any smooth $1/n$ law. T105 admissibility, re-measured in the scaled frame, sets a resolution floor $\nu \geq 4$ (at $\nu = 1$, 3/10 test zones are inadmissible). The flatness verdict is honest and double-edged: the ratio is *not* flat, but the exponent is *frame-invariant* — the components shift massively (m_k: $n^{-1.93} \to n^{-0.95}$; need109: $n^{-0.98} \to n^{+0.03}$, i.e. practically n-independent in the scaled frame), yet the difference stays -0.974.

The three walls, scaled. The **ladder wall is structurally gone**: with $D_k = g_k/(2\nu)$ the nested step has exactly ν cells per end *by construction* — twin pairs included — verified as an exact arithmetic lemma over all zones; the atom entry is the exact zero matrix by construction rather than by arithmetic accident; and the T111 killer pair $461 \to 463$ now certifies *spectrally* ($h = 1418$, retention 1.000000), with the reserve *opening*: $f_{\text{crit}} = 8.1 \times 10^{-4}$–$4.2 \times 10^{-3}$ instead of 1.00 — roughly a factor 1000. (The pair $521 \to 523$ would need $h = 1634$, above the dimension cap: arithmetically co-certified, spectrally declared honestly as not run.) The ω **wall is no longer a depth wall**: the depth coefficient is $n^{-0.079\pm0.121}$, compatible with zero — ω_{cert} is driven by the resolution and the local gap, not by the depth of the ladder; a residual requirement-vacuum tail of 5 zones on the deepest ladder remains and is not booked as "gone". The **margin wall stays**, practically unmoved: the first sub-1 zone sits at $n = 449$ against the frozen frame's 461–463 (a factor 1.07) — it is not geometry but the substance of the T109 requirement itself.

The limit object. Overlaying the scaled windows, the archimedean part (0.657) and the pole part (0.558) contract Cauchy-like with amplitude laws $D^{-0.26}$ and $D^{+0.37}$; the atom part does *not* contract (0.935) — it is the prime-gap statistics itself, written in scaled coordinates. The spectral shape does not drift: $\lambda_2/\lambda_1 \sim n^{-0.029\pm0.052}$. The resulting formulation is a split object: *[a deterministic limit shape: arch + rank-one pole] + [the atoms as a controlled, non-convergent perturbation]*. New and uncomfortable is the **regrid object**: the zone frames are not nested, so Rayleigh–Ritz transfers nothing between them; the regrid cost is measured at rate $(D'/D)^{+2.93}$ — an object that simply does not exist in the frozen frame, disclosed here because any limit argument has to pay it.

The trade — and the disclosed prime-gap dependence

Three arithmetic walls are exchanged for two analytic demands:

[P1] **Positivity of the limit operator** — *one* operator instead of a sequence of matrices; the deterministic part (arch + rank-one pole) contracts, the atom part is the perturbation to be controlled.

[P2] **A convergence rate below the step reserve** — the Cauchy rates and the regrid cost against the opening reserve f_{crit}.

The prime-gap inputs are disclosed rather than hidden. Upward, Bertrand–Chebyshev 1852 ($g_k \le \log 2$; used and verified in-run) and the trivial even-gap bound suffice. The only remaining arithmetic parameter is the normalised gap $\theta_k = g_k\, n/\log n = 0.18$–$4.12$ — and downward, θ is *provably* unboundedly small: by Zhang 2014 / Maynard 2015 (infinitely many bounded gaps), $\theta \to 0$ on subsequences, so any gap-coupled construction must be uniform in $\theta \to 0$, at the price of the $1/\theta$ cost explosion.

T113 subsequently answered the currency question (next subsection): the wall is currency-invariant — and what it measures is the discretisation, not the spectrum.

7.12 The currency question: the wall is substance, and it measures the discretisation (T113)

T113 (`limit_operator_probe.py`, contract `LIMIT.OPERATOR`, 27/27, SUBSTANCE-CONFIRMED) answered the question T112 left open — is the margin fall in the scaled frame a normalisation artefact? — and the answer reinterprets the program's hardest number.

The currency table. The floor/requirement ratio carries the *same* exponent, $n^{-1.168\pm0.259}$, in all five currencies tested (raw, per $\lambda_{\max}$, per trace density, per D, per D^2) — and the invariance is structural, not accidental: the piecewise-constant basis is L^2-orthonormal (Gram matrix exactly I, so there is no free D-power to spend), and the zone matrices are restrictions of *one* quadratic form to subspaces of the same L^2 space (cross-grid form identity at relative 1.2×10^{-14}). The floor's homogeneity degree in the cell width is exactly 1.0; the requirement's is 1.0 under every legitimate rescaling. No flat currency exists. The wall is currency-invariant: substance, not bookkeeping.

Two surprises about what that substance is. First, the limit object is *not* "arch + pole with the atoms as a perturbation": the atom-free window form is deeply negative ($\lambda_{\min} = -3.08$ to -19.36, growing like $n^{+0.60}$), the atom block is equally large (norm 3.21–19.44), and the positive floor survives only as a *cancellation* of relative size 1.3×10^{-7}–9.7×10^{-5} — on the floor vector, -11.64 against $+11.64$. The Weyl bound is saturated: norm perturbation theory is five orders of magnitude too coarse for this object. Second, there is *no plateau under refinement*: at a fixed window, on nested grids over 8 levels (a factor 128 in resolution), $\lambda_1 \sim D^{1.83}$ and $\lambda_2 \sim D^{1.76}$ — the *same* power, with λ_2/λ_1 flat. **The continuum window form has no gap; m_k measures the gap of the discretisation.** This resolves the T112 tension exactly: λ_2/λ_1 constant while the floor falls is what it looks like when both eigenvalues carry the same D-power.

The atom part is almost invisible at the floor. The entering (deepest) atom contributes almost nothing to the coupling (8.3×10^{-11}–1.3×10^{-6}) — which is why handover steps certify at retention 1.000000 while the margin tears. The floor is almost pure geometry: $\log m = \mathrm{const} + 1.87 \log D - 2.07 \log \alpha$, with an arithmetic residual of a factor 1.13 and no $\Lambda(n)$ dependence. The $\theta \to 0$ stress test separates cost from substance: the operator norm does *not* diverge ($\theta^{-0.005}$); an extra atom at the window edge shifts the floor by only 0.2–0.4%, while the same atom in the interior shifts it 10^7 times more — the $1/\theta$ explosion of T112 is purely a *cost* statement ($h \sim \nu n/\theta$ cells), not an operator instability.

Regrid, re-read. The raw regrid rate $(D'/D)^{3.85}$ is mostly the floor's own D-power — a normalisation jump, not an instability; the reserve certifies $f_{\mathrm{crit}} = 9.5 \times 10^{-2}$ (better than the T112 estimate), and after subtracting the D-power the reserve buys 15 steps (against 0 on the raw jump).

The new hardness balance — replacing [P1]/[P2]

The T112 trade is restated by measurement; this balance supersedes the earlier two-demand form.

[P1] **Semidefiniteness of the balanced form without a gap** — the continuum window form is semidefinite at best; strict limit-floor formulations demand what refinement demonstrably refuses.

[P2] **Certify the D-power itself** — a Grenander–Szegő-type consistency statement for $\lambda_{\min}$ of the scaled form; the power, not an abstract rate, is the object.

[P3] **New, and now leading: the requirement homogeneity** — need109 carries $D^{0.63}$ against the floor's $D^{2.38}$: the T109 chain divides by an *artifact* margin. The repair the measurements point at is a *margin-free step certificate* — a Schur/Loewner formulation certified from semidefiniteness alone, never dividing by a margin.

[P4] $\theta \to 0$ **is a cost-uniformity demand only** — the frame needs $h \sim \nu n/\theta$ cells, but the operator itself is θ-stable.

Disposition after T114 (next subsection): [P3] is **discharged** — the margin-free step certificate exists and runs past the wall; [P4] stands as a pure cost statement; a new **[P5]** appears (the (R) demand is grid-bound and is deleted, not transported); and [P1]/[P2] are restated as the two-object core — ε relative to κ, and Schur transport between grids.

T114 (`margin_free_step_probe.py`, contract `MARGIN.FREE.STEP`) then rebuilt the chain without the margin division — and the wall dissolved.

7.13 The wall dissolves: the margin-free step (T114)

T114 (`margin_free_step_probe.py`, contract `MARGIN.FREE.STEP`, 22/22, WALL-DISSOLVES) built the certificate that [P3] demanded, and the answer to the preregistered question — does the margin wall disappear as a requirement artefact? — is yes.

The margin-free step. The structural fact that makes it possible was already on the table (T110/T111): the restriction of the grown window's form to the old window is bit-exactly Q_{old} — the entering atom's block is the exact zero matrix — so margin-freeness *is* that property, once the step stops asking for more. Route (i) is Albert 1969: the generalised Schur criterion with the Moore–Penrose inverse,

$$Q' = \begin{pmatrix} A & C \\ C^{\mathsf{T}} & X \end{pmatrix} \succeq 0 \iff X \succeq 0, \quad \operatorname{ran}(C^{\mathsf{T}}) \subseteq \operatorname{ran}(X), \quad A - CX^{+}C^{\mathsf{T}} \succeq 0$$

(the range condition is Douglas 1966) — no margin appears anywhere. It certifies 27/27 ladder steps, *eleven of them beyond the old wall* (467, 479, 512, 529, 547, 577, 661, 773, 887, 1129, 1331; deepest $h = 1495$). The exact 4×4 Schur complement is an $O(0.1)$ object with *no* cancellation: $\lambda_{\min}(S) = 0.068$–$0.154$, i.e. 42–67% of the block scale. Route (ii), the T110 graded minorant in its degenerate limit $x_{\text{in}} = 0$, dies structurally on 27/27 — which retypes T110's margin: it was not paying for positivity but for an *abandoned direction.* Route (iii), unshifted Cholesky, passes 27/27 against the declared backward-error floor. Negative controls are sharp (5/5), and the certificate is frame-invariant at $\nu = 8$ (3/3). All seven zones at which the T109 chain tears — including the wall zone $n = 449$ itself, where need109$/m = 1.241$ — are certified by the margin-free step.

The wall in one line. The same quantity, bounded through norms (Weyl: $\lambda_{\min}(A) - \|C\|^2/\lambda_{\min}(X)$), is *negative* by a factor of 2.4×10^5–9.6×10^7: an $O(1)$ numerator divided by the 10^{-6} artifact floor. Every norm route had to fail — and the exact Schur complement passes through none of it.

The favourable power direction. On nested refinement chains at a fixed window, $\varepsilon \sim D^{1.77\text{–}2.01}$ and $\kappa \sim D^{3.64\text{–}4.84}$ (fits) — the preregistered "same power" expectation is refuted *in the favourable direction*: κ falls faster, so the scale-free ratio $r = \kappa/\varepsilon \sim D^{1.6\text{–}3.1}$ shrinks under refinement. The ratio closure $r = 2 \times 10^{-4}$–0.103 holds on 27/27 steps (11/11 beyond 463). The circle [base semidefiniteness (a declared hypothesis input) + margin-free step + ratio closure] closes 27/27; ten multi-step chains run (21 certified steps, the longest of length 4); and *every* chain ends at the cap

$h \leq 1500$ — a window *cost* — never at a step. Regrid honesty: 0/464 gap ratios are dyadic, so the non-nested transport object of T112/T113 remains real.

[P5]: the (R) demand is grid-bound. Holding the physical demand $(\mu/2)P_{-}$ fixed and refining the grid, ω grows monotonically (5/5) and crosses 1 (4/5): the (R) formulation is *false in the continuum* — it is to be *deleted*, not transported. The exact Schur complement needs no demand surrogate in its place.

Cancellation robustness. The 10^{-7}-relative cancellation of T113 has 5.3–8.8 decimal digits of headroom above the Cholesky backward-error floor (Wilkinson/Higham), and it sits almost entirely in the rank-one pole — whose positivity is exactly the T108 Szegő identity ε.

The new core — two objects

The four-part balance collapses. [P3] is discharged (the margin-free step certificate exists and runs past the wall), [P4] was only ever a cost statement, and [P5] deletes (R). What is left:

[P1] $\varepsilon > 0$ **with controlled size relative to** κ — the exact Szegő/Levinson identity of T108, now the single positivity object of the program.

[P2] **Transport of the exact Schur complement between non-nested grids** — an $O(0.1)$-sized object with no cancellation, sharpened from the earlier abstract D-power demand.

Disposition after T115 (next subsection): [P2] **splits** — the transport across a real regrid certifies only mild refinement ($\rho \lesssim 1.8$), for a principled reason (the Schur floor itself falls under refinement), while a two-scale compression breaks the compute cap; [P1] **sharpens** to one scalar inequality, the classical Szegő–Levinson prediction-error bound. The core is restated as the three-point list at the end of the next subsection — only one point of which is an inequality.

T115 (`schur_transport_probe.py`, contract `SCHUR.TRANSPORT`) then ran the two last bricks — and split them.

7.14 Transport blocked, cap broken: the two-scale compression (T115)

T115 (`schur_transport_probe.py`, contract `SCHUR.TRANSPORT`, 26/26, TRANSPORT-BLOCKED — with a large compression gain) took the two-object core apart: transport of the exact Schur complement across the ladder's non-nested grids, and a multi-resolution compression of the already-certified interior.

The transport splits at $\rho^* \approx 1.8$ — for a principled reason. Haynsworth's partial minimisation $y^{\mathsf{T}}Sy = \min_z\,[y;z]^{\mathsf{T}}Q'[y;z]$ makes $\lambda_{\min}(S)$ available without any inverse and without a margin; the two-sided transport bracket built from it is an *identity* (bookkeeping), with the evidence sitting in the consistency defect η and in the lower end. On the eleven real regrid pairs of the ladder ($\rho = 1.023$–3.425) the lower end is positive on 7/11, with a clean split: every pair with $\rho \leq 2.061$ certifies and none with $\rho \geq 2.291$ does; a synthetic scan puts the threshold at $\rho^* = 1.83$, against a median real refinement ratio of 2.001 — 39% of the ladder's 7686 refinement pairs are covered. The blocker is *not* the transport machinery: the exact cell-overlap projection beats three deliberately wrong transfer maps on 9/9 (by factors up to 270), and on *nested* ladders — where the transport error is exactly zero — $\lambda_{\min}(S)$ itself falls like $\rho^{-1.73}$ to $\rho^{-1.65}$. The drop is real, so no bound can undo it. Two sharpenings came free: non-nestability is a matter of *integrality*, not dyadicity (0/14 807 gap ratios are integer, the closest miss 3.4×10^{-5}), and the measured defect $\eta = 2.7 \times 10^{-2}$–$8.1 \times 10^{-2}$ is 10^3–10^6 times smaller than its a-priori Céa/Strang norm surrogate — which is exactly where the margin division would have re-entered.

The compression breaks the cap. The two-scale window (interior merged into blocks, the boundary strip kept fine, the merge anchored at the centre) preserves $X_{\text{mixed}} = Q_{\text{old,mixed}}$ *exactly* (rel 0.0) — the compressed step is still margin-free — and Albert certifies 66/66 (zone, q) combinations up to $q = 64$, with the compression error certified *one-sided and second order* in the projection defect (Rayleigh–Ritz plus stationarity of the fine minimiser; Céa/Strang; 66/66) and savings down to $m/h = 0.043$. The deep run then breaks the T114 cap: the margin-free step certifies at $n = 155\,921$ (117× T114's 1331), on a fine lattice of $h = 93\,470$ cells (62× the hard cap $h \leq 1500$) compressed to $m = 1490$; re-coarsening inside a chain is free (Rayleigh–Ritz, rel $\leq 2.1 \times 10^{-12}$); the longest chain is now *ten* steps (T114: four), with 33 certified steps over four chains; the stopper is the cost cap on 3/4 runs and a failing step on 0; and every certified quantity sits 10^5–10^{11} times above the declared Cholesky backward-error floor. Stated honestly: $\lambda_{\min}(S_{\text{mixed}}) \approx 5.1$ is $52\times$ the uniform value because the coarser Galerkin space sees fewer soft directions — a weaker statement about the continuum, declared as such, and flat across the whole deep band. The contiguous reach an induction needs moves from $n \leq 125$ to $n \leq 5437$ (factor 43.5): the $n \log n$ cost barrier is *divided*, not broken.

The remaining list — three points, one inequality

The shortest list the program has produced. From T114's balance, the margin (dissolved), the (R) demand (grid-bound, deleted), the floor/ε transport (the object to transport is $\lambda_{\min}(S)$, the slowest-falling of the three) and the compression risk (it does not break the step) are all *struck*. What is left:

(1) **[P1] as one scalar inequality** — given $T = Q + \tilde{t}\tilde{t}^{\mathsf{T}} \succ 0$, positivity of $Q|_{\text{odd}}$ is *equivalent* to $\varepsilon = 1 - \tilde{t}^{\mathsf{T}} T^{-1} \tilde{t} \geq c(D) > 0$: the classical Szegő–Levinson prediction-error bound for one explicit symbol (the Haynsworth double-Albert device: $T \succ 0 \wedge \varepsilon \geq 0 \iff Q \succeq 0$). This is the target-sentence candidate. Measured: $\varepsilon \sim \rho^{-1.71}$, falling slightly faster than $\lambda_{\min}(S) \sim \rho^{-1.69}$ (fits).

(2) **An a-priori bound on the transport defect** η — a regularity statement about the minimiser z^*; it would turn the measured transport into a certificate for $\rho \leq 1.8$.

(3) **A boundary formulation** (suggested by the exact-zero lemma) — never represent the old interior at all; it would remove the compute cap *and* the regrid at once.

Only (1) is an inequality; (2) is a regularity statement, (3) is a reformulation.

Disposition after T116 (next subsection): (3) is **run and splits** — the boundary recursion itself is exact (the pole rides in the state with no truncation, and the march cost is flat), but the state cannot be *bounded*: the prime comb refuses compression. (2) is **devalued** — a single boundary march needs no regrid, so the transport defect no longer sits on the critical path. (1) **sharpens** into a two-variable law with a named classical proof technique: ε is a Galerkin error, Aubin–Nitsche duality predicts the measured D-exponent, and the jumps at prime-power entries become an explicit side-condition.

T116 (`boundary_formulation_probe.py`, contract `BOUNDARY.FORMULATION`) then ran the reformulation — and it held exactly where the contract feared it would break, and broke exactly where it seemed free.

7.15 The boundary recursion: exact pole, flat cost, and the prime comb (T116)

T116 (`boundary_formulation_probe.py`, contract `BOUNDARY.FORMULATION`, 33/33, RICCATI-PARTIAL) executed point (3) of the list: never represent the old interior at all — run the induction as a pure boundary process.

The recursion stands, and the pole is free. The state is the triple (Σ, r, σ) obtained by partially minimising the pole-free form over the interior (Haynsworth 1968): the identity $t^{\mathsf{T}} T^{-1} t = r^{\mathsf{T}} \Sigma^{-1} r + \sigma$ holds on six zones at rel 6.7×10^{-10}, the Haynsworth double-Albert equivalence certifies 6/6 with sharp negative controls, and the Riccati-type chaining of two eliminations is *exact* (synthetic 1.4×10^{-16}; bit-exact on real windows). The first feared obstruction — pole bookkeeping — dissolves for free: the rank-one pole is genuinely *global* (its boundary value at half depth is 38.6%, against 0.2% for the archimedean part), but the pair (r, σ) carries it *exactly* through bordered elimination, with no truncation anywhere.

The tail bookkeeping breaks — on the prime comb. The second feared obstruction was the archimedean kernel tail, and it is innocent: the arch part decays cleanly, $\exp(-1.73\,\omega)$. What does not decay is the *full* symbol — it stays at $\geq 93\%$ of its maximum beyond every tested width, and 100% of the off-band mass sits in the stripes of the two *prime combs*, carried by the reflection comb: every incoming cell couples through a Hankel term to the interior cell at $\alpha - \log n_j$, for *every* prime power $\sqrt{n} < n_j < n$. A truncation of the state would have to discard 1.3×10^2–1.1×10^6 times the margin ε it must respect, a budget that grows like $n^{2.17}$ ($\mu_j \sim n^{-1/4}$ against $\varepsilon \sim n^{-1.8}$; fits); 11/18 real truncations break outright; a comb-aware state that keeps the stripes still spans 83.6% of the window and reaches only rel 20; and the ceiling of the whole idea is a $\Theta(\log^2 n)$ gain. There is no sparse faithful state — the stripes drift with fill-in.

The deep boundary run — a cost-geometry demonstration, declared as such. One unbroken chain of 169 236 prepends ran to a window of $h = 1\,354\,088$ cells — 903× the old cap — with the entire state $12 \times 12 + 12 + 1$ numbers and the cost *flat*: 76 μs per step, first-to-last-decile ratio 1.00. Stated honestly, this is *not* a Weil certificate: at the fixed cell width $D = 1.9 \times 10^{-6}$ the band contains no atom, the stopper is $\varepsilon < 0$ at $h = 1.35 \times 10^6$ — the *band model* loses positivity there, not the machine — and the atom reach stays at $n = 173$. What the march certifies is the cost geometry of the boundary formulation.

The one inequality, sharpened into a law with a proof technique. The measured law is $\varepsilon \approx 8.34\, D^{1.790}\alpha^{-6.04}$ (jackknife errors ≤ 0.041, rms ≤ 0.097; φ from a fixed-D sweep, ± 0.08; all fits). ε is *not* smooth in α: it jumps by factors up to 120 at prime-power entries, so any smooth ansatz is wrong from the start. (T117 subsequently resolved the same sweep cell by cell and *corrected* this reading: the entries themselves move ε *up*, and the factor-120 falls are the accumulated smooth bordering decay between sweep points — see the T117 subsection. The criticality and D-law statements of this paragraph are unaffected.) The continuum form is *exactly critical* — $\tilde{t}^{\mathsf{T}} T^{-1} \tilde{t}$ reaches 0.999975745, and Szegő–Kolmogorov says the pole is representable in the limit — with every measurement $\geq 1.1 \times 10^5$ above the declared Cholesky floor. The classical hit: in the Galerkin reading, $\tilde{t}$ represents the D-independent function $2\sinh(x/2)$, ε is the error of a piecewise-constant Galerkin method for it, and Aubin–Nitsche duality predicts $\varepsilon \sim D^{2s}$: the measured $\theta = 1.79$ gives $s \approx 0.90$, exactly the regularity a log singularity plus window edges would produce. The target inequality is now formulated: $\varepsilon(\alpha, D) \geq c_0\, D^{\theta} \alpha^{\varphi}$, uniformly across the jumps.

The remaining list after T116 — one inequality, one obstruction

Won: the exact pole (no truncation), the exact Riccati chaining, and flat cost per step. Broken: **[SUPPORT]** — the prime comb; there is no sparse faithful state, and the stripes drift with fill-in. Devalued: **[P2]** — a single boundary march needs no regrid, so the transport defect leaves the critical path. The basis is uncritical. What remains on the critical path:

(1) **[P1] as a two-variable law** — $\varepsilon(\alpha, D) \geq c_0\, D^{\theta} \alpha^{\varphi}$ with $\theta = 1.790$ and $\varphi = -6.04$ (fits), to be proved *across* the factor-120 jumps at prime-power entries; the proof technique is named (piecewise-constant Galerkin error for $2\sinh(x/2)$, Aubin–Nitsche duality, $s \approx 0.90$).

(2) **The comb as the honest support obstruction** — any bounded-state formulation must either carry the reflection stripes ($\Theta(\log^2 n)$ ceiling) or find a representation in which the comb is diagonal.

Disposition after T117 (next subsection): (1) is **theorem-shaped** — the Galerkin reading becomes four exact identities, the lower-bound direction acquires a certified two-level chain that loses no power of D, and the jumps acquire exact closed forms with the *opposite* sign: entries *raise* ε, the factor-120 falls were a sweep artefact, and the jump side-condition is *removed* from the ansatz. What is left of (1) is three named analytic lemmas about one symbol, two of them constants. (2) is untouched.

T117 (`epsilon_theorem_probe.py`, contract `EPSILON.THEOREM`) then took the inequality apart — keeping the one honest asymmetry visible everywhere: the classical Galerkin machinery produces *upper* bounds for approximation errors, and the target needs a *lower* bound.

7.16 The one inequality becomes theorem-shaped (T117)

T117 (`epsilon_theorem_probe.py`, contract `EPSILON.THEOREM`, 23/23, THEOREM-SHAPED) converted the T116 reading into identities and a certified chain. The direction problem is stated once and kept visible: Céa 1964, Aubin 1967/Nitsche 1968 and Bramble–Hilbert 1970 all bound a Galerkin error from *above* by an interpolation error; the missing lower-bound ingredient carries a classical name — the *saturation assumption* (Bank–Smith 1993; Dörfler–Nochetto 2002) — and is known to be an assumption, not a theorem. Every classical anchor below is therefore cited with the direction it actually gives.

The Galerkin identity, exact. Four identities, verified on six windows to 10^{-10} relative or better, none of them asymptotic: (i) $\tilde{t}$ is the cell functional of the *D-independent* function $f(x) = 2\sinh(x/2)$ (independent quadrature, rel 5.2×10^{-16}); (ii) the family is *exactly nested* — with P the block-averaging prolongation, $T_c = P^{\mathsf{T}} T_f P$ and $\tilde{t}_c = P^{\mathsf{T}} \tilde{t}_f$ (rel 2×10^{-14}) — so ε is a Galerkin best-approximation error of *one* bilinear form on a nested family, and its monotonicity under refinement is a *theorem*, not a fit; (iii) $\varepsilon = \min_v [\,1 - 2\tilde{t}^{\mathsf{T}} v + v^{\mathsf{T}} T v\,]$ with the two-level Pythagoras $\varepsilon_c - \varepsilon_f = \|u_f - P u_c\|^2_{T_f}$ (rel 6.2×10^{-11}); (iv) the dual-norm residual form $\varepsilon_c - \varepsilon_f = \|\tilde{t}_f - T_f P u_c\|^2_{T_f^{-1}}$, whence $\varepsilon > 0 \iff \tilde{t} \notin T(V_c)$: positivity is a *non-membership* statement — which is exactly why the quantitative version is hard, since a non-membership statement carries no rate by itself.

The lower bound, honestly. The crude psd-minorant route ($\varepsilon \geq 1 - \tilde{t}^{\mathsf{T}} G^{-1} \tilde{t}$ for $0 < G \leq T$) is measured dead: criticality forces any useful minorant to be sharp to relative 8.3×10^{-4} in the one direction $u = T^{-1}\tilde{t}$ — which is the same as knowing u. What survives is the *two-level chain*

$$\varepsilon_c \;\geq\; \varepsilon_c - \varepsilon_f \;=\; y^{\mathsf{T}} S y \;\geq\; \lambda_{\min}(S)\,\|y\|^2 \;\geq\; \lambda_{\min}(S)\,D_f^3 \max_j g_j^2/2 \;>\; 0,$$

with y the oscillation of the fine dual solution, S the T-Schur complement onto the oscillation space (Haynsworth 1968), and the last step the one-cell Payne–Weinberger 1960 cell-variance inequality used in its *lower* form — the only place where a classical approximation statement points the right way. Every link is an identity or a completed Cholesky: certified on all 19 nested pairs, bound/$\varepsilon \in [0.111, 0.185]$, everything at least 6.7×10^5 above the declared backward-error floor. The rate: $\theta'(\text{sum}) = 1.74$ against $\theta = 1.76$ (jackknifed fits) — a loss of 0.04 powers of D, i.e. *none*, because $\lambda_{\min}(S)$ *grows* under refinement (+0.20 per halving of D; log-versus-small-power undecidable on an 8× lever, and nothing in the chain rests on deciding it): the chain costs a constant, not a power (bound/ε spread 1.195 over the whole lever). The one-cell version costs a further 0.93 powers — the classical price of replacing a sum over cells by its largest term.

The jumps, exactly — and a correction to T116. Growing the window by one cell per end at fixed D is a pure prepend in frame A, so the bordered inverse gives the *exact* drop $\varepsilon(h+1) = \varepsilon(h) - r_0^2/s_0$ (Levinson 1947; rel 3.7×10^{-12} over consecutive steps). A prime-power entry at fixed window is a *corner* update supported on the two outermost Hankel anti-diagonals, of rank exactly $k_0 + 1$ (k_0 the corner distance), and Woodbury/Sherman–Morrison reproduces its effect on ε in closed form on all 23 measured entries (rel 2.1×10^{-11}). The direction *corrects* T116: all 23 entries move ε *up* (factors 1.0003–1.0506; for $k_0 = 0$ the update is $+\beta e_0 e_0^{\mathsf{T}}$ with $\beta > 0$, so an entry can never by itself lower ε at that corner distance). The T116 factor-120 falls were a *sweep artefact*: resolved cell by cell over 400 steps, the largest single-cell drop is a factor 0.9555, the entry share of the total log-drop is −0.3% *with the opposite sign*, and the falls are the accumulated smooth Levinson product between sweep points spaced 150–300 cells apart. The jump side-condition disappears from the ansatz; what a bound must survive is a smooth $\alpha^{-6.07\pm0.03}$ decay (fit; the α lever is short and bound by the compute cap $h \leq 1400$, declared as such).

The theorem candidate — and the three-lemma missing list

Candidate. Let $V_c \subset V_f$ be the reflection-odd piecewise-constant spaces on $(-\alpha, \alpha)$ with M and $2M$ cells, T the pole-free odd Weil form, $\tilde{t}$ the pole functional of $f(x) = 2\sinh(x/2)$. Assume (H1) $T_f \succ 0$ [certified per window]; (H2) $\lambda_{\min}(S) \geq \kappa > 0$ on the oscillation space [certified per window; measured to grow under refinement]; (H3) one coarse cell carries a non-vanishing discrete slope of the dual solution [measured]. Then the four-line chain above gives $\varepsilon_c \geq \kappa\, D_f^3 G^2/2 > 0$; at fixed D the α-decrease per added cell pair is exactly r_0^2/s_0; and prime-power entries only raise ε — so the certified bound carries the full measured rate. Line-by-line attribution: the identities are exact algebra, (H1)/(H2) are completed Choleskys, the cell step is classical, the exponents are jackknifed fits.

What is still missing — three named analytic lemmas about one symbol, each with a classical address:

(1) a uniform lower bound on *one* local slope of the dual solution $u_f = T_f^{-1}\tilde{t}_f$ (H3 as analysis): the corner asymptotics of T^{-1} for a symbol with a log singularity — the Kac–Murdock–Szegő 1953 / Widom 1974 circle of results;

(2) a lower bound for $\lambda_{\min}(S)$ at *one* level (H2 as analysis): S is the form seen by oscillations, i.e. the symbol at the Nyquist frequency π/D, where the archimedean weight is largest — the measured monotonicity then carries it;

(3) the saturation constant: dropping ε_f costs the factor $1-\varepsilon_f/\varepsilon_c$, measured in $[0.675, 0.744]$ — literally the Bank–Smith / Dörfler–Nochetto saturation assumption, bounded away from 0 by a constant, so it costs no rate.

Two of the three are *constants*, not rates. None of this touches the T116 [**SUPPORT**] comb obstruction — that is a separate question about *state compression*, not about ε.

Status after T118 (next subsection): (3) is **classically done** — saturation is an *identity* here, $\varepsilon_c - \varepsilon_f = y^\mathsf{T} Sy$, plus the one hypothesis Bank–Smith themselves assume; (2) is **certified on windows and arithmetically reduced** — not through the symbol of S itself, which is a weighted *harmonic* mean of the aliasing pair and vacuous on the comb, but through the strengthened Cauchy–Schwarz shift onto the oscillation Gram; (1) is **corrected and open** — the corner exponent does not settle, because a log singularity is the boundary of all powers, and (H3) must be upgraded to a mean-square statement, which also repairs the chain's last step.

T118 (`symbol_lemmas_probe.py`, contract `SYMBOL.LEMMAS`) then measured the three lemmas against their classical addresses, one by one — and two of the three stand.

7.17 Two of the three lemmas stand (T118)

T118 (`symbol_lemmas_probe.py`, contract `SYMBOL.LEMMAS`, 36/36, TWO-OF-THREE) put each missing lemma against its classical address. The outcome, stated before the detail: the saturation lemma (3) is an *identity* here; the Schur-floor lemma (2) acquires a fully certified classical route on half the measured windows and an arithmetic reduction on all of them; the corner lemma (1) does *not* contradict its address — the address simply does not cover this symbol — and its failure *corrects* both an old exponent estimate and the last step of the T117 chain. What the theorem still needs is exactly one genuinely new analytic statement.

The harmonic versus the arithmetic aliasing mean. The two-level oscillation Schur complement of a Toeplitz form has an *exact* two-grid symbol (re-derived in closed form; local Fourier analysis, Brandt 1977/Hackbusch 1985; multigrid for Toeplitz, Fiorentino–Serra 1991), and it is the weighted *harmonic* mean of the symbol at a low frequency and at its Nyquist partner: $1/\sigma_S = \cos^2(\theta/2)/f(\theta+\pi) + \sin^2(\theta/2)/f(\theta)$. A harmonic mean needs *both* entries positive, and f is negative on a comb of dips at the window's own resolution scale: $\inf \sigma_S = 0$ on 14/14 (zone, level) rows — every pointwise route through σ_S is *vacuous* on the real symbol, not merely weak (a perturbative split recovers at most 2.6×10^{-5} of $\lambda_{\min}(S)$; the comb is non-perturbative). The classical reason is computed rather than asserted: $\lambda_{\min}$ at finite M is a Fejér–Riesz *moment* problem, not a pointwise infimum — the ground state of S *concentrates* on the negative set (3.3–43× the uniform share), and positivity is a cancellation of relative size 10^2–10^4 between the positive and negative parts of the symbol integral. Coarse-graining shows where the mass sits: the one-cell-averaged symbol is positive on 14/14 windows (a heuristic, declared as such). The growth is a *logarithm*, not a power: on an FFT-only lever of up to 128× in D the log model beats the power model (at $b = 0.44$–0.93 per $\log(1/D)$), exactly what the log-singular kernel demands (Kac–Murdock–Szegő 1953; Widom 1974). And the pole mass in the oscillation space is $\le 2.1\times10^{-5}$ — an identity via the DTFT factor $|\sin(\theta/2)|$ — which is why the finite-section effect is absent there.

Saturation is an identity here — and the CBS shift rescues Lemma B. For a nested piecewise-constant pair, saturation is not an assumption but the *identity* $\varepsilon_c - \varepsilon_f = y^\mathsf{T} Sy$ (rel 4.9×10^{-11}), so β is *computed*: $\beta \in [0.252, 0.336]$ over 20 triples, with the monotonicity in the favourable direction; what remains of Lemma C is the one hypothesis Bank–Smith 1993 themselves assume. Their strengthened Cauchy–Schwarz constant γ (Yserentant 1986) is then the lever that rescues Lemma B: the identity $\lambda_{\min}(L_z^{-1} S L_z^{-\mathsf{T}}) = 1-\gamma^2$ (rel 2.1×10^{-15}) moves the symbol question off S and onto the oscillation Gram, whose symbol $\sigma_z(\phi) = \sin^2(\theta/2) f(\theta) + \cos^2(\theta/2) f(\theta+\pi)$ is the *arithmetic* mean of the same aliasing pair — the low-frequency negativity that poisoned the harmonic mean is suppressed *quadratically* by $\sin^2(\theta/2)$. The route is then certified: $\inf \sigma_z > 0$ on 7/14 windows (systematically: every window with $\alpha \le 1.30$ and $h \ge 200$), with the certified bound recovering up to 52% of the measured $\lambda_{\min}(S)$; on *all* 14 windows the reduction $\lambda_{\min}(S) \ge (1-\gamma^2)\,\lambda_{\min}(T_h(\sigma_z))$ holds at 32–76% sharpness — Lemma B is reduced to the positivity of a plain Toeplitz section of an explicit symbol; and on the long FFT lever (M up to 15 680, no matrix ever formed) the certified floor $\inf \sigma_z$ *rises* logarithmically and *crosses zero* on 3/5 zones — the failing windows were *under-resolved*, not obstructed.

The corner lemma corrects itself — and repairs the chain. The classical statement is an edge exponent: for a symbol vanishing like $|\theta|^{2s}$ the finite-section solution vanishes like $\mathrm{dist}(\cdot,\mathrm{edge})^s$. The estimator is calibrated on a model class where this is exact ($|2\sin(\theta/2)|^{2s}$, lags in closed form; the Kac–Murdock–Szegő 1953 closed-form inverse as the $s=0$ control). On the *real* symbol the corner exponent *drifts*: $+0.238$ per halving of D (3–7.4σ) — $\log(1/|\theta|)$ is the boundary of all powers, so there is no asymptotic (s,κ) to cite. (The old T108/T109 24-cell exponent was doubly mis-estimated; its apparent stability was the cancellation of two errors.) The (H3) correction: the single-cell form is mis-dimensioned — the concentration factor grows like $h^{0.99}$, the slope mass sits on a fixed *fraction* of cells — and the repair is at the T117 chain's last step: the rate lives on $\|y\|^2$ (the stage-two bound carries $D^{1.751}$ against $\varepsilon \sim D^{1.770}$ — lossless; the single-cell stage costs $D^{2.74}$), so (H3) must be a mean-square statement.

The theorem after T118 — five unconditional links, two hypotheses

Candidate (V2). Five unconditional links — the saturation identity $\varepsilon_c - \varepsilon_f = y^\top S y$, the CBS identity $\lambda_{\min}(L_z^{-1} S L_z^{-\top}) = 1-\gamma^2$, the isometry ordering onto $T_h(\sigma_z)$, Grenander–Szegő in the lower direction, and the certified Cholesky floors — give, for every window the probe touched,

$$\varepsilon(\alpha, D) \;\ge\; (1-\gamma^2)\,\lambda_{\min}\big(T_h(\sigma_z)\big)\,\|y\|^2.$$

Two hypotheses close it: **(H1)** γ bounded away from 1 uniformly in (α, D) — the standard CBS hypothesis of the two-level literature, measured $1-\gamma^2 \ge 0.181$ with the favourable trend; **(H2)** $\|y\|^2 \ge c\,D^{1.75}$ uniformly — (H3) upgraded to mean square, and *the* one genuinely new analytic statement the theorem needs, not importable from the Kac–Murdock–Szegő/Widom power-law corner theory because this symbol is log-singular with atoms.

The per-lemma missing lists. **L-B** (two points, both *arithmetic*): $\inf \sigma_z > 0$ for all $D < D_0(\alpha)$ — D_0 set by the comb depth against the Nyquist-band mass, an atom-counting question — and narrow-dip positivity of finite sections. **L-C** (one point): γ-uniformity, i.e. (H1). **L-A** (two points): the mean-square form (H3′), and the Fisher–Hartwig extension of the corner theory to log-singular symbols with atoms.

Disposition after T119 (next subsection): the first L-B point is now a **theorem** with an explicit $D_0(\alpha)$ and a universal constant; the second is diagnosed as a moment problem, with one window newly certified by the large sieve; (H2) is proven *not derivable* from the links above — genuinely new content, as suspected — and reduced, through the mean-square form (H3′), to a single scalar that is itself an exact identity, $\kappa_{\mathrm{end}} = 1/(1+R)$; what remains of the whole list is one discrete Harnack inequality plus two constants.

T119 (`oscillation_mass_probe.py`, contract `OSCILLATION.MASS`) then took the two halves that remained — the arithmetic half of Lemma B and the oscillation-mass half (H2)/(H3′) — and closed the first as a theorem.

7.18 The arithmetic half closes; one Harnack inequality remains (T119)

T119 (`oscillation_mass_probe.py`, contract `OSCILLATION.MASS`, 27/27, ARITHMETIC-DONE) is the part where the comb, the standing obstruction since T116, becomes an explicit and *bounded* object. The outcome, stated before the detail: the arithmetic half of Lemma B is a theorem with an explicit $D_0(\alpha)$; the narrow-dip point is diagnosed as a moment problem, with one window newly certified by the large sieve; (H2) is proven *not derivable* from the existing identities — and the sharpest identity of the run reduces the whole oscillation-mass half to one discrete Harnack inequality with a classical address.

The D_0 theorem, with a universal constant. The lag sequence splits exactly as $c = c_{\mathrm{arch}} + c_{\mathrm{comb}}$, and the comb half has a *closed-form* symbol: every prime-power atom (u_j, μ_j) contributes exactly one cosine at the frequency of its own lag, split over the two nearest integer lags by linear interpolation — and under the aliasing average σ_z the *parity* of the lag index decides everything (an even lag survives at full strength, an odd one is multiplied by $-\cos\theta$). Two consequences, both elementary and both new: the whole comb is bounded *uniformly in* D by the atom count, $|\sigma_z^{\mathrm{comb}}| \le \Xi(\alpha) = \sum_{n\le e^{2\alpha}} 2\Lambda(n)/\sqrt{n} \sim 4e^{\alpha}$ (Chebyshev); and the archimedean growth is an *identity*, $\inf \sigma_z^{\mathrm{arch}} = \log(1/D) + B$ with slope exactly one and a *universal* $B = -1.0474$ (drift $\le 3.1\times10^{-4}$ over $\alpha = 0.98$–4.70; measured slope 1.0030 ± 0.0012). Their conjunction is the theorem:

$$\inf \sigma_z > 0 \quad \text{for all} \quad D < D_0(\alpha) = \exp\big(-(\Xi(\alpha) + B)\big),$$

verified deep on a 1024× FFT-only lever (M up to 65 536, sixteen zones to $n = 11\,981$, no matrix ever formed).

The two-regime correction, and the large sieve's one win. The deep run corrects T118's reading: there are two regimes. Narrow zones ($\alpha \le 2.00$, 5/16) cross zero *certified* at the frame's own resolution; wide zones ($\alpha \ge 2.22$, 11/16) also cross — the theorem is not in doubt — but at $M \sim 10^{13}$–10^{187}: a defect of the *constant*, not of the theorem. The narrow-dip point (L-B.2) becomes a certifiable criterion through the large sieve (Montgomery–Vaughan 1974, used in the honest direction: an upper bound on the L^2 mass a polynomial of degree $< h$ can place on the dips, which is exactly what a lower bound on $\lambda_{\min}$ needs). It wins *once* — a certified positive section floor $+0.396$ on a window where $\inf\sigma_z = -0.349$, the series' first certificate of section positivity *without* pointwise positivity — and it dies structurally for $\alpha \ge 2.5$, where a single Fejér bump fits inside one dip: the obstruction is a moment problem, not a localisation problem. A by-product is proved exactly: coarse-graining can only raise the section floor, $\lambda_{\min}(T_h(\sigma * K)) \ge \lambda_{\min}(T_h(\sigma))$ by unitary averaging — the direction *opposite* to T118's heuristic reading, which is thereby retired.

(H2) is genuinely new — proven, not suspected. $\|y\|^2$ is the squared L^2 distance of the fine piecewise-constant solution from the coarse space, so (H2) is an *inverse* approximation statement. Its structure is dissected: the mass is distributed (corner share 0.18–0.42, middle 0.19–0.38, inner half 0.36–0.48 — no single-cell carrier), the exponent bookkeeping closes ($\varepsilon \sim D^{1.761}$, $\|y\|^2 \sim D^{1.961}$, $\lambda_{\min}(S)\|y\|^2 \sim D^{1.757}$ — the second stage loses no power), and the Levinson structure of the solution is verified in full (the corner identity $u_0 = -\sqrt{2}\,\rho/E$ to 2.8×10^{-8}, the mass budget $1-\varepsilon = \sum_n \rho_n^2/E_n$ to 2.5×10^{-13}, exactly one negative pivot). The derivability question is answered *no*: the energy route is valid and empty — it returns $\varepsilon \ge r\,\varepsilon$ with $r < 1$, a tautology — so (H2) is genuinely new analytic content, exactly as T118 suspected.

The constant becomes an identity. The sharpest result of the run removes the constant from (H3′): Cauchy–Schwarz is nearly sharp, the corner increments are monotone on 100% of the cells, and the end-cell share $\kappa_{\mathrm{end}} = 0.4906$–$0.5093$ is flat over 50 rows and 15 ladder pairs and *exponent-blind* — it drifts by ≤ 0.0019 per level while the corner exponent itself creeps by $+0.046$–0.242 per level, a factor 128 less. And it is not a measured constant at all: for a monotone corner profile,

$$\kappa_{\mathrm{end}} = \frac{1}{1+R}, \qquad R = \frac{\sum_j |a_j|}{\sum_j |w_j|},$$

exactly (verified to 1.1×10^{-16}; a telescoping sum), with w_j the within-cell and a_j the across-cell increment of u and R measured 0.9634–1.0349. The one hypothesis left inside the chain is therefore a *discrete Harnack inequality* for the increments of a log-singular Toeplitz inverse — the profile does not oscillate at the mesh scale — and not a bare constant.

The theorem after T119 — fourteen links, three defects

Candidate (V3). Fourteen links close the chain $\varepsilon_c \ge (1-\gamma^2)\,\lambda_{\min}(T_h(\sigma_z))\,\|y\|^2 > 0$ for every window the probe touched: eight identities (among them, new in T119, the closed-form comb symbol, the parity rule, the Levinson corner value with its mass budget, and $\kappa_{\mathrm{end}} = 1/(1+R)$), one arithmetic bound (the atom count $\Xi(\alpha)$, uniform in D), one certified growth law (the archimedean identity with universal B), one theorem (the explicit D_0), the isometry, Cauchy–Schwarz, and the chain itself — five of the fourteen new in T119.

The defect list, shortest form — three items:

(D1) the discrete Harnack inequality $\sum_j |a_j| \le R_{\max} \sum_j |w_j|$ on the corner cells — the one genuinely new analytic statement, with a classical address (Widom 1974; Trench 1964 for the explicit inverse; Böttcher–Silbermann); *any* finite $R_{\max}$ gives $\kappa_{\text{end}} \ge 1/(1 + R_{\max}) \approx 0.49$;

(D2) the D_0 constant for wide α — the theorem is unconditional, but at the frame's own resolution the pointwise floor is negative for $\alpha \gtrsim 2$, and the section route that would cover it is a Fejér–Riesz moment problem;

(D3) γ-uniformity — (H1), inherited from T118 unchanged.

Distance to a paper, honestly: as a *conditional* lemma paper — under (D1) and (H1), $\varepsilon \ge c\, D^{\theta}$ with an explicit c — the material is essentially complete, and the missing work is writing, not discovery; an unconditional version needs exactly the Harnack statement. The chain is classical numerical analysis end to end and contains no zeta input anywhere.

Disposition after T120 (next subsection): (D1) is *explained* — $R \approx 1$ is parity symmetry, and the summed inequality follows unconditionally from one sign of the corner increments by a two-line pairing argument, while the per-cell version is provably false — so the one hard statement is replaced by a strictly weaker and far more classical hypothesis; but (D2) is enlarged (the frame consumes broad windows by construction, and the unconditional D_0 route covers not one handover of the real ladder) and (D3) loses ground as a uniformity claim ($1 - \gamma^2$ drifts down with α): the defect count goes three → four.

T120 (`harnack_probe.py`, contract `HARNACK`) then attacked that defect list head on — and explained the Harnack core while sharpening the side defects.

7.19 The Harnack core is proof-shaped; the wide-window defects sharpen (T120)

T120 (`harnack_probe.py`, contract `HARNACK`, 21/21, HARNACK-EXPLAINED) is the part where the one hard statement of V3 stops being an inequality about two families and becomes a two-line consequence of a symmetry. The outcome, stated before the detail: $R \approx 1$ is *parity symmetry* — the two increment families are one sequence offset by a single fine cell — with an unconditional $|R - 1| \le 0.047$ certificate; two structural routes to the remaining hypothesis are refuted, and the per-cell version of the inequality is provably *false*; and the honest defect count rises from three to four, because the α-audit shows the unconditional D_0 route covers not one handover of the real ladder and the CBS constant decays with the width.

The parity symmetry, and the certificate. The two families of (D1) are not two objects: with $v_i = u[i+1] - u[i]$ the single increment sequence of the *same* solution $u = T^{-1}\tilde{t}$, the within-cell and across-cell increments are the even and the odd half, $w_j = v_{2j}$ and $a_j = v_{2j+1}$ — offset by exactly one fine cell. If v has one sign on the corner cells (measured at 100% of them on every one of 3915 rows), the difference of the two parity sums is a sum over *disjoint* adjacent pairs, so

$$|R - 1| \;\le\; \frac{\sum_j |v_{2j+1} - v_{2j}|}{\sum_j |v_{2j}|} \tag{C1}$$

unconditionally — no monotonicity, no smoothness. Measured over the whole sweep, (C1) gives $|R - 1| \le 0.04745$ against a measured 0.03485, and it dominates the measurement on all 3112 sign-pure rows, as it must. If v is moreover monotone the left side telescopes to a boundary term over a bulk sum, $|R - 1| \le |v_{2n_C} - v_0| / \sum_j |v_{2j}| \le (C - 1)/n_C$ (C2) — recorded as *conditional* (the monotone share is only 0.50–0.96) and kept as the explanation of the rate: R sits at 1 because a shift by one fine cell is a boundary perturbation, and R is resolution-blind because refining raises n_C and shrinks the boundary term at the same rate. The certificate itself confirms the rate: (C1) decays like $n_C^{-0.58\pm0.05}$ (a fit). The brute-force sweep (155 windows × resolutions × corner fractions, 3915 rows) keeps $R \in [0.9098, 1.1011]$ in the corner regime with sign purity exactly 1.0000; the single excursion above 1.1 sits at the clamped minimal corner $n_C = 4$, exactly where the boundary-term mechanism predicts it, and for $n_C \ge 8$ the window certificate is $\kappa_{\text{end}} \ge 0.480827$. One negative result belongs to this positive one: the *per-cell* Harnack inequality is *false* — $|a_j/w_j|$

reaches 4.8×10^3 — so no cellwise comparison can be true; only the summed form holds, which is exactly the form (C1) proves.

Two structural routes refuted. Both obvious routes to the one remaining sign hypothesis are now closed, each by a measurement. First, the 40×40 corner block of T^{-1} is entrywise positive on 18/18 (zone, level) rows without exception — and the checkerboard alternative is refuted exactly (0.5000, a coin flip) — but it is *not* TP2 (the share of non-negative 2×2 minors is only 0.51–0.88), so the Gantmacher–Krein oscillation-matrix route is closed; nor is T an M-matrix. Second, differencing $Tu = \tilde{t}$ once gives an *exact* discrete boundary-value problem for the increments alone, $Kv = s - u_0 G\mathbf{1}$ with $K = D_1 TL$ (relative residual 2.4×10^{-12}), with a smooth, positive right-hand side — the correct classical shape for a maximum-principle argument — but K^{-1} is *not* entrywise positive (positive share 0.44–0.51, minimum entry -333): the discrete maximum principle is not unproved, it is *false*. The reason is visible in the solution itself: u is monotone at the corner and oscillatory in the bulk — that is the prime-power comb. The corner block of K^{-1} *is* entrywise positive (0.9961–1.0000), so a localised statement is not excluded — but the corner components of v contract full rows of K^{-1} carrying a genuine cancellation (no-cancellation ratio 10^{-7}–10^{-3}), so what the sign hypothesis needs is a corner-localised Fisher–Hartwig decay estimate (Widom 1974; Böttcher–Silbermann), not a sign pattern.

The α-audit: (D2) is not vacuous but total. The frame's window is not a free parameter: in frame A the handover at the atom $u_k = \log n_k$ uses $D = g_k/(2\nu)$ and $M_o = \lceil u_k/D \rceil + 1$, hence $\alpha_o = u_k/2 + O(D)$ *exactly* (maximum deviation 1.5×10^{-2} over all 732 admissible handovers) — the consumed half-width *is* half the log of the zone, and only 1.4% of the handovers have $\alpha \le 2$. Because $B = -1.0474$ is a property of the archimedean kernel alone and hence universal (spread 1.2×10^{-6}), the criterion $\log(1/D_{\text{frame}}) \ge \Xi(\alpha) + B$ can be evaluated on the whole zone table: it holds on exactly 3 of 1492 zones ($n = 2, 3, 4$), i.e. for $\alpha \le \alpha^* = 0.693$ — *below* the first admissible handover $n = 7$. Not one step of the induction is covered by the unconditional D_0 route (the resolution it would need explodes to $M \sim 10^2$–10^{188}). The fallbacks, measured and labelled: the comb supremum is only 0.038–0.603 of its certified cap Ξ — an incoherence gain of up to 1.7×, but a grid supremum, hence a measurement and never a certificate (a certified version would be an exponential-sum estimate over prime powers) — and the large-sieve narrow-dip floor (Montgomery–Vaughan 1974) is positive on 2/6 surveyed zones, at $n = 32$ where $\inf \sigma_z = -0.45$ — the dip route genuinely buys something — and dies beyond $\alpha \approx 2$.

The γ formula, and the wrong direction. The CBS constant has an exact local symbol identity: with $f_1 = f(\theta)$, $f_2 = f(\theta + \pi)$, $\sigma_c \sigma_z - \tau^2 = f_1 f_2$ (relative residual 3.9×10^{-16}), hence $1 - \gamma(\theta)^2 = f_1 f_2/(\sigma_c \sigma_z)$ — (D3) is no longer a bare measured constant but a symbol infimum, the same class of object as $\inf \sigma_z$. Two honest caveats, both measured: the naive symbol infimum is 25–210× *below* the matrix constant (the finite odd section never sees the worst aliasing pair — a finite-section transfer is needed, not the infimum), and the measurement itself moved: $1 - \gamma^2$ *falls* with α (fitted slope -0.297 ± 0.013, range 0.079–0.283, flat in the resolution at a fixed zone) — T119's ≥ 0.181 is reproduced on its own α-range but is *not* uniform.

The theorem after T120 — eighteen links, four defects

Candidate (V4). Eighteen links (7 identities — among them, new in T120, the increment system $Kv = s - u_0 G\mathbf{1}$ and the symbol identity $\sigma_c \sigma_z - \tau^2 = f_1 f_2$ — 4 certified, 2 classical, 2 measured, and the chain), and the one hard statement of V3 is *replaced*: (D1) is no longer a bare inequality but a proved two-line implication — the pairing certificate (C1), unconditional given one sign of the corner increments. What a lemma paper could claim today: *let T be the finite odd Weil form on a window, $u = T^{-1}\tilde{t}$, and suppose the increments of u have one sign on the outer n_C coarse cells; then $|R - 1| \le \delta(n_C)$ with the explicit pairing ratio δ, hence $\kappa_{\text{end}} \ge 1/(2 + \delta)$ and $\|y\|^2 \ge (u_{\text{end}} - u_0)^2/(2n_C(2 + \delta)^2)$* — everything after "suppose" is proved; measured, $\delta \le 0.0964$ for $n_C \ge 8$, falling like $n_C^{-0.58}$.

The defect list — four items (was three):

(E1) *one sign of the corner increments* — the single hypothesis of (C1)/(C2), measured on 100% of the corner cells on all 3915 rows; the two obvious structural routes are refuted (the corner of T^{-1}

is positive but not TP2; the global discrete maximum principle is false), so what remains is a corner-localised Fisher–Hartwig estimate (Widom 1974; Böttcher–Silbermann);

(E2) *a uniform bound on the pairing ratio* δ — a discrete gradient estimate for the increments of $T^{-1}\tilde{t}$, strictly weaker than the old Harnack statement (any finite δ suffices, $\kappa_{\text{end}} \geq 1/(2+\delta)$);

(E3) D_0 *for the whole ladder* — (D2), enlarged: the frame consumes $\alpha = u_k/2$ by construction and the criterion fails at $\alpha^* = 0.693$, below the first admissible handover; the open routes are a certified comb-incoherence bound or the large-sieve dip geometry, which works on the first zones and dies beyond $\alpha \approx 2$;

(E4) *the* α*-dependence of* γ — (D3) with the direction now known: $1 - \gamma^2$ shrinks like $e^{-0.30\alpha}$ (a fit), so uniformity is not merely unproved but questionable in the direction that matters.

Honest movement: the one hard statement got much easier — a two-line implication with one classical hypothesis — and the side defects got sharper and both got worse; the count went three → four, and that is the honest direction.

Disposition after T121: (E3) is retired *as stated* — the chain link is a section eigenvalue, not a symbol infimum, and the section stayed positive on the whole real ladder — and is replaced by the section lemma (E3*) with the Hankel term; (E4) is replaced by one named poly-log, (E4*); (E1) and (E2) keep their kind, with the corner certificates restated at $n_C = h/16$ (T121 found the $h/8$ convention used here too wide); and the symbol-to-section transfer this list implicitly relied on is refuted (the odd sector's Hankel term). The next subsection reports the measurements.

7.20 Section positivity survives the real ladder; the net balance names one poly-log (T121)

T121 (`wide_window_probe.py`, contract `WIDE.WINDOW`, 21/21, WIDE-RESTRUCTURED) asks the question the T120 defect list left sharpest: does the σ_z route survive the *real* ladder — the one whose window grows with the zone, $\alpha_o = u_k/2$? It does, but not as the statement T120 wrote down. The route survives as a *section* statement and dies as a *symbol* statement; the net α-balance of the whole chain loses only a single poly-log to a demand that itself falls like α^{-6}, and that poly-log splits without residue into two repairable steps; one link of the chain, filed by T120 as classical, is honestly refuted on the way.

Section positivity instead of symbol positivity. The chain never needed $\inf \sigma_z > 0$; it needs $\lambda_{\min}$ of the *section* $T_h(\sigma_z)$ at the frame's own resolution. Over 16 frame windows of the real ladder ($n = 7 \ldots 283\,303$, $\alpha = 0.985 \ldots 6.277$, $M_{\text{frame}} = 118 \ldots 431\,162$, $h = 29 \ldots 107\,790$) the section is *positive on* 16/16 while the symbol infimum is negative on 8 of them — and where it is negative, the section eigenvalue sits at 0.64–3.23 $\times\, |\inf \sigma_z|$ on the *positive* side. Direction kept honest: the dense rows ($h \leq 1700$) carry a shifted-Cholesky certificate with the Wilkinson–Higham floor; the large rows are matrix-free FFT-Lanczos *measurements* (a Ritz value is an upper bound for $\lambda_{\min}$, so it can refute positivity but never prove it). The mechanism is quantified at its classical address — the Christoffel function (Szegő 1939; Grenander–Szegő 1958, Ch. 5): a degree-h polynomial cannot concentrate below the Fejér bandwidth $2\pi/h$, and every comb dip is only 0.156–0.720 of one Fejér width. The certified per-dip moment budget $\lambda_{\min} \geq \text{thr} - \sum_j (\text{thr} + \delta_j)\, w_j\, (h+1)$ is positive on 8/16 rows (up to $\alpha = 3.446$, sharpness 0.20–0.76) and tears above, for a computed reason: the dip *count* grows like $e^{2.41\alpha}$ (a fit) against the budget's factor h. So (E3) is reduced from a symbol-infimum requirement that no ladder step satisfies to a *section lemma* the measurement satisfies everywhere: a Christoffel lower bound surviving $O(e^{\alpha})$ Fejér-subresolved dips — a Szegő/Fisher–Hartwig question about the comb, no longer a symbol question.

The net α*-balance — the core number of the part.* Every factor of the chain is measured on one two-dimensional (D, α) lever (14 zones $\times$ 3 resolutions, 42 rows). The individual links do fall with α, exactly as T120's (E4) recorded: $1 - \gamma^2 \sim \alpha^{-0.79}$, $\lambda_{\min}(A_z) \sim \alpha^{-1.07}$, $\|y\|^2 \sim \alpha^{-5.63}$, so the assembled supply falls like $\alpha^{-7.49}$. But the *demand* falls almost as fast ($\alpha^{-5.93}$ on this lever; T116 quoted -6.04): 79% of the decay (E4) complained about is the demand's own. What is left is the only α deficit of the chain,

$$\mathrm{SUPPLY}/\varepsilon_c \;\sim\; D^{-0.177\pm0.096}\,\alpha^{-1.565\pm0.135}$$

(fits; the measured ratio is 4.9×10^{-3}–0.15 over $\alpha = 0.98$–6.31) — net-negative in α by one to two powers (the refit per resolution lands in $-1.88\ldots-1.62$ — a range, not a decimal) but *uniform in D*: the D exponent is zero within its own band, so refining the resolution neither helps nor hurts. The deficit is then decomposed *exactly*: the identity $y^{\mathsf T}Sy/\mathrm{SUPPLY} = r_{\mathrm{ray}}\,r_{\mathrm{cbs}}/(1-\gamma^2)$ holds to 2.8×10^{-14}, so $\alpha^{-1.57}$ splits without residue into $\alpha^{-1.02}$ from the *Rayleigh step* (using $\lambda_{\min}(A_z)$ on a y that is nowhere near the minimal eigenvector; measured slack 1.29–13.19), $\alpha^{-0.51}$ from the CBS step (measured slack 3.42–16.67) and $\alpha^{-0.04}$ from discarding ε_f — *nothing* of it sits in the value of γ. The named rebuild is therefore structural, not a new constant: bounding $y^{\mathsf T}A_zy$ directly (legitimate, because y is the oscillation part of the solution, not an arbitrary vector) recovers 65% of the deficit by itself, and the CBS step's own slack covers the rest.

Link (5) refuted: the Hankel term. The section comparison $\lambda_{\min}(A_z) \ge \lambda_{\min}(T_{h_c}(\sigma_z))$ — which T120's assembly used as a classical Grenander–Szegő transfer — is *false* on 42/42 rows (ratio 0.357–0.916, drifting with the window). The reason is structural and was hiding in the odd coordinates: $A_z = Z^{\mathsf T}(\mathrm{Toeplitz}-\mathrm{Hankel})Z$, and Grenander–Szegő carries the even Toeplitz part only; the reflection Hankel term is not sign-definite. A defect made visible rather than created: the section lemma (E3*) must be about A_z, not about $T_{h_c}(\sigma_z)$.

The corner sign, and the inconclusive Fisher–Hartwig comparison. The one-sign run of the increments covers the corner region the (C1)/(C2) certificates need on 24/24 (zone, resolution) rows at $n_C = h/16$ (run share 0.21–0.42 of the section against the required 0.125 — a safety factor of 1.68–3.38, the sign uniformly negative); at the $h/8$ convention T119/T120 used it holds on only 21/24, so the certificates are restated at $h/16$ — which costs nothing, because δ grows like $n_C^{+0.29}$ (a fit): halving n_C *improves* κ_{end}. The α-slope of the run share is -0.0258 ± 0.0042 (a fit); extrapolated linearly, the $h/16$ margin would be exhausted only near $\alpha\approx 11.7$ (an extrapolation, not a prediction). The Fisher–Hartwig comparison itself is *inconclusive, and that is the result*: the local exponent of the corner row of T^{-1} drifts by $-0.008\ldots+2.934$ between dyadic windows where the log-corrected FH shape would allow only -0.120 — the corner is in no asymptotic power regime at any resolution a factorisation reaches, and an FH exponent quoted from these numbers would be a fit masquerading as a structure. What the block does deliver: the no-cancellation ratio 4.6×10^{-8}–7.1×10^{-5} settles that (E1) can *never* be a sign-pattern argument, and the minimal sufficient statement (E1′) is written out as an explicit head-versus-tail inequality (head share $1/(1+1.63)$ to $1/(1+11.73)$ of the mass at $J = h/16$), with the tail exponent as the one unknown a Widom-type estimate must supply.

δ over the full ladder. The pairing quotient stays in 0.0126–0.1331 over 360 rows spanning $n = 7\ldots299\,983$ and three corner fractions — always below 1, no outlier — so $\kappa_{\mathrm{end}} \ge 1/(2+\delta) \ge 0.4688$ everywhere measured, and $\delta\sim n_C^{+0.29}$ confirms the $h/16$ restatement is free.

The theorem after T121 — the count stays four, every entry changes kind

Candidate (V5). Nineteen links: the identities (Schur/Haynsworth, CBS/Yserentant, the T120 symbol identity, the aliasing form of σ_z, $\kappa_{\mathrm{end}} = 1/(1+R)$), the certified pieces (the pairing certificate (C1); the shifted-Cholesky section floors; the Christoffel/Fejér per-dip budget up to $\alpha = 3.446$), the classical steps (CBS + Rayleigh, lower; Cauchy–Schwarz, lower), the measured/fitted entries (section positivity on the ten Lanczos rows; the one-sign run; the net balance) — and one link *refuted*: (5), the symbol-to-section transfer, killed by the odd sector's Hankel term.

The defect list — four items, each with a changed kind:

(E1) *one sign of the corner increments* — unchanged in kind, sharpened twice: the certificates move to $n_C = h/16$ (clean on all rows there, only 21/24 at $h/8$; the move is free since $\delta\sim n_C^{+0.29}$), and (E1′) is now an explicit head-versus-tail inequality whose tail exponent a corner-localised Fisher–Hartwig/Widom estimate must supply — the FH comparison is inconclusive at reachable h, and no sign-pattern argument can ever do the job (no-cancellation ratio 4.6×10^{-8}–7.1×10^{-5});

(E2) *a uniform bound on the pairing ratio δ* — unchanged: no outlier over 360 rows to $n = 299\,983$,

$\delta = 0.0126$–0.1331, any finite δ suffices;

(E3*) *the section lemma, with the Hankel term* (replaces (E3)) — $\inf \sigma_z > 0$ at frame resolution is *retired as a requirement*: the chain link is a section eigenvalue, positive on 16/16 frame windows up to $\alpha = 6.277$, $h = 118\,115$, including 8 windows with a negative symbol infimum. What replaces it is strictly smaller but has two parts: (a) a Christoffel-function lower bound surviving $O(e^{\alpha})$ Fejér-subresolved dips (the certified budget delivers it for $\alpha \le 3.446$ and tears above); (b) the odd sector's Hankel term, which refutes link (5) as stated — the lemma must be about $A_z = Z^{\mathsf{T}}(\text{Toeplitz} - \text{Hankel})Z$, not about $T_{h_c}(\sigma_z)$;

(E4*) *one residual poly-log in α* (replaces (E4)) — the net balance carries $\alpha^{-1.565\pm0.135}$ and $D^{-0.177\pm0.096}$: net-negative by one to two powers of α (per-resolution refits $-1.88\ldots-1.62$), *uniform in D*. Strictly better than (E4) as T120 wrote it: 79% of the supply's α-decay is the demand's own, and the deficit sits in the Rayleigh step ($\alpha^{1.02}$) and the CBS step ($\alpha^{0.51}$), none of it in γ — the address is a structural lower bound for $y^{\mathsf{T}}A_z y$, which alone recovers 65%.

Honest movement: the counter stays at four — the target of getting back below three was not met, because the residual α deficit is real — but every entry changed kind, and one link recorded as classical was found refuted, a defect made visible rather than created.

Disposition after T122: both named repairs landed. Link (5) is retired by an *identity* — $A_z = W^{\mathsf{T}}\text{Toeplitz}_M(c)\,W$ with $W = BZ$ isometric, so the Hankel term is the reflection half of an isometry and not an error term — and the certified band bound that replaces it is non-vacuous to $\alpha = 6.31$, where the Christoffel budget tore at 3.45; the Rayleigh step is made structural and is *sharp* (residual slack 1.00–1.03, flat in α); the certified deficit halves to $\alpha^{-0.73}$, exactly uniform in D, and with the measured CBS coupling the balance closes to within the chain's own ceiling. (E3*)/(E4*) are thereby absorbed into one band-margin estimate (F3) plus one structural CBS estimate (F4). The next subsection reports the measurements.

7.21 The Hankel term is an isometry half; the certified deficit halves (T122)

T122 (`net_balance_repair_probe.py`, contract `NET.BALANCE.REPAIR`, 20/20, NET-IMPROVED) attacks the two worst-case steps the T121 decomposition named — the Rayleigh step and the link-(5) transfer — and both repairs land. The headline is structural: the odd sector's Hankel term, the object that refuted link (5), was never an error term to be estimated.

The Hankel repair is an identity, not an estimate. The compressed Hankel term has a closed form, $(Z^{\mathsf{T}}HZ)_{jk} = b_{j+k}$ with $b_s = (c_{K-2s} - 2c_{K-2s-1} + c_{K-2s-2})/2$ — half a second difference of the lag sequence, exactly like the Toeplitz-side lags — verified to 7.0×10^{-16} on all 36 (zone, resolution) rows. So the oscillation injection Z suppresses *both* halves by the same DTFT factor $|2\sin(\theta/2)|^2$: the measured norm ratio is 0.123–0.781, i.e. $O(1)$ — the T118 mechanism does not make the Hankel term small. The honest anatomy (the archimedean half numerically of rank 10–14, the Hartman/Peller smoothing; the comb half corner-localised on few anti-diagonals) is structure, not a bound: the valid Weyl floor $\lambda_{\min}(T_{h_c}) - \|Z^{\mathsf{T}}HZ\|$ is positive on only 12/36 rows (up to $\alpha = 2.44$) and nowhere near sharp. The valid replacement for link (5) is an *identity*: with $W = BZ$,

$$A_z \;=\; W^{\mathsf{T}}\,\text{Toeplitz}_M(c)\,W, \qquad W^{\mathsf{T}}W = I \;\;\text{(exact to 0.0)}$$

— the Hankel term *is* the reflection half of an isometry. From this and a certified cell envelope ($g_{\text{thr}} = (\text{thr} - \ell)_+ \ge 0$; Parseval is exact because $|Wv|^2$ has degree $< M$) follows the certified band bound

$$\lambda_{\min}(A_z) \;\ge\; \text{thr} - \lambda_{\max}(G), \qquad G = W^{\mathsf{T}}\text{Toeplitz}_M(g_{\text{thr}})\,W \succeq 0,$$

non-vacuous on 36/36 rows up to $\alpha = 6.31$ — where the T121 Christoffel budget tore at $\alpha = 3.45$ — recovering 0.429–0.995 of the true $\lambda_{\min}(A_z)$; the $e^{2.41\alpha}$ dip count no longer enters anywhere (measured gain over the per-dip Nikolskii budget: 8.4–1.1×10^2). The honest caveat is where the threshold sweep puts its optimum: $\text{supp}(g) = 0.95$–1.00 of the circle — one certified end of a one-parameter family whose other end (small thr) is the dip budget that tore.

The structural Rayleigh step is sharp. "y is high-band" is a statement about Wy at the window resolution (99.3–100.0% of its mass above $|\theta| = \pi/2$, with $\|Wy\|^2 = \|y\|^2$ exact), not about y itself

(0.1–1.7% high-band in the coarse coordinates — the oscillation lives in the injection, not in the coefficients). The same certified band identity, evaluated on y instead of over the whole oscillation space — $y^\top A_z y \geq \mathrm{thr}\|y\|^2 - y^\top G y$ — is positive on 36/36 rows, never exceeds the truth, and beats the worst-case product $\lambda_{\min}(A_z)\|y\|^2$ by a factor 1.28–18.90: the slack it removes is exactly T121's r_{ray}, growing like $\alpha^{+0.83}$, and the residual slack of the structural bound is $q_{\mathrm{str}} = 1.00$–1.03 with drift $\alpha^{-0.002\pm0.001}$ — flat, i.e. *sharp*: no α-dependence of the chain passes through the Rayleigh step any more. The CBS step is equally pessimistic: the actual coupling is $\kappa_y = 0.0067$–0.5311 against the worst case $\gamma^2 = 0.7173$–0.9647 (a factor of up to 100+, growing like $\alpha^{+1.52}$), and $1 - \kappa_y = r_{\mathrm{cbs}}$ exactly — T121's slack, now with a name. One lever limit declared: at $M \leq 2048$ the coarse envelope of σ_z is nonnegative on 13/36 rows — the deep dips T118/T121 chased appear only at the frame's own resolution.

The new net balance. Five supplies on one (D, α) lever (12 zones × 3 resolutions, 36 rows), all sound (36/36 below the truth $S_4 = y^\top S y$). The T121 baseline reproduces exactly ($S_0/\varepsilon_c \sim \alpha^{-1.562\pm0.163}$); repairing link (5) alone does not change the exponent at all ($\alpha^{-1.565}$ — a uniform floor cannot beat $\lambda_{\min}(A_z)$; V1's contribution is that the floor is certified over the whole ladder instead of tearing at $\alpha \approx 3.45$); with both repairs the *certified* balance is

$$S_2/\varepsilon_c \;\sim\; D^{-0.003\pm0.075}\,\alpha^{-0.729\pm0.090}$$

(fits; measured 0.033–0.207): the deficit *halves*, and it is exactly uniform in D. Adding the *measured* CBS coupling gives $\alpha^{-0.113\pm0.063}$ — statistically indistinguishable from the chain's own ceiling $S_4/\varepsilon_c \sim \alpha^{-0.116\pm0.062}$ (the cost of discarding ε_f, untouched by both repairs). The residual decomposition closes to 2.9×10^{-16}: $-0.729 - 0.002 + 0.615 = -0.116$. In one sentence: V1 buys the ladder, V2 buys the exponent, and the chain closes *up to its own ceiling* as soon as the CBS step is no longer a worst case.

The theorem after T122 — four defects, consolidated

Candidate (V6). The chain renumbered: the Haynsworth identity; the exact CBS split $y^\top S y = (1-\kappa_y)\,y^\top A_z y$ with $\kappa_y \leq \gamma^2$ classical (Yserentant); the new two-grid identity $A_z = W^\top \mathrm{Toeplitz}_M(c) W$ ($W = BZ$, $W^\top W = I$) — the line that retires link (5); the certified cell envelope; the certified band bound (5R) with its uniform, y-free variant (Gershgorin); and the quoted demand fit.

The defect list — four items (T119 = 3, T120 = T121 = V6 = 4; the target < 4 not met, but the kinds changed twice over):

- **(F1)** *one sign of the corner increments* — unchanged from T120/T121 and untouched by this part: 24/24 clean at $n_C = h/16$, no sign-pattern argument possible, the minimal statement is the head-versus-tail inequality (E1′) with an open tail exponent;
- **(F2)** *a uniform bound on the pairing ratio δ* — unchanged and untouched: 0.0126–0.1331 over 360 rows, no outlier, a discrete gradient estimate still missing;
- **(F3)** *an analytic (D, α) bound for the band margin* $1 - y^\top G y/(\mathrm{thr}\|y\|^2)$, measured 0.024–0.363 over $\alpha = 0.98$–6.31 with fitted drift $D^{-0.042}\alpha^{-0.525}$ — the one *new* item, and it absorbs link (5), the Nikolskii budget and both readings of (E3)/(E4): a Fejér/Christoffel question about how the comb dips of the window symbol overlap the high-band mass of the two-level residual, not an eigenvector question. An α-flat band margin would carry the certified chain to within the CBS step of the truth;
- **(F4)** *a structural CBS estimate* — $\kappa_y \leq \gamma^2$ is the last worst-case step in the chain (worth $\alpha^{+0.615}$ in the balance), and the actual coupling is only 0.009–0.551 of it.

Honest movement: link (5) is not repaired but *retired* — it was a false comparison between an isometric compression and one of its two halves — and the per-dip Nikolskii budget went with it; what is left of the old (E3)/(E4) is a single scalar estimate about dips versus high-band mass, plus the CBS item.

Disposition after T123: the two scalar items were run against the same band machinery and they *split.* (F3) closes almost for free — the reduction to the band profile of Wy costs under 1%, and the oscillation block gains uniform certified positivity, (5R*) — while (F4) resists structurally: every certified route to κ_y needs the *coarse* form from below, whose near-null direction is smooth and lives on a cancellation that no pointwise symbol minorant can see. The remaining $\alpha^{+0.500}$ gap to the ceiling is thereby identified with the ε_f lid itself — the recursion base case. The next subsection reports the measurements.

7.22 The two-isometry completion: the margin closes, the coupling resists (T123)

T123 (`structural_cbs_probe.py`, contract `STRUCTURAL.CBS`, 20/20, CBS-RESISTS) attacks the two scalar estimates the V6 list left — the structural CBS step (F4) and the band margin (F3) — with the same band machinery that made the Rayleigh step sharp in T122. The margin closes almost for free; the coupling does not, and the reason *why* is the content of the part.

The coupling anatomy, in closed form. The coarse block is the *other* half of the T122 construction: with $V = BP$ (P the piecewise-constant prolongation), V is an isometry too — $V^{\mathsf{T}}V = I$ and $V^{\mathsf{T}}W = 0$ exact — so the coarse, oscillation and coupling blocks are three compressions of *one* window Toeplitz form,

$$A_c = V^{\mathsf{T}}\mathrm{Toeplitz}_M(c)\,V, \qquad B_x = V^{\mathsf{T}}\mathrm{Toeplitz}_M(c)\,W, \qquad A_z = W^{\mathsf{T}}\mathrm{Toeplitz}_M(c)\,W$$

(identities to 3.4×10^{-15} on all 24 rows). Two new closed forms complete the T122 anatomy: the coupling is half a *first* difference of the lag sequence, $(B_x)_{jk} = \beta_{j-k} - (c_{K-2(j+k)} - c_{K-2(j+k)-2})/2$ with $\beta_m = (c_{|2m+1|} - c_{|2m-1|})/2$, verified to 1.8×10^{-15} (A_z was half a *second* difference), and in symbol form

$$b(\varphi) \;=\; -\tfrac{i}{2}\,\sin(\varphi/2)\,\big[\sigma(\varphi/2) - \sigma(\varphi/2+\pi)\big]$$

— the *aliasing difference* of the window symbol times $\sin(\varphi/2)$, verified to 5.7×10^{-16}: the coupling weight vanishes exactly where the coarse space lives. That explains κ_y quantitatively for the first time: Wy carries 99.64–100.00% of its mass at $d = |\theta - \pi| < \pi/2$ and only 8.9×10^{-6}–1.5×10^{-3} in the cell containing $\theta = 0$, while the worst-case y^* that attains γ^2 is a *middle*-band vector (81.22–100.00% of its mass exactly there) — the factor 2–109 between $\kappa_y = 0.0067$–0.4622 and $\gamma^2 = 0.7248$–0.9467 *is* that mismatch.

The certified κ bound: both routes fail for one reason in two guises. Route 1, band-split Cauchy–Schwarz (per frequency band separately, the strengthened-CBS mechanism of Axelsson–Vassilevski, including the exact two-grid factorisation that puts the $\sin(\varphi/2)$ zero into the certificate), is correct on 24/24 rows and non-vacuous on 0/24: it costs exactly one coarse condition number ($\mathrm{cond}(A_c) = 2.1 \times 10^4$–$3.1 \times 10^7$). Route 2 substitutes the certified minorant $T_M(\ell) \preceq T_M(\sigma^{(M)})$ into the Schur complement's minimum characterisation and delivers two new links:

$$(5\mathrm{R}^*)\quad y^{\mathsf{T}}A_z y \;\ge\; y^{\mathsf{T}}E_z y, \qquad (7\mathrm{R}^*)\quad y^{\mathsf{T}}Sy \;\ge\; y^{\mathsf{T}}E_z y - (E_x y)^{\mathsf{T}}E_c^{-1}(E_x y),$$

with $E_c = V^{\mathsf{T}}T_M(\ell)V$, $E_x = V^{\mathsf{T}}T_M(\ell)W$, $E_z = W^{\mathsf{T}}T_M(\ell)W$. (5R*) holds with $\lambda_{\min}(E_z) = +0.303\ldots{+5.017} > 0$ on *every* row of the ladder — uniform certified positivity of the oscillation block, at slack 1.0007–1.007, subsuming the whole T122 threshold sweep for free (since $\min(\ell, \mathrm{thr}) \le \ell$ for every threshold) — while (7R*) is certified as soon as $E_c \succ 0$, which holds on 0/24 rows ($\lambda_{\min}(E_c) = -3.44\ldots-0.043 < 0$ everywhere). The asymmetry is the result of the part: $\lambda_{\min}(A_z) = 0.32$–$5.08$ is well conditioned, while $\lambda_{\min}(A_c) = 1.7 \times 10^{-5}$–$2.9 \times 10^{-4}$ sits three to five orders below the certified envelope margin (8.4×10^2–$4.0 \times 10^5 \times \lambda_{\min}(A_c)$, rising like $\alpha^{+4.202}$; against $\lambda_{\min}(A_z)$ the same margin is only 0.037–17). The near-null direction of the window form is *smooth*, so it lives in the coarse space, and its eigenvalue comes from a cancellation *inside* the form that no pointwise symbol minorant can see. The classical worst-case step $\kappa_y \le \gamma^2$ (Yserentant) therefore *stays* in the chain — and V7 now states why it must.

The band margin closes; the certified balance; the ε_f lid. (F3) reduces by the same machinery: $y^{\mathsf{T}}Gy$ splits *exactly* over dyadic depth layers of the certified gap (closure 1.1×10^{-15}), and the two-factor bound [certified symbol depth] × [Parseval mass of Wy per band] overshoots by only 1.20–3.51 — the reduction to the band profile of Wy, the same object the coupling analysis needs, costs under 1%. The certified margin is positive on 5/24 rows (+0.0024–+0.3290 against the exact margin 0.070–$0.560 \sim \alpha^{-0.449}$); the honest lever limit: $\min \ell = -2187\ldots-9.7$ at $M \le 2944$ — the deep dips of the frame's own resolution stay out of reach. The certified balance — the core number: the best fully certified supply is

$$S_6/\varepsilon_c \;\sim\; D^{-0.069\pm0.099}\,\alpha^{-0.544\pm0.113}$$

(fits; $\alpha = 0.98$–6.31, uniform in D and stable across the three resolutions) against the ceiling $S_4/\varepsilon_c \sim \alpha^{-0.044\pm0.076}$, so the gap is $\alpha^{+0.500}$ — and the residual split against S_6 has exactly *two* terms, closing to 2.8×10^{-17}: band profile $\alpha^{+0.000}$ (F3, now negligible) plus coupling $\alpha^{+0.499}$ (F4, all of it). The ε_f question then identifies the lid: $\varepsilon_c = \varepsilon_f + y^\mathsf{T} S y$ is an identity ($\varepsilon_f/\varepsilon_c = 0.177$–$0.546 \sim \alpha^{+0.030}$, flat), and an additive chain that carries ε_f instead of discarding it would have ceiling exactly 1 — but ε_f needs its bound one level down, 4–8 recursion levels to an exact base case, and that base case is precisely (F4). The lid is not a modelling step; it is the *same object* as coarse-level positivity.

The theorem after T123 — four defects, restructured

Candidate (V7). The chain rewritten with the two-isometry completion: the Haynsworth identity; the exact CBS split $y^\mathsf{T} S y = (1-\kappa_y)\, y^\mathsf{T} A_z y$ with $\kappa_y \le \gamma^2$ classical (Yserentant; Axelsson–Vassilevski) — the *only* worst-case step left in the chain, and V7 states why it stays: every certified route to κ_y needs coarse-level positivity from below; the two-grid identities $A_c = V^\mathsf{T} T V$, $B_x = V^\mathsf{T} T W$, $A_z = W^\mathsf{T} T W$ with $V = BP$, $W = BZ$ two orthogonal isometries ($V^\mathsf{T} W = 0$); the certified cell envelope; the uniform oscillation positivity (5R*); the certified band-margin bound; and the quoted demand fit.

The defect list — four items (T120 = T121 = V6 = V7 = 4), restructured:

(F1) *one sign of the corner increments* — unchanged, quoted from T120/T121; together with (F2) it is the *Harnack pair*, untouched by the band machinery;

(F2) *a uniform bound on the pairing ratio δ* — unchanged: 0.0126–0.1331 over 360 rows, any finite value suffices;

(F3) *one band-profile lemma* — the band margin reduces to the band profile of Wy at a cost of 0.7%, and (5R*) supplies the uniform certified positivity of the oscillation block for free; what remains is one analytic statement about how the mass of Wy distributes over frequency bands;

(F4) *coarse-level positivity* — *identified*, not estimated: every certified route to the coupling needs $\lambda_{\min}$ of the coarse form from below, that eigenvalue is a smooth-direction cancellation invisible to symbol minorants, and it is the same object as the ε_f lid that defines the chain's own ceiling.

Honest movement: the counter stays at four, but the terrain changed: the whole $\alpha^{+0.500}$ gap to the ceiling now sits in one named obstruction whose resolution cannot be a sharper two-level estimate. The next step is not sharper banding but the *recursion* — the level telescope $\varepsilon_\ell = \varepsilon_{\ell+1} + y_\ell^\mathsf{T} S_\ell y_\ell$ with certified per-level contributions.

Disposition after T124: the recursion was built, and the direction flipped. Written as a telescope, the rung contribution is a residual in the *inverse* norm — a maximum, not a minimum — so it is certifiable from *above*, where the certified envelope works: (F4) collapses from a quantitative estimate to the sign a coarse-to-fine induction already carries, the base case becomes pure semidefiniteness, and $\lambda_{\min}(A_c)$, $\mathrm{cond}(A_c)$ and γ^2 leave the chain entirely. The next subsection reports the measurements.

7.23 The direction flip: the telescope carries, and (F4) collapses to a sign (T124)

T124 (`multilevel_telescope_probe.py`, contract `MULTILEVEL.TELESCOPE`, 28/28, TELESCOPE-CARRIES) builds the recursion T123 demanded, and the exit turns out to be not the recursion's *length* but its *direction*: written as a telescope, the quantity each rung must supply is not a Rayleigh quotient (a minimum, needing the form from below) but a residual in the inverse norm (a maximum, needing the form from above) — and from above, the certified envelope works, because Parseval is two-sided.

The telescope, exact and geometric. On the level chain $D_0 > D_1 > \cdots > D_L$ (halvings, α fixed) the ladder is *one* window form on nested spaces:

$$P^\mathsf{T} A^{(\ell+1)} P = A^{(\ell)}, \qquad P^\mathsf{T} b^{(\ell+1)} = b^{(\ell)}$$

exactly (2.4×10^{-14}; the pole-vector consistency is the identity $\sinh(a-d)+\sinh(a+d) = 2\cosh d \sinh a$) — the coarse block of the window form at $D_{\ell+1}$ *is* the window form at $D_\ell = 2D_{\ell+1}$, so the T122/T123 two-level system is literally one rung of the ladder. Every rung is the T118 saturation identity, $\varepsilon_\ell = \varepsilon_{\ell+1} + \delta_\ell$ with $\delta_\ell = y_\ell^\mathsf{T} S_\ell y_\ell$, and Galerkin orthogonality (Céa) makes it the energy of the level correction, $\delta_\ell = \|u_{\ell+1} - Pu_\ell\|^2_{A^{(\ell+1)}}$ (verified to 3.7×10^{-8}) — nonnegative, hence the exact telescope $\sum_\ell \delta_\ell = \varepsilon_0 - \varepsilon_L$ (9.2×10^{-10}). The level distribution is geometric on average: the top rung carries 0.2722–0.8806 of ε_0, consecutive rungs fall by a median factor 0.316 (quartiles 0.229–0.473, against $2^{-\theta} = 0.289$ from the demand law), and at $L = 5$ the ceiling of the additive chain is 0.9911–0.9987 — against T123's two-level ceiling $\alpha^{-0.044}$. Rung law (a fit): $\delta \sim D^{+1.598}\alpha^{-5.900}$.

The certified rung contribution — the direction flip. The rung has two dual descriptions. Primal, $\delta_\ell = \min_{w \in V_\ell} \|u_{\ell+1} - w\|^2_A$: a *minimum*, every test vector an upper bound — the wrong direction for a supply. Dual: the level residual $r_\ell = t_{\ell+1} - A^{(\ell+1)}Pu_\ell$ satisfies $V^\mathsf{T} r_\ell = 0$ (Galerkin), so $r_\ell = W\hat{s}_\ell$ lives purely in the oscillation space, and

$$\delta_\ell \;=\; \|r_\ell\|^2_{A^{-1}} \;=\; \hat{s}^\mathsf{T} S^{-1} \hat{s} \;=\; \max_v \frac{(\hat{s}^\mathsf{T} v)^2}{v^\mathsf{T} S v}$$

— a *maximum*: every test vector gives a lower bound, and the denominator wants S from *above*. That is where the certified machinery works: $S \preceq A_z$ needs only $A_c \succeq 0$ (Haynsworth monotonicity), $A_z \preceq U := Z^\mathsf{T} T_M(\mathrm{up}) Z$ is Parseval against the certified majorant $\mathrm{up} \geq \sigma^{(M)}$ (the *mirror* of T123's $E_z = W^\mathsf{T} T_M(\ell) W$, on the block where it works), and Loewner inversion turns the upper bound on S into the lower bound on the rung,

$$\text{(8R)} \qquad \delta_\ell \;\geq\; \hat{s}^\mathsf{T} U^{-1} \hat{s},$$

with the optimal test vector $v^* = U^{-1}\hat{s}$ in closed form — no coarse inverse, no $\lambda_{\min}(A_c)$, no CBS constant. (8R) is valid on 400/400 rungs (80 chains $\times$ $L = 5$, 20 zones), recovering $\mathrm{cb}/\delta = 0.4739$–$0.9938$ at drift $\alpha^{-0.080\pm0.007}$ — seven times weaker than T123's $\alpha^{-0.544}$; of the loss, the coupling accounts for 0.5036–0.9967 and the envelope only 0.7895–0.9994. T123's obstruction remains and is *irrelevant*: $\lambda_{\min}(E_c) < 0$ on 399/400 rungs, and (8R) never forms E_c. The Levinson bordering route honestly *falls*: the innovation ledger $q = \sum_n \rho_n^2/E_n$ is exact per level (9.6×10^{-12}; Levinson–Durbin–Delsarte–Genin), but refinement is not a bordering — the two filtrations are transversal (head/$q_\ell = 0.61$–1.00, tail/$\delta_\ell = 13$–863) — so the rung must be read in the (8R) form.

Self-similarity, and why the recursion closes. The near-null direction of the window form stays *smooth* at every level (oscillation fraction 4.3×10^{-6}–1.2×10^{-1}) — T123's diagnosis, confirmed level by level — and it is self-similar along the ladder: prolonged, w_ℓ has Rayleigh quotient exactly $\lambda_{\min}(A^{(\ell)})$ at the finer level (an identity, from nesting), i.e. within a factor 1.30–10.11 of that level's own minimum, at subspace overlap $\|Q_K^\mathsf{T} P w_\ell\| = 0.944$–$1.000$. But (8R) asks only for the *sign* of A_c, and by the nesting identity that sign is the previous rung's own conclusion: positivity propagates fine $\to$ coarse for free (so the opposite direction would be circular — stated explicitly), and the non-circular induction runs coarse $\to$ fine, with base case $\varepsilon_L \geq 0$ — pure semidefiniteness of the smallest window form.

The balance; the honest negative. Telescope reading: $\sum_\ell \mathrm{cb}_\ell/\varepsilon_0 = 0.6404$–$0.9729$ against the ceiling 0.9911–0.9987 (fit $D^{-0.066}\alpha^{-0.084\pm0.033}$). Two-level comparison, like-for-like against T123's certified balance: $\mathrm{cb}/\varepsilon_c \sim D^{-0.090}\alpha^{-0.100}$ against T123's $D^{-0.069}\alpha^{-0.544}$ — the α exponent moves by $+0.444$ of the $\alpha^{+0.500}$ gap. The honest negative, recorded as (F5): the *solution-free* version of the rung fails — the coupling term cancels the data high-pass rather than perturbing it ($|r_{\mathrm{corr}}| = 0.88$–$2.61$ of the explicit part), so any norm bound kills the certificate to exactly 0; what would be needed is a bound on a *direction*, not a size. (8R) therefore carries the coarse solution along — on the same bookkeeping standard as the whole chain: $\hat{s}$ is a function of the carried coarse solution exactly as y is in T122/T123's own certified supplies.

The theorem after T124 — (F4) collapses to a sign

Candidate (V8). The telescope chain: the nesting identity $P^\mathsf{T} A^{(\ell+1)} P = A^{(\ell)}$ (the ladder is one window form on nested spaces); the saturation telescope $\varepsilon_0 = \varepsilon_L + \sum_\ell \delta_\ell$ (exact); the dual residual form of the rung with the certified per-rung bound (8R) $\delta_\ell \geq \hat{s}^\mathsf{T} U^{-1} \hat{s}$, needing only $A_c \succeq 0$ (Haynsworth) and the

Parseval majorant $U \succ 0$; and the coarse-to-fine induction whose hypothesis at rung ℓ is the conclusion of rung $\ell - 1$, with base case $\varepsilon_L \geq 0$.

The defect list — the counter reads differently now:

(F1) *one sign of the corner increments* — unchanged, quoted from T120/T121; with (F2) the *Harnack pair*, untouched by telescope and band machinery alike;

(F2) *a uniform bound on the pairing ratio δ* — unchanged: 0.0126–0.1331 over 360 rows, any finite value suffices;

(F3) *closed* — the T123 reduction to the band profile of Wy plus the uniform oscillation positivity (5R*); the telescope no longer needs it;

(F4) *collapsed* — no longer a quantitative estimate: (8R) needs only the sign of A_c, which the coarse-to-fine induction already carries; $\lambda_{\min}(A_c)$, $\mathrm{cond}(A_c)$ and γ^2 leave the chain entirely, and the base case is pure semidefiniteness;

(F5) *new, a bookkeeping convention, honestly stated* — the certified rung carries the coarse solution along (the solution-free version provably kills to zero: the coupling is a cancellation, not a perturbation); this is the same standard the whole chain already uses, but it is a convention, not a theorem, and it is listed.

Honest movement: the two-level worst case is gone — the last quantitative obstruction of the certification arc is replaced by a sign that the induction itself supplies, at the price of one named convention. What remains is the Harnack pair, one constant-not-rate loss per rung, and the assembly of the full chain end to end.

7.24 The grand assembly (T125): the chain composes

T125 (`grand_assembly_probe.py`, contract `GRAND.ASSEMBLY`, 34/34, ASSEMBLY-GREEN) is the series finale, and it does exactly one thing: it mounts every certificate of the arc on one ladder, end to end, and then states what the resulting object is. Five stages per zone — [**A**] the base-case Cholesky, [**B**] the telescope ascent with the (8R) rungs, [**C**] the ε lower bound on the target level, [**D**] the κ_{end}/parity chain, [**E**] the margin-free Albert handover — complete on 52 of 52 ladder zones, plus ten deep zones on stage [E] alone up to $n = 1331$ (h_{new} up to 1495).

The composing run. The construction that makes this an assembly rather than a list is ladder 1: 30 *consecutive* prime-power atoms $n = 2 \ldots 79$ on *one* common resolution $D = \min_k g_k/(2\nu) = 3.472446 \times 10^{-3}$ (with $\nu_{\text{eff}} = 4.0$–$58.4 \geq 4$, so every zone of the run is admissible in the T105 sense). Because the resolution is common, the new window of zone k *is* the old window of zone $k+1$: $M_n(k) = M_o(k+1)$ on all 29 consecutive pairs as an *integer* identity, the incoming atom's triangle restricted to the old window is the exact zero matrix (bit-exact, $\max|\text{entry}| = 0$), so $X = Q_{\text{old}}$ with no truncation, and the leading $h_o \times h_o$ block of the next zone's balanced form equals that X to 2.9×10^{-14}. The output of every step *is* the input of the next. This is a constructive way *around* T114's [P2] seam rather than through it: 0 of 84 consecutive log-gap ratios are dyadic — per-zone frames have no common refinement — so the run chooses one frame for all of it, and pays for that with its length (30 zones, $n \leq 79$, because D is set by the smallest gap inside the run). Ladder 2 keeps the per-zone frames and reaches 22 further zones to $\alpha = 2.885$, with the inter-zone seam declared rather than closed.

The load-bearing finding. Reading the assembly stage by stage, the weakest stage of the *whole* mounting is [D] at WINDOW-CERT on every zone — but [D] is the second, independent route. The *load-bearing spine* [A][B][C][E] is at CHOLESKY-CERT on all 52 zones, its weakest stage being [A]. In other words: **the Harnack pair (F1)/(F2) no longer carries anything**. It survives only inside conclusion (iv) of the theorem below, which (i)–(iii) and (v) do not use.

The numbers, stage by stage. The four identities the assembly stands on hold on all 88 rungs of all 52 zones (nesting to 3.0×10^{-14} and 6.2×10^{-16}; the saturation/dual-residual/Galerkin trio to 4.7×10^{-7}; the exact piecewise-constant lag formula to 2.2×10^{-14}; the telescope $\sum_\ell \delta_\ell = \varepsilon_0 - \varepsilon_L$ to 3.3×10^{-7}). (8R) is certified on 88/88 rungs by its three completed Choleskys, at retention $\mathrm{cb}/\delta = 0.9071$–$0.9971$ — T124's band 0.4739–0.9938, reproduced on a *different* ladder (frame windows, not the T124 M-chain) — and the direction control holds: the primal minimum form returns values *above* δ on 88/88 rungs, i.e. it is an upper bound and can never be a supply. The certified ε lower bound lands on 52/52 zones, retaining 0.0013–0.9694 of the measured ε_0 but 0.9140–0.9929 of the L-level *ceiling* $1 - \varepsilon_L/\varepsilon_0 = 0.001295$–$0.993502$ that an L-level chain can supply at all, and it exceeds the *declared* floating-point bound $c_h u\,\mathrm{cond}(A_0)$ by a factor 3.2×10^1–8.7×10^{10}. The base case is an *equivalence*, not a bridge: $\operatorname{sign}\lambda_{\min}(Q^{(L)}) = \operatorname{sign}\varepsilon$ on 52/52 zones at $h = 416$–1384, with $\lambda_{\min}(Q^{(L)})$ sitting 4.4×10^5–5.6×10^{10} times its own floating-point floor. The margin-free handover certifies 62/62 steps ($n = 2 \ldots 1331$) with the exact $g_c \times g_c$ Schur complement at $\lambda_{\min} = 1.484 \times 10^{-3}$–$1.253 \times 10^{-1}$, a factor 3.0×10^{10}–4.6×10^{13} above its own Cholesky floor — while the norm-perturbation surrogate of the *same* Schur complement, the one that divides by $\lambda_{\min}(X)$ and *was* the T113 wall, is negative on 62/62 steps. In total: 430 completed Cholesky certificates and 440 verified identities.

Three gates corrected in the open. (a) The κ band: the unconditional pairing certificate (C1) dominates the measurement on 52/52 rows, but its worst case over the ladder is $C_1 = 0.20927$ — and that worst case is a *single* zone, the shallowest one ($n = 2$, $\alpha = 0.3507$, only $n_C = 12$ corner cells, too short for the pairing to average); drop that row and $C_1 \leq 0.06721$ on the remaining 51, against T120's 0.04745 measured on deep zones only. (b) The head-room minimum is 31.6, at the thinnest zone ($n = 243$, $\alpha = 2.7492$, $h_0 = 679$), and it is thin because that zone's retention 0.6054 is itself pinned to its small L-level ceiling 0.629042 — not because the arithmetic is marginal; the factor stands over a *rigorous* bound, so any value above 1 certifies the sign. (c) The balance is fitted at *rung* level rather than on the collinear zone fit: $\mathrm{cb}/\delta \sim D^{-0.010}\alpha^{+0.001}$ (bands ± 0.004, ± 0.005, rms 0.012) — flat in *both* parameters, the strongest form of T124's quoted $\alpha^{-0.100}$ drift — and at zone level on the composing run, where D is common and only α moves, $\sum \mathrm{cb}/\sum\delta \sim \alpha^{-0.012}$: the certificate does not thin out along the run. The ratio reading $\mathrm{cb}/\varepsilon_c$ comes out at $D^{+0.345}\alpha^{+1.566}$ here and is explicitly *not* comparable with T124's $D^{-0.090}\alpha^{-0.100}$ (a different denominator: a 2–5-level frame chain against $M = 32$–2560 pairs) — reported, not claimed.

Five seams, named and measured. (S1) the inter-zone frame seam — *closed* on ladder 1 by the common-frame construction; (S2) the direction of the ε bound — structural and declared: the certified bound lands on the *coarse* level, so the target window must be the coarsest level of its chain and the base case the finest (a bound at the fine level would need certified *upper* bounds on the rungs, i.e. the CBS constant T123 proved resists); (S3) the matrix cap — cost, not a defect; (S4) the (F5) accounting standard — open, declared inside the theorem, with the solution-free variant measured to return exactly 0 of the rung ($|r_{\mathrm{corr}}| = 0.3888$–$14.8018$: a cancellation, not a perturbation); (S5) finiteness — open, *and it is the gap.*

Theorem V-final — the certified multilevel margin chain (T125)

Setting. Fix $\nu \geq 4$; let $n_i < \cdots < n_j$ be *consecutive* prime-power atoms with $u_k = \log n_k$ and log-gaps g_k, and put $D = \min_k g_k/(2\nu)$ (one common resolution), M_k the smallest even integer with $M_k D > u_k + D$, $\alpha_k = M_k D/2$, $A^{(k)}$ the reflection-odd window form on $h_k = M_k/2$ coordinates built from the exact Weil window lags, $b^{(k)}$ the odd pole vector, $\varepsilon_k = 1 - b^{\mathsf{T}} A^{-1} b$ and $Q^{(k)} = A^{(k)} - b^{(k)} b^{(k)\mathsf{T}}$; for each k let $D = D_0 > \cdots > D_{L_k}$ be halvings at fixed α_k.

Hypotheses. **(H-base)** $\varepsilon^{(k)}_{L_k} \geq 0$ at the *finest* level, equivalently $Q^{(k,L_k)} \succeq 0$ (Albert 1969) — pure semidefiniteness, no rate, no margin. **(C-F5)** an accounting convention, *declared open*: the rung numerator $\hat{s}_\ell = Z^{\mathsf{T}} b^{(k,\ell+1)} - B_x^{\mathsf{T}} u^{(k,\ell)}$ is a function of the *carried* coarse solution, the standard T122/T123 already used; a solution-free rung is *not* claimed — it is measured to fail. **(H-F1)/(H-F2)** the Harnack pair, needed *only* for conclusion (iv).

Conclusions. (i) [IDENTITY] $P^{\mathsf{T}} A^{(k,\ell+1)} P = A^{(k,\ell)}$, $P^{\mathsf{T}} b^{(k,\ell+1)} = b^{(k,\ell)}$, hence $\varepsilon^{(k)}_0 = \varepsilon^{(k)}_{L_k} + \sum_\ell \delta_\ell$ exactly. (ii) [CERTIFIED, given $A^{(k,\ell)} \succeq 0$] (8R) $\delta_\ell \geq \hat{s}_\ell^{\mathsf{T}} U_\ell^{-1} \hat{s}_\ell =: \mathrm{cb}_\ell > 0$; the rung is a *maximum*, so its denominator wants the form from above, and $v^* = U_\ell^{-1}\hat{s}_\ell$ is optimal in closed form — no A_c^{-1}, no $\lambda_{\min}(A_c)$, no CBS constant. (iii) [CERTIFIED, given (H-base) and (C-F5)] $\varepsilon^{(k)}_0 \geq \sum_\ell \mathrm{cb}_\ell > 0$, i.e. $Q^{(k)} \succ 0$. (iv) [WINDOW-CERT, second and *independent* route] the κ_{end} chain, with $|R-1| \leq 0.20927$, hence $R \leq 1.20927$ and $\kappa_{\mathrm{end}} \geq 0.45264$ on this run. (v) [CERTIFIED, margin-free] $X = Q_{\mathrm{old}}$ exactly, and $Q^{(k+1)} \succeq 0$ follows from Albert 1969 plus the Douglas range condition — nothing refers to the *size* of ε_k, only to its *sign*, which is (iii). (vi) [THE CHAIN] therefore (i)–(iii) and (v) compose: $Q^{(i)} \succeq 0 \Rightarrow \cdots \Rightarrow Q^{(j)} \succeq 0$, each step a completed Cholesky above its declared floating-point floor; (iv) is not used.

Attribution (25 lines). 10 IDENTITY, 9 CHOLESKY-CERT, 3 WINDOW-CERT (all three inside (iv)), 1 HYPOTHESIS ((C-F5)); every eigenvalue, condition number and retention is labelled MEASURED, every exponent law a FIT.

Verified instance. $D = 3.472446 \times 10^{-3}$, 30 consecutive zones $n = 2 \dots 79$, $\alpha = 0.3507$–2.1494, $h_k = 101$–619, 88 telescope rungs, plus 22 non-composed reach zones to $\alpha = 2.885$ and 10 handover-only zones to $n = 1331$; all 430 Cholesky certificates completed.

What is not claimed. (a) No statement about all zones — the run is *finite* and its length is limited by $\min_k g_k = 0.027780$. (b) (H-base) is a Cholesky fact about *one* finite matrix, not a theorem. (c) (H-F1)/(H-F2) are window certificates, measured and bounded on the measured windows, with no uniform proof. (d) (C-F5) is a convention, not a solution-free bound. (e) No zeta function, no zero data, no explicit-formula positivity, *in either direction*: this is a statement about a given quadratic form at a given resolution.

8 The kill list

A program that only reports what survived is not reporting. These are the readings and routes that were refuted *by the program's own probes*, each with the measurement that killed it. They are listed as a section rather than as footnotes because they are the more transferable output: each one tells a later attempt where not to go.

K1. The essential-singularity reading of the onset (T96, refuted T102). T96 read the counterfactual onset as super-polynomial, compatible with $\exp(-c/\delta')$, i.e. an essential singularity. T102 showed the onset is *anchored* at the crossing $\delta_c = 2w_k$ of two finite quantities ($R^2 = 0.968$), that exponential versus power behaviour is not separable at this resolution, and that the scatter of c (CV 103%) collapses to $c' = 1.52$ in the scale-free variable. The essential-singularity reading is *compatible but not distinguished* — and it is no longer used.

K2. The C/g law as a causal law (T101, refuted T102). Triple negative: g_k is causally impossible (Q_{k-1} is blind to atom $k+1$), statistically dispensable (causal law $C \cdot u$: CV 23.3% versus 27.6%), and arithmetically only a ceiling ($C_k \le g_k \mu_k$ exactly; the 16-zone extrapolation violates its own ceiling from $k = 69$; checked on 18 120 prime-power atoms to $n = 200\,000$). *It was a proxy.*

K3. The naive margin route (T104). $M \ge m \cdot \mathrm{Id}$ holds in 0/16 zones; the coupling-mass to margin ratio is 10^3–10^6; $m \sim M^{-1.7}$ vanishes in the continuum (fit). The genuine form is nearly singular on the induction window and no margin rescues it. (The prolate wing basis recommended by T96/T103 was tested in both arms and also rejected.)

K4. The density chain (T106). It would need a hard gap $w_0 \ge 0.55$–2.44; measured 5.3×10^{-6}–1.1×10^{-3} — short by a factor 5×10^2 to 3.6×10^5. Dead for good.

K5. The Landau–Widom anchor (T106). The wrong classical anchor for this problem (rms 0.50–1.55). The right one is Grenander–Szegő on the Planck-coarsened symbol — averaging over the mode-spacing cell π/M — which hits 0.92–2.43 and is better on 67/72 pairs. This also explains why a negative symbol (dip $-21.3 \dots -3.4$) still gives $Q \succeq 0$: the atoms oscillate exactly at the window's Planck scale.

K6. The accumulated invariant with self-propagation (T106). A no-go *with a measured mechanism.* The wing demand gives $\beta_{\text{wing}} = 2.21$–$14.43 > 1$ on 16/16, but inherited demands give $\beta_{\text{int}} \sim 10^{-2}$ on 15/15: an accumulated invariant with absolute anchoring is false. The reason is transport — in a larger window an old wing pair becomes an interior pair and draws 99.7–99.95% of its demand from the 64 softest modes (coupling 751–2653× stronger). No self-propagation either ($\beta_{\text{on}} < 1$ on 15/15, loss 13–320×, $\theta_k = 0.9866$–0.9998). *The wing margin is an edge property of the window, not a property of the pair.* What survives is the window-wise family $\sigma_k(\delta_0) \geq 1.31(\mu_k/2)$, uniform in M with drift $\leq 2.01\times$.

K7. The symbol route for the odd channel (T107). Structurally dead, not merely weak: the symbol lower bound for T_{odd} is certified but *negative* ($\lambda_{\text{cert}} = -2.54 \ldots - 0.81$) against a measured $\lambda_{\min}(T_{\text{odd}}) = 4.7 \times 10^{-5} \ldots 6.4 \times 10^{-2}$ and a symbol infimum of $-22 \ldots -6$: five orders of magnitude and a sign change. The soft edge of T_{odd} is a pure finite-section effect that no symbol can see. Coarse certified chains (Cauchy–Schwarz against $\lambda_{\min}$) close 0/16 — and fail even with the *measured* $\lambda_{\min}$, so it is a category error, not a precision problem.

K8. Three certificate candidates for the avoidance norm (T108). Killed *quantitatively*, which is the useful form: the cancellation route ($\sum |k| = 401$ against $\sum k = 1$), the Bessel route (off by 10^7) and the pole Cauchy–Schwarz route (off by 10^6). ω_{cert} is blocked by the compression Schur distance alone (0.18–0.52). What replaced them is not a fourth bound of the same kind but a different object class: a boundary-decay statement about one explicit vector. T109 then killed the natural fourth candidate as well: **the Combes–Thomas resolvent bound at the boundary**, evaluated with the *exact* Green row so that no constant can rescue it, closes 1/16 — the mechanism at the edge is cancellation (44–4.6×10^4 on 15/16 zones), which no absolute-value bound survives. What works instead is the residual certificate that carries the cancellation (Section 7).

K9. The scalar step law for the margin (T110). Killed structurally, in three pieces: bordered Weyl closes 0/15 handovers; the Schur/Friedrichs scalar cap is vacuous on 15/15 (its scalar-only bound degrades back to Weyl); and the strictly scalar graded minorant ($n_{\text{soft}} = \dim$) closes 0/15, with the strictly scalar *induction* compounding to a Schur factor of 2.7×10^{-36} by $k = 3$. The reason is a boundary layer: the new cells of a grown window attach at the edge where the pole source sits, so a one-number summary of the bordering is blind to the deciding direction. What survives is the graded ($n_{\text{soft}} = 1$) Loewner minorant — one soft direction is exactly the amount of geometry the step needs.

K10. Two earlier structural kills, recorded here because they still bound the design space: the cross-block $\sqrt{\lambda_0}$ law is Douglas range inclusion and therefore circular (T98), and the cone-library route saturates at 5/24 coverage and is dead as a covering strategy (T72, promoted into v540 as a named limit). Likewise the seam route to the archimedean place (T20–T25) and the Kohnen-plus route to the central values (T44, promoted into v537 as a scope fence).

The finale re-counted this list against the whole certification arc and made it quantitative: 24 **refuted routes across** 18 **parts**, every one of them structural — an identity (the T123 ceiling, the T124 primal direction), an exact sign ($\lambda_{\min}(E_c) < 0$, the negative norm surrogate, the non-dyadic gap ratios), or a measurement that tuning cannot move (the vacuous band split, the empty energy route, the cancelling solution-free rung). The counting matters because *four* of the 24 died of the *same* cause — a bound that needs the coarse form from *below* — which is why the exit was a direction reversal and not a sharper estimate. T125 added one entry of its own: a certified ε bound at the *fine* level of a chain is the CBS wall again, which is why V-final mounts the target window as the *coarsest* level.

9 The series balance (T102–T125)

The finale's second half is bookkeeping about the series itself: what the hard core was at each stage, how much of the final chain is a certificate rather than a hypothesis, and how far that chain is from any infinite statement. All three are measurements against the same source, and they are reproduced here because they are the part a later reader can check without re-running anything.

9.1 The reduction cascade, seven stations

Each arrow is a strict reduction in dimension or in kind, and every one of them was paid for by a refutation in the list above:

1. **Matrix wall** (T104–T106) — a Loewner statement $M \succeq m\,\mathrm{Id}$ on $M/2$ dimensions, killed as a margin route and localised by the parity split to one statement (R) in the odd channel.
2. **Scalar** (T107–T108) — Sherman–Morrison/Woodbury reduce (R) exactly to one scalar ratio $r = \kappa/\varepsilon \le 1$, and ε itself becomes an identity (the square of the last Cholesky pivot).
3. **Ratio** (T108–T109) — the ratio splits into $\omega < 1$, certified unconditionally by a graded matrix cap, and one statement at an explicit vector.
4. **Boundary value** (T109) — what is left is literally one boundary value, carried by a residue certificate that *keeps* the cancellation instead of estimating it away.
5. **One inequality** (T110–T117) — the frame removes the ladder and ω walls, the margin wall turns out to measure the discretisation, the margin-free Albert step removes the division that was the wall; what remains is one lower bound on ε.
6. **Harnack pair + one sign** (T118–T124) — the ε bound becomes an identity plus a supply, and the supply needs either the corner route ((F1)+(F2)) or the telescope route ((8R), whose only requirement is the *sign* of A_c).
7. **One sign, one convention** (T125) — assembled: the spine [A][B][C][E] needs (H-base), a sign on *one* finite matrix, plus the (C-F5) accounting convention. **The Harnack pair survives as the second, independent route (iv) and is no longer a bottleneck of the spine** — this last station is what T125 added.

9.2 The certification degree

Every link of V-final counted once, classified by what actually establishes it:

Class	Links	%	Spine	%
IDENTITY	14	45.2	11	42.3
CHOLESKY-CERT	14	45.2	14	53.8
WINDOW-CERT	2	6.5	0	0.0
HYPOTHESIS	1	3.2	1	3.8
Total	31	100.0	26	100.0

90.3% of the whole chain is identity or completed Cholesky; on the load-bearing spine it is **96.2**%, with *zero* window certificates and exactly one hypothesis — the (C-F5) accounting convention. The three window certificates are confined to conclusion (iv).

9.3 The honest RH distance, in six checkable statements

(R1) No zeta input. The chain consumes exactly two arithmetic facts, both classical and both only for *admissibility* of the frame: $g_k \le \log 2$ (Bertrand–Chebyshev 1852) and $g_k \ge \log(1+1/n)$. No zeta function, no zero data, no zero-derived quantity — the probe's own AST firewall rejects the tokens, the imports and write-mode file access in its source.

(R2) RH is not used, in either direction. "RH $\Rightarrow$ window Weil positivity" appears nowhere; (H-base) is a completed Cholesky of one explicit matrix of size $h = 416$–1384.

(R3) No infinite statement. V-final quantifies over a *finite* run of consecutive atoms whose length is bounded by $\min_k g_k = 0.027780$: here 30 zones, $n = 2 \dots 79$.

(R4) A measurement ladder is not all zones. 62 of 85 admissible handovers in the table, and 48 of 34 034 prime-power atoms up to 400 000 touched at all. Everything outside is untested, not implied.

(R5) The open items are uniformity, not size. What is missing for any infinite statement is a uniform-in-k version of stage [E]'s $g_c \times g_c$ Schur complement, a uniform lower bound on cb_ℓ, a proof rather than a measurement of (H-F1)/(H-F2) — classical addresses named (Widom 1974 / Fisher–Hartwig for the corner regularity of the inverse, Ostrowski 1937 / Varga 1962 for the sign) — and removal of the (C-F5) convention, which T124 measured to be a cancellation.

(R6) What it is. A machine-certified, finite, multilevel *lower* bound on the Galerkin prediction defect of a log-singular Toeplitz form with a prime-power comb, together with a margin-free bordering induction that transports its sign along a nested window chain. That is a statement in numerical analysis. It is not a statement about zeros.

The defect counter closes accordingly: (F1)/(F2) window-certified and now *outside* the spine, (F3) closed in T123, (F4) collapsed to a sign in T124, (F5) declared as (C-F5) — and one new entry, **(F6) uniformity**, the only thing between V-final and an infinite statement.

10 Phase 2: toward uniformity (T126–T163)

The series closed at T125 with a mandate fulfilled and a distance named: what separates V-final from any infinite statement is *uniformity in the zone index*, not size. The first post-series part — T126 (`uniformity_seams_probe.py`, contract `UNIFORMITY.SEAMS`, 31/31, verdict SEAMS-CERTIFIED) — opens a second phase of the diary by attacking exactly that gap in its three components: the *segment seams* (consecutive runs carry different common resolutions, and the seam between them is one re-gridding), the *uniformity of the spine constants* (the (F6) entry of V-final, audited one constant at a time), and the *continuum argument* (finite-section positivity at every resolution $\Rightarrow$ positivity of the continuum window form on a test class). All three move; what remains at the end is stated in Section 10.4 and is deliberately short.

10.1 The seam architecture: the partition question dissolves

The natural first question is arithmetic: can the prime-power axis be cut into segments such that each segment carries one common resolution $D_{\text{seg}} \propto$ min-gap and neighbouring segments have a resolution ratio the certified transport reaches? Since D_{seg} is set by the segment's minimum log-gap, this is a question about *minima of windows of the log-gap sequence* and nothing else — so it is decided *exactly*, by a dynamic programme over cut positions on the real gap table (9 700 zones, $n \le 10^5$; 33 DP runs in 2.71 s). The answer is a refutation with a replacement:

- **At T115's synthetic threshold $\rho^* = 1.83$ no partition exists.** The DP dies at zone $n = 127$, at the local gap pattern $[0.0684, 0.0325, 0.0159, 0.0078, 0.0232, 0.0448]$ — a small gap directly behind a large one with no intermediate gap in the admissible window: the twin-after-a-large-gap mechanism, named exactly. The true requirement is $\boldsymbol{\rho_{\text{crit}} = 2.0258}$, identical on $n \le 10^4$ and $n \le 10^5$ — not an aggregate but essentially *one* quotient of neighbouring gaps, $g(125)/g(127) = 2.0238$, reproducing ρ_{crit} to 2.0×10^{-3}. A partition at ρ_{crit} is *constructed* (306 segments; the T112 frame lemmas hold on all 9 700 zone steps, $\min \nu_{\text{eff}} = 4.00$, zero violations).
- **A refinement-only partition exists at no ratio** ≤ 64, so the tight-cap partition forces *coarsening* seams — and coarsening fails for real: on real pairs the transported certificate fails at ratios 0.309 and 0.531 while the target floor itself stays positive on 9/9 (what fails is the certificate, not the positivity), and ladders of intermediate resolutions make it *worse* ($k = 1, 2, 3$ at $n = 47$: $-2.54 \times 10^{-2} \to -2.52 \times 10^{-2} \to -4.63 \times 10^{-2}$) — the obstruction is the *destination grid* on which every ladder must land; a finer subdivision cannot cure a bad target.

- **The decisive new measurement:** holding the worst-case ratio 2.0256 *fixed* and sweeping the zone, the transported certificate survives on only 12 of 22 zones. The seam dies at the *zone*, not at the ratio — so no ratio threshold can ever close the programme, and the lemma the seam architecture needs is not "a larger ρ" but a *zone-uniform* η/τ bound at ratio ≤ 2.026 (U5 below, sharpened).

What carries is the free-resolution route. Releasing the resolution from the per-segment cap and taking the running minimum $D_k = \min(\mathrm{cap}_k, D_{k-1})$ makes monotonicity hold *by construction* — no DP, no gap theorem, not one coarsening seam anywhere. The largest single refinement drop over $n \leq 10^5$ is 2.0256 (zone $n = 8191$ — the same binding quotient, now read in the good direction), and re-gridding happens at only **764** of 9 700 boundaries (7.88%): a drop needs a *record-small* gap, and every other boundary is a no-op at ratio 1. All 12 affordable actual seams of that route ($n = 3 \dots 125$, $h \leq 1222$, largest real ratio 1.943) are built and transported: 11 carry a positive certified floor in *one* re-gridding, the one at $n = 31$ (transported mass $\tau = 0.363$, unusually poor) closes through a three-stage ladder with composed floor $3.56 \times 10^{-2} > 0$; median retention 0.578, all 12 brackets valid. The seam device itself is sound throughout: the Haynsworth partial minimisation reproduces $\lambda_{\min}(S)$ to 2.8×10^{-14} on both grids and the transport bracket contains the fine value on 30/30 real pairs; and a *full* seam is certified end to end at the ratio the arithmetic demands — spine on grid A, one re-gridding (lower bracket end $8.671 \times 10^{-3} > 0$, retention 0.103), spine on grid B — at $\rho = 2.251 \geq 2.0256$ on 3 of 4 attempts (the failures are the zone dependence above, printed and not hidden). The price of the route is honest and structural: the resolution overhead $D_{\mathrm{free}}/\mathrm{cap}$ has median 0.201, i.e. the *running minimum* of the gaps sets the window cost instead of the local gap — window size, not an assumption; certified seam by seam the route covers $n \leq 125$, an evidence limit rather than a gap in the argument.

10.2 The uniformity audit (U1–U6)

Every constant of the load-bearing spine is measured over zones *and* telescope depth (35 rungs, 14 zones, (8R) valid on 35/35 while audited; every fit jackknifed; the status rule declared before the numbers, and "measured-flat" is never upgraded to "uniform"):

Constant	**Finding**	**Status**
U1 envelope margin	forced by the construction's own doubling loop	self-certifying
U2 oversampling depth L/M	needs one α-uniform Bernstein/Markov bound	classical in form
U3 rung quality cb/δ	measured-flat, but the reason is *direction*, not spectrum: the worst pencil direction of (S, U) does not explain the flatness (alignment 0.866–0.987, median 0.975; $\mu_{\min}$ itself falls with the window)	genuinely new
U4 base-case head-room	a compute wall over the declared fp floor	no proof obligation
U5 Albert–Schur floor	$\lambda_{\min}(S)/\|A\|$ *falls* in the window direction — the one load-bearing gap; sharpened by the seam scan to a zone-uniform η/τ bound at ratio ≤ 2.026	genuinely new
U6 resolution monotonicity	the running minimum is monotone by construction	resolved

The audit's own finding is that (F6) is not one lemma but three — U2 (classical in form), U3 (a direction lemma on $\hat{s}$), U5 (a zone-uniform Schur floor) — of which exactly two are genuinely new mathematics.

10.3 The continuum step is a fractional Dirichlet energy

The chain one would write down first — density of piecewise-constant functions plus a continuity constant — is *refuted*: the archimedean kernel is Cauchy-type $\sim 1/(2|s|)$, not log-type, hence not locally integrable; its L^1 mass over the window and $\lambda_{\max}$ both diverge like $\log(1/D)$, so neither an L^2 nor a sup-norm continuity constant exists (measured). What makes the divergence harmless is an *identity*: with row sums w_r,

$$v^{\mathsf{T}} A v \;=\; \sum_r w_r v_r^2 \;+\; \sum_{r<s} (-A_{rs})\,(v_r - v_s)^2,$$

with 86.1% of the off-diagonal entries measurably negative and the row sums bounded — the window form *is* a fractional Dirichlet energy, and the divergent diagonal is the $\frac{1}{2}$-energy of a step function, a property of the test class rather than a defect of the kernel. The limit argument is therefore quantitative, and it is measured: on the dyadic tower (5 levels, $h = 63 \ldots 1008$, a $16\times$ spread in D) the Galerkin values converge at order $D^{1.798\pm0.057}$ for a C^∞ bump and $D^{1.802\pm0.065}$ for a Lipschitz hat — superlinear in both cases, exactly as the energy identity predicts — with every level strictly positive and Richardson limit $2.1256 \times 10^{-4} > 0$. What is classical here is the consistency estimate for a Cauchy-kernel Galerkin scheme (Calderón–Zygmund; Bramble–Hilbert 1970); what is measured is its rate on this kernel; the abstract density step is not needed and not used. The statement that would stand at the end is **positivity of the Weil window form on Lipschitz test functions supported in the reached window** ($\alpha = 1.5046$ on the tower, 1.6882 along the seam chain) — a finite-window statement whose bridge to RH is Weil's criterion (Weil 1952; Bombieri 2000), cited here and used in neither direction.

10.4 The full-proof map: sixteen points, two genuinely new

The definitive fail list after the three attacks has 16 points: 5 closed or classical, 9 conditional (constructed here, classical in form, or a machine limit), and exactly *two* genuinely new pieces of mathematics:

The two remaining inequalities (T126)

(U3) The direction lemma. Why the rung's data residual $\hat{s}$ sits in the good subspace of the pencil (S, U): the measured flatness of cb/δ is provably *not* the pencil's worst-case behaviour (alignment 0.975 while $\mu_{\min}$ falls), so a proof must control the *direction* of $\hat{s}$, not the spectrum.

(U5) The zone-uniform seam floor. A zone-uniform η/τ bound for the re-gridding transport at ratio ≤ 2.026 — the constant every seam consumes. Ladders provably cannot repair a bad destination grid, so the reach must be won per *zone*, not per ratio.

S4, the coarsening bracket — which this part opened and which would have been a third genuinely new statement — is *dissolved* by the free-resolution route rather than proved. Still open and unchanged: the (C-F5) accounting convention, and the all-α step — the one thing no computation can close (at the tight cap, 6 boundaries in 9 700 would need a double seam; for all n one needs a gap theorem, or the free-resolution relaxation, which is unconditional in the arithmetic). T127 (`two_inequalities_probe.py`, contract `TWO.INEQUALITIES`, 28/28, verdict BOTH-SHAPED) then took both inequalities head-on. The result — both come back proof-shaped, and *neither is the statement T126 named* — is the remainder of this section.

10.5 The separating variable is geometric, not arithmetic (T127)

The fixed-ratio sweep reproduces exactly: at the worst-case ratio 2.0256 the transported certificate survives on 12 of 22 zones, T126's list — and what dies is the *certificate*, not the positivity: the true target eigenvalue $\lambda_{\min}(S_f)$ is positive on all 22 zones. Of 29 candidate separating variables, exactly 5 separate the surviving from the dying zones perfectly (22/22; the rank statistic is a description of a 22-point sample and is never called a test), and all five measure the same *geometric* quantity: the right-hand protrusion $s_{l,r}$ — the two grids quantise the same window independently, their border blocks have equal width but sit offset by one rounding, and the transported mass τ_{dn} loses exactly what protrudes past the coarse block's inner edge (correlation $+0.997$ with τ_{dn}). Every arithmetic candidate fails (gap, α, atom weight, alias phase; best separation 0.727) — the separation is *not* the atom and *not* the gap. Deep arithmetic ($n \leq 300\,000$, 1 082 record seams): the fine border block has 4 cells everywhere, 97.60% of real seams have ratio ≤ 1.05 with median

1.00140 — real seams are *near-identity re-griddings* — and the 11 seams with ratio > 1.2 *all* have next-atom distance ≤ 2: Mersenne $(3, 7, 31, 127, 8191)$, Fermat-like $(4, 16, 256, 65536, 131071)$, one twin $(101, 103)$. Mihăilescu 2004 (Catalan) is cited as the classical address of that list and not used.

10.6 U5 as posed is refuted; the U5 band replaces it

As a zone-uniform certificate at ratio ≤ 2.026, U5 is *dead as posed.* The replacement is **U5-band**: all 22 zones survive for every ratio $\rho \leq 1.49531$, and at least one dies for every $\rho \geq 1.49766$ (bisected to 2.3×10^{-3}; binding zone $n = 19$) — a band that covers 99.26% of the route's real seams, the remainder being the 8 enumerated pairs above. The split of the form defect is an *identity* ($\eta = \eta_{\mathrm{cons}} + \eta_{\mathrm{proj}}$, exact), and η does *not* become small as $\rho \to 1$ (max $|\eta|/\lambda = 1.146$ at the *tightest* ratio): the band buys $\tau_{\mathrm{dn}} \approx 1$, not small η. The floor is measured at $c \geq 0.0090$ over 307 in-band transports with a positive single-step floor up to $n = 1331$, $h = 1496$. **The sharpest new finding:** the near-identity census (43 deep zones, all at ratio 1.00140) fails *once* — $n = 479$, $|\eta|/\lambda = 1.472$, within 0.14% of the identity — so the ratio band is only a *proxy* for the border geometry; the failure is repaired by a three-stage ladder (retention 0.5027), exactly as T126 repaired $n = 31$ — the fallback is part of the statement. And the risk is recognisable *before* any factorisation: the indicator $\tau_{\mathrm{dn}}/\tau_{\mathrm{flat}}$ ranks the margin at Spearman +0.749, and the one failure is its argmax (τ_{flat} is a predictor, not a bound).

10.7 U3 defuses to a coarse floor: the harmonic-mean identity

Three identities restructure U3. The pole part of the rung data has the closed form $(Z^{\mathsf{T}}b)_j = -\frac{16}{\sqrt{2D}} \sinh^2(D/4) \cosh(\bar{x}^{\mathrm{c}}_j/2)$ (to 3.2×10^{-13}); the rung quality is a *harmonic mean*, $\mathrm{cb}/\delta = 1/\sum_i w_i/\mu_i$ (to 1.5×10^{-7}); and the weights split by Parseval (to 6.1×10^{-16}). The residual's mass on the good half of the pencil spectrum $\{\mu \geq \frac{1}{2}\}$ is $0.9665 \ldots 0.99998$ over 42 rungs (flat), so $\mathrm{cb}/\delta \geq 1/(2m + (1-m)/\mu_{\min})$ — a floor as *crude* as $\mu_{\min} \geq 1.62 \times 10^{-2}$ (one part in 62) already gives $\mathrm{cb}/\delta \geq \frac{1}{4}$, where T126 had to measure 0.192–0.307 sharply. The reading "$\hat{s}$ avoids the bad direction" is *refuted*: the flatness is generic — every smooth, flat, alternating or quasi-random direction has good-band mass above 0.50 — and exactly *one* direction type fails, a *boundary layer* at the window edge (concentration 3.1× there, against 0.03× for $\hat{s}$ and 0.001× for its closed-form pole part). The bad pencil directions are *not* edge-localised in the ℓ_2 sense (1.23×) — only the U-metric concentration separates.

10.8 The map V2: three inequalities and one enumeration

The map V2 has 16 points: 10 carried, 2 conditional-classical (one Bernstein/Markov bound, the declared fp floor), and the 2 genuinely new ones of T126 *reformulated*:

The remaining mathematics (T127, map V2)

(T) The border-retention bound. A lower bound for the transported mass τ_{dn} under the edge-geometry hypothesis — from predictor to bound, via the support of the S_f eigenvector; the ratio band $\rho \leq 1.50$ is only a proxy for it.

(E) The $|\eta|$ upper bound. Both parts of the form defect $\eta = \eta_{\mathrm{cons}} + \eta_{\mathrm{proj}}$, to be bounded *jointly* with (T): the two rank together at +0.749.

(M) The pencil-mass bound. The residual's good-band mass plus the crude floor $\mu_{\min} \geq 1.62 \times 10^{-2}$; the one direction type to exclude is named (a boundary layer at the window edge), and the closed-form pole part provably is none.

(L) The enumeration. The 8 out-of-band seams up to $n = 300\,000$ (Mersenne/Fermat pairs and one twin), a finite list; where a single step fails, a three-stage ladder composes — the fallback is part of the statement.

The craft judgement, labelled as one: the three remaining inequalities are of the *same craft* as the proven links of the chain — finite-dimensional inequalities between constructed matrices

— and the residual risk is concentrated in the word *uniform*, which no amount of this probe's arithmetic can discharge. T128 (`teml_probe.py`, contract `TEML`, 27/27, verdict THREE-OF-FOUR) then worked that four-point list in order of cost, cheapest first. Three of the four points stand at their preregistered bars; the fourth — the protrusion concentration inside (T) — missed its own bar by 3.6%, systematically in the re-gridding ratio, which is exactly the kind of near-miss that distinguishes a bound from a fit. The remainder of this section is the four points in order.

10.9 (L) is derived and closed: the out-of-band table (T128)

The out-of-band class is no longer quoted but *derived*: applying the U5-band threshold $\rho > 1.49531$ (T127's number, quoted and never re-bisected) to the free-resolution record schedule yields *exactly eight* seams up to $n = 300\,000$, all at next-atom distance 1 and all with identical border geometry ($\mathrm{gc}_c = 2$ coarse border cells, $\mathrm{gc}_f = 4$ fine) — each is a pair $(2^k{-}1, 2^k)$ or $(2^k, 2^k{+}1)$ whose log-gap halves the previous record. Every seam carries a status and no residue:

seam n	ratio ρ	class	status	retention / required h_f
7	1.671	Mersenne	CERTIFIED-1STEP	0.179
16	1.943	Fermat-like	CERTIFIED-1STEP	0.156
31	1.910	Mersenne	CERTIFIED-LADDER (2 rungs)	0.091
127	2.024	Mersenne	BUDGET-OPEN	$h_f = 2\,476$
256	2.012	Fermat-like	BUDGET-OPEN	$h_f = 5\,694$
8191	2.026	Mersenne	BUDGET-OPEN	$h_f = 295\,253$
65536	2.001	Fermat-like	BUDGET-OPEN	$h_f = 2\,907\,297$
131071	2.000	Mersenne	BUDGET-OPEN	$h_f = 6\,177\,926$

The five open seams exceed the hard factorisation cap $h \le 1500$ by 1.7× to 4119×, and each is open for *one* reason only: a single Cholesky of size $h_f - \mathrm{gc}_f$ is missing — nothing conceptual. Each is characterised arithmetically and geometrically without any factorisation (ratio, both border cell counts, both window roundings, the flat transport factor), so the characterisation is available at any depth. "Closed list" means every item is enumerated and labelled, not that every item carries a Cholesky — that distinction is the point of the block. Mihăilescu 2004 (Catalan) remains the cited, unused address of the class; over a finite range a list is enumerated, not conjectured.

10.10 (T) becomes a bound, and its bar is missed honestly

The step from predictor to bound is an exact identity plus two explicit majorants. With y the transported S_f eigenvector, m_{prot} its mass on the protrusion, $m_{\mathrm{fill}} \ge 0$ the interior mass on the sliver the coarse block adds, and $V_{\mathrm{bord}} \ge 0$ the intra-coarse-cell variance of the transported function,

$$\tau_{\mathrm{dn}} \;=\; \|y\|^2 - m_{\mathrm{prot}} + m_{\mathrm{fill}} - V_{\mathrm{bord}}, \qquad \text{hence} \qquad \tau_{\mathrm{dn}} \;\ge\; 1 - m_{\mathrm{prot}} - V_{\mathrm{bord}},$$

verified to 9.0×10^{-13} against an independent full projection on all 126 transports. The bound is valid on 126/126 (slack $= m_{\mathrm{fill}}$, median 0.000); with the two explicit majorants — the outer-cell majorant for m_{prot} and the discrete oscillation majorant $\rho(\lceil\rho\rceil{+}1)/2 \cdot \sum(\Delta y)^2$ for V_{bord} — it stays valid on 126/126 and positive on 125/126 (the exact form is positive on 126/126); total slack median 0.275, and the loose part is the oscillation majorant (median 7× overshoot), so what (T) still needs is a Bernstein/Markov-type statement about the discrete oscillation of one $\mathrm{gc}_f \times \mathrm{gc}_f$ eigenvector. **The honest miss:** the protrusion concentration $\kappa = m_{\mathrm{prot}}\,\mathrm{gc}_f/(t_l + t_r)$ — eigenvector mass on the protrusion against what a flat border vector would put there — measures 0.086–2.071 over the 120 transports with a nonzero protrusion, and *misses the preregistered bar* $\kappa \le 2$ *on* 1/120 *by* 3.6%. The excess is systematic in the ratio, not accidental: the group maxima rise monotonically, 1.889/1.965/2.071 at $\rho = 1.001/1.250/1.495$, with the argmax at the band edge. The bar is *not* moved — a bar moved after the numbers would make everything else here worthless; the shape the next preregistration should declare is $\kappa \le 2 + C(\rho - 1)$, the shape the U5 band already has. A side finding with a side: κ is 1.61–2.07 when the protrusion sits at the *inner* edge of the border block and 0.09–0.25 at the window edge — (T) is an *eigenvector* statement, not a rounding statement, contrary to what the +0.997 correlation invites one to think.

10.11 (E) and (T) jointly: the ℓ_2 route dies, the energy route carries

The ℓ_2 route to η_{proj} is structurally lossy where it matters: the norm bound overshoots the true defect by a median factor of 55, and since the available margin relative to λ_c is smaller still, it certifies 0 of 12 real seams. What carries is the *energy route*, an exact Cauchy–Schwarz chain in the Q-inner product: with Q_f certified PSD and $d = w_f - Pv$ the projection residual,

$$\sqrt{\lambda_f} \;\geq\; \sqrt{\lambda_c\,\tau^* - |\eta_{\text{cons}}|} \;-\; \sqrt{e_d}, \qquad e_d = d^\mathsf{T} Q_f d,$$

with $e_d/\lambda_c = 0.041$–0.154 on the real seams. The joint chain certifies a positive floor on 9/9 **in-band seams** (floor ≥ 0.0850) and 11/12 overall. The one tear — $n = 31$, already on the (L) list — is η*-limited*, not geometric: $|\eta_{\text{cons}}|/\lambda_c = 0.392$ exceeds the whole geometric floor $\tau^* = 0.363$, so no sharper (T) would have helped; and shrinking the ratio is no alternative either, since the consistency defect does not vanish as $\rho \to 1$ (median 0.089 at $\rho = 1.0014$ over 43 zones). The tear is repaired by the declared two-rung ladder in which *every rung closes the chain by itself* — the fallback is part of the statement, not a patch.

10.12 (M) is earned: the exclusion is a proof, the floor is constructed

Three parts. *(a)* The boundary-layer exclusion for the pole part of $\hat{s}$ is now a *proof* from the closed form $(Z^\mathsf{T} b)_j = -\frac{16}{\sqrt{2D}} \sinh^2(D/4)\cosh(\bar{x}^c_j/2)$: the adjacent-component ratio lies in $[e^{-D/2}, 1]$ and the outer-k-cell share is bounded by $kD(1+\cosh\alpha)/\sinh\alpha$ — pole edge mass $\Theta(D)$ against a layer's $\Theta(1)$, separation at least 84× on all 32 (α, D) pairs, by elementary inequalities rather than a finite sample. *(b)* The crude floor is *earned*: $\mu_{\min}(S, A_z) = 1 - \gamma^2$ exactly (to 5.9×10^{-15}; the strengthened Cauchy–Schwarz constant of the two-grid splitting, Eijkhout–Vassilevski 1991), and $U \preceq (1 + \varepsilon_{\text{env}})A_z$ with $\varepsilon_{\text{env}} \leq 0.038$ Cholesky-certified, so the constructed floor is a true lower bound on all 31 rungs: **0.188** against the 1.62×10^{-2} the harmonic-mean argument demands — an 11.6× margin, where T127 could only measure. *(c)* The comb residual stays measured, but it is measurably *no* layer: half of its mass needs 12–39% of the half-window against one cell for a genuine boundary layer, and it vanishes with the cell width as $D^{+1.54\pm0.09}$ (a fit with a jackknife band).

10.13 The map V3: twenty-one points, five genuinely open

The map V3 (T128): what is left, by kind

21 points: **10 done** (T127's map still had the (M) floor, the (T) identity and the (L) list among its open items; all three are now closed), **3 certified on real objects** — valid on every seam, transport and rung the hard cap admits, wanting only the word "uniformly" — and **3 conditional-classical** (Bernstein/Markov for the border oscillation, Bramble–Hilbert for the consistency defect and the Q-invisible residual, and the continuum leg). **5 genuinely open**, and no two of the same kind:

M9 the κ bar — missed by 3.6%, systematically in ρ; next hypothesis $\kappa \leq 2 + C(\rho - 1)$.
M17 the pencil mass of $\hat{s}$ on $\{\mu \geq \frac{1}{2}\}$ — one measured scalar per rung, no derivation.
M18 the comb residual is not a cell-scale layer — measured, not yet bounded.
M19 uniformity over *all* zones — the word "for all", which a finite probe structurally cannot supply.
M21 the RH address itself — Weil positivity as the address of the limit statement, cited and never used.

The honest distance, in the probe's own three sentences, quoted:

> *(1) After this block the finite-window chain has no step left that is merely plausible: every inequality it uses is either a completed certificate, an identity verified to machine*

> *precision, or an explicit inequality that has been evaluated — and found valid — on every seam, transport and rung the hard cap $h \le 1500$ admits, so the remaining mathematical content is three named classical estimates (Bernstein/Markov on one small Schur complement, Bramble–Hilbert twice) plus two measured scalars (the pencil mass, the comb's delocalisation). (2) The distance is nevertheless not small, and it is not small in a way no larger computation can shrink: 5 of the 21 map items are open because a finite list of objects cannot produce the word "for all zones", and one of them — the protrusion concentration κ — has just been measured outside its own preregistered bar by 3.6 per cent, systematically in the re-gridding ratio rather than accidentally, which is exactly the kind of near-miss that distinguishes a bound from a fit. (3) And even with all four TEML points closed, what would stand is a positivity statement about a finite window at the α actually reached: the continuum leg (M20) and the passage to Weil's criterion (M21) are untouched by anything in this probe, so the honest distance to RH is not "one uniformity lemma" but "one uniformity lemma, three classical estimates, a continuum limit, and a theorem nobody has".*

T129 (`kappa_deep_seams_probe.py`, contract `KAPPA.DEEP.SEAMS`, 28/28, verdict KAPPA-WILD) then did what the map demanded: it froze the constant of the κ law, tested the law on fresh transports, brought the two affordable deep seams under the cap, and bounded the comb. The verdict is the most productive break of the phase — *the fitted law falls, and the theorem underneath it stands.* The remainder of this section is that part.

10.14 The honest break: the κ law falls once, and the theorem underneath stands (T129)

The constant was frozen *before* the test: $C^* = 0.3506$, the least-squares slope of the T128 group maxima against $\rho - 1$ over the three calibration groups (jackknife SE 0.0955, intercept 1.8909, rms 0.0096) — a fit, labelled a fit, not rounded up, not padded, not refitted after the test; on the calibration set itself the law is violated 0/105, reported for completeness because it is *not* the test. On **331 new transports** at 14 ratios none of which is in the calibration set ($\rho = 1.05 \dots 4.00$, $h_f = 29 \dots 1450$, $n = 5 \dots 1331$), the frozen law $\kappa \le 2 + C^*(\rho - 1)$ is violated **once** ($n = 7$ at $\rho = 3.00$, slack -0.0334), where the bare unmoved bar $\kappa \le 2$ would be violated 78 times; the sharper per-side law on κ_r alone is violated 1/293; and the T128 side asymmetry reproduces on entirely new data ($\kappa_r = 1.697$–2.913 against $\kappa_l = 0.054$–0.205). The bar is not moved — hence the verdict.

What makes the break productive is that the *structural ground* of κ is now solved, with identities rather than fits. Three pieces, in the order they stack. *(a)* The interval identity $\sum_i (1 - g_i) = t_l + t_r$ (verified to 7.1×10^{-13} on all 105 calibration transports) makes κ a *probability average of the border density*: the weights sum to 1 exactly, so the **flat** border vector sits at $\kappa = 1$ *exactly* — T128's bar 2 was *twice* the flat value, not the flat value. *(b)* The profile identity $p_{N-1} = 2 - p_0 + 2E$ (to 6.7×10^{-16}) makes a **linear** density profile with vanishing edge give exactly $\kappa = 2$: the bar 2 is the *linear-density limit*, and everything above it *is* the convexity of the border density ($E > 0$ on 219 of 436; Chebyshev's correlation inequality gives the sign). *(c)* The chain

$$\kappa \;\le\; 2 - p_0 + \tfrac{N-2}{N} \sum_i \left|\Delta p_i - \overline{\Delta p}\right| + (p_{\max} - p_{N-1})$$

is a *theorem with an explicit constant* — Hölder for the mean, telescoping for the profile identity, Abel summation for the majorant — and it holds on **all 436 transports** (calibration plus test; slack 0.0184–2.3126, median 0.3438). On 392 of 436 the density peaks at the inner end of the border block, the last correction vanishes, and the bound is the clean form $2 - p_0 + \frac{N-2}{N}\,\mathrm{CURV}$. The ρ-systematics T128 found is thereby *named*: $2E > p_0 \iff p_{N-1} > 2$ *exactly*, confirmed transport by transport (114 of 436 satisfy both, no exception) — a competition between the convexity term and the edge value, not a one-sided effect; the majorised curvature grows as $0.1445 + 0.1634(\rho - 1)$ (a fit) while p_0 shrinks with ρ, and both pull the same way. **The depth block:** 14 graded transports at $h_f = 1656 \ldots \mathbf{17\,669}$ — up to 12× the hard factorisation cap 1500, reached matrix-free through the two-scale assembly ($m/h = 0.064$–0.512), $\rho = 1.25$–2.50, $n = 79$–1019 — show **zero** violations of the frozen law, the curvature chain holds on 14/14 (slack 0.042–1.998), and κ does not drift upward with depth: whatever residual risk remains is a *ratio* effect, not a resolution effect, which is the one thing a finite computation can actually settle.

10.15 The deep seams carry — on the graded space, honestly downgraded

The machinery first, verified against the uniform reference at $n = 71$ ($h = 619$): the matrix-free two-scale assembly equals $J^{\mathsf{T}}QJ$ to 2.7×10^{-16} (and at $q = 1$ it equals the uniform form to 0.0, so the graded path *contains* the uniform path and no separate code is trusted); the compressed step identity $X = Q_{\text{old}}$ holds to 2.4×10^{-17}, so the step stays margin-free (Albert 1969); and the graded interval overlap reproduces the prolongation J itself to 6.4×10^{-14}. Then the calibration, printed *before* the seams: over 48 (zone, q) pairs at 12 zone/ratio settings the graded Schur floor is 1.534–5.181× *larger* than the uniform one (median 2.646) — Rayleigh–Ritz, the wrong direction for a proof, with no exception — and on **4 of 48** pairs the graded floor is *positive where the uniform floor is not*: demonstrated **false positives** of the compression, a measured 8% rate at depths shallow enough that both sides can be computed. Both deep seams are then built and both fit: $n = 127$ and 256 ($h_f = 2476$, 5694) compressed to $m_f = 1256$, 1452 ($m/h = 0.507$, 0.255; $q_f = 2$, 4), every factorised block at or below the cap and the fine square matrix never formed; **both carry a complete graded seam certificate** (X positive definite by Cholesky, both Schur floors above the Wilkinson floor, transport bracket positive), retentions 0.3070/0.4165; as a κ test on *genuine record seams* ($\rho = 2.0238/2.0118$, outside the calibration range) they give 0 violations (chain slack $+0.079/+0.178$); and the floor survives 6 further coarsening rungs (0.166–0.883) — measured evidence against a one-merge artefact, which does *not* repair the Rayleigh–Ritz direction. The honest ledger, stated as the probe states it: $(L) = [\,3$ uniform-certified (T128), 2 GRADED-CERTIFIED here with the declared false-positive rate, 3 budget-open$\,]$ — emphatically *not* “5 certified, 3 open”. The two deep seams moved from “no certificate at all” to “a complete certificate on a coarser Galerkin space, with the direction of the error known and its size calibrated” — strictly more than T128 had and strictly less than a uniform certificate.

10.16 The comb is bounded: the operator majorant works, the row bound is vacuous

Both candidate majorants for the comb’s outer-k-cell share are majorants on all 40 zone × depth points (5 zones, depths ×1/2/4, $h = 64$–946; the measured share itself is 1.96×10^{-3}–0.573, median 0.038). The unconditional Cauchy–Schwarz row bound is **vacuous**: below the bar 1 on 0 of 40, overshooting the truth by a median factor of 116 524 — reported as the failure it is, not as a footnote. The **S-procedure operator bound works**: below 1 on 40 of 40, running 0.170–0.774 and only 1.3–182× above the truth (median 7.8) — a statement about *every* coarse vector in the measured smoothness cone, not merely the one that happens to occur. Status CERTIFIED-CONDITIONAL: conditional on one measured cone parameter with a known law ($\theta_1 \sim D^{1.219}$, a fit with jackknife SE 0.088 — exactly the scaling of a fixed smooth profile sampled at cell width D). The honest limit: the bound *saturates* at $(kD_c)^{0.263}$ (a fit) instead of decaying like the pole part’s closed-form $O(kD)$, because the cone still contains vectors far more edge-loaded than the coarse solve that occurs.

10.17 The map V4: two irreducibles, three new points, and the promotion assessment

The map V4 (T129): nine entries, and the two irreducibles named

3 reduced or bounded by this probe: **M9** the κ bar — *reduced*, not merely retested: κ is now an identity plus a theorem, and what remains open is a bound on the *curvature* of the border density on $\lceil t \rceil$ cells, a strictly smaller object than the original mass hypothesis; **L2** the two affordable deep seams — *graded*-certified with the direction of the error known and its 8% false-positive rate measured; **M18** the comb — *bounded* by the conditional operator majorant. **1 untouched**: **M17** the pencil mass (T128's measured 0.9665 and the constructed floor 0.188 stand). **2 irreducible here**: **M19** the word "for all" — the curvature chain is a theorem *per transport*; making it a theorem *for all* transports needs a zone-uniform curvature bound no finite computation can supply; **M21** the RH address itself — Weil 1952 / Bombieri 2000 / Connes 1999 cited and used for nothing. **3 new, created and named rather than buried**: **M22** the fitted constant C^* (it stays a fit until the curvature term is bounded in closed form — the same open question in different clothes); **M23** the graded-to-uniform conversion (needs the Céa/Strang defect $\|(I - \Pi)z^*\|^2$ of the fine partial minimiser; Q has an FFT matvec, so this is an *arithmetic* obstruction with a known route, not a conceptual one); **M24** the per-side law (κ averages a large right protrusion with a small left one, so T128's quantity is systematically milder than the mechanism it stands for).

Promotion readiness (T129): ready in principle, as a narrow module

The load-bearing content of this probe that would survive a `verification/vN_*` gate is exactly the part that is identity or theorem and needs no sampling: *(1)* the interval identity $\sum(1 - g_i) = t_l + t_r$ and hence $\kappa_{\text{flat}} = 1$; *(2)* T128's four-term (T) identity; *(3)* the profile identity $p_{N-1} = 2 - p_0 + 2E$; *(4)* the Abel majorant $\kappa \le 2 - p_0 + \frac{N-2}{N}\,\text{CURV}$; *(5)* the assembly identities (two-scale assembly $= J^{\mathsf{T}}QJ$, graded overlap reproduces J). Those five are checkable at fixed small sizes in seconds and are true per instance. **Not ready**: the constant C^* (a fit), the graded seam floors (a weaker Galerkin statement with a demonstrated false-positive rate), the comb-law exponent (a fit), and anything containing the word "uniform". A module built only on (1)–(5) would be load-bearing and small; a module that quoted C^* or a graded floor as a certificate would not be, and the ledger would be right to refuse it.

The honest distance, in the probe's own three sentences, quoted:

> *After K1 the retention bound (T) is no longer a hypothesis about eigenvector mass but a theorem reducing it to the curvature of the border density on $\lceil t \rceil$ cells, with $\kappa = 1$ the flat limit, $\kappa = 2$ the linear limit and an explicit Abel majorant for the rest. After K2 the (L) list has no item left that is open for a conceptual reason: two of the five deep seams are certified on a coarser space with the error direction known and its size calibrated, and the other three need one Cholesky each at a size that is a pure arithmetic fact. After K3 both parts of $\hat{s}$ — the pole part in closed form and the comb part by a certified majorant — are excluded from being boundary layers, so (M)'s one remaining direction type is gone; what is left is what was left before: the word "for all", and the distance from any finite window to the criterion of Weil, which nothing in this probe touches.*

T130 (`curvature_bridge_probe.py`, contract `CURVATURE.BRIDGE`, 30/30, verdict ONE-OF-TWO) then attacked the two named open pieces T129 left — the curvature bound (M22/M24) and the graded-to-uniform bridge (M23) — and exactly one of the two stands: *the bridge stands, the curvature bound does not.* The remainder of this section is that part.

10.18 The curvature term becomes a total variation — and the frozen shape band breaks (T130)

The structural gain of the curvature front is a new *exact* rewrite of what the term is:

$$\sum_j |\Delta p_j - \overline{\Delta p}| \;=\; \mathrm{TV}(P), \qquad P_k \;=\; p_k - (p_0 + k\,\overline{\Delta p}), \quad P_0 = P_{N-1} = 0,$$

the total variation of the density profile's own *chord deviation*, certified per transport by $\mathrm{TV}(P) \le (2n_{\mathrm{run}} - 2) \cdot \mathrm{sag}$ with $\mathrm{sag} = \max_k |P_k|$ — equality when P has a single monotone hump, in which case the curvature term is *one* geometric number; on 83 of the battery's 765 transports that is literally the case. The border eigenvector never changes sign inside the border block on 765/765 transports (the T109 cancellation geometry lives outside the block), and the shape family is decided: the power profile $|w_i| \sim (i+1)^a$ wins on 635/635 transports against linear, exponential and reciprocal-pole candidates ($a = 0.703$–0.999, median 0.856 — T109's linear border profile, independently recovered). The Levinson coupling supplies the *structure*, not the bound: the T108 corner identity $u_0 = -\sqrt{2}\,\rho_{M-1}/E_{M-1}$ is independently reproduced (24 zones up to $M = 1236$, to 4.2×10^{-9}, exactly one negative pivot per zone), but the pole solve as a profile model is only a *proxy* (direction cosine 0.992–0.996, curvature ratio 1.5–5.3) — no majorant deployment. And the honest core: the shape band frozen on the calibration half ($a \in [0.734, 0.897]$, $S^* = 1.1926$) **breaks on 13 of 545 disjoint test transports** — the bar is not moved; with the per-transport measured exponent the same S^* holds on 520/545. **M22 is thereby reduced to a zone-uniform bound on *one* exponent — reduced, named, and not closed.**

10.19 The bridge stands as an identity: the matrix-form Céa/Strang defect

The graded-to-uniform bridge — the object M23 named — is implemented and it is an *identity*:

$$S_{\mathrm{graded}} - S_{\mathrm{uniform}} \;=\; R^{\mathsf{T}} X^{-1} R,$$

verified to 1.3×10^{-13} and evaluated entirely matrix-free (an FFT matvec for the fine form, a BPX-preconditioned CG, one graded Cholesky at or below the cap — the fine square matrix is never formed). The decisive distinction is scalar against *matrix* form: the scalar bound $\lambda_{\min}(S_{\mathrm{graded}}) - \lambda_{\max}(G)$ is vacuous (it loses a factor 3–28), while the matrix form reproduces the *directly computed* uniform Schur floor on **84 compression pairs at 21 zones to 2.5×10^{-11} relative, without a single overshoot** — so GRADED-CERTIFIED becomes a one-sided statement about the *fine* space. T129's measured 8% false positives are thereby explained *completely*: 16/84 reproduce at the bracket level but **0/84 at the floor level** — they sat in the transport bracket (graded test vectors), an object the bridge makes unnecessary, not in the Schur floor. **Both deep seams carry**: $n = 127$ ($h_f = 2476$) $\to \lambda_f = 0.2255$ and $n = 256$ ($h_f = 5694$) $\to \lambda_f = 0.2614$, with positive bridged transport brackets, at 1.7× and 3.8× the factorisation cap, reached through FFT matvec and CG only. The status is honest: BRIDGED-CERTIFIED-CONDITIONAL — exactly *one* measured number per grid enters, a positive lower bound for $\lambda_{\min}(X_{\mathrm{fine}})$, currently supplied by Lanczos (which bounds $\lambda_{\min}$ from *above*: a measurement, never a certificate); the Grenander–Szegő symbol route for that number is *dead* (the Weil lag symbol is indefinite at these windows, rigorous symbol floor $-76 \ldots -27$); and the robustness is benign: the bridge conclusion does not move when the assumed floor is shrunk by 100× or 10 000×.

10.20 The theorem battery, the map V5, and the promotion check list

The κ battery against the *theorem* chain rather than the fitted law: **zero violations of the exact chain on all 765 transports** of the new battery — the chain is proved per transport, so this is implementation verification, not a test. The fully explicit chain (the D1 candidate substituted for the curvature term) closes 635/635 of the shape transports — but under the bare bar 2 it closes *none*: measured $\kappa > 2$ on 130/765. Read honestly: the chain's own slack survives the 13 shape-band misses — that is robustness, not repair; the bound that would close M22 still does not exist.

The map V5 (T130): twelve items, one new gap, and the self-supply candidate

Certified: **I1** the interval identity $\sum(1 - g_i) = t_l + t_r$ (ω a probability measure, $\kappa_{\mathrm{flat}} = 1$); **I2** κ as a weighted mean of the border density (Hölder); **I3** the profile identity with $2E > p_0 \iff p_{N-1} > 2$; **I4** the Abel chain; **I5 new: the TV rewrite** with the one-hump collapse and the $(2n_{\mathrm{run}} - 2) \cdot \mathrm{sag}$

bound. **Certified-conditional**: **I6** the Céa/Strang bridge and **I7** the two deep seams (one measured number per grid). **M22** reduced to a zone-uniform bound on one exponent; **M23** *conditionally closed* (the bracket-level false positives remain as a test-vector artefact; the bridge replaces the bracket rather than repairing it); **M25 new: a certified positive lower bound for** $\lambda_{\min}(X_{\text{fine}})$ — the only input I6 lacks, and it is small (the 10^4 robustness above); its elegant candidate is the *self-supply*: the chain's own telescope ε floor could deliver it, which is exactly what T131 tests. **M19** ("for all") and **M21** (the RH address) are untouched.

The promotion check list (T130): what a narrow vN module would carry

Exactly the certified block: *(1)–(5)* T129's five identity/theorem pieces (interval identity with $\kappa_{\text{flat}} = 1$; the mean representation with the Hölder step; the profile identity and its equivalence; the Abel chain; the TV rewrite with the sag bound), plus new from this probe *(6)* the defect identity $S_{\text{graded}} - S_{\text{uniform}} = R^{\mathsf{T}} X^{-1} R$ with its matrix bridge — pure Schur/Galerkin algebra, checkable on a random symmetric pair in three lines. **Not in**: the frozen shape band and S^* (a calibrated hypothesis, M22), the Lanczos inner floor (M25), and both deep-seam certificates (they inherit M25). The three remaining (L) seams are pure scale questions: only $n = 8191$ ($h_f = 295\,253$) admits a graded frame under the cap at all ($q = 256$, $m/h = 0.0042$ — far below the calibrated range), the other two at no merge factor $q \le 2048$.

T131 (`self_supply_probe.py`, contract `SELF.SUPPLY`, 25/25, verdict SUPPLY-PARTIAL) then tested exactly that candidate — the self-supply of $\lambda_{\min}(X_{\text{fine}})$ through the chain's own telescope ε floor, the hardening of the four M22 properties, the M17 composition, and a map V6 with the final promotion list — and the answer is: *the self-supply loop is built, one number short of closed.* The remainder of this section is that part.

10.21 The secular sandwich: the self-supply becomes a theorem (T131)

The $\varepsilon \to \lambda_{\min}$ chain is no longer a candidate but a *theorem*, derived from the secular equation of the rank-one downdate (Sherman–Morrison):

$$\frac{\varepsilon}{\|u\|^2 + \varepsilon/\mu_1} \;\le\; \lambda_{\min}(Q) \;\le\; \frac{\varepsilon}{\|u\|^2}, \qquad \varepsilon > 0 \;\wedge\; A \succ 0 \iff \lambda_{\min}(Q) > 0,$$

with $\mu_1 = \lambda_{\min}(A)$ — the upper half is convexity of the secular function, the lower half the Sherman–Morrison denominator, and the sign half is an *equivalence* (Albert 1969). On 10 dense windows ($h = 26$–434, $n = 3$–31): **zero violations**, predicted relative width $1 + \varepsilon/(\mu_1 \|u\|^2) = 1.2663$–$1.3894$, and the true value sits at 1.06–1.31× the lower bound (median 1.22) — the bound is not merely valid, it is *sharp*. The ε side is supplied *without* a fine factorisation: the same Céa/Strang defect that carried the bridge carries ε (a CG-tightened certified lower bound), valid on 10/10 and retaining 1.0000 of the true ε. Two side findings, both named: *(a)* the **direction obstruction** — the mirror rung lacks a Loewner *minorant*, $T(\ell)$ is indefinite on 4/4 windows ($\min \ell < 0$ wherever the lag symbol dips: the same indefiniteness that killed the Szegő route), so the telescope runs coarse-from-fine only; *(b)* a **correction to T130**: the Lanczos Ritz value overestimates $\lambda_{\min}(X)$ by a factor of up to **7.9** — it is an upper bound by construction (Rayleigh–Ritz), so T130's conditional hung on a number that is systematically *too large*; a chain floor below the Lanczos value is not a loss but the difference between an estimate and a bound. The bridge test: on **84 bridge pairs the chain floor replaces the Lanczos floor with zero brackets lost** (bar 0, declared), retaining 0.766–1.059 (median 0.875) against the genuine certified Cholesky floor; **both deep seams are chain-positive** ($n = 127$: $\varepsilon \ge 2.63 \times 10^{-7} \to$ floor 1.08×10^{-6}, $\lambda_f = +0.2255$; $n = 256$: $3.83 \times 10^{-8} \to 2.52 \times 10^{-7}$, $\lambda_f = +0.2614$); and the residual input is robust — μ_1 may be underestimated by **nine decades** before the floor dies, because μ_1 enters only as a correction. **M25 is thereby reduced-to-pole-free**: what was "certify $\lambda_{\min}$ of the near-singular inner block at $h = 5694$" is now "certify positivity of the *pole-free* odd Toeplitz–Hankel section, with any floor within nine decades of the measured value".

10.22 The Perron sign theorem, the honest one-hump break, and the M17 vacuity (T131)

On 575 transports, one of the four M22 properties becomes a theorem and one honestly breaks. *(i)* **Sign constancy is a theorem** — not via Metzler (the border block has all off-diagonals nonpositive on only 11/575, so that route fails as a hypothesis) but via **Perron–Frobenius on** S^{-1}, which is entrywise positive on 575/575 transports (Perron root $= 1/\lambda_{\min}(S)$); the eigenvector is strictly sign-constant *and* $\lambda_{\min}$ is simple — exactly what T130's 765/765 could only record. *(ii)* The **one-hump property breaks at depth**, revising T130: on the deeper sweep P has more than two monotone runs on 530/575 transports, so the curv $= 2 \cdot$ sag identity is not generally available; what carries instead is the certified $(2n_{\text{run}} - 2) \cdot$ sag majorant, valid on 575/575 with no hump hypothesis at all, plus the run-count chain $n_{\text{run}} \leq s_2 + 2$. *(iii)* The exponent band stays a *measurement* ($a = 0.727$–0.999 over 492 transports, 55/492 outside the old calibration band; the drifts are fits, not bounds). *(iv)* S^* stays a measurement — and the deeper sweep *raises* it: the frozen 1.1926 is exceeded 16 times, and the constant this sweep would need is **1.8472**. M22 thereby collapses onto *one* object — the border profile exponent — but by a different route than expected: (i) is proved, (ii) is dropped rather than reduced. And M17 is **assembled — and vacuous, with the loss attributed**: the mass bound is mounted from its two parts (the pole part in closed form to 1.8×10^{-13}, the harmonic-mean identity to 2.0×10^{-11}), majorises the true bad mass on 27/27 rungs — and reaches the bar on 0/27. The dominant loss is the **U-metric mismatch** 2.18 (the assembly measures in the U-metric but bounds the comb in the Euclidean one); the named fix is M18 run in the whitened basis — not more measurement. Status: ASSEMBLED-VACUOUS.

The map V6 (T131): thirteen items, and the nine-item promotion list

Theorem/identity (3): the $\varepsilon \to \lambda_{\min}$ sandwich (M25a), the Perron sign theorem (M22a), the Céa/Strang bridge identity (M23). **Certified(-conditional)** (2): ε at a deep grid without a fine factorisation (M25b), the chain floors on the 84 bridge pairs and both deep seams (M25c). **Measurement/hypothesis/fit** (4): $\lambda_{\min}(A) > 0$ for the pole-free section (M25d — the one number the whole loop now rests on), the exponent a (M22c), the constant S^* (M22d), the quantifier (M19). **Dead/broken** (3): the mirror rung (M25e), the one hump (M22b), the RH address (M21 — cited, never used). The **promotion list grows from six to nine items**, each a per-instance certificate, none zone-uniform — new here: *(7)* the $\varepsilon \to \lambda_{\min}$ sandwich, *(8)* sign constancy from $S^{-1} > 0$ (Perron–Frobenius), *(9)* the run-count chain $n_{\text{run}} \leq s_2 + 2$. The shortest honest remaining list has four points: *(1)* $\lambda_{\min}(A) > 0$ pole-free (nine decades of slack), *(2)* a zone-uniform bound on the exponent and the run count, *(3)* $S^{-1} > 0$ structurally, *(4)* the quantifier.

10.23 The reverse flow: the toolkit pays the theory side back (T132/T133)

For the first time the direction of the program inverts. Two *reverse-flow* probes carry the certification toolkit built for the prime front back to the theory side of the compiler, and both land.

T132 (`bd_seam_probe.py`, contract `BD.SEAM`, 21/21, verdict SPECTRUM-ONLY) reads the seam Dirichlet-to-Neumann operator through the Beurling–Deny triad (jump kernel, killing measure, diffusion part) and uses it as an *operator* discriminator where the spectrum alone is blind. The Beurling–Deny identity is exact to 2.4×10^{-16} at every N; the triad reconstructs the operator, not just its form (1.4×10^{-14} at $d = 256$); the diffusion part vanishes identically (a finite atomic state space carries no strongly local part, and the continuum DtN is the pure-jump $\frac{1}{2}$-Laplacian, Caffarelli–Silvestre 2007, fitted decay exponent 1.980); and the killing measure *is* the μ_4 mark-sum curvature profile pointwise (1.3×10^{-14}) — the marks enter the operator exclusively through the killing term. The Markov remainder is not small but *exactly zero*, with a closed jump-weight form $J_{rs} = 1/(d \sin^2(\pi(r - s)/d))$ for $r - s$ odd and exactly 0 for $r - s$ even, which is what explains T126's 86.1% count fraction (the even lags are the zeros, not defects). The discriminator then separates: the model DtN and the μ_4-graded free contraction share their spectrum to 7.5×10^{-13} but are *not* the same operator, and the gap sits in the killing measure at 0.1746 at every N (spread 1.9×10^{-14} relative) while the jump gap shrinks monotonically ($7.7 \times 10^{-3} \to 3.5 \times 10^{-4}$) — so the difference is not a truncation artefact. The mark-preserving lattice symmetries (the clock rotation and the sheet reflection, generating D_4) leave the triad exactly invariant (1.6×10^{-14}), while a random orthogonal or a generic μ_4-equivariant conjugation destroys it: the triad is canonical relative to the *marked boundary coordinate*, not to the marks alone — which is a coupling statement, `QGEO.MARKS.01` ↔ `QGEO.KERNEL.01`. **Ceiling, stated:** this is the *model* DtN, not the raw RP seam, so `QGEO.KERNEL.01` is *not* closed.

T133 (`cert_floor_probe.py`, contract `CERT.FLOOR`, 23/23, verdict MIXED, 840 control validations on the certificate machinery itself) turns the toolkit on the load-bearing suite's own PSD rows and produces one finding that changed a module in this same round. The 10×10 Hankel matrix v379 builds is, *as a matrix of doubles*, certifiably *not* positive semidefinite — a certified negative direction $x^{\mathsf{T}}\widehat{M}x = -7.75 \times 10^{-18}$ at 60 digits — and the module's own tolerance ($\lambda_{\min} > -10^{-10}$) passes it. The reason is priced, not guessed: cond $\approx 1.8 \times 10^{18}$ against a certifiability ceiling of 1.0×10^{13} at $n = 10$, and the entry bar 7.6×10^{-12} from evaluating a transcendental kernel in binary64 dominates every floor the floating-point route could produce; the blind band is exhibited explicitly (a matrix with a certified negative direction passes the test). The *mathematical* matrix is fine — at 40 digits it carries a certified floor 2.7×10^{-25}, while rounding its entries to doubles moves it by 3.5×10^{-17}, eight orders more — and the exact repair needs no precision at all: the same object is a *positive-mixture Gram* $M = V^{\mathsf{T}}\mathrm{diag}(1/1600)V$ with $V_{ki} = e^{-\varepsilon_k \tau_i}$, so $x^{\mathsf{T}}Mx = \sum_k w_k(v_k \cdot x)^2$ is a sum of squares and $M \succeq 0$ holds in exact arithmetic with no bar. The probe's recommendation table asks for method and wording, explicitly *not* a marker move (v171's Vandermonde Gram is exactly certified already; v527/v519 certify at the 40 digits they already run). **That repair is executed in this same change:** v379 now asserts the exact mixture identity (1.1×10^{-15}, at the rounding level) plus a sum-of-squares certificate, the eigenvalue check is retained and relabelled a diagnostic *measurement*, and the ledger row `SEAM.S3.RP.01` is reworded accordingly — marker **[E]** unchanged.

The narrow identity module of this phase is promoted: v542, Section 4.8.

10.24 The pole-free floor: existence closes, and every cheap route fails by sign (T134)

T134 (`pole_free_floor_probe.py`, contract `POLE.FREE.FLOOR`, 21/21, verdict PARTIAL; sub-verdicts FLOOR-CERTIFIED-PER-WINDOW / CLASS-IDENTIFIED / MASS-VACUOUS-METRIC-FIXED) attacks the three items of T131's rest list, and nothing else. The first answer closes the existence question: the floor is *never* an existence problem. Every Cholesky pivot of the *pole-free* form is positive on 79/79 windows — T119's famous negative pivot belonged to the form *with* the pole — and the per-window Cholesky certificate carries R4 on 79/79 and R2b on 68/68. But the route table is unanimous: every cheap lower-bound route fails by *sign*, not size, so the nine decades of slack T131 measured are worthless to them — they die at a sign that no margin can heal.

Cheap route	Certifies	Failure mode
Dirichlet + Gershgorin	0/68	sign
global Gershgorin	0/68	sign
Jacobi dominance	0/68	sign
Ostrowski (1937)	0/68	sign
path comparison	infeasible	congestion up to 4.7×10^5

The anatomy names the culprit. The good half $\mathrm{diag}(w) + L_N$ of the T126 fractional Dirichlet split is positive definite on 68/68 windows, with $\lambda_{\min} = 0.13$–3.06 — that is 8–85,000× *above* the target; the *entire* loss is the positive-off-diagonal Laplacian L_P at the long lags: $\lambda_{\max}(L_P)/\mu_1$ up to 4.7×10^5 at effective rank $k_{\mathrm{eff}} = 24$–397, so L_P is neither small nor low-rank, and T131's rank-one trick does not generalise. R5 is measured rather than hoped: the compression *and* the Schur floor both sit *above* $\lambda_{\min}$ on 56/56 pairs — no coarse recursion can ever supply a lower bound here — and the certified floor degrades like $D^{2.26}$ (a fit, labelled a fit). The second answer gives the inverse-positivity its *class*: the naive split stays definite on only 659/900 blocks, but the **lumped Stieltjes comparison matrix** $S_B = S + L_\Delta$ (the positive off-diagonals lumped onto the diagonal) is Stieltjes *and* positive definite by construction, with Jacobi spectral radius $\rho = 0.709$–$0.990 < 1$ on 900/900 blocks — so S_B^{-1} is a convergent nonnegative Neumann series, a discrete Green function: the M-matrix ground of the sign constancy that T131 could only record via Perron. The naive killing reading is refuted (row sums negative on 900/900), the Neumann margin is a valid majorant on 558/558 (a-priori coverage 26.9%), entrywise positivity of S^{-1} is now *certified* a posteriori on 900/900 (residual $\leq 2.6 \times 10^{-11}$ against smallest entries $\sim 10^{-2}$), and the weakening "reflection positivity on the floor eigenspace only" is *refuted* (535/900): the full statement is needed, the weaker one is false. The third answer honestly corrects T131's diagnosis: the 2.18 U-metric mismatch was *not* the dominant loss — in the one honest norm the *comb dominates the pole* by 4.7–81×, invisible to T131 because its comparison mixed two metrics. With the exact normaliser the whitened mass bound gets a *finite* number for the first time: 2.81–5.04 (sharp 2.52–4.26) against the bar 0.50 — a factor 5.6–10, no longer orders of magnitude — and $\sigma_b = 0.77$–1.00 exhausts the budget alone, so what is missing is a *localisation* statement about the bad pencil subspace, not more measurement.

After T134: fifteen promotion items, five remaining points

The promotion list grows from nine to **fifteen** per-instance items (new: the per-window Cholesky certificate of the pole-free form, the lumped Stieltjes construction, the ρ(Jacobi) < 1 series, the a-posteriori $S^{-1} > 0$ certification, the row-sum refutation of the killing reading, the whitened mass identity). The remaining list has **five** points: *(1)* D-uniformity of the floor via the M-matrix question (under attack in T136), *(2)* a zone-uniform bound on the exponent and the run count, *(3)* $\rho(K)$ a priori rather than measured, *(4)* the M17 localisation, *(5)* the quantifier.

10.25 The reverse flow, continued: a bounded seam state where the Weil window has none (T135)

T135 (`comb_compressibility_probe.py`, contract `COMB.COMPRESS`, 13/13, verdict BOUNDED-STATE (MATCHED)) is the third reverse-flow trace: does the T116 boundary-state machinery, which *failed* at the Weil window because the off-band mass sits in the prime comb, *succeed* on the seam DtN of T132/v210, whose comb is an arithmetic progression? The port is verbatim — the T116 Riccati recursion runs unchanged on Λ_Σ (bordered pole identity 1.5×10^{-16}, the banded m-march reproducing the dense state to 8.2×10^{-16}), with the pole candidate the rank-8 Calderón polarisation of v113 on the 16 residue classes, documented as a *model* (v210's DtN carries no rank-one defect of its own). The census is not a kill: 100.00% of the off-band mass sits on the mod-4 stripes (the complement is machine zero), and the stripe count grows exactly like $\lfloor W/4 \rfloor = \Theta(W)$ (fitted exponent 1.0000 ± 0.0000) where the true prime comb grows like $\pi(W)$ (0.726 ± 0.018). The core number: $\mathbf{m}_{\mathrm{cert}} = \mathbf{12}$ — the recursion error is 3.3×10^{-16} of the margin exactly ($h = 64$–1500) and 2.1×10^{-17} certified on the deep march ($h = 1000$–$100{,}000$, *falling* in h) — against T116's 1.3×10^2–1.1×10^6 *above* the margin at the Weil window; growing the state $m = 4 \to 48$ divides the error by 4.7×10^8, where T116 saw no decay at all. The match is honestly labelled: 12 lies in the pre-declared look-elsewhere set $\{8, 12, 16, 24, 32\}$ (the search ladder contains five non-member values), so the verdict is MATCHED, not "derived" — and the value is bar-fragile, with only the band $8 \leq m_{\mathrm{cert}} \leq 16$ defensible. The controls name the driver: an artificial mod-3 seam *also* closes ($m = 8$), so the causal driver is **weight summability**, not the comb period as such; and the *true* prime comb *fails* under identical machinery (no h-uniform floor, error/margin 0.16–0.27 and rising, 12× more state buying only 1.69×) — non-vacuity established: the same recursion separates the two combs. **Sober, three times over:** `QEC.SEAM.01` is *not* advanced (the driver is not the period), m_{cert} is partly circular (the 16 channels are set by hand), and this is the model DtN, not the raw reflection-positive seam. What is new and non-circular: the finite-dimensionality of the seam is visible in a *recursion* for the first time, not only in a spectrum, with the Weil window as the proved contrast.

10.26 The M-matrix pair: one item closes, and the bookkeeping is exact (T136)

T136 (`m_matrix_pair_probe.py`, contract `M.MATRIX.PAIR`, 30/30, verdict ONE-CARRIES) attacks the M-matrix question T134 named, and one of the three goals carries cleanly. The classical row-sum criterion — the thing one would normally invoke to get $\rho(\text{Jacobi}) < 1$ — is *vacuous* at the border block (the row sums are negative on 900/900, the same sign failure that killed T134's cheap routes). It is not needed: for the lumped Stieltjes comparison, **Varga's regular splitting turns convergence into an identity**, $\rho(J) = \tau/(1+\tau)$ with $\tau = \rho(A_B^{-1}N_0)$, and the Collatz–Wielandt bound at the M-matrix anchor vector reproduces the measured $\rho = 0.709$–0.990 with a gap shortfall of only 1.00–1.03, *flat* in D and in the zone on 900/900 blocks (a priori $\rho \le 0.7160$–0.9903 against the declared bar 10). Two solves replace a spectrum, and rest item *(3)* — "$\rho(K)$ a priori rather than measured" — is **closed**.

The other goal resists, but its anatomy is now exact. The pair is the right tool and runs the right way; the anchor half is certifiable and loses at most 1.51. What kills it is the margin: $1-\rho(E)$ is only 2.5×10^{-6}–2.5×10^{-2} wide while *every* cheap upper bound on $\rho(E)$ sits at 1.82–3.11. The shifted ladder is empty *by monotonicity of M-matrix inverses*, not by measurement. And the three-way bookkeeping splits the degradation exactly:

$$\lambda_{\min}(A) \;\sim\; D^{2.27} \;=\; \underbrace{D^{-0.56}}_{\text{anchor}} \times \underbrace{D^{2.72}}_{\text{margin gap}} \times \underbrace{D^{0.12}}_{\text{slack}} \qquad \text{(exact split; the exponents are fits).}$$

The anchor *improves* with depth; the slack is flat; the entire loss is the gap. Finally M17 closes *negatively*: the bad pencil subspace is **delocalised** (participation ratio 0.54–0.98, 91% of the cells carrying 90% of the mass, flatness 0.98 on every rung), neither a border layer nor the comb stripes, with a dimension tracking the *atom count* rather than the grid ($n_{\text{bad}}/n_{\text{at}} = 1.64$ median). Hence $\sigma_b(k) \ge 0.646$ on the whole ladder and no choice of cut brings M18 under its 0.50 bar (best 2.76–7.00): what M18 needs is a localisation *statement*, and there is none. The identity block of this part is promoted: `v543`, Section 4.9.

10.27 The long lags: the support is arithmetic, and the absolute value is the culprit (T137)

T137 (`long_lag_probe.py`, contract `LONG.LAG`, 22/22, verdict BOTH-RESIST) names the perturbation, and the name is arithmetic. The archimedean half of the kernel is nondecreasing outside its near field and its odd section carries *no* positive off-diagonal (0 of 40 windows), so the whole bad half L_Δ is **comb-generated**, supported on the anti-diagonal stripes $r+s = M-1-\text{round}(u_j/D)$ addressed by the prime-power atoms of the window (1.00–1.00 of the positive entries within one grid cell of an atom, 4–192 stripes on 3–99 atoms, 1.33–1.97 stripes per atom), with the certifiable amplitude law $\Delta \le c_{\text{comb}}(\text{anti})$ — while the single-atom form $\mu_j/2$ is only *typical* (worst ratio 2.268). Each stripe is a perfect matching, so L_Δ is explicit.

Half the question then closes, and it closes as a *certificate of death.* Gershgorin bounds, row-sum norms and Collatz–Wielandt quotients at positive weights are all upper bounds for $\rho(|E|)$, so a single *lower* bound decides them all: the Collatz–Wielandt lower end gives $\rho(|E|) \ge 1.3218$–2.2963, above 1 on 35/35 windows. **No member of the absolute-value envelope family can ever produce a floor** — not "has not yet". The second family (edge/Gram bounds built on the *second differences* of the Green function) is alive but does not beat the margin: 1.1530–2.3502 against a true ρ of 0.9752–1.0000 and a target of 0.998879–0.999999, clearing it on 0 of 35 windows (overshoot 24–7.1×10^5 gaps). The coarsening ladder says why it fails structurally — the bound drops below 1 only when the entire Gram is a single block, so *the inter-stripe coupling is the object* — and the measured decay of those second differences is algebraic ($\text{dist}^{-0.63}\pm0.02$, a fit) where a Demko–Moss–Smith argument would need a geometric one, A_B being dense so that theorem does not apply at all.

Rest item *(2)* is delivered without being closed. The Neumann series of the regular splitting gives $(S_B^{-1})_{rs} \ge \max_{k\le6}(J^k)_{rs}/d_s$ *with no error term* — only a discarded nonnegative tail — the partial sum recovering 0.011–0.406 of the true minimum entry, and the anchor-weighted ceiling overshooting the maximum by only 2.49–12.24: the Green function is two-sidedly bracketed by path quantities. But the certificate for $S^{-1} > 0$ still stops at 563/900 blocks (716/900 with the sign-aware k-exact split, 66/900 from the structural bracket alone) because $\rho(|F|) \geq 1$ on 77 blocks — the same wall. **The diagnosis is one sentence:** the absolute value destroys the cancellation that the seam lives on, and what is missing on both sides is a *sign-preserving* bound. The structure block of this part is promoted: `v544`, Section 4.10.

After T137: thirty promotion items, and a rest list of three

The promotion list grows from fifteen to **thirty** per-instance items (7 new in T136, 8 in T137); thirteen of them are now load-bearing as `v543` and `v544`. Of T134's five remaining points, *(3)* $\rho(K)$ a priori is **closed** (Varga plus Collatz–Wielandt) and *(4)* the M17 localisation is **closed negatively** (the bad subspace is delocalised, so no cut works); the Green lower bound asked for under *(1)* is delivered two-sidedly. What is left is *(1)* D-uniformity of the floor — now sharpened to a single named requirement, a *sign-preserving* bound on $\rho(E)$, every absolute-value route being certifiably dead — together with *(2)* the zone-uniform exponent and *(5)* the quantifier.

10.28 The sign law is interval geometry, and the m-paired certificate removes the arithmetic wall (T138)

T138 (`sign_compensation_probe.py`, contract `SIGN.COMPENSATION`, 26/26, verdict PAIR-EXACT) keeps the cancellation instead of majorising it, and finds the mechanism. The sign of an inter-stripe coupling is decided by the **interval geometry** of its two edges — nested pairs positive on 0.76–0.94 of the area, crossing pairs on 0.59–0.63, disjoint pairs on only 0.06–0.36 (i.e. mostly *negative*) — and *not* by the lag distance (the alternation reading is false, flip rate near 0.5). The couplings cancel globally to $|\Sigma|/\Sigma|\cdot| = 0.29$–$0.70$: that one number is the reason for every absolute-value overshoot of T134–T137. The honest negatives stand beside it: Karlin total positivity *fails* (only 0.54–0.65 of the TP_2 minors nonnegative), and the Green function is rank-one dominated but *not* semiseparable — the Gantmacher–Krein one-pair form stays a candidate, not a theorem.

The best *certified* signed bound (band-exact by completed Cholesky, tail by block row sums) reaches 0.9758–1.1020 against a target of 0.998556–0.999979 — clearing 1 of 16 windows, overshoot 0.02–565 gaps against T137's 27–8×10^5: three orders of magnitude better, and still no floor. The clamp is measured, not conjectured: the band part alone crosses the target at bandwidth $b = 2$–8 while the certified tail there is still 0.46–0.70 — the two terms are never small together; the far tail is *net negative* (truncation *raises* ρ on 59/142 rungs, so the tail is no small remainder — it carries compensation); the Leibniz pairing buys nothing; and the exact two-stripe blocks are thin ($\leq 0.6\%$ gain, three-body terms up to $2.5\times$): **the compensation is a many-body effect**.

The gain of the part is the m**-paired Neumann certificate**. From the identity $(I - F)^{-1} = \left(\sum_{j<m} F^j\right)(I - F^m)^{-1}$ only the *outer* factor is majorised, so the condition is $\rho(|F^m|) < 1$ — genuinely weaker than $\rho(|F|) < 1$, because the signs inside the m-fold product are kept ($m = 1$ is verbatim T137's certificate). It removes the arithmetic wall on *all* 77 dead blocks and certifies 52 of them; the pool rises from 563 to 875/900 (mechanism: $\max |F^m|/|F|^m$ falls to median 0.295). And M17 gets its characterisation: whitened, $\mathrm{St} = I - W_S$ with $W_S \succeq 0$, so $\lambda(\mathrm{St}) \leq 1$ structurally, and the *direct* Kantorovich harmonic-mean route needs no bad subspace anywhere, relaxing the required margin from 0.5 to 0.8284 (854/900) — HARMONIC-PARTIAL. The honest conclusion: the signed route does *not* inherit the absolute-value wall — the m-ladder is a real structural gain — but the margin question returns one level down, as an upper bound on $\rho(W_S)$.

10.29 The decay lemma is refuted at its hypothesis, and the sign law is derived (T139)

T139 (`green_decay_probe.py`, contract `GREEN.DECAY`, 30/30, verdict DENSE-RESISTS, bars declared in the docstring before any number) asks whether the classical dense-matrix literature supplies the decay lemma T138 named as the one missing input — and the answer is an *arithmetic* refusal. The lemma fails **at the hypothesis**: the certified kernel envelope of the lumped pair decays only like $d^{-0.15}$–$d^{-0.04}$ where Jaffard 1990 needs $p > 1$ — the archimedean lags sit *on* the borderline ($\sim 1/d$) and the prime-power atoms lay $O(1)$ spikes across the whole window, so no entrywise envelope can see the thinness of the comb. The constant chain is built constructively rather than asserted: Demko–Moss–Smith is *verified* where it applies (75/75 banded rungs — the failure on the dense object is a statement about the object), the operator gate reads 0.97–1.83 (1/75 below 1), the algebra gate 3.4–1.6×10^3 (0/75); and the exponent is *not* inherited (kernel $d^{-1.5\ldots-1.1}$ against inverse $d^{-0.52\ldots-0.37}$). The positive replacement object is an identity, not a fit: the exact double telescoping $b_e^\mathsf{T} G b_{e'} = \sum_{\text{box}} H$ (4×10^{-15}), with the mixed second-difference kernel H negative on 0.71–0.87 of the off-diagonal and positive on 100% of the diagonal — **T138's entire sign law follows from one kernel plus box geometry** (disjoint pairs have boxes missing the diagonal), derived instead of measured; the Gantmacher–Krein one-pair form is quantitatively dead.

The never-truncating layer series is then killed *from below*. It delivers 3.388–6.171 against a target of 0.999182–0.999972 (0/15, overshoot 681–46900×), and triple direction pedantry pays: no layer may be dropped (zero trace forces $\lambda_{\max} > 0$, and a negative *mean* is not a Loewner sign), Gram $\preceq$ Band$_b$ holds *never* (85/85), and by Weyl every layer series is $\geq \max_b \text{Rayleigh}(\text{Band}_b) =$ 1.0303–1.1116, above the target on 15/15 windows — **no decay lemma of any strength can rescue the family**, and the overshoot *is* the effective layer count, growing with the window. The clean-up localises the rest: on a rebuilt pool of 620 border blocks the m-paired ladder out to $m = 48$ converges on all 620 and certifies **619/620** — the one open block (need 2.15) has its tightness at index distance 15, far-carried 1.000: the single item for which an H-decay statement would be the right tool. One level down, $\rho(W_S) = 0.667$–0.939 inherits the structure verbatim; the three-body cumulant is negative on 0.93–1.00 of 4200 stripe triples — certified *benign* (it pushes upper bounds down). The three-sentence verdict, quotable: the lemma does not exist classically for this object, for an arithmetic reason; the right object is now identified exactly (the sign law is derived), but every absolute-value envelope built from it is certified dead; **the measured core has shrunk from "a decay law is missing" to "one signed inequality at stripe distance $b \leq 16$ is missing"**.

After T139: forty-seven promotion items, and a rest list led by one signed inequality

The promotion list grows from thirty to **forty-seven** per-instance items (7 new in T138, 10 in T139). The sign-preserving bound T137 asked for is now *located* rather than merely named: everything that fails, fails at stripe distance $b \leq 16$ — the far layers are certifiably harmless (the tail is net negative, the three-body remainder certified benign, the border pool at 619/620) — so *(R1)* is **one signed small-bandwidth inequality** that keeps the cancellation among the first few stripe layers (a signed block-Rayleigh or interlacing step, not a norm). Behind it: *(R2)* a replacement for the Weyl step (any $O(nb)$ triangle-inequality route is dead by construction), *(R3)* an optional certified decay statement for H (which by construction cannot close R1 — every entrywise envelope is an absolute-value route), the same margin question one level down ($\rho(W_S)$), and, unchanged, the zone-uniform exponent and the quantifier.

10.30 The signed inequality acquires an exact finite core, and the D-dependence is located (T140)

T140 (`signed_band_probe.py`, contract `SIGNED.BAND`, 31/31, verdict FINITE-CORE, bars declared in the docstring before any number) attacks exactly the one signed inequality at stripe distance $b \leq 16$ that T139 left — and nothing else. The central move lifts the H-telescope from the *entries* of the Gram to its *form*: Gram $= CHC^\mathsf{T}$ exactly (3×10^{-15}), with $C = \text{diag}(\sqrt{\Delta})M$ and M the edge-interval incidence matrix, so $x^\mathsf{T} \text{Gram}\, x = u^\mathsf{T} H u$ with $u = C^\mathsf{T} x$ the occupation function of the edge system — Abel summation over the box structure, with no absolute value and no factorisation. Three exact consequences follow. First, rank(Gram) $\leq h - 1$ (measured 44–202 against 128–1499 edges, a 2–10× reduction): the whole spectral question lives on the index grid.

Second, **the exact finite-core reduction** $\rho(W) = \lambda_{\max}(K^{1/2}HK^{1/2})$ (6×10^{-14} on every window), with $K = M^\top\Delta M$ the *covering kernel* in closed geometric form $K_{rs} = W([r\wedge s,\ r\vee s])$ — entrywise nonnegative, monotone by inclusion, Green-like but *not* one-pair (the off-tridiagonal defect of K^{-1} is 0.08–0.43). The signed inequality is thereby a statement about two matrices of size $h-1$ per prime-power zone, reached with no truncation, no envelope and no series anywhere. Third, the energy reordering $H = \mathrm{diag}(s) + L_N$, exact to 5×10^{-16}: T139's sign law read as a mass term plus a long-range *Dirichlet form* with positive weights on 0.71–0.87 of the off-diagonal. The direct certified chain (every Loewner step a completed Cholesky) then locates the failure exactly: it **tears at step (i)** — dropping the 13–29% *positive* part of H's off-diagonal alone costs the whole margin (6.4–$239\times\rho(W)$), because that positive part is precisely the compensation the sign law lives on; the discrete Hardy step adds only 6.4–9.0× on top, with the first-moment input $\max_k Q_k = 18.5$–339; below the target on 0/16 windows for all four routes.

The checkerboard split solves R2. With stripe groups of length b the band condition forces adjacent groups, the adjacent pairs two-colour into two families of disjoint supports, and $\mathrm{Band}_b = D + A_{\text{even}} + A_{\text{odd}}$ holds *entrywise exactly* — so **three D-independent Weyl steps replace the $O(nb)$ ones** whose count T139 identified as the overshoot itself (value 0.76–$1.79\times\rho(W)$). And the zone question is answered by measurement: the stripe block norms are *zone-uniform* (group diagonal $\sim D^{0.13\pm0.01}$, cross blocks $\sim D^{0.02\pm0.01}$, fits) while $\lambda_{\max}(K)\sim D^{-2.99}$, $\lambda_{\max}(H)\sim D^{+0.33}$ and $\max_k Q_k \sim D^{-2.20}$ against a flat $\rho(W) = 0.9962$–0.9999 — **the entire D-dependence of the core sits in the geometry, not in the stripe blocks**, so only a *joint* bound on the pair can be D-uniform. Floquet is not applicable (the stripes sit at atom positions; group spread 0.15–0.69, quasi-periodic); Bendixson certifies 1.07–$1.08\times\rho(W)$ without any Gram (the spectrum of W is real), below the target on 0/16. Two hard negatives close the additive readings for good: the band *from below* exceeds the target on 16/16 windows (Rayleigh 0.3894–1.1116 on 229 (window, b) pairs — the mechanism is *rank inflation*: truncation inflates the rank by 1.2–10.1×, cutting exactly the directions in which the boxes compensate), and the new index-grid band family — the only one in which a truncation could have been legitimate — fails the Loewner test too: the discarded far part is certified Loewner-*positive* ($\lambda_{\max}\ge1.14\times10^{-1}$) on 256/256 pairs. The clean-up inherits and localises: of 420 rebuilt border blocks the m-paired ladder to $m = 24$ certifies 402; the 18 open ones carry need 1.01–10.99 with argmax at index distance 11–15 and *far mass fraction* 0.635–0.912 — genuinely far-carried, so a decay statement is the right tool *there*, unlike R1; and the finite core is inherited one level down ($\rho(W_S) = \lambda_{\max}(K_S^{1/2}H_SK_S^{1/2})$ to 4×10^{-15}; $\rho(W_S) = 0.541$–0.925).

After T140: the rest list is led by one zone-uniform discrete Hardy inequality

The map V12 has 18 rows and the promotion stock grows from forty-seven to **sixty-four** per-instance items (17 new). The rest list is the shortest of the series: *(R1′)* a **joint** bound on $\lambda_{\max}(K^{1/2}HK^{1/2})$ — not factor-wise (the D-exponents -2.99 and $+0.33$ cancel against each other), not after dropping the positive part of H (costs 6.4× alone), not after truncating H in either grid (both far parts are Loewner-positive); *(R1″)* the ingredient, *named and located*: a **zone-uniform discrete Hardy inequality** comparing the long-range Dirichlet form of $N = -\mathrm{offdiag}(H)$ against the form K^+, with the first-moment weight $\max_k Q_k = 18.5$–339 as its natural input; *(R3′)* a first-moment bound on $(-H)_+$ rather than a decay exponent — only that enters the certified chain; *(R4)* the 18 far-carried border blocks; *(R5)* nothing for the layer / envelope / Jaffard / one-pair families — certified dead, not to be retried.

10.31 The Hardy ingredient resists, and the resistance has an address (T141)

T141 (`discrete_hardy_probe.py`, contract `DISCRETE.HARDY`, 22/22, 51.2 s on 60 windows with $h = 50\ldots1173$, zones $n = 11\ldots529$ and $D = 2.31\times10^{-3}\ldots2.60\times10^{-2}$, verdict HARDY-RESISTS, bars declared in the docstring before any number) attacked exactly the ingredient T140 named — the zone-uniform discrete Hardy inequality for the pair (Dirichlet form of N, form K^+) — and nothing else. It did not close, and the way it fails is the content of the part.

Four exact identities put the question in classical shape. The reduction of T140 becomes a **Rayleigh quotient against the pseudoinverse**, $\rho(W) = u^\top Hu/u^\top K^+u$ at $u = K^{1/2}v$; the

Dirichlet part is a **signed crossing kernel**, $L_N = D^\mathsf{T} B D$ with $B_{kl} = \sum_{r \le k \wedge l,\, k \vee l < s} N_{rs}$ — the same closed covering formula as K, one grid further in; T140's Cauchy–Schwarz weight is *identified*, $Q = B_+\mathbf{1}$, so that step is the Gershgorin diagonal step on B_+ and its price is attributable; and $DKD^\mathsf{T} = L_\Delta$ on the interior nodes, whose diagonal is the **endpoint edge mass** — a *local* quantity, not $\lambda_{\max}(K)$. Together these give the two-weight Muckenhoupt quantity in closed form, $A_{M0} = \max_k Q_k(\Delta\mathbf{1})_{k+1}$, computed by two independent routes. The identity block of T140 and T141 is now load-bearing as v545, Section 4.11.

The separation is the finding. The certified Muckenhoupt constant is *not* zone-uniform: it grows like $D^{-0.366\pm0.036}$ (a fit) against a preregistered bar of 0.25. But the object it is supposed to bound — the exactly computed Dirichlet share of the finite core — is **strictly uniform on the same surface**, $D^{-0.229\pm0.007}$. The growth therefore does not sit in the inequality being false; it sits in the *diagonal profile* of the certificate, i.e. in the weight one puts on the endpoint mass. That is a repairable defect in a way a growing target would not be.

Both bound shapes are typed, and both types are negative. The **additive** shape is dead *as a shape*, not by a weak constant: its own exact Weyl floor — the best any mass-plus-Dirichlet splitting can ever reach — already sits at 1.694–3.855× the target, so no sharpening of the Hardy step can rescue it. The **joint** shape $\Lambda(H, Y) \cdot \Omega(Y)$ survives the eigenvalue side and fails at the *normalisation* alone, $\Omega = 20.7$–2724: Gantmacher–Krein returns as the obstruction one grid down, and the whole question is now a question about the profile Y.

After T141: R1 collapses to R1b, a single conductance profile

The rest list contracts from "a joint bound" to **R1b: exhibit a closed conductance profile Y with $Y \preceq K^+$ in the Loewner order and $\Omega(Y) \approx 1$.** Both requirements are per-instance checkable and both are already met separately — Y positive definite with the geometric profile, and $\Omega \approx 1$ for profiles that lose the Loewner step. The classical addresses are named, not invoked as authority: Muckenhoupt 1972 and Bradley 1978 for the two-weight criterion, Opic–Kufner 1990 for the profile calculus, Miclo's spectral-gap weights for the conductance reading. *(R3′)* is now shaped as well: what the Hardy route consumes is a **first moment** of $(-H)_+$ and not a decay exponent — a strictly weaker thing to have to prove — and keeping the signs in the crossing kernel is worth a factor 31.95–15561.41 against the 6.4–239× the T140 Loewner drop costs. Border pool: of 256 blocks the m-paired ladder to $m = 24$ certifies 245; the 11 open ones are far-carried (far mass fraction 0.835–0.902 at index distance 13–15), unchanged in kind from T140 — a rebuilt pool, so the anatomy transfers and the tally does not.

Verdict integrity. The negative half of this part rests on a comparison across zones, so the surface matters: the finite-core reduction of T140 is what makes an 11.21× span in D affordable at all, and on a narrower surface the separation $D^{-0.366}$ against $D^{-0.229}$ would have been within its own error bars. The verdict HARDY-RESISTS is recorded with that span attached. Thirteen statements of T140/T141 were mature enough to promote (v545); the sandbox stock stands at seventy-five items.

10.32 The optimal weight is found exactly, and the comparison path closes (T142)

T142 (`conductance_profile_probe.py`, contract `CONDUCTANCE.PROFILE`, 24/24, 155.4 s on 26 windows with $h = 210 \ldots 878$, zones $n = 71 \ldots 317$ and $D = 2.90 \times 10^{-3} \ldots 1.08 \times 10^{-2}$, verdict PROFILE-RESISTS, bars declared in the docstring before any number) does not guess a fourth profile. It *constructs* the optimum, from the classical address (Miclo 1999; Maz'ya's capacity criteria; Muckenhoupt 1972 and Opic–Kufner 1990 as addresses, not as authority) — and the construction settles the search half of R1b outright. With $J = DKD^\mathsf{T} = L_\Delta$ the endpoint Laplacian, $x = K^{-1}\mathbf{1}$ the equilibrium charge and cap $= \mathbf{1}^\mathsf{T} K^{-1}\mathbf{1}$ the capacity,

$$K^{-1} \;=\; D^\mathsf{T} J^{-1} D \;+\; \frac{xx^\mathsf{T}}{\mathrm{cap}}\,,$$

the capacity decomposition of the optimal Hardy weight — verified as an identity to 5.5×10^{-12} on 26/26 windows, never asserted. Its Dirichlet half is an *orthogonal projection* under the congruence ($X = J^{-1/2}DK^{1/2}$ has $XX^\top = I$), so $\Omega(D^\top J^{-1}D) = 1$ **exactly** (certified 1.0000001, the excess being the declared Cholesky floor) — precisely the object T141 was missing, in closed form, against the guessed profiles' $\Omega = 20.7$–2724. The weight was never a free choice: it is the Green function of the endpoint Laplacian plus the equilibrium rank one, and the exact capacity Rayleigh form follows with no inequality taken anywhere, $\rho(W) = \sup_u u^\top Hu/[(Du)^\top J^{-1}(Du)+(x^\top u)^2/\mathrm{cap}]$.

The failure is a constant, not a power. The variational optimum over the strict conductance class (a heuristic search whose every reported number is re-certified by Cholesky; 11 windows) reaches $\Lambda \cdot \Omega = 1.3160$–$1.4235 = 1.32$–$1.42 \times \rho(W)$ with a fitted D-exponent of -0.029 ± 0.014 — inside the uniformity bar, so the class fails by a *bounded factor at every depth*, a different (and harder) situation than T141's $D^{-1.87}$ blow-up. The stable shape is found: $c_k \sim 1/J_{kk}$, the reciprocal endpoint mass (log-log correlation 0.988–0.995), and the optimum wants essentially *no* mass — a pure Dirichlet form. The certified chain of the best closed member reads $\Lambda \cdot \Omega = 2.2671$–$2.4536$ against a target of 0.999983–0.999999 (0/26, median shortfall 2.3376, flat in D at $D^{0.006\pm0.004}$); the control run with T141's guessed profile on the same machinery gives 74.84–320.03, so the capacity construction beats guessing by 31.9–137×. The single-direction class barrier is saturated ($\beta = 1.000000$, the argmax being the top mode itself) and is recorded as a dead route with its reason.

The rank ladder closes the comparison path. On the anchored ladder — the diagonal minorant plus the r dominant residual modes, a certified Loewner minorant at every rank with $\Omega \leq 1$ throughout — the chain stalls at 1.408 ($r = 1$) … 1.391 ($r = 128$) and first crosses the hurdle 1 only at $r^*/m = 0.995$–0.998: the tautology $Y = K^{-1}$ itself (end-to-end anchor: at $r = m - 1$ the chain collapses to $\rho(W)$ to 1.7×10^{-11} relative). The structural consequence deserves its own sentence: since $\rho(W) = 1 - \Theta(D^3)$ and any comparison bound is $\geq \rho$ with equality only at $Y \propto K^{-1}$, a class bound would have to reproduce the optimal weight to relative accuracy $O(D^3)$ — **no comparison argument can deliver D-uniformity**. The next move is forced, not chosen: the *sharp* route, estimating the exact capacity Rayleigh form directly, with no Loewner step taken at all.

After T142: the comparison branch is closed, and the rest list is led by the sharp route

(R1c) **the sharp capacity–Rayleigh route**, which *replaces* the comparison rather than continuing it: estimate $\rho(W) = \sup_u u^\top Hu/[(Du)^\top J^{-1}(Du) + (x^\top u)^2/\mathrm{cap}]$ directly (Miclo 1999, Maz'ya capacity — a Hardy inequality for the Green form of the endpoint Laplacian, no profile choice left). What sends the next probe there is measured here: the best closed minorant is short by 2.27–2.45, the whole conductance class by 1.32–1.42, and no truncation below $r = m/2$ ever helps. *(R1d)* a multi-direction dual certificate for the class (LP/SDP over the conductance cone), without which every negative statement about the class stays *measured*. *(R3″)* the corrected R3′: T141's line was backwards — only 0.03% of the first moment of $(-H)_+$ sits *inside* index distance 8 (support radius 204–873), so R3′ survives as a *long-range* inequality (the Loewner step $L_{N_-} \preceq T_{Q^-}$ is certified with the full weight on every window) whose constant $\max_k Q_k^- = 3.20 \times 10^2$–$3.46 \times 10^3 \sim D^{-1.73}$ is not D-uniform. *(R4)* border pool rebuilt: 72 blocks, 68 certified, 4 open and far-carried — the required far shrink factor, 0.656–1.052, is the one number a decay statement has to beat. Map V14; nine statements ripe (stock 75 → 84), 0 promoted here. The verdict, stated precisely: a resistance of the comparison *method* at a known optimum, not a failure to identify the weight. T143 (`sharp_capacity_probe.py`) is running at exactly R1c.

10.33 The sharp route carries: D-uniformity is reduced to one named inequality (T143)

T143 (`sharp_capacity_probe.py`, contract `SHARP.CAPACITY`, 24/24, 18.5 s on 26 windows with $h = 142 \ldots 1173$ (core $m = 141 \ldots 1172$), zones $n = 31 \ldots 529$ and $D = 2.54 \times 10^{-3} \ldots 1.28 \times 10^{-2}$, verdict SHARP-CARRIES, bars declared in the docstring before any number) estimates the exact form instead of comparing it — exactly R1c, with no Loewner step taken anywhere. The T140/T142 identities assemble into one quotient, verified as an *identity* on all 26 windows (numerator to 1×10^{-11}, denominator to 5×10^{-12}, eigenvalue reproduction 3×10^{-6} relative under the explicitly declared 1/gap conditioning factor):

$$1-\rho(W) \;=\; \inf_{v}\; \frac{d^{\mathsf T}(J^{-1}-B)\,d \;+\; (x^{\mathsf T}v)^2/\mathrm{cap} \;-\; \sum_k s_k v_k^2}{d^{\mathsf T}J^{-1}d \;+\; (x^{\mathsf T}v)^2/\mathrm{cap}}\,, \qquad d = Dv\,.$$

The ingredient bookkeeping at the exact minimiser is structurally new: the minimiser is *orthogonal* to the equilibrium charge (to 10^{-5} — the capacity rank one carries *nothing*); the mass share is *negative* (-0.0698 to -0.0029 — the site masses *help* the gap); and the gap itself is the cancellation between the Green share (exactly 1) and the crossing share (1.0029–1.0698). The naive split $\rho \le \rho_{\mathrm{mass}} + \rho_{\mathrm{long}}$ is certified dead at 6.19–66.63, i.e. 6–67 $\times\,\rho$ itself: any capacity criterion must be applied to the *gap* form, where its factor of four costs the coefficient and not the scale.

The core number: Maz'ya's criterion lands inside its window. Brought to the gap form by a congruence (the capacities in closed form, $\mathrm{cap}_E(A) = \mathbf{1}^{\mathsf T}[(E^{-1})_{AA}]^{-1}\mathbf{1}$, verified against the constrained minimisation itself), Maz'ya's ratio gives $\Phi_{\sup}\lambda = 0.5438$–$0.6457$ on 26/26 windows — inside the window $[1/4, 1]$ the criterion forces for a Dirichlet form — although this form is *not* one (72–84% positive off-diagonals). The lower bound $\lambda \ge 1/(4\Phi_{\sup})$ captures 0.3872–0.4597 of the true gap, and the route's **loss factor is zone-uniform**: $D^{-0.048\pm0.010}$ against the preregistered 0.25 bar (the gap's own scatter, $D^{0.000\pm0.079}$, belongs to the arithmetic and not to the criterion); as a coefficient of $D^{2.829}$ the bound reads $c = 4.95$–67.70 against the envelope constant 12.04. The supremum is not an oracle: the closed families (Green diagonal, transported potentials, index intervals) recover 0.9606–1.0000 of the best value found, a full add/remove local search improves them by at most 1.0410×, and in the *node* coordinates the near-optimal sets are **intervals** of window lags, dominating the extremal level sets by a factor 8.3–129.5 — the one-dimensional structure Muckenhoupt's 1972 closed form needs. The honest half: Miclo's constructive capacitary *chain* loses a factor 46–2561 ($\kappa = 19.9$–1144.9, drifting like $D^{-2.13}$), because the top dyadic level sets are small sets of $O(1)$ capacity against an eigenvalue of 10^{-5} — the criterion's *conclusion* holds while its classical *proof mechanism* does not, and that is precisely where the remaining work sits.

The rest, typed. R1d is negative with a reason: the LP dual over the conductance cone (39 Λ and 35 Ω directions, a hand-verified dual point on 4/4 windows) certifies a class floor of $\rho(W)$ itself to a factor 1.000000 — saturated at the trivial floor, exactly as T142's single-direction barrier was; a finite direction sample of Ω cannot bind the cone, so R1d needs the full SDP constraint $K^{1/2}YK^{1/2} \preceq I$ (recorded so the sampling version is never retried — and after the core number above, no longer on the critical path). R4: 36 border blocks rebuilt, 32 certified by the ladder, 4 open; on the 6 largest blocks *all* 255–16383 subsets were enumerated, so $\Phi_{\sup}$ is *exact* there — $\Phi_{\sup}\lambda = 0.6087$–$0.9016$, again inside the window, the only non-family-restricted supremum of the probe — and the capacity argmax set is *entirely* the far tail (1.000 of its indices at block index ≥ 8), the exact object a decay statement has to act on, with T142's far shrink 0.656–1.052 still the number to beat.

After T143: what is missing is a proof, not a number — one named inequality

(S1) **the one named inequality**: $\mathrm{cap}_E(A) \ge |A|\,\lambda_0/c_0$ for *all* sets A with an absolute c_0 — equivalently $\Phi_{\sup}\lambda \ge 1/c_0$. Measured here at 0.5438–0.6457 on 26 windows and *certified by enumeration* on the small border blocks; for a Dirichlet form this is Maz'ya's theorem with $c_0 = 4$, and what is needed is the criterion for a *non-Markovian* form ($\sim$ 80% positive off-diagonals). *(S2)* **the Muckenhoupt route to S1**: in the node coordinates the near-optimal sets are intervals, so a comparison of the gap form to a one-dimensional path form *for the purpose of the capacity only* reduces the sup over all sets to a closed two-weight sup (Muckenhoupt 1972) — and T142's obstruction does *not* apply, because the capacity only has to be right to a bounded factor, not to $O(D^3)$. *(S3)* a closed formula for $\Phi_{\sup}$ or a closed lower bound on $\mathrm{cap}_E(A)$ (the families already recover 0.9606–1.0000: bookkeeping, not a search). *(R1d′)* the class dual as an SDP, no longer on the critical path. *(R4)* the 4 open border blocks, their capacity argmax entirely far. Map V15; ten statements ripe (stock 84 → 94), 0 promoted here. The honest state: D-uniformity is reduced to a capacity verification, and the verification is a Muckenhoupt-type statement about *intervals*, not another profile. T144 (`capacity_inequality_probe.py`) is running at exactly S1/S2.

10.34 The interval route carries: the chain has exactly one unproven input (T144)

T144 (`capacity_inequality_probe.py`, contract `CAPACITY.INEQUALITY`, 31/31, 91.9 s on 34 windows with core $m = 142 \dots 1393$, verdict INTERVAL-CARRIES, bars declared in the docstring before any number) attacks exactly the first three items of T143's rest list — S1, the interval route S2, and the closed capacity bound S3. The probe's single comparison licence is the *Jacobi whitening* — a denominator comparison, which T142's obstruction does not touch — certified at $\kappa_{\mathrm{up}} = 1.1195$–$1.3162$ per window (Gershgorin would have allowed ≤ 1.983); it puts the criterion in coordinates where the measure is the counting measure. Two different interval-reduction statements then separate, and only one survives. The *pointwise hull comparison is false*: $\sup_A \Phi(A)/\Phi(I(A))$ reaches 425 and drifts like $D^{-1.137}$, with T143's scattered level sets as the offenders — so S2 cannot be proved by transporting each set to its convex hull; it must be a *testing condition.* The *best-interval comparison holds*: $c_{\mathrm{glob}} = \Phi_{\mathrm{sup}}/\Phi_{\mathrm{int}} = 1.0000$–$1.6876$, zone-uniform at $D^{-0.068\pm0.024}$ against the preregistered 0.25 bar. The exhaustion behind these numbers is honest: 136 blocks of size $g = 14$ with *all* 16383 subsets each, 2.24 million sets in total; and capacity *monotonicity* holds on all of them (measured), although the form is not Dirichlet.

The two-weight sup, exact rather than sampled. The interval class is exhausted *exactly* on every window — 11 390 676 intervals via a Cholesky nestedness, $\mathrm{cap}_E([a,b))$ a prefix sum of squared triangular-solve entries, verified to 10^{-11} against an independent solve. The core number of the part: $B_{\mathrm{res}}\hat{\lambda} = 0.6694$–$0.7813$ inside the Maz'ya window $[1/4, 1]$ on the whole surface, fitted $D^{0.013\pm0.005}$ — zone-uniform and coordinate-robust (T143 measured 0.5438–0.6457 in the unwhitened coordinates). The free half passes its falsification test, $\Phi_{\mathrm{sup}}\hat{\lambda} \le 1$ on all 2.24 million sets, as the theorem requires. The chain then closes modulo one constant,

$$\lambda \;\ge\; \frac{1}{c_0\,\kappa_{\mathrm{up}}\,c_{\mathrm{glob}}\,B_{\mathrm{res}}}\,,$$

certified per window at 0.1002–0.2653 of the exact gap — with the remaining D-dependence sitting in the Jacobi *transfer* ($D^{-0.178}$), not in the capacity calculus. Dead with a reason: the naive Muckenhoupt *transcription* through trial-function energies misses the interval supremum by 89–6995 ($D^{-1.885}$ — the equilibrium potential spreads over the whole window, and no local trial function sees it); closed rankings bound the value but do not point at the argmax. And the honest counter-side, computed exactly: $\Phi_{\mathrm{int}}\hat{\lambda} = 0.0928$–$0.3578$ drifts like $D^{0.654\pm0.016}$ — the intervals saturate S1 only from *above*, no interval-only argument will produce the constant; what carries is the closed resistance sup above them, a two-weight statement about the Green function and not about a family of sets.

The rest, typed. (i) The Markov perturbation route is *certified dead*: $\lambda_{\min}(E-P) = -0.340\dots-0.274 < 0$ on every window, and the largest certified t with $E - tP \succ 0$ shrinks like $t^* \sim D^{3.96}$; only 3–12% of the off-diagonal *entries* are positive, but they carry 36.7–64.5% of the *mass* — the positive couplings are load-bearing, so S1 must be proved without the Markov hypothesis. (ii) S3 returns *stronger than asked*: the Cauchy–Schwarz floor $\mathrm{cap}_E(A) \ge |A|^2/(\mathbf{1}^{\mathsf{T}} R_{AA}\mathbf{1})$ makes Φ_{sup} a *maximum density subgraph* value for the Green kernel $R = E^{-1}$ (Charikar 2000; Goldberg 1984), so the sup over *all* 2^m sets is bounded with a cited absolute constant: $\Psi_{\mathrm{all}}\hat{\lambda} = 0.7399$–$0.8515$ at $D^{0.015\pm0.005}$ — the flattest number of the probe, with no family restriction, no interval reduction and no Markov hypothesis anywhere; and the argmax density set lives on *intervals* too, an independent confirmation of the reduction, with B_{res} only a factor 1.078–1.129 below. (iii) R4: 22 border blocks (a new count), 19 certified, 3 open; on 5 blocks *all* subsets were enumerated, so $c_{\mathrm{glob}} = 1.0000$ *exactly* there.

After T144: one unproven input — the absolute Maz'ya constant

(S1′) **the leading object, the sharpest available shape of S1**: it suffices to prove the capacitary strong-type inequality for forms whose Green function satisfies a *mean-density bound* ($\Psi_{\mathrm{all}}\hat{\lambda} \le 0.8515$ measured here) — a hypothesis of *Muckenhoupt* type rather than of Markov type. *(S1)* unchanged in

statement, reduced in scope: the family restriction is gone (the density bound covers all 2^m sets), the Markov hypothesis is unrestorable (the positive couplings are load-bearing), and the coordinates are fixed at a certified $O(1)$ cost. *(S2′)* the best-interval comparison $c_{\text{glob}} \le 1.6876$ is measured, not proved; a proof would be a testing condition, and the exhaustive R4 blocks show it is not vacuous. *(R4)* the 3 open border blocks. *(R5)* the whitening sandwich $\lambda/\hat{\lambda} = 1.213$–$2.175$, the last $O(1)$ factor of the chain, removable by a two-sided estimate of κ. Map V16; nine statements ripe (stock 93), 0 promoted here. The honest state, in one sentence: the *only* unproven input of the certified chain is the absolute constant c_0. T145 (`mazya_proof_probe.py`) is running at exactly S1′.

10.35 The proof attempt: one step missing, and it has a name (T145)

T145 (`mazya_proof_probe.py`, contract `MAZYA.PROOF`, 33/33, 8.0 s on 64 windows $m = 26 \dots 1173$, verdict ONE-STEP-MISSING, the verdict rule hard-wired and checked *before* the verdict is printed) attacks exactly S1′: it does not look for a better chain, it takes the classical *proof* of the capacitary strong-type inequality apart, transcribes it onto the non-Markovian form line by line, and reads off what survives. The step list, declared before it is measured:

Step	Content	Status here	Factor
M1	dyadic level decomposition of the minimiser	theorem, verified	1.649–2.670 (≤ 3)
M2	$\lvert A\rvert \le \Phi_{\text{sup}}\,\text{cap}(A) \le \Psi\,\text{cap}(A)$ (Green floor)	theorem (Cauchy–Schwarz)	$\times 1$
M3	$\text{cap}(A_{k+1}) \le 4^{-k} E(f_k)$, truncation admissible	theorem, verified	$\times 4$, slack ≤ 1.036
M4	$\sum_k E(f_k) \le E(f)$ — *the* Markov line	**void** on this form	measured

M1–M3 are theorems for *any* positive definite form (verified here in the direction they are used, on 168 levels of 24 windows for M3). M4 is the only step that uses the Markov property, and it uses it twice — the sign of the conductances and the sign-coherence of the truncation increments. On this form it is *void*, not merely unproved: on all 64 windows 3.2–14.9% of the off-diagonal entries of E are positive and carry 18.3–58.7% of the off-diagonal mass, the conductance part of the minimiser's energy is itself *negative* on 61 of 64 windows ($-0.380 \dots 0.593$ of $E(\psi)$) — and the same code on the Stieltjes surrogate $E - P$ reproduces the classical step, so the death belongs to the form and not to the measurement. T143's Miclo chain loss of 46–2561 is the same death, now localised in *one line* of the proof instead of in a construction.

But M4 splits, and only one half is Markovian. By the pair (Beurling–Deny) decomposition the step is the sum of a *conductance* half, which needs a nonnegative weight and fails above, and a *mass* half, which rests on the pointwise theorem $\sum_k f_k^2 \le \psi^2$ for the dyadic truncations (measured ratio 0.2639–$0.2995 \le 1$) — and the mass half *dominates* (0.407–1.380 of $E(\psi)$), so the full step $\sigma_{\text{tot}} = \sum_k E(f_k)/E(\psi) = 0.2145$–$0.4425$ is below 1 on every window although its Markov half is dead. That one exact number per window is all the classical proof ever needed. Two routes then make the constant *explicit*: the **energy route** (the classical proof with M4 replaced by its exact value at the minimiser) gives $c_0 = 12\,\sigma_{\text{tot}} = 2.5734$–$5.3099$; the **sign-free Green route** (entrywise domination $x^{\mathsf{T}} R x \le |x|^{\mathsf{T}} |R| |x|$ for the first use of Markov, nested-set domination of a nonnegative matrix for the second — no sign hypothesis anywhere) gives $c_0 = G_{\text{dy}} = 2.1559$–$6.3937$, the geometric level constant of the minimiser's own layer cake. The better route:

$$c_0 \;=\; 2.2480\text{–}4.2265, \qquad \text{fit } D^{0.028\pm0.017} \text{ (zone-uniform)},$$

against Maz'ya's classical Dirichlet value 4 and the minimum 1.067–1.352 the measured density sup allows. And S1′ stands as a *certified window inequality* on all 64 windows: cert_lam_max$(R) \le c_0\,\Psi$, with the certified side independent of whether the computed vector is the minimiser. The price of dropping the sign, as a number: the density hypothesis must be stated for $|R|$ rather than R^+, a factor 1.12–1.41 — and the direction error this fences ($x^{\mathsf{T}} R x \le |x|^{\mathsf{T}} R^+ |x|$ is *false*) carries its own built-in witness, $R = \left[\begin{smallmatrix} 1 & -1/2 \\ -1/2 & 1 \end{smallmatrix}\right]$ at $x = (1, -1)$, where $3 > 2$.

The no-go, and why the hypothesis list is tight. The "direct spectral argument" is not an independent bypass: $\Psi(A)$ is a Rayleigh quotient of R at $\mathbf{1}_A$, so $\Psi_{\text{all}}\hat{\lambda} \le 1$ is a theorem with *no* *content* for S1′ — the load-bearing direction is the lower one, a statement about the minimiser's *profile* (participation ratio 0.3026–0.4588, $K = 35$ levels, delocalised and sign-structured). And the explicit form $R = aa^{\mathsf{T}} + \varepsilon I$ with $a_i = i^{-1/2}$ — positive definite, entrywise *nonnegative*, with the *proved* density ceiling $4 + \varepsilon$ (prefixes by rearrangement) — has $\lambda_{\max} \sim \log m$: $\Psi_{\text{all}}\hat{\lambda}$ decays $0.7023 \to 0.4882$ along $m = 64 \to 1500$ while G_{dy} grows $7.6 \to 36.8$ (flat 1.69–4.16 on the Markovian control, 2.25–4.23 on the TFPT windows). So positivity, a nonnegative Green function and a bounded density sup do *not* imply S1′ with any absolute constant: a hypothesis on the minimiser's *level profile* is necessary, and the step list is not merely sufficient but tight.

The verdict discipline, as its own statement. The verdict rule is checked before the verdict is printed: PROOF-SHAPED requires every step factor to be a theorem or a certified bound, the replacement of M4 is a *measured* constant at the minimiser, no a-priori bound for it is available here — so exactly one step of the transcribed proof is open and the verdict is ONE-STEP-MISSING. Boundedness (max $4.2265 \le 16$, the preregistered ceiling) and flatness ($D^{0.028\pm0.017}$ against the 0.25 bar) of that constant are *evidence* for the missing lemma and are reported as such, never as the lemma. The rest, typed: S2′ is *retired as a requirement* (the chain never mentions an interval, so c_{glob} is not an input; the finite test family brackets the all-sets sup with $c_{\text{test}} = 1.0782$–1.1327 against Charikar's cited 2); R5 is certified on *both* sides per window but the sandwich width $\kappa_{\text{up}}/\kappa_{\text{lo}} = 2.668$–$28.532$ drifts like $D^{0.543}$ — downgraded, not retired, the used side $\kappa_{\text{up}} = 1.1491$–$1.4230$ staying flat ($D^{0.065}$); R4: all 24 border blocks of this pool close directly and inherit exactly the one missing input. Map V17, 11 new statements (stock 104). The identity spine of the whole cycle T142–T145 — the capacity decomposition, the exact capacity–Rayleigh identity, the Cholesky nestedness, the whitening certificate and the M4 split with the certified chain — is load-bearing as `v546` [`verification/v546_capacity_chain_identities.py`] (Section 4.12).

After T145: one named lemma — L1, the level lemma

(L1) **the only mathematical item left in the S1′ branch**: an a-priori bound, uniform in D, for *one* constant of the minimiser's own dyadic layer cake — either $\sigma_{\text{tot}} = \sum_k E(f_k)/E(\psi) \le C$ on the energy route (measured 0.2145–0.4425) or $G_{\text{dy}} \le C$ on the sign-free route (measured 2.1559–6.3937). Either one alone closes S1′ with an explicit c_0; the no-go proves some such bound is *necessary*; and both are delocalisation statements about the minimiser (participation ratio 0.303–0.459). *(L2)* the visible route: the T143 profile (orthogonal to the charge, masses help) plus a perturbation estimate, with the level count $K = 35$ and the sign share 0.348–0.672 as the profile's address. *(R4)* bookkeeping only. Retired with reasons: S2′ (the chain has no interval step), the Markov perturbation search (T144 certified it dead, and the proof needs Markov in exactly one line with two replacements), and the reading of $\Psi_{\text{all}}\hat{\lambda} \le 1$ as evidence (a content-free theorem). The honest state, in one sentence: T144's unproven input was an absolute constant with no address — after T145 it is a bounded, flat, exactly computable quantity attached to a named object with a necessity proof behind it, which is a strictly better place to stand and still not a theorem. T146 (`level_lemma_probe.py`) is running at exactly L1.

10.36 The level lemma stands (T146)

T146 (`level_lemma_probe.py`, contract `LEVEL.LEMMA`, 30/30, 7.6 s on 64 windows $m = 26 \ldots 1173$, verdict LEVEL-CARRIES, the verdict rule hard-wired on 9 preregistered conditions: a constant *measured at the minimiser* counts as *not* a priori, and the rule cannot be argued out of that) attacks exactly L1 — and closes it on the measurement surface as a chain of theorems and certified window inequalities, with no step reading the minimiser.

The profile defeat is the point. Four hand-written closed profiles — the endpoint sine mode, the linear ramp, the frame's own pole vector, the equilibrium potential of the lumped partner — lose against the true minimiser by *five to nine orders of magnitude* in the Rayleigh quotient: T143's cancellation anatomy seen from the other side, since every guessed profile sees $O(1)$ while the gap is $O(D^3)$. What survives is the *Green column* itself — one solve of the closed endpoint-Laplacian geometry, a legal test vector (Rayleigh–Ritz) with $\lambda_{\text{up}}/\hat{\lambda} = 1.0497$–$1.3764$, tens of percent and not orders. The transfer step one would write next is instrumented and then *discarded*: Davis–Kahan is informative on 0 of 64 windows (bound 1.28–87.4), because the bottom of the whitened spectrum is a near-degenerate *block* ($\lambda_2/\hat{\lambda} = 1.2542$–$3.4471$ — the separation sits in the denominator); and even where informative it would not help, since transferring a level-set functional needs an anti-concentration hypothesis, a second unproved input. L1 bypasses the transfer entirely because $\psi = \lambda R\psi$ is an *identity*. The step classification, declared before it is run and then earned: S1 a certified window object, S2 a theorem (Rayleigh–Ritz), S3 a theorem (the counting lemma, form-independent), S4 a theorem (the resolvent identity) plus certified column norms with the solve residual paid for — and S5, the transfer, *not needed*.

Delocalisation, read off the form. The lever of the whole part is one display:

$$\|\psi\|_\infty \;\le\; \lambda_{\mathrm{up}} \max_j \|Re_j\|\,,$$

verified on every window within a factor 1.055–1.426 of the truth — the $\Theta(D^3)$ smallness of the gap, everywhere else in this line the obstacle, is here the *tool*: the minimiser is λ times the Green function applied to itself, so it cannot concentrate where the Green columns do not. Made quantitative: the trivial ceiling $\max_j \|Re_j\| \le 1/\lambda_{\min}$ would give only $\Gamma = \sqrt{m}\,\lambda_{\mathrm{up}} \max_j \|Re_j\| \le \sqrt{m} = 5.1$–$34.2$ (useless); measured, $\Gamma = 1.8356$–2.2797 — the column norms sit a factor 2.7–16.0 *below* the trivial ceiling, and that factor *is* the delocalisation, a functional of the form and never of the minimiser. The localisation contradiction (restricted Gershgorin over all $\binom{m}{l}$ sets, $\ell^* = 2$–5) stays a theorem with no quantitative content (leak tolerance $\le 2.38 \times 10^{-3}$ — it forbids only exact localisation), and Perron–Frobenius applies only to the lumped partner (sign purity 1.0000), not to E (0.348–0.672 positive entries): delocalisation does not come from positivity on this surface.

The composite constant, and the two structural findings. The a-priori level constant

$$c_0^{\mathrm{ap}} \;=\; 2b^2\,\Gamma\,\min(1,\Gamma_1) + \varepsilon \;=\; 3.9042\text{–}4.8488 \qquad (b = 1.031,\ \ 64/64 \text{ windows})$$

dominates T145's *measured* 2.156–6.394, and the chain $\lambda \ge 1/(\kappa_{\mathrm{up}}\, c_0^{\mathrm{ap}}\, \Psi_{\mathrm{abs}})$ holds with loss factor 0.0422–0.1586 of the exact gap — every factor a functional of the form alone. Finding (a): *the cake base is free.* The chain consumes only the lower domination $|\psi| \le \psi_t$, so Maz'ya's classical dyadic 8 was never a price of the theorem, only of the convention: $2b^2 \to 2$ as $b \to 1$, a gain of 3.76 at zero cost, because the union identity makes the level constant an $O(m)$ closed form and the levels are never materialised. Finding (b): the spectrum bottom is a *block*, which kills every perturbation route and costs the identity route nothing — the resolvent identity holds for every eigenvector at the bottom. Remarkably, c_0^{ap} lands at the size of Maz'ya's classical Dirichlet value 4. The stress tests discriminate in both directions: on T145's no-go form Γ grows $3.676 \to 13.800$ ($m^{0.420}$, *unbounded* — the preregistered criterion) while the theorem half survives there (slack 1.087–1.242), and on the Dirichlet path-Laplacian control Γ stays flat (1.7183–1.7315); κ_{up} counts as a certified building block with its qualification stated (per-window completed Cholesky, maximum 1.4230; the D-flatness is a fit, so what is certified is a finite list — the same status the level lemma has, and the chain is homogeneous). The border pool of this part: 24 blocks, the direct Schur route closes 24/24 and the a-priori chain is positive on all 24 ($c_0^{\mathrm{ap}} = 2.855$–$3.176$); T145's 3 open blocks are not reproducible from a rebuilt pool — the anatomy transfers, the count does not. The four a-priori pieces — the union identity, the counting lemma at a general base, the resolvent delocalisation bound and the composite level lemma with its no-go discriminator — are load-bearing as v547 [`verification/v547_level_lemma_identities.py`] (Section 4.13).

After T146: the level lemma stands — one remainder, all-D

(G1) **the one genuine remainder**: D-uniformity for *all* D rather than on a finite surface — Γ is a certified per-window number (fit $D^{-0.016\pm0.005}$, a fit and never load-bearing), and turning the finite list into a theorem needs an asymptotic bound on $\max_j \|Re_j\|\,\lambda_{\min}$, i.e. the delocalisation of the *Green columns* themselves. *(G2)* σ_{tot} on the energy route stays measured — *not* a hole (either constant alone closes S1′, and the Green route is the certified one), recorded so nobody mistakes the asymmetry for a gap. *(G3)* the *actual* 3 open R4 border blocks from the T134 assembly, rather than a proxy pool — bookkeeping. The honest state, in one sentence: T145's missing lemma is now a theorem chain on the measurement surface with an explicit a-priori constant of classical size, and what separates it from a

statement for all D is exactly one asymptotic delocalisation bound. T147 (`green_asymptotic_probe.py`) is running at exactly that asymptotics.

10.37 The identity and the named mechanism (T147)

T147 (`green_asymptotic_probe.py`, contract `GREEN.ASYMPTOTIC`, 39/39, 27.2 s on 85 windows $m = 26 \ldots 1491$ spanning a factor 15 in D, verdict ASYMPTOTIC-SHAPED, the verdict rule wired constructively: "proved" cannot be returned in any branch, because a finite surface proves no statement for all D) attacks exactly the one remainder of T146 — and the last open object becomes an *exact identity*,

$$\Gamma \;=\; \sqrt{Q^\star}\cdot S_w\,, \qquad S_w = \lambda_{\rm up}\|R\|_F\,, \qquad Q^\star = m\,\max_j (R^2)_{jj}/\,\mathrm{tr}(R^2)\,,$$

verified to 2.3×10^{-16} on all 85 windows and on both stress forms: the D-uniformity of the level constant is *equivalent* to the boundedness of two named scalars and to nothing else — S_w purely spectral (the effective bottom multiplicity $n_{\rm eff} = \sum_k(\lambda_{\rm up}/\lambda_k)^2 = 1.251$–$2.831$, certified $S_w \le 4.6438$ at a certified cut of a preregistered ladder, the count confirmed by a completed LDL^T inertia and no eigenvector read), and $Q^\star$ purely geometric and scale-free (one for a perfectly flat Green diagonal, m for a localised column; certified ≤ 2.8634 against the localised extreme $m = 1491$).

The classical route dies first, by computation. Demko–Moss–Smith 1984 gives the decay rate $q = 0.994815$–0.999999 on this surface, so the envelope $q^{m/2} = 0.9346$–0.9995 sits above the 0.05 bar on *every* window — the culprit is the condition number $1/\Theta(D^3)$, not the form (E itself decays off-diagonally with Jaffard exponent 0.06–0.82) — and the Green columns are *delocalised* (half-mass radius 0.048–0.105 of the window width), so no decay statement can be the mechanism: the delocalisation itself *is* the bound. The norm anatomy locates it — 0.9864–0.9996 of $\|Re_j\|^2$ at the largest column comes from the bottom block alone ($|B| = 3$–6, certified by inertia) — and the equidistribution constant $Q_B = m\,\max_j(\Pi_B)_{jj}$, whose exact floor is $|B|$, measures $Q_B/|B| = 1.375$–1.839: the naive alternative $Q_B \sim m$ is wrong by a factor 5–258, and T146's unexplained factor 2.7–16.0 is *identified* as $\sqrt{m/Q_B}$ up to $\lambda_{\rm up}/\lambda_{\rm lo}$ — growing like $\sqrt{m}$, exactly the rate that makes Γ bounded. The honest negative is reported with it: the crude free-tail split (drop the mid-spectrum with $\|\Pi^\perp e_j\| \le 1$) is valid but its tail piece grows like $m^{0.538\pm0.020}$ — on record, *not* used.

The mechanism, named and made quantitative. The form is a *Toeplitz-minus-Hankel section* ($A_{rs} = c_{|r-s|} - c_{M-1-r-s}$), whose bottom eigenvectors are the Fourier modes at the symbol's near-zero by Szegő/Widom theory (Kac–Murdock–Szegő 1953; Widom 1958; Basor–Ehrhardt 2009) — and a Fourier mode is *exactly* equidistributed. The sharp prediction $Q_B \le 2|B|(1+o(1))$ *holds* (1.375–$1.839\times|B|$), and the chain down to it is pure arithmetic: $Q_B \le \sum_{k\in B} m\|\psi_k\|_\infty^2 \le 2\sum_B \|a_k\|_1^2$, with sparsity ≤ 8.773 per bottom mode and 0.999+ of each mode's energy in the lowest $4|B|$ Fourier modes. The surface is stratified in four D-strata with their own certified maxima; six trend exponents are all flat against the preregistered 0.25 bar and all labelled fits, never load-bearing. Discarded with a reason: the congruence structure of the zones does not carry (scatter 0.465 — the equidistribution is a property of the frame, not of the zone arithmetic).

Clean-up, in both directions. $\sigma_{\rm tot}$ on the energy route does *not* fall, but now has an address: truncation is a nonlinear operation and leaves the invariant subspace (block leakage up to 0.9627 against the 2.7×10^{-7} that boundedness through the split would need) — not a hole, the Green route is the certified one. **The 3 open R4 border blocks close**: rebuilt in this probe's coordinates, 8/8 border windows carry a valid positive lower bound through the T146 chain plus the new spectral bound ($c_0^{\rm ap} = 2.9135$–3.0383) — the same mechanism, no border-specific argument; what stays open there is the same all-D step, not a border gap. The mandatory stress separates the forms: on T145's no-go form both constants *diverge* as required ($Q_B = 13.49 \to 190.10$ over $m = 64 \to 1500$) and hit the closed-form prediction $Q_B = m/H_m$ to 1.0000, while on the Dirichlet path-Laplacian control — whose bottom modes are half-sines, exactly the Fourier modes the mechanism predicts — both stay flat ($Q_B/|B| = 1.4142$–1.4376 against the closed sine value 2). A bound that stayed finite on the no-go would be too strong and therefore wrong.

After T147: an exact identity, two certified factors — one lifting statement

(H1) **the one lifting statement**: a Szegő-type theorem for the *diagonally reweighted* Toeplitz-minus-Hankel section — that the bottom block of $\Lambda^{-1/2}A\Lambda^{-1/2}$ is Fourier-sparse with an m-independent ℓ^1 norm. That single statement lifts the whole chain from the stratified surface to all D (address: Widom 1958; Basor–Ehrhardt 2009; the measured obstacle is that the mechanism's hypothesis is translation covariance, which the Jacobi whitening breaks — Toeplitz defect 0.21–0.54). *(H2)* an a-priori bound for the spectral half S_w, i.e. for the eigenvalue counting function of the form near zero: the factorisation pays $S_w \geq 1$ whatever $Q^\star$ does, so a bottom block whose width grows with D would break the bound even at perfect equidistribution — a separate question from the shape of the eigenvectors. *(H3)* the σ_{tot} classification — either an energy-side analogue of the split or the explicit statement that the Green route is the only certified one. The honest state, in one sentence: what stands after T147 is a theorem-shaped task with two addresses — the all-D claim remains an extrapolation over 85 stratified windows and a factor 15 in D until the two statements are proved. T148 (`szego_bottom_probe.py`) is running at exactly the Szegő statement.

10.38 The lifting statement dissected (T148)

T148 (`szego_bottom_probe.py`, contract `SZEGO.BOTTOM`, 31/31, 13.0 s on 48 windows $m = 96\ldots1077$ in four D-strata, stress ladders to $m = 1500$, verdict ONE-INPUT-MISSING, the verdict rule wired constructively: "proved" cannot be returned in any branch) attacks exactly T147's two rests — and closes one of them, at the price of an honest negative, while the other resists with its hypothesis reduced to a single named scalar.

The step classification, every sub-statement typed. The passage from the pure Toeplitz-minus-Hankel section to the reweighted one $E = \Lambda^{-1/2}A\Lambda^{-1/2}$ runs in seven steps: *(1)* the order-2 model ladder $4k^2/\pi^2 \leq \lambda_k/\hat\lambda \leq k^2$ is a THEOREM (Kac–Murdock–Szegő 1953), verified exactly with $C \leq \pi^2/4$; *(2)* the Fourier sparsity of the *pure* section's bottom mode is a THEOREM (Widom 1958; Basor–Ehrhardt 2009), measured flat on the model ladder ($\nu = 4.236$–4.257); *(3)* the ℓ^1 *ceiling* — the lever — is CERTIFIED: from the identity $\mu_k^p a_k = \langle s_k, L_0^p\psi\rangle$ the coefficient bound $|a_k| \leq \min(1, \|s_k\|_\infty\|L_0^p\psi\|_1/\mu_k^p)$ and, with $\mu_k \geq 4k^2/(m+1)^2$, the *m-free* closed form $\|a\|_1 \leq 2\sqrt{\nu}+1$, $\nu = (m+1)^{3/2}\|L_0\psi\|_1/(2\sqrt{2})$ — every power of m cancelled, ν alone carries the m-dependence (largest excess -2.6×10^{-6}, i.e. slack); *(4)* the chain $Q^\star \leq 2\langle\|a\|_1^2\rangle_w \leq 2\langle\text{ceil}^2\rangle_w$ is CERTIFIED on 48/48; *(5)* the commutator/Davis–Kahan route is DEAD BY COMPUTATION (perturbation norm from *below*, the correct direction for a vacuity claim: factor 5.4×10^2–5.3×10^5, vacuous by 2.7–5.7 orders — the reweighting is not a perturbation); *(6)* the live route is weighted Szegő via the *Liouville transform* — $E\psi = \lambda\psi \Leftrightarrow A\varphi = \lambda\Lambda\varphi$ with $\varphi = \Lambda^{-1/2}\psi$, an exact change of variables — $m\|\psi\|_\infty^2 \leq 2\kappa_\Lambda\,\text{ceil}(\hat\varphi)^2$ with $\kappa_\Lambda = \max\Lambda/\min\Lambda = 1.733$–$3.933$ certified (a ratio of two diagonal entries of the form, a priori), buying a factor 50.6 in the smoothness functional of the Green-weighted modes; *(7)* ν bounded by a functional of the form: THE OPEN INPUT. *The hypothesis isolated*: a controlled weight-class experiment on a model section with a genuine order-2 zero shows both bounded-variation weight classes flat (ν trends $x^{0.000\pm0.001}$ and $x^{-0.007\pm0.002}$) while the rough class (TV $\sim m$) grows as $x^{1.994\pm0.002}$ — at *identical* $\kappa_\Lambda = 4$: it is the roughness, not the conditioning, and $\mathbf{TV}(\log\boldsymbol{\Lambda})$ is the named scalar. On the real surface the a-priori $Q^\star$ ceiling measures 81.4–469.0 and grows $m^{0.615\pm0.016}$; ν_L would need $\lesssim m/(8\kappa_\Lambda) \approx 34$ at the largest window, measured 282 (trend $x^{0.166\pm0.025}$); $\text{TV}(\log\Lambda) = 3.63$–$11.93$, trend $x^{0.444\pm0.006}$, *not* flat — the real whitening is rough in the experiment's sense.

S_w closes — and the symbol falls. A layer cake over LDL^T inertia counts at a preregistered threshold ladder (no sorted eigenvalue list; Sylvester 1852) certifies $\mathbf{S_w \leq 1.3905}$**–**$\mathbf{1.9587}$ against the true 1.2820–1.6825, trend $x^{0.007\pm0.007}$ *flat*, certified per D-stratum (4×12); $n_{\text{eff}} = S_w^2 = 1.644$–2.831. The honest negative, asserted rather than hidden: the Toeplitz symbol of the arithmetic lag sequence has $\mathbf{f(0) < 0}$ **on 48/48 windows** ($-87.3\ldots-22.5$, curvature 1.9×10^5–9.7×10^7) — the positive definiteness of the section is a *finite-section effect* of the minus-Hankel part and the Stieltjes lumping, not a symbol property; the KMS order-2 hypothesis is not available at the symbol level, and the bottom scaling survives only as a *measurement* ($\alpha = 1.643$–1.989 inside the

preregistered band $[1.5, 2.5]$). The mandatory stress separates exactly where it must: the no-go violates precisely the smoothness hypothesis ($\nu = 48.7$–4.28×10^3, divergent; its spectral half is innocent, $n_{\mathrm{eff}} = 1.0000$), and the Dirichlet control is exact ($\nu \to \pi$, $m\|\psi\|_\infty^2 \to 2$).

The whole chain, end to end, with the big number. The chain $\lambda \geq 1/(\kappa_{\mathrm{up}} c_0^{\mathrm{ap}} \Psi)$ with $\Gamma = \sqrt{Q^\star} S_w$ (T147's identity, re-verified in this file's code path to 4.4×10^{-16}) is a certified *lower* bound on all 48 windows and delivers **8.36–15.86% of the true gap** (median 11.74%, no window losing more than a factor 12); the same chain with the a-priori-shaped factors substituted is still a lower bound everywhere at a factor 10.5 (fraction 0.72–1.56%) — the price of a-priority is a number, and the loss sits entirely in the looseness of the ℓ^1 ceiling, not in the architecture. The uniformity balance over all ten factors: six CERTIFIED, two MEASURED (ν_L, TV), two FITS — and the fits enter no inequality; the load-bearing distinction is now certified-per-window versus certified-*and*-m-free, and exactly one factor sits on the wrong side. σ_{tot}: RETIRED AS A ROUTE — the Green route is the only certified one, no compiler statement waits on it, and later passes should not re-open it as a gate. The identity/certificate core of T147+T148 — the factorisation identity, the Q_B floor, the ℓ^1 ceiling, the layer-cake S_w certificate and the Liouville transform with its weight-class discriminator — is load-bearing as v548 [`verification/v548_green_szego_identities.py`] (Section 4.14).

After T148: S_w closes on the surface — one scalar input remains

(I1) **the load-bearing rest**: a bound $\nu_L \leq F(\text{form})$ with F m-free for the bottom block of the *reweighted* section — equivalently bounded $\mathrm{TV}(\log \Lambda)$, or any weaker smoothness of the multiplier that still controls $\|L_0 \hat{\varphi}\|_1$; the target is $\nu_L \lesssim 34$ at the largest window, measured 282. That single scalar statement is the whole rest of the lifting statement Z1. *(I2)* cosmetic: a theorem behind the order-2 bottom ladder directly from the minus-Hankel structure (Basor–Ehrhardt 2009 is the address) — the S_w certificate does not need it (it is an inertia count), but without it the m-freeness of the KMS constant rests on a measurement. The honest state, in one sentence: what stands after T148 is a certified end-to-end chain that delivers 8.4–15.9% of the true gap on a stratified surface of 48 windows, with exactly one scalar — ν_L, equivalently the roughness of the whitening weight — between the finite list and a statement for all D. T149 (`weight_smoothing_probe.py`) is running at exactly that scalar: whether a smoothed whitening $\tilde{\Lambda}$ with a certified sandwich removes the TV roughness.

10.39 The smoothing freedom (T149)

T149 (`weight_smoothing_probe.py`, contract `WEIGHT.SMOOTHING`, 30/30, 61.4 s on 44 prime-power windows $m = 277 \ldots 1393$ in four D-strata, 9 preregistered gauges, verdict PARTIAL-SMOOTHING, the verdict rule wired constructively: "proved" cannot be returned in any branch) attacks exactly T148's one open input, through the observation that the whitening diagonal is a *free gauge*: for any positive diagonal $\tilde{\Lambda}$ the chain runs unchanged — an identity plus one Rayleigh step, licensed and verified before use — so every gauge gives a valid lower bound and the maximum over the family is valid too.

The three numbers, under the preregistered winner rule (8 admissible of 9 candidates; the winner is `const`, the geometric mean of the Jacobi diagonal): the sandwich is CERTIFIED at $\sigma \leq 5.5789$ (trend $x^{0.154\pm0.022}$, bar 8.0 held); $\mathbf{TV}(\log \tilde{\boldsymbol{\Lambda}})$ **is exactly 0.0000** — T148's blocking number, 11.93 growing as $x^{0.444}$, is not reduced but *eliminated*; and $\nu_{\tilde{L}} = 222.70$ ($x^{0.170\pm0.047}$) meets the target $m/(8\kappa)$ only on **21/44** windows ($\rho = 1.347$ at $m = 277$ falling to 0.485 at $m = 1393$) — PARTIAL: the small-m end is missed. The price of the gauge: $\tilde{\kappa}_{\mathrm{up}} \leq 2.3146$ against Jacobi's 1.2647 (a factor 1.83, paid once and linearly, certified). *The real finding is the two-regime decomposition, and it refutes the hypothesis*: the flutter-smoothing gauges (flutter removed, macro profile kept) are *cheap* (sandwich 1.129–1.572, TV $\to$ 1.27–1.73) yet move $\nu_{\tilde{L}}$ by at most **0.9**%; only the macro-removing gauges improve $\nu_{\tilde{L}}$ at all (factor 1.27) — the roughness the smoothness functional responds to was never in the multiplier: $\mathrm{TV}(\log \Lambda)$ *was the mislocated hypothesis*, withdrawn and relocated — the roughness sits in the *form* itself.

The lifted chain. All 178 assembled chains are valid lower bounds; $\Gamma = \sqrt{Q^\star} S_w$ holds gauge-invariantly at 4.4×10^{-16} over 89 assemblies. Certified, the family-maximum chain delivers **9.52–18.45% of the true gap** (median 12.68%; T148's 8.36–15.86% reproduced on a longer surface), a priori 0.800–1.566% (an improvement of up to 1.284×); on the certified side the Jacobi gauge still wins (44/44 — its sharper $\tilde{\kappa}_{\rm up}$ outweighs the smoother $Q^\star$ ceiling: the gauge freedom is a tool for the a-priori side, not a free win). The $Q^\star$ ceiling falls 697.9 → 149.1 with its exponent falling 0.551 → 0.330 — bounded yes, *flat no*, and the driver is named: $\nu_{\tilde{L}} = 265.2$ ($x^{0.772\pm0.048}$) on the Green-weighted modes *above* the bottom block.

Stress, and the corrected question. The mandatory no-go's own multiplier is already smooth (best sandwich 1.2629 over the whole family), and its ν diverges as $x^{1.420\pm0.004}$ under *every* gauge — the break stays visible in the form, smoothing rescues nothing and cannot manufacture equidistribution (passed, preregistered). The Dirichlet control has a constant diagonal, all 9 candidates are the identity on it (5.9×10^{-14}), $\nu\to\pi$ exactly, $m\|\psi\|_\infty^2 \to 2$ — and the control *corrects the question*: classically $\nu_k = \pi k^2$ *exactly*, so the right form is the ladder $\boldsymbol{\nu_k \le Ck^2}$ **with** $\boldsymbol{m}$**-free** C (measured in the `const` gauge: $C = 18.66$–44.61, $x^{0.272\pm0.021}$, attained at $k=1$). The TV anatomy of the Jacobi diagonal locates the cheapness: the macro part is 0.78–1.52 (8.5–14.2% of the total), the flutter part 6.42–13.02 — many *small* steps (amplitude 0.064–0.182, *flat* at $x^{-0.007\pm0.028}$, largest single log-step ≤ 0.193) spread over the whole window, with the frame edges not the source (enrichment 0.75–1.20): TV grows through the step count while the sandwich pays only the amplitude. The order-2 bottom ladder from the minus-Hankel structure stays NAMED OPEN, cosmetic ($C_{\rm bot} = 5.6$–14.1; Basor–Ehrhardt 2009 does not cover a sign-changing symbol; the chain does not wait on it — S_w is certified by an inertia count). Seven promotion candidates are named and stay unpromoted.

After T149: TV eliminated, the missing input relocated — a ladder remains

(J1) **the load-bearing rest**: an m-free constant C in the ladder $\nu_k \le Ck^2$ for the modes of the smoothed section (measured $C = 18.66$–44.61, $x^{0.272\pm0.021}$) — with the constant gauge this is *exactly* a question about the deep modes of the pure Toeplitz-minus-Hankel section, no multiplier, no whitening, no arithmetic weight; it makes the $Q^\star$ ceiling m-free and the whole measurement-surface chain a priori. *(J2)* **the second lever, which would close (J1) outright**: an a-priori bound on the flutter *amplitude* of $\log\operatorname{diag}(A_B)$ — measured 0.064–0.182 and *flat* ($x^{-0.007\pm0.028}$); it would make $\kappa_{\tilde{\Lambda}} = 1$ and $\tilde{\kappa}_{\rm up} \le 2.3146$ a priori rather than certified per window. *(J3)* cosmetic: the order-2 ladder theorem from the minus-Hankel structure (unchanged from T148). The honest state, in one sentence: after T149 the smoothing freedom has retired $\mathrm{TV}(\log\Lambda)$ as a hypothesis — eliminated exactly, at a certified price — and the one input between the finite list and a statement for all D is now the smoothness of the deep modes of the pure section, in ladder form. T150 (`mode_ladder_probe.py`) is running at exactly that ladder, through the flat flutter amplitude.

10.40 The mode ladder and the parity mechanism (T150)

T150 (`mode_ladder_probe.py`, contract `MODE.LADDER`, 36/36, 48.0 s on 72 prime-power windows $m = 50\ldots1393$, verdict ONE-TERM-MISSING, the verdict rule wired constructively: "proved" cannot be returned in any branch) attacks the ladder $\nu_k \le Ck^2$ through the flat flutter amplitude — T149's rest (1) via rest (2) — and comes back with the mechanism's *name*.

The parity compression. The reflection-odd form that every probe since T106 works with is *exactly* the compression of the full symmetric Toeplitz section onto its antisymmetric parity sector:

$$U_-^{\mathsf T}\,T_M(c)\,U_- \;=\; T-H\,, \qquad U_+^{\mathsf T}\,T_M(c)\,U_+ \;=\; T+H\,, \qquad U_+^{\mathsf T}\,T_M(c)\,U_- \;\le\; 4.0\times10^{-16}$$

also on real windows (Basor–Ehrhardt 2009: the parity sectors are the natural objects of the Toeplitz+Hankel algebra) — and the inertia *localises*: **the entire negative inertia of the sign-changing symbol sits in the even sector** (odd 0, even 1 on 72/72, $\mathrm{LDL}^{\mathsf T}$ certified). T148's honest negative $f(0) < 0$ is thereby *explained rather than worked around*: it is a statement about the *other* parity sector. The matched parity sines $t_k(r) = \frac{2}{\sqrt{N}}\sin(2\pi k(r{+}1)/N)$, $N = 2m{+}1$ (exact eigenpairs of the parity Laplacian, corner 3 forced by the antisymmetry; control model exact, $\nu_k = \pi k^2$ to 0.0002) give a third valid ceiling, improving the ladder constant by 1.000–1.130 over the Dirichlet+Neumann minimum of T148/T149.

The zone diagonal, and who carries the positivity. The whitening diagonal is an *explicit zone function*, $\Lambda_r = c_0 - c_{M-1-2r} + \sum_{s\ne r}\max(c_{|r-s|} - c_{M-1-r-s}, 0)$ (residual 0.0 on 72/72), and the lag vector splits exactly as $c = c^{\text{arch}} + c^{\text{atom}}$ with $c^{\text{atom}} \le 0$. The flutter amplitude becomes a certified *form* functional, no fit anywhere: $\text{amp}_{\text{loc}} = 0.0606$–$0.1993$ at trend $x^{-0.165\pm0.015}$ (T149's fit-based 0.064–0.182 reproduced fit-free), with the certified chain $\text{amp}_{\text{loc}} \le \text{half} \cdot d_1^{\log} \le \text{half} \cdot d_1^{\lambda}/\min\Lambda$ (1.345–3.954) and the closed atom budget $\|c^{\text{atom}}\|_1 \le 4B\sqrt{N}$. The honest negative beside it: the archimedean diagonal is *not* the macro profile — the level shift $\|\log(\Lambda/\Lambda^{\text{arch}})\|_\infty = 0.734$–$2.647$ grows as $x^{0.244}$; the flutter is small and a priori, the *level* is not. And the perturbation step is precisely dead, for two independent reasons: the multiplicative (Loewner) step transfers scale-free with $\exp(\text{amp}) \le 14.11$ — the gauge step *closes* — while the additive step (Weyl 1912 / Bauer–Fike 1960 / Davis–Kahan 1970) compares $\|A^{\text{atom}}\| = 4.3$–$12.3\%$ of the section against a $\Theta(D^3)$ bottom and misses by 2.4–5.8 *orders of magnitude*; moreover the archimedean-only section is itself *indefinite* (2–7 negative eigenvalues, LDL^T certified) while the full section has none — **the atoms are co-responsible for the positivity**, not a perturbation of it.

The one gate, with its number. The chain delivers 0.0680–0.1845 of the true gap certified (the best window reproduces T149's 0.1845 exactly), a priori 8.16×10^{-3}–1.74×10^{-2}; the ladder-shaped $Q^\star$ ceiling is valid by monotonicity (38.76–261.3 against the certified minimum 36.98–144.6, price $\le 2.555\times$ for the reduction to *one* constant). But $C = 14.24$–43.39 ($= 13.81\pi$) with trend $x^{0.258\pm0.009}$: the gate "ladder constant non-growing" falls — the only one of all. The maximum sits at $k = 1$ (67/72), the growth is carried by the deeper modes, and the gap $\pi \to 13.81\pi$ is exactly the *arithmetic leakage* of the bottom mode (ℓ^1 mass 1.41–1.79, $m\|\psi\|_\infty^2 = 2.45$–$3.41$ against the model's 2). The stress is now localised: because the amplitude and the ladder are separate statements, the T145 no-go can be asked *which* it violates — the answer is the ladder alone (its multiplier is smooth, $\text{TV} \le 0.2334$; its ladder constant grows $x^{1.840\pm0.130}$), while the Dirichlet and parity controls reproduce $C = \pi$ to 0.0004/0.0002. The identity/certificate core of T149+T150 — the gauge freedom with the const gauge and its certified sandwich, the two-regime discriminator, the explicit zone diagonal with the closed atom budget, the parity compression with the inertia localisation, and the parity-sine calibration with the per-window ladder certificate — is load-bearing as v549 [`verification/v549_gauge_parity_identities.py`] (Section 4.15).

After T150: the mechanism is named — one inequality remains

(K1) **the load-bearing rest**: $\nu_k \le Ck^2$ with an m-free C for the *odd parity sector* of a sign-changing-symbol Toeplitz section — *the* one inequality; measured $C \le 43.391 = 13.81\pi$ certified per stratum, trend $x^{0.258\pm0.009}$ against the flatness bar 0.25. *(K2)* a closed lower bound on $\min\Lambda^{\text{arch}}$ from the smooth kernel — it would make the flutter-amplitude ceiling closed rather than per-window certified. *(K3)* the extension of Basor–Ehrhardt 2009 to a symbol with $f(0) < 0$, or an explicit local model at the symbol minimum inside the odd sector. The honest state, in one sentence: after T150 the mechanism has a name — parity — T148's honest negative is explained (the negative inertia lives in the even sector), the flutter amplitude and the zone diagonal are certified form objects with the atoms co-responsible for positivity, and the one input between the finite list and a statement for all D is a single named inequality about the low modes of one explicitly described finite matrix. T151 (`odd_ladder_probe.py`) is running at exactly that odd-sector ladder.

10.41 The odd ladder and the Sobolev reroute (T151)

T151 (`odd_ladder_probe.py`, contract `ODD.LADDER`, 27/27, 43.4 s on 72 prime-power windows $m = 96 \dots 1491$, verdict ODD-CARRIES, the verdict rule wired constructively: "proved" cannot be returned in any branch) attacks the one remaining inequality — and closes it by *rerouting off the ladder*.

The grid step-over, and the matrix-versus-symbol lesson. The positivity of the odd sector is a *grid* fact: the parity sines live on the shifted grid $\theta_k = 2\pi k/N$ with $\mu_k^P = 2 - 2\cos\theta_k$ exactly, and the symbol's negative window (width $\theta_c = 7.06 \times 10^{-4}$–$1.32 \times 10^{-2}$) lies *below* the first odd grid point θ_1 on every window ($\theta_c/\theta_1 = 0.328$–$0.407$, trend $x^{-0.057}$): the grid steps over it on 72/72 — stepped over rather than cancelled. But the pointwise symbol *language* fails at exactly this resolution: $f''(0) = 3.8 \times 10^5$–6.8×10^8, f moves by a relative 0.47–0.85 across *one* mode spacing, the section form averages the symbol against a Fejér kernel of exactly that width instead of sampling it (mode-by-mode deviation 1.39–1.83 at $k = 1$), and the one-parameter symbol margin is vacuous ($\rho \geq 1$) on 72/72. **T150's rest item 3 — extending Basor–Ehrhardt 2009 to $f(0) < 0$ — is thereby answered in the negative**: the local model at the symbol minimum exists, but as a *matrix* statement, not a symbol one — $\lambda_k(A) \leq S\mu_k^P$ for $k \leq 8$ certified by completed LDL^T counts (Sylvester 1852) with $S = 1.1019$–2.3870: the odd effective minimum sits at θ_1, order 2, $O(1)$ prefactor, $f(0) < 0$ *absorbed* by the Rayleigh floor $L_P \geq \mu_1^P I$. The leakage T150 left as a bare number is anatomised: a *low-k rotation* of the bottom mode away from t_1 ($\sin\angle(\psi, t_1) = 0.8156$–$0.8943$, band shares 0.355 at $k = 1$, 0.687 at $k = 2$–4, 0.005 beyond $k = 64$; sideband peak at $l^* = 2$) — not an arithmetic sideband.

The Sobolev step, and the linear bound. The multiplicative mode comparison T150 asked for *is* the discrete Sobolev step: an odd-sector vector vanishes at the virtual node, so $\|\psi\|_\infty^2 \leq 2\|\psi\|_2\|\nabla\psi\|_2$; the certified pencil floor $\kappa = 0.2273$–0.4502 (*one* completed Cholesky per window, floor carried, trend $x^{0.047\pm0.009}$) bounds the gradient energy by λ_k/κ, and

$$b_k \;:=\; m\|\psi_k\|_\infty^2 \;\leq\; 2m\sqrt{\frac{K_{\mathrm{bot}}\,\mu_k^P}{\kappa}} \;\leq\; C_S\,k$$

with $C_S = 11.5137$–19.5731 (median 13.82, trend $x^{0.020\pm0.007}$): **linear in k and non-growing** where T150's quadratic ladder grew ($C = 13.81\pi$ at $x^{0.258}$) — nothing additive anywhere, so the $\Theta(D^3)$ bottom that killed Weyl / Bauer–Fike / Davis–Kahan never enters, and the $\sqrt{m}$ that kills every rotation route (the ν conversion costs $4\sqrt{m}$ and overshoots π by 4.2×10^3–1.0×10^6) is never paid; at the bottom the gain over the ν route is a factor 7.5–34.9. The layer cake buys only 1.00–2.37 and does not change the trend. The honest negative beside it: the *global* pencil ratio grows $x^{2.695}$ (useless — L_P saturates at 4 while the section's top does not); everything is carried by the *bottom* ratio $R = K_{\mathrm{bot}}/\kappa = 3.3634$–$9.7108$ at $x^{0.037\pm0.015}$ — the *one* remaining fit in the chain.

The large numbers, and the localised stress. T150's rest item 2 is closed *and attained*: $\min_r \Lambda_r^{\mathrm{arch}} = c_0^{\mathrm{arch}} - c_{M-1}^{\mathrm{arch}} = 2.5928$–$4.8981$ at relative deviation 0.0 (two per-window-checked shape properties). $Q^\star_{\mathrm{sob}} = 12.39$–$23.94$ ($x^{0.029\pm0.008}$) beats the ℓ^1 ceiling 43.66–158 by 2.69–10.13; end to end moves from 7.88×10^{-3}–1.74×10^{-2} to 2.01×10^{-2}–3.52×10^{-2} of the true gap (gain 1.64–3.18×), and the dominant loss is now $\Psi = 4.7 \times 10^3$–6.1×10^6 — never before the bottleneck. The stress localises exactly where it must: the T145 no-go breaks *in the pencil and only there* (R grows $x^{1.986}$ to 3.0×10^7; its localised bottom mode has parity Dirichlet energy $\sim 1/\log m$ instead of μ_1^P), while the controls are exact (parity pencil $(1,1)$ to 3.73×10^{-12}, $C/\pi - 1 \leq 2.5 \times 10^{-4}$; the model Sobolev price is *exactly* π — the honest, m-free cost of leaving the ℓ^2 world), and the Dirichlet-versus-parity pencil is 1.985–1.999: one rank-one corner mismatch, bounded. Verdict ODD-CARRIES: all six preregistered gates met, every layer typed — *theorem* (parity compression, Weyl minimax, Sylvester, exact μ^P), *certified* (κ, K_{bot}), *elementary-verified* (the Sobolev step) — and the one fit named. The identity/certificate core of T151 — the grid step-over with the certified symbol vacuity, the certified bottom ladder, the Sobolev step with the π-price calibration, the closed-and-attained archimedean minimum, and the linear per-mode bound with the no-go discriminator — is load-bearing as v550 [`verification/v550_odd_sector_identities.py`] (Section 4.16).

After T151: the bound is linear — one scalar remains

(L1) **the load-bearing rest**: the m-freeness of the bottom pencil ratio $R = K_{\mathrm{bot}}/\kappa$ — *one* scalar, no eigenvector, the only non-certified link left in the chain; measured 3.3634–9.7108, trend $x^{0.037\pm0.015}$, and the T145 no-go breaks exactly there ($x^{1.986}$). *(L2)* a closed lower bound on $\kappa = \lambda_{\min}(A, L_P)$ from the lag vector — it would make the Sobolev route closed rather than per-window certified; the symbol route to it is *vacuous* and the reason is now known, so this needs the arithmetic of the atoms, not an asymptotic. *(L3)* Ψ, the new dominant end-to-end loss now that $Q^\star$ has been improved. The honest state, in one sentence: after T151 the one inequality closes by rerouting off the ladder — the odd grid steps over the symbol's negative window, the bottom spectrum is certified against the parity Laplacian, the Sobolev step makes the per-mode bound linear with a non-growing constant — and the whole distance between the finite list and a statement for all D is the m-freeness of a single printed scalar. T152 (`pencil_ratio_probe.py`) is running at exactly that scalar.

10.42 The pencil ratio and the cancellation (T152)

T152 (`pencil_ratio_probe.py`, contract `PENCIL.RATIO`, 37/37, 33.2 s on 60 prime-power windows $m = 149 \ldots 1445$, verdict ONE-TERM-MISSING, the verdict rule wired constructively: "proved" cannot be returned in any branch) attacks the one scalar T151 left as a fit — the m-freeness of the bottom pencil ratio $R = K_{\mathrm{bot}}/\kappa$ — and returns the sharpest structural fact of the phase.

Both hoped-for gifts refuted: positivity is a cancellation. A^{arch} is *not* positive in the odd sector on any of the 60 windows ($\lambda_{\min} = -2.8132$–-1.8396, certified negative by a completed Cholesky of $A^{\mathrm{arch}} - tI$ with the floor carried): the odd sector removes the constant mode but not the negative archimedean mass near the origin. In pencil normalisation the failure is catastrophic, not marginal — $\kappa_{\mathrm{ar}} = \nu_{\min}(A^{\mathrm{arch}}, L_P) = -5.28 \times 10^5$–$-4.00 \times 10^3 = O(-m^2)$, because the lowest parity sine has L_P-energy $\mu_1^P = O(m^{-2})$ while it sees an $O(1)$-negative form. And the atom part is negative too ($\nu_{\min}(A^{\mathrm{atom}}, L_P) = -1408$–$-25.6$ on 60/60): *neither* summand of the exact split is positive, and the sum is — **the positivity of the odd section is a cancellation between the archimedean kernel and the prime-power atoms, not a property of either half** (exhibited exactly at t_1: an archimedean part -5.26×10^5–-3.98×10^3 plus an atom part 8.19×10^3–2.88×10^6 add to a near-$O(1)$ quotient — the atoms even *overshoot* the archimedean hole). The additive Weyl route misses κ by a measured 1.5×10^4–1.8×10^6 and the closed ℓ^1 budget route by a further 2.3×10^5–9.9×10^7 — the $O(m^2)$ conversion $1/\mu_1^P$ kills both. But the archimedean diagonal is negative on only an *m-independent* number of parity modes ($k \leq 4$–7 of up to 1445), and above them its pencil quotient is ≥ 0.989 on 60/60: **the Schur two-block criterion (Schur 1917 — an equivalence, with both Cholesky floors subtracted) with a fixed 16-mode low block certifies $\kappa \geq t = 0.2250$–0.2500 on every window**, at $m^{0.006\pm0.036}$ with identical quartile medians — the floor is no longer a fit. It consumes *one* unproven inequality — $B_{HH} \succeq tI$ on the bulk block — and entrywise routes are provably insufficient for it: Gershgorin returns -8.99×10^5–-1.3×10^3 on the full whitened parity block and still -7189–-58 after cutting at the last arch-negative mode, the median row coupling being 3.9–19.6 against a diagonal of 0.36–0.89. What is needed is a *block* statement, not a bound on individual prime-power sums.

The ratio, certified on both ends. The archimedean part does not carry the *ceiling* either: $K_{\mathrm{bot}}^{\mathrm{arch}} = 24.6$–$1326$ at $m^{1.519}$, one to two orders of magnitude above the true $K_{\mathrm{bot}} = 1.102$–2.118 and growing where the true one is flat — the atoms do not perturb an archimedean alignment, they *create* it. The new number: $R \leq K_{\mathrm{bot}}/t = 4.408$–$8.471$ **at $m^{0.099\pm0.167}$, certified on both ends** (inertia ladder above, Schur floor below), with a quartile walk of 1.269 over a $4.54\times$ m-range. Two structural m-free routes to the ceiling lose four to six orders of magnitude: Courant–Fischer on the parity basis (the atom overshoot is $O(m^2)$), and the two-sided bulk block — whose KMS ratio step *is* a theorem ($\sin a/\sin b \leq a/b$, hence $\mu_{k+16}^P/\mu_k^P \leq 289$, verified 285.95–$288.97 \leq 289$ on the table) but whose ingredient $T_{HH} = 26.8$–3116 grows $m^{2.044}$: the section's low *eigenvalues* track the model's while its low *eigenvectors* do not. The no-go reconstruction breaks exactly at the unclosed step (the floor survives; ceiling and R grow $m^{1.994}$) — the gap is real, not a tool loss; the control (L_P against itself) is $(1,1,1)$ to 3.5×10^{-11}.

The Ψ map, and the honest dual reading. $\Psi \leq 8.70 \times 10^3$–$6.24 \times 10^5$ *is* $1/\lambda_{\min}(E)$ to within 1.12–1.19 — the spectral bottom of the whitened section and essentially nothing else; it grows $m^{2.18}$ because $\mu_1^P = O(m^{-2})$, not because κ degrades. The reserve: 89–92% **of** trace(E^{-1}) **sits on the eight lowest modes — exactly those the certified ladder already controls**; rebuilding the subset supremum into a ladder sum is worth 3.46–$5.29\times$ end to end (it would lift T151's 2.01×10^{-2}–3.52×10^{-2} into the 8×10^{-2}–1.4×10^{-1} range) — now the dominant loss, outranking any further sharpening of R. The verdict is kept honest by printing both readings: on the band estimator alone gate B would fail and the verdict would read RATIO-RESISTS; the quartile-median reading (15 windows per stratum, high variance of the band fit) is the declared one, and both are in the output. Also on the record: T151's R exponent 0.037 was fitted against the zone variable (not monotone in m) — the same quantity refit in m is 0.099, so part of that flatness was the fit variable rather than the object. Ten map entries are named; *none is promoted* — the strongest candidates consume the unproven block positivity $B_{HH} \succeq tI$ and stay pending with exactly that reason.

After T152: the floor is structural — one term and one rebuild remain

(M1) **the block inequality**: $B_{HH} \succeq tI$ on the bulk block — the single unproven input the Schur certificate consumes; entrywise/Gershgorin bounds provably cannot deliver it (-8.99×10^5–-58 even after the cut), so it must be a block statement. *(M2)* **the load-bearing rest**: an m-free ceiling on K_{bot} — every structural route loses four to six orders of magnitude because the low eigenvectors are not the model's, and the T145 no-go breaks exactly there ($m^{1.994}$). *(M3)* the Ψ rebuild, subset supremum $\to$ ladder sum — worth 3.46–5.29× end to end, now the dominant loss and ahead of any further work on R. Only after these: frames beyond A, zones beyond prime powers. The honest state, in one sentence: after T152 the positivity of the odd sector is known to be a cancellation between geometry and arithmetic rather than a property of either half, the pencil floor is certified by a fixed-size Schur block instead of a fit, the ratio is certified on both ends with one m-free ceiling missing — and the largest remaining loss sits on modes the certified ladder already controls. T153 (`psi_ladder_probe.py`) is running at exactly that rebuild and that block inequality.

10.43 The Ψ pinning and the two remaining terms (T153)

T153 (`psi_ladder_probe.py`, contract `PSI.LADDER`, 33/33, 7.6 s on 18 prime-power windows $m = 96 \ldots 1430$, block anatomy on 11 and the ceiling on 12 of them, verdict REBUILD-RESISTS, the gates *evaluated* rather than chosen) carries out the rebuild T152 mapped — and refutes it by exactly its own factor.

The pinning: the reserve was never there. The rebuild itself is real and comes in three rigorous forms, each an upper bound on the same Ψ: the mode-resolved ladder sum (cost 1.195–1.761× against the T149 route), the arithmetic-free operator form $(\mathrm{const}/t)\,\Psi_P$ against the pure Kac–Murdock–Szegő model (cost 3.73–9.70×), and the theorem-grade collapse $\Psi \le \mathrm{const}/(t\,\mu_1^P)$ (cost 3.73–9.70×). But the decisive measurement is a *lower* bound: for the subset that maximises the first mode, every term of $\sum_k \lambda_k^{-1}\langle v_k, \mathbf{1}_S\rangle^2$ is nonnegative, so $a_1/\lambda_1 \le \Psi \le 1/\lambda_1$ with $a_1 = 0.7695$–0.8306 — exactly the $8/\pi^2 = 0.8106$ of the first parity sine. **Ψ is pinned to within $1/a_1 = 1.204$–1.300, so T152's mapped 3.46–5.29× reserve never existed**; the head/tail replacement T152 proposed lies 2.31–6.12× *below* the hard lower bound and is not an upper bound at all — the maximising subset is aligned with v_1 itself, so the head *is* the value of Ψ, not a redundancy. The bookkeeping is exact: the rebuild retires the level constant $c_0^{\mathrm{ap}} = 4.157$–$4.870$ and pays the collapse cost 3.73–9.70; on this surface the two cancel, the net end-to-end gain is 0.50–1.11× (median 0.62), and the certified fraction moves from 2.01×10^{-2}–3.52×10^{-2} to 1.01×10^{-2}–3.92×10^{-2} — not the targeted 8×10^{-2} band. The collapse cost is named exactly: $\lambda_1(A)/(t\,\mu_1^P) = 3.25$–$8.40$ — the pencil minimum is not attained at the bottom mode, so a floor uniform over all modes is lossy for mode 1 alone. What the rebuild does buy is *uniformity*: the collapse form contains nothing beyond an explicit arithmetic sum, the one open block inequality t, and the exact μ_1^P — it retires Charikar's licence and every per-window diagonalisation from the Ψ step. **Ψ is closed as a term.** The balance is clean: t is flat ($x^{-0.004}$, certified 0.2400–0.2500 on all 18 windows) and the gain is flat ($x^{-0.039}$).

The sign reversal, the four dead candidates, and the live route. $B_{HH} \succeq tI$ is the original pencil inequality restricted to $\mathrm{span}\{t_k : k > 16\}$ — an identity, not an analogy: the Schur route relocates the question onto a subspace, it does not ease it. The find of the zone reverses a T152 sign: **on the bulk parity block the archimedean part is positive** ($\lambda_{\min}(B_{HH}^{\mathrm{arch}}) = 1.0362$–$1.4143 > t = 0.25$), after an m-free peeling of at most 8 modes on every window — the archimedean negativity lives entirely in the lowest parity modes, and on the bulk it is the *atoms* that are negative (-484.6–-14.95): the obstruction has moved from a negative smooth kernel to an arithmetic remainder. Four block candidates die: entrywise Gershgorin (-1929–-21.35 on the bulk), block Gershgorin (Feingold–Varga 1962) on a dyadic partition (-333.4–-7.68; the coupling is long-range), a mode-index Toeplitz-plus-Hankel fit (residual 0.70–0.95 of the Frobenius mass — no second-level Szegő/Widom argument exists), and a recursive Schur step, which certifies the same $t = 0.25$ at levels 0/1/2 — *scale-invariant*, so recursion alone can never terminate. The one route that produces positivity at all is a relative (Kato-type) bound in the archimedean metric: $1 + \lambda_{\min}(C) = 8.3 \times 10^{-4}$–$2.9 \times 10^{-2}$, certified positive on every window where all three additive candidates return a negative number — but it shrinks as $x^{-1.778}$: positivity per window, not a uniform floor. The whole of t sits in how far inside -1 the arch-whitened atom operator lies, and that distance is of the same order as μ_1^P itself.

The inverse-iteration find. No family fixed in advance carries the K_{bot} ceiling: the fixed-size band replacement grows exactly like the bulk ceiling it replaces (T_{band} at $x^{2.017}$ against T_{HH} at $x^{2.019}$), the parity sines and the Hann taper grow at $x^{2.47}$ and $x^{2.46}$, and the Fejér family degenerates (a sine is an eigenvector of every convolution). The atom overshoot is located exactly: the Toeplitz half of the atom response of the first parity sine is -1.99 (the atoms *help*) while the Hankel reflection half is $+9.45$ — a boundary-reflection effect, not a bulk-atom effect. But **two inverse-iteration steps** $(A^{-1}L_P)^2 t_k$ **give a flat** $K^F = 1.432$–2.369 ($x^{0.116}$) against the certified true $K_{\mathrm{bot}} = 1.102$–1.896, on a certificate of fixed size 8: the ceiling *is* carried by a fixed-size family — just not by one written down in advance. Nine map entries are named (v580–v588); *nothing is promoted* — the probe is exploration only, and the T149–T152 candidates stay pending.

After T153: Ψ is closed — two terms remain, and they are one object

(N1) **the block inequality** $B_{HH} \succeq tI$ on the bulk parity block — after T153 its sharp form is *one* number: the distance of the arch-whitened atom operator inside -1 (8.3×10^{-4}–2.9×10^{-2}, shrinking as $x^{-1.778}$), to be bounded against the m-free archimedean reserve 1.04–1.41 the bulk block now carries. *(N2)* **the Green/alignment estimate**: a structural description of $(A^{-1}L_P)^2 t_k$ for $k \le 8$ — it closes the ceiling (the flat $K^F = 1.43$–2.37 is already certified *by* that family) *and* recovers the collapse factor 3.25–8.40 that the uniform floor throws away: **both remaining terms lose to the same missing object** — where the bottom eigenvector of the section sits. The Ψ rebuild is off the list: the theorem-grade collapse needs no certificate beyond t. The honest state, in one sentence: after T153 the hoped-for reserve is known to have been a mismap — Ψ is pinned to within 1.204–1.300 by the first parity sine and closed as a term — the archimedean part is positive on the bulk block after an m-free peeling of at most 8 modes, and the two remaining terms are one and the same missing object, a Green estimate for the bottom of the section. T154 (`green_align_probe.py`) is running at exactly that estimate.

10.44 The ceiling closes at fixed size, and the obstruction becomes geometry (T154)

T154 (`green_align_probe.py`, contract `GREEN.ALIGN`, 29/29, 5.0 s on 12 prime-power windows $h = 50 \ldots 1077$, the Kato anatomy on 10 of them, verdict ALIGN-RESISTS, the gates *evaluated* rather than chosen) attacks the Green/alignment estimate both T153 terms lose to — and the ceiling half closes, exactly.

The ceiling, closed with a direction lesson. The part opens with a correction to its own commission, stated explicitly because it is easy to get backwards: the ceiling needs *no residual argument at all.* Ritz values are *upper* bounds for the eigenvalues of the *same index* (Courant–Fischer 1920 / Cauchy 1829 interlacing), so a Ritz quotient is a ceiling by a theorem; Temple 1928 / Kato 1949 is a *floor* device and is needed only there. The certificate: $\mathrm{span}\{t_1..t_8\} + A^{-1}L_P\,\mathrm{span}\{t_1..t_8\}$ — sixteen columns, *fixed* size, the whole certificate 8 completed Choleskys of matrices $\le 8 \times 8$ — carries $K^F = 1.1019$–1.9964, flat ($x^{0.094\pm0.058}$), and **agrees with the inertia-certified** K_{bot} **to a relative** 5.17×10^{-7} **on every window**: the sixteen-column certificate does not merely bound the true bottom ladder, it *attains* it, and the eight $\mathrm{LDL}^{\mathsf{T}}$ counts of size m per window are retired from the ceiling step [`verification/v551_ritz_ceiling_certificate.py`]. The parity sines alone overshoot as $K^F = 11.2$–2.1×10^4 at $x^{2.47}$ (T152's $O(m^2)$ reproduced), and the geometry says why the Green step repairs them: the true bottom eight of A lie inside the twice-iterated subspace to $\le 0.33°$, the Green columns have only 1.8–4.9% of their mass outside the sine block, and their pencil Rayleigh quotients sit at 0.287–0.681, just above the certified floor $t = 0.25$ — precisely what an inverse-iteration step should do (Kaniel 1966 / Paige 1971, quoted as the explanation, never used as a bound).

The floor half, refuted at fixed size and recovered at full size. The Temple/Kato residual argument — with the separation $\beta = t\,\mu_9^P$ that the containment of the parity sines makes free,

$\mu_9^P/\mu_1^P \to 81$ exact — fails by $\|R\|^2/(\beta\lambda_1) = 6.3\times10^3$–$5.6\times10^9$, growing as $x^{4.50}$: any sine-containing subspace has residual $O(\|A\|) = O(1)$ against a target $\lambda_1 = O(\mu_1^P) = O(m^{-2})$, for *every* fixed containment depth (the preregistered scan over $k_c = 8\ldots48$ closes 1 of 12 windows, and that one needs $k_c/m = 0.64$ — not fixed size in effect; the scalar Temple form is vacuous on 12/12). **The obstruction is named**: seven of the eight bottom directions of A agree with the bottom of L_P to 0.15–1.35° (median), *one* sits at 82.93–89.79° — and that single misalignment *is* the collapse price. One completed Cholesky of $A-\gamma I$ per window recovers it *in full*, 4.408–7.985 — certified per window, size m, not m-free, labelled as such. The missing fixed-size ingredient reduces to *one arithmetic-free number*: the L_P floor on the complement of the eight bottom Ritz directions — a question about the tridiagonal L_P and one eight-dimensional subspace, with no prime-power input anywhere — measured flat at 5.93–$8.25\,\mu_1^P$, and feeding the measured value into the fixed-size Temple step returns 4.408–7.617 of the price 4.408–7.986: worth 91–100%, conditionally.

The Kato anatomy, and the double refutation. The loss of the live R1 route is *not* the floor: $\lambda_{\min}(B_{HH}) = 0.266$–$0.415 > t = 0.25$, yet the Kato product delivers only a factor 1.97–121 less ($x^{1.635}$), because the two minima are attained in different directions, 24.9–89.8° apart — the minimiser that ruins the relative bound is a perfectly harmless direction for the block inequality itself. Its mass sits 96.5–100% on the eight modes *directly above* the peel cut (only 0.9–11.9% at the boundary), so T153's Hankel-reflection culprit is refuted *twice*: at the minimiser both halves hurt (Toeplitz $-100.3\ldots-0.248$, Hankel $-72.69\ldots-0.90$), and removing the reflection destroys positive definiteness altogether ($\lambda_1 = -7.49\ldots-2.64$) — the reflection half is load-bearing, not a nuisance term. Boundary peeling buys 1.08–1.38×, the norm route is dead by two to three orders, and the only repair that moves the number needs a peel of 0.40–0.69 of the window — m-proportional, for a scale-free reason: the minimiser always sits on the lowest eight modes of whatever block is left.

Two end-to-end numbers, never conflated. m-free in shape: *unchanged* at 1.01×10^{-2}–3.92×10^{-2} — the ceiling closure buys uniformity and cost (eight size-m factorisations retired), not size, because T153 already used the certified K_{bot} and the certificate attains exactly that number. Per window certified, with the collapse price recovered by one Cholesky: $\mathbf{4.45\times10^{-2}}$–$\mathbf{3.13\times10^{-1}}$ — inside the target band 3×10^{-2}–3×10^{-1}. The honest reading: the *size* target is met and the *uniformity* target is not, and those are different targets. The obligatory stress separates cleanly: on the T145 no-go reconstruction the floor survives ($t = 0.05$) while K_{bot} explodes at $x^{1.99}$ and the certificate explodes *with* it ($x^{2.29}$, overshooting the true constant by 1.12–2.19) — so the exact attainment is a property of the *real* section, not of the instrument — and the collapse price separates the families by orders of magnitude (1.8×10^3–2.1×10^5 against 4.4–8.0); the parity control is exact to 3.5×10^{-6} and the Courant–Fischer direction is violated on 0/26 positive-definite cases.

After T154: the ceiling is closed — two uniformity floors remain

(O1) **R1′, the m-free block floor**: an m-free lower bound for $\lambda_{\min}(B_{HH})$ on the bulk parity block, now *direction-aware* — every relative route is dead with a number (the Kato number $1+\lambda_{\min}(C) = 3.3\times10^{-3}$–$2.3\times10^{-1}$ shrinks as $x^{-1.69}$ against a requirement 0.19–0.30, and the loss is the misalignment, not the floor), and the anatomy says where to look: 96.5–100% of the offending direction sits on the eight modes immediately above the cut, wherever the cut is put. *(O2)* **R2′, the m-free bottom-mode floor**: an m-free lower bound for $\min_{v\perp\text{bottom}} v^\mathsf{T}L_Pv/v^\mathsf{T}v$ — arithmetic-free, only the tridiagonal L_P and one eight-dimensional subspace, measured flat at 5.93–$8.25\,\mu_1^P$ and worth 91–100% of the collapse price at fixed size. Both are now *uniformity* terms: the numbers are certified per window, and what is missing is the m-free argument. The verdict rule was applied honestly: ONE-TERM-MISSING was deliberately *not* taken — the two terms are of the same type but not the same statement (the pencil minimum demonstrably does not sit at the bottom mode, which is exactly what the collapse price measures), and merging them would be a silent upgrade. The honest state, in one sentence: after T154 the ceiling of the bottom ladder is closed at fixed size and closed exactly, the collapse price is a named geometric object recovered per window by one Cholesky, and what stands between the per-window band 4.45×10^{-2}–3.13×10^{-1} and an m-free statement is the uniformity of two floors. T155 (`bottom_floor_probe.py`) is running at exactly those two floors.

10.45 The complement floor becomes exact at fixed size, and the block floor excludes every symbol argument (T155)

T155 (`bottom_floor_probe.py`, contract `BOTTOM.FLOOR`, 27/27, 9.0 s on 16 prime-power windows $h = 142 \ldots 1293$, the block surface on 14 of them, verdict FLOORS-RESIST, the gates evaluated from the numbers) attacks the two uniformity floors T154 left — and reduces the first to its smallest form, exactly.

The exact 12×12 *reduction.* The single m-sized object left in the bottom-mode chain was the complement floor $L_c = \min_{v \perp Q_8} v^\top L_P v / v^\top v$. The new certificate is one identity plus one Rayleigh bound, a theorem for every m and every subspace W with orthonormal basis Q_W:

$$\min_{v \perp W,\ \|v\|=1} v^\top L_P\, v \ \geq\ \mu^P_{K+1} - \lambda_{\max}\big(M^{1/2}\,(I - GG^\top)\,M^{1/2}\big), \qquad M = \mathrm{diag}\big(\mu^P_{K+1} - \mu^P_k\big)_{k \leq K},$$

with $G = T_K\, Q_W$ and T_K the first K exact parity sines — the direction is checked at every window (the Cholesky-certified $\lambda_{\max}$ is *subtracted* from a lower bound, and the certificate never exceeds the exact value). It loses nothing: the ratio certificate/exact is **0.999999783–0.999999958** on 16 of 16 windows, and $K = 12$ suffices everywhere (preregistered ladder $8 \ldots 64$) — a 12×12 eigenvalue problem in the exact Kac–Murdock–Szegő numbers plus one 12×8 overlap matrix replaces the m-sized eigenproblem T154 had to solve. Both closed-form controls are hit: $W = \mathrm{span}\{t_1..t_8\}$ returns μ^P_9 to 1.05×10^{-12}, and $W = \mathrm{span}\{t_2..t_9\}$ — the configuration in which the certificate is *worthless* — returns μ^P_1 to 8.89×10^{-11}: the instrument reports its own worthlessness rather than something flattering. (This certificate is now load-bearing: it is promoted, together with T157's three instruments, as v552, Section 4.18 — [`verification/v552_angle_instruments.py`].)

The defect is localised, and the 83–90° *direction is demystified.* If Q_8 covered the eight lowest L_P modes, the floor would be $\mu^P_9 = 80.74\text{–}81.00\, \mu^P_1$ (m-explicit, KMS); measured it is 5.82–7.86, a factor 10.31–13.92 lost — and the loss is *not* spread over the eight modes: the largest uncovered fraction $1 - \|\mathrm{row}_k(G)\|^2$ sits at $k = 1$, the bottom mode itself, at **0.539–0.611** (flat, $x^{-0.063 \pm 0.016}$). The reason is visible in the same object: T154's misaligned direction (82.08–89.70° on this surface) carries only 2.8×10^{-5}–1.9×10^{-2} of its mass on modes 1–8, has L_P Rayleigh quotient $81.2\text{–}103.4\, \mu^P_1$ and peaks at mode 9–10 — the certificate spends one of its eight directions on modes 9–12 and pays for it at mode 1, which is exactly why it needs $K = 12$ and not $K = 8$. The payoff: feeding the fixed-size floor into the unchanged Temple step recovers 3.250–7.226 of the collapse price 3.250–8.471 — **78.8–100.0%** *at fixed size* (in full on 12/16 windows, at least ninety percent on 14/16; the two remaining windows are residual-limited, not floor-limited), with 32 Green columns, an 8×8 Ritz block, one 12×12 certificate and a Temple bisection — nothing in the chain has a size depending on m, where T154 needed one Cholesky of size m. Two repairs are refuted with their mechanism: the pencil-ceiling chain — a theorem in every step (projector-split identity, Davis–Kahan 1970 transfer $s = 5.2 \times 10^{-4}$–1.0×10^{-2}) — dies on its one new input $\kappa = \lambda_{\max}(B) = 3.1 \times 10^3$–$2.7 \times 10^6$ growing as $x^{2.82}$ (the two-sided pencil is maximally ill-conditioned exactly at the bottom the floor is about; peeling does not cure it), and widening the floor subspace to $r = 12/16/24$ raises the certified floor ×1.19–3.71 while the residual rises ×12.4–8514 — the Temple step returns a floor on 0 of 32 widened configurations: floor and residual are bought from the same account, and at $r = 8$ the account is balanced.

The block floor, and the impossibility of a symbol argument. $\lambda_{\min}(B_{HH}) = 0.2430\text{–}0.4249$ is certified positive on every window (the inequality holds against $t = 0.25$ on 13/14; on the rest the Schur ladder backs off to a smaller certified t — reported, not smoothed). First, a correction *of target*: T154's anatomy (the minimiser sits 95.3–99.6% on the eight modes above the cut, confirmed here) belongs to the minimiser of the arch-whitened operator C — the minimiser of B_{HH} *itself* puts only 1.5×10^{-5}–0.69 of its mass there: a decomposition aimed at the C-minimiser is aimed away from $\lambda_{\min}(B_{HH})$. The direction-aware 2×2 decomposition dies on the coupling (both diagonal blocks healthy at 0.3518–0.5245 and 0.2431–0.4273, but $\|B_{wd}\| = 5.81\text{–}389.41$ against the allowed 0.292–0.456 — too large by 16.8–908, growing $x^{1.94}$: the atoms couple the mode index long-range); the local atom norm on the eight-mode window exceeds the local arch reserve by

1.50–5.12 on every window; and the deeper cut does not terminate (the deep block's own floor is 2.1×10^{-3}–$6.0 \times 10^{-2} < t$, with a Kato factor of the same order as at the original cut — moving the cut up by 8 modes moves the problem up by 8 modes). The one positive result is a mechanism: **the arch reserve *is* the symbol infimum** — $\min_{\theta \geq \theta_{k_b+1}} f^{\mathrm{arch}}(\theta)/(4\sin^2(\theta/2))$ equals $\lambda_{\min}(B_{HH}^{\mathrm{arch}})$ to 1.000022–1.000521 on 14/14 windows (band 1.0004–1.4049, a factor 4.0–5.6 above the target) — a theorem *candidate* (Szegő 1915 / Widom 1958 / KMS 1953): proving it means proving an inequality about one function of one variable. And the same instrument delivers the strongest negative statement of the series: **the full symbol infimum over the same bulk range is** $-714.2\ldots-7.6$**, negative on every window** (the atom symbol alone reaches -1148.5), while the section floor is positive — any Szegő/Widom-type argument returns the symbol infimum and therefore a negative number here, so *no symbol argument can ever produce R1′*: whatever carries it must be a genuinely finite-section mechanism (the Fejér averaging of the atom spikes against the section), and the diagonal already shows the section is doing the work ($\min_k (B_{HH})_{kk} = 0.3840$–$0.7444$, a factor 1.33–1.89 above the true minimum).

The honest miss, and the stress. The end-to-end fraction reaches 3.28×10^{-2}–2.83×10^{-1} with the label *certified at fixed size* (T154 left 1.01×10^{-2}–3.92×10^{-2} m-free in shape / 4.45×10^{-2}–3.13×10^{-1} certified per window with a size-m Cholesky). The declared bar was 4×10^{-2} and the bottom of the band is 3.28×10^{-2} — **missed by a factor** 1.22, reported rather than repaired: the pairing of two band minima from different and wider surfaces is deliberately pessimistic, and the like-for-like number — fixed size against size m, same window, same Temple step — is the share 78.8–100%, which is what this probe moved. Fixed size is *not* m-free: the certificate does not grow, but its value is measured on 16 windows, not bounded by an argument. Balance: 5 theorem, 3 certified-fixed-size, 2 certified-per-window at size m, 1 measured, 1 theorem candidate — T154's balance had zero fixed-size entries below the ceiling row. The obligatory stress breaks on two axes: on the T145 no-go family the ceiling explodes at $x^{2.010\pm0.003}$ (flat on the real family), and the new instrument separates too — the mode-1 uncovered fraction reads 0.999–1.000 (real: 0.539–0.611), the misalignment is exactly 90.00°, and the Temple recovery is 0/8 against 16/16 real. The parity sines are exact L_P eigenvectors to 4.85×10^{-14}, and the Dirichlet negative control hits its own rank-one prediction $2/\sqrt{N}$ to 4.1×10^{-7}.

After T155: both floors resist — two smaller open terms remain

(O1) **R2″, the m-free overlap defect**: an m-free *upper* bound for $\lambda_{\max}(M^{1/2}(I - GG^{\mathsf{T}})M^{1/2})$ at $K = 12$ — equivalently, a bound on how much of t_1 the eight bottom Ritz directions of A may miss (measured 0.539–0.611, flat $x^{-0.063\pm0.016}$). The term has shrunk to *one* 12×8 matrix: everything else in the bottom-mode chain is a theorem or a fixed-size certificate, and the one m-free-shaped route this probe could construct — the two-sided pencil — is refuted by a factor 10^{-3} for a structural reason. *(O2)* **R1″, the m-free atom part on the bulk**: excluded are relative routes (T154), direction-aware two-block splits (coupling too large by 16.8–908), local atom norms (short by 1.50–5.12), deeper cuts (scale-free), and *every symbol argument* (the full symbol infimum is negative while the section floor is positive); the one surviving shape is a finite-section Fejér cancellation, and the arch half is a theorem candidate (it *is* its symbol infimum to 5×10^{-4}). The verdict rule was evaluated from the numbers: the R2′ gate reads 10^{-3} against a required 0.5 and the R1′ gate reads -5.5 against $t = 0.25$, hence FLOORS-RESIST; merging the terms would be false — R2′ is a question about L_P and a subspace with the arithmetic compressed into one 12×8 matrix, R1′ a question about prime-power sums against a finite section, now provably outside the reach of the method that settles the first. The honest state, in one sentence: after T155 the complement floor — the one m-sized object of the bottom-mode chain — is reproduced exactly by a 12×12 certificate and the collapse price is recovered 78.8–100% at fixed size, while the block floor has acquired an obstruction that closes off a whole family of methods; this probe bought size and structure, not uniformity. T156 (`twelve_by_eight_probe.py`) is running at exactly those two objects.

10.46 Both terms become single scalars: the t_1 loss is one angle, the block positivity is alignment (T156)

T156 (`twelve_by_eight_probe.py`, contract `TWELVE.EIGHT`, 37/37, 6.1 s on 16 prime-power windows $h = 142 \ldots 1293$, the block surface on 12 windows $h = 50 \ldots 1077$, verdict TERMS-RESIST, the gates evaluated from the numbers: TERMS-CLOSE needs zero measured rungs, ONE-TERM-MISSING exactly one — there are three) attacks the two rest objects T155 left, and reduces both to single scalars.

The exact $F(P, r)$ reduction. The subspace S_2 contains t_1 *and* $y_1 := A^{-1}L_P\, t_1$, and the bottom Ritz direction of A mixes them — that mixing is where t_1 is lost. On $\mathrm{span}\{t_1, y_1\}$ the mixing closes *exactly*, because $L_P t_1 = \mu_1^P t_1$ turns the coupling into an identity with no A in it:

$$t_1^\mathsf{T} A\, y_1 \;=\; t_1^\mathsf{T} L_P\, t_1 \;=\; \mu_1^P,$$

so the t_1-loss of the two-dimensional Ritz pair is a *closed* function $F(P, r)$ of exactly two dimensionless scalars, both Rayleigh quantities of A at the single vector t_1: the Kantorovich product $P = (t_1^\mathsf{T} A t_1)(t_1^\mathsf{T} A^{-1} t_1) \geq 1$ and the inverse moment ratio $r = \|A^{-1}t_1\|^2/(t_1^\mathsf{T} A^{-1} t_1)^2 \geq 1$ (the reduction agrees with the four-scalar form to 2.2×10^{-16} and with the true 2×2 Ritz pair to 4.4×10^{-16} — an identity, not a fit). Measured: $P = 5.6 \times 10^2$–1.0×10^6 (enormous — A and L_P differ by orders of magnitude at t_1, so F is essentially $F(\infty, r)$) and $r = 2.7158$–3.4089, flat ($x^{-0.047\pm0.020}$). The box route over the admissible (P, r) region is *vacuous*, despite a new m-free bound: the exact spectral-defect identity yields $P - 1 \geq L^2 s^2 (r - 1)$ (with $L = \lambda_1(A)/\mu_1^P$ and $s = \mu_1^P\, t_1^\mathsf{T} A^{-1} t_1$; it holds on 16/16 windows with slack 2917–4.87×10^6) and cuts the region — but $\sup F$ over what remains is 1.0000 where the chain needs at most 0.632, attained at $P \to 1$ with $r - 1$ of the same order: a genuine configuration the pencil does not exclude. The best route is a two-line theorem,

$$r \;\leq\; \frac{1}{Ls} \;\leq\; \frac{1}{p_1}, \qquad Ls = \sum_j p_j\, \frac{\lambda_1}{\lambda_j} \in [p_1, 1],$$

and its ceiling $1/(Ls) = 2.930$–3.967 is *tight* to a factor 1.03–1.16: **R2″ is one m-free lower bound on the overlap** $p_1 = \cos^2 \angle(t_1, e_1(A)) =$ **0.2010–0.3282** (55.05–63.36°, flat, $x^{0.054\pm0.029}$). The loss at the cruder $1/p_1$ ceiling is 0.6718–0.7990 against the model's 0.6318–0.7067 against the truth d_1(8 Ritz directions) $= 0.5392$–0.6316 — the chain survives the cruder of the two ceilings. The model *dominates* the truth on 16/16 windows (ratio 0.833–0.917) — measured, no monotonicity theorem, and the stress below is exactly about that. The interlacing route is refuted in both of its forms: at $K = 12 > \dim W = 8$ the matrix $I - GG^\mathsf{T}$ carries the eigenvalue 1 *exactly* (a rank hole — the $\sigma_{\min}$ form returns μ_1^P and nothing more, identically, for every m), and at $K = 8$ it is not vacuous but empty: $\sigma_{\min}(T_8 Q_W) = 1.7 \times 10^{-2}$–$1.4 \times 10^{-1}$ yields a floor of 1.023–2.579 μ_1^P against the 5.78–7.83 the *same* overlap matrix gives through the T155 certificate. T155's anatomy is reproduced (object ratio 0.9533–0.9654, $d_1 = 0.5392$–0.6316, the defect direction peaking on modes 9–10).

The J2a correction — an honest negative. The one-variable statement T155 expected is *false*: the infimum of $f^{\mathrm{arch}}(\theta)/(4\sin^2(\theta/2))$ over the bulk range is 0.8226–1.3973 ($x^{0.171\pm0.009}$), **below 1 on 3 of 12 windows** — T155's quoted 1.000022–1.000521 was the upper surface of the arch-reserve-equals-symbol-infimum identity (reproduced here to 3.6×10^{-5}–1.8×10^{-3}), not the floor of a ≥ 1 statement. What survives, and what the chain actually needs, is the *weaker* one-variable statement $\inf \geq t = 0.25$: a factor 3.29–5.59 above target, non-decreasing in h, grid-stable to 1.4×10^{-5} under fourfold refinement, with the minimum at $\theta/\pi = 0.990$–1.000 — the alternating lag sum, which is where a proof would have to work. Status: a certified-per-window grid fact about a function of one variable, not a theorem.

The mechanism of the block floor: alignment, not mass. The lag decomposition $v^\mathsf{T} A v = \sum_l c_l\, a_l(v) - \sum_k c_{M-1-k}\, b_k(v)$ is exact to 6.9×10^{-15} on every window (Fejér 1915 — the weights below are the section's true weights, not a model). The Fejér damping is real but worth *one order of magnitude*: on the minimiser the realised atom form is 3.2×10^{-2}–1.9×10^{-1} of the naive ℓ^1

budget ($x^{-0.501\pm0.024}$), a factor 5–31, where the split needs 3–1981 *and growing*; and the additive split is *divergent*: $\|B_{HH}^{\text{atom}}\| = 2.2$–$2673.4$ grows like $h^{2.31}$ ($x^{2.314\pm0.082}$) and exceeds the arch reserve by 2.7–1.98×10^3, closing 0 of 12 — no bound of the shape $\lambda_{\min}(B_{HH}^{\text{arch}}) - \|B_{HH}^{\text{atom}}\|$ can close R1″ for *any* damping of the atom mass, because what is being subtracted is not bounded at all. The mechanism, identified: **on the block minimiser an arch part of** 1.4685–91.7130 **and an atom part of** $-91.2694\ldots-1.1151$ **cancel to** $\lambda_{\min}(B_{HH}) = 0.2661$–$0.4436$**; the minimiser sees only** 6.16×10^{-3}–5.03×10^{-1} **of the atom operator (**$x^{-1.044\pm0.109}$**), at** 52.7–$90.0°$ **from the atom-extremal vector — the positivity is an *alignment* fact, not a size fact**, and diagonal dominance fails as well ($\min_k(B_{HH})_{kk} = 0.3997$–$0.8186$; the off-diagonal atom coupling eats 38–63% of the diagonal).

The balance, the stress, and the unchanged end to end. The end-to-end fraction is 3.28×10^{-2}–2.83×10^{-1}, *unchanged*: J1 and J2 are statements about what is missing, and neither adds a certified inequality to the chain. The recovery, paired like for like ($K = 12$ fixed by contract, a wider surface, the pairing declared before the number): 49.5–100.0%, median 99.8%, at or above 75% on 12/16 windows. The balance: 9 theorem rungs (3 new: the $F(P, r)$ reduction, the chain $r \le 1/(Ls) \le 1/p_1$, the lag decomposition), 6 certified, **3 measured** — the zero-measured target is *not* met, two of the three are the open terms themselves, and the third (the model domination) is a new debt created by the reduction, counted rather than hidden. The obligatory stress breaks on *three* axes, two of them new and both J1's own: the ceiling explodes at $x^{2.047\pm0.009}$ (T154 confirmed); p_1 — the one scalar J1 reduces to — *collapses* to 7.4×10^{-15}–6.6×10^{-10} on the no-go family ($x^{-4.818\pm0.080}$) against 0.2010–0.3282 real, so the one number J1 asks for is exactly the number that separates the families; and the r-ceiling tightness separates too (12.5–24.6 against 1.03–1.16). The floor survives ($t = 0.05$), so a floors-only probe would have seen nothing. **And the stress kills J1's one measured step**: the domination of the eight-dimensional defect by the 2×2 model holds on 16/16 real windows and *fails* on 8/8 no-go sizes (model 0.977–0.982 against truth 0.9990–0.9996, a narrow undershoot of 1.7×10^{-2}–2.2×10^{-2} — which is the point): it is not a theorem and cannot become one by more measuring on the real family. Controls exact: parity 4.8×10^{-14}; μ_9^P to 2.3×10^{-13}; μ_1^P to 2.0×10^{-11} in the *worthless* configuration $W = \operatorname{span}\{t_2..t_{13}\}$ (the certificate reports its own worthlessness); the Dirichlet negative control hits its own $2/\sqrt{N}$ prediction to 7.4×10^{-5}.

After T156: both terms resist — two angles remain

(O1) **R2″, one m-free lower bound on p_1**: the squared overlap of the bottom parity sine with the bottom eigenvector of the section, measured 0.2010–0.3282 (55.1–63.4°), flat. Everything above it is now a theorem: the 2×2 reduction to $F(P, r)$ is an identity, $r \le 1/(Ls) \le 1/p_1$ is two lines, and F is explicit — a bound $p_1 \ge p^*$ would give the m-free loss $F(P, 1/p^*)$ directly (0.672–0.799 at the measured p_1). Separate second debt: the step from the two-dimensional model to the eight Ritz directions, which holds on 16/16 real windows and fails on 8/8 no-go sizes — a good separation instrument, not a theorem. *(O2)* **R1″, one m-free alignment statement**: a lower bound for the arch Rayleigh quotient *on the subspace where the atom form is large*, equivalently an m-free angle between the two extremal vectors — *not* a mass bound of any kind (refuted with numbers: symbol arguments, additive splits, ℓ^1 budgets, diagonal dominance); the arch half is ready for it (a one-variable grid infimum 0.8226–1.3973, non-decreasing, factor 3.29–5.59 over t). The verdict was evaluated from the numbers: 2 open terms and 3 measured rungs, hence TERMS-RESIST. T155's 12×12 certificate is strengthened as an *instrument* (exactness controls to 2.0×10^{-11}, self-reported worthlessness, the no-go breaking on three axes) and weakened as a *uniformity claim* (its value is governed by the single scalar p_1, and the load-bearing inequality of the reduction step fails on the no-go family) — the honest recommendation: promote it as a fixed-size certified instrument with its per-window numbers, and promote nothing about uniformity. The honest state, in one sentence: after T156 both remaining terms are single scalars — one angle each — whose m-freeness is not a matter of measuring more windows but of proving one inequality each. T157 (`angle_floor_probe.py`) is running at exactly those two angles.

10.47 Neither angle falls, both change shape: the p_1 floor becomes a theorem times one number, the arch half is certified (T157)

T157 (`angle_floor_probe.py`, contract `ANGLE.FLOOR`, 32/32, 3.9 s on 16 prime-power windows $h = 142\ldots1293$, the alignment surface on 12 windows $h = 50\ldots1077$, verdict ANGLES-RESIST, the gates evaluated from the numbers: p_1 m-free FALSE, p_1 non-trivial TRUE, arch half certified TRUE, alignment m-free FALSE — terms = 2) attacks the two angles T156 left, and neither falls — but both change shape. The instrument cores below are promoted as [`verification/v552_angle_instruments.py`].

The best p_1 route is the resolvent, and it splits into a theorem times one number. Write $\gamma = Te_1(A)$ for the parity coordinates of the bottom eigenvector and split the modes at $k_b = 16$. The tail is a *theorem*, one line from the certified pencil floor $A \succeq tL_P$ and the certified ladder ceiling $\lambda_1 \le S\mu_1^P$ alone:

$$\|\gamma_H\|^2 \;\le\; \frac{\lambda_1}{t\,\mu_{17}^P} \;\le\; \frac{S/t}{\rho_{17}} \;=\; 0.0165\text{–}0.0293, \qquad \rho_{17} = \frac{\mu_{17}^P}{\mu_1^P},$$

so $e_1(A)$ lives 97.1–98.4% inside the first sixteen parity sines, *m-freely in shape* — T146's measured "98 per cent in the sine block" now has a one-line proof, and the lesson is the ceiling: substituting the pencil ceiling κ (3.1×10^3–2.7×10^6, growing $x^{2.821\pm0.038}$) for the flat ladder $S = 1.1676$–2.1178 ($x^{0.010\pm0.092}$) would make the bound vacuous (43.5–38632). What remains measured is *one* number: $\hat{g}_1^2 = 0.2010$–0.3282 (flat, $x^{0.054\pm0.029}$), the first coordinate of the bottom vector of a 16×16 Schur null problem (solved to a relative residual 8.4×10^{-15}), and the resulting floor $p_1 \ge \hat{g}_1^2(1 - \text{tail}) = 0.1968$–$0.3228$ is 97.9% of the measured p_1. Three other routes are refuted with numbers: the classical Rayleigh angle bound $p_1 \ge (\lambda_2 - \mathrm{RQ})/(\lambda_2 - \lambda_1)$ is *empty already in its measured form* (16/16 negative — $\mathrm{RQ}(t_1)$ exceeds λ_2 by 818–519458, and the bottom is nearly degenerate, $\lambda_2/\lambda_1 = 1.73$–$5.87$: the shape fails, no certification can rescue it); its block version answers the T156 question — the chain *does* admit the block, and the block mass is spectacular, $p_{\text{blk}}(2) = 0.9369$–$0.9993$ (t_1 lives almost entirely in the bottom *pair*) — but the block Rayleigh bound stays empty at every J of the ladder; and the angle-free Cauchy–Schwarz route $Ls \ge L/\hat{a}$ is a theorem that loses exactly the Kantorovich product $P = 5.6 \times 10^2$–1.0×10^6, beating the p_1 floor on 0 of 16 windows.

The structural gem: the whole first term is one diagonal entry. There is an identity nobody had used: $s = \mu_1^P\, t_1^{\mathsf{T}} A^{-1} t_1 = (B^{-1})_{11}$, hence $1/s$ is carried by the $(1,1)$ entry of the *same* fixed-size 16×16 Schur complement $S_L = B_{LL} - B_{LH}B_{HH}^{-1}B_{HL}$ that the T152..T157 chain already forms for its floor certificate — and $1/s = 2.3359$–6.2049 is *flat* ($x^{-0.088\pm0.008}$), an $O(1)$ number, against $\hat{a} = 2.8 \times 10^3$–$2.4 \times 10^6$ for the same entry *without* the Schur subtraction. R2$''$ is thereby an m-free *upper* bound on one diagonal entry of a fixed-size matrix instead of a lower bound on an angle between two m-sized vectors. The obvious certificate fails for a named reason: the inversion-free Cauchy–Schwarz ceiling on $b^{\mathsf{T}} B_{HH}^{-1} b$ is off by 5.4×10^2–5.2×10^5 — the cancellation in that one bilinear form is nearly complete, and Cauchy–Schwarz cannot see it.

The arch half is now certified, and the mechanism is localised. The adaptive Lipschitz ceiling is *executed*, not asserted: interval bisection with the local Lipschitz constant (the global θ_{lo} constant would need up to 7.8×10^6 uniform grid points, $x^{3.365\pm0.147}$) certifies $\inf f^{\text{arch}}/(4\sin^2(\theta/2)) \ge t$ on 12/12 windows — certified floor 0.2500–0.5353 in 255–3139 symbol evaluations, cost growing only like $h^{0.850\pm0.097}$: the arch half carries a certificate for the first time (T156's 1.4×10^{-5} grid stability was evidence; this is a certificate). And the extremals are located exactly: the *atom* extremal sits at $\theta/\pi = 0.0158$–0.3762, the *arch* extremal at 0.9901–0.9995 — opposite ends of the band (separation 0.61–0.98), with the bulk minimiser going with the atom. The strategy turn this forces: the arch Rayleigh quotient is *smallest* at $\theta = \pi$ and *grows* as θ falls, so a proof of R1$''$ must use the *θ-growth* of $f^{\text{arch}}/(4\sin^2(\theta/2))$, not its infimum at π — the one-variable infimum is the wrong number to feed it. The shape $\lambda_{\min}(B_{HH}) \ge \inf_{\text{arch}} \cos^2\varphi - \|\text{atom}\|$ is refuted as a shape (on the minimiser the arch form is 1.47–91.71, nowhere near its infimum); what replaces it is the domination form $B_{HH}^{\text{arch}} - tI \succeq (-B_{HH}^{\text{atom}})_+$, which holds per window with quotient 1.0003–1.0907 — but the margin at the largest window is 7.3×10^{-4} with a shrinking trend ($x^{-0.018\pm0.015}$): a certificate, not a uniform statement.

The balance, the stress, and the price of the substitution. Substituting the resolvent bound into the first row of G costs a factor 1.00–1.62 (direction-clean and conservative: the substituted complement floor 4.37–6.20 μ_1^P never exceeds the measured 5.84–7.88). The balance: 9 theorem rungs, 2 certified-uniform (the ladder S; the executed arch ceiling), 2 certified-per-window (the pencil floor t; the R1$''$ domination), **3 measured** ($\hat{g}_1^2$, the eleven other rows of G, the T156 model domination) — the target of one is missed, but all three measured steps are now *fixed-size in their statement*, which is the change T157 bought. The obligatory stress breaks on *five* axes: p_1 collapses ($x^{-4.818\pm0.080}$, T156 reproduced); the ladder S explodes ($x^{2.289\pm0.079}$) against flat on the real family; the confinement dies ($(S/t)/\rho_{17} = 7.9$–1.6×10^3, $x^{2.253\pm0.120}$ — the no-go bottom eigenvector is *not* confined, and the m-free bound correctly reports vacuum); the resolvent floor is *identically zero* there (the route reports its own vacuum); and $(S_L)_{11}$ explodes ($x^{2.268\pm0.041}$) against flat. The floor survives ($t = 0.05$), so a floors-only probe would have seen nothing. Controls: parity 4.8×10^{-14}; μ_9^P to 2.3×10^{-13} and μ_1^P to 2.0×10^{-11} in the worthless configuration; $s = (B^{-1})_{11}$ to 1.9×10^{-13} with $s = 1$ exactly on $A = L_P$; the Dirichlet negative control hits its own $2/\sqrt{N}$ prediction to 7.4×10^{-5}.

After T157: both angles resist — three named rests remain

(O1) **an m-free upper bound on one number**: $(S_L)_{11} = 1/s = 2.3359$–$6.2049$, the $(1,1)$ entry of the 16×16 Schur complement of B — equivalently an m-free lower bound on $\hat{g}_1^2 = 0.2010$–0.3282. Cauchy–Schwarz on $b^\mathsf{T} B_{HH}^{-1} b$ is refuted (off by 5.4×10^2–5.2×10^5), so what is needed is a bound that *sees* the cancellation in that one bilinear form. *(O2)* **an m-free R1$''$ domination with a margin that does not shrink**: the certified quotient is 1.0003–1.0907 with trend $x^{-0.018\pm0.015}$ and a margin of 7.3×10^{-4} at $h = 970$; the pointer is the mechanism finding — use the θ-growth of $f^{\text{arch}}/(4\sin^2(\theta/2))$ as θ falls, not its infimum at π. *(O3)* the T156 debt T157 did not touch: the two-dimensional model dominating the eight-dimensional defect — measured on the real family, refuted on the no-go family, and the only remaining measured step whose *statement* is not fixed size. What stands a priori on the measurement surface, precisely: the KMS spectrum and the exactness of the parity sines, the ratio floors ρ_k, the identity $F(P, r)$ with $r \le 1/(Ls)$, the identity $1/s = (S_L)_{11}$, the sine-block confinement, the classical inequalities the chain is assembled from, and the arch one-variable infimum with an *executed* Lipschitz certificate per window — and nothing else: neither p_1 nor the alignment margin is available before the window is built. The four instrument candidates of T155/T157 (the 12×12 complement certificate, the sine-block confinement, the Schur-entry identity, the adaptive Lipschitz ceiling) are promoted as `v552` — per-instance instruments, no angle closed, both open terms typed open. T158 (`schur_entry_probe.py`) is running at exactly the cancellation-seeing bound.

10.48 The cancellation-seeing bound exists, and it is a theorem: the entry as a Dirichlet maximum, the growth pointer refuted (T158)

T158 (`schur_entry_probe.py`, contract `SCHUR.ENTRY`, 36/36, 8.2 s on 28 prime-power windows $h = 254 \ldots 1393$, the band surface on 21 windows $h = 254 \ldots 1031$, verdict ENTRY-RESISTS, the gates evaluated from the numbers: entry ceiling m-free FALSE — the cancellation depth grows $x^{2.135\pm0.021}$ where m-free needs non-growing; entry ceiling sharp-and-uniform TRUE — tightness ≤ 1.2738, flat; domination margin non-shrinking FALSE — trend $x^{-1.649\pm0.643}$ — terms = 2) attacks exactly the cancellation-seeing bound T157 demanded, and it exists — as a theorem.

The entry as a Dirichlet maximum, and the ladder of positive terms. The object that makes trial spaces legitimate in the right direction is the dual (Thomson) form of the entry (Maz'ya 1985; Miclo 1999 — s is a *conductance* of mode 1 into the bulk, and every admissible flow is a floor):

$$s \;=\; (B^{-1})_{11} \;=\; \max_x \,(2x_1 - x^\mathsf{T} B x),$$

so *every* trial vector is a lower bound on s and hence an upper bound on $1/s$ — the entry R2$''$ must ceiling. This is exactly the variational structure Cauchy–Schwarz lacks: CS evaluates

$b^\mathsf{T} B_{HH}^{-1} b = \max_z(2b^\mathsf{T} z - z^\mathsf{T} B_{HH} z)$ at the *single* direction $z \parallel b$, and it missed by 3.13×10^3–5.18×10^5 (reproduced with its direction intact on 28/28: what fails is sharpness, not sign — the cancellation is spread over the low modes, and one direction cannot see it). The Cholesky ladder $g_K = \sum_{j \le K} y_j^2$ with $y = L_K^{-1} e_1$ (the nested-Schur identity) writes s as a sum of *strictly positive* terms with monotone partial sums on 28/28 windows, and $1/g_1 = \hat{a}$ is exactly T157's route: the ladder starts where T157 ended. At $K = 16$ — the same low block the T152..T157 chain already forms — the certified ceiling is $1/g_{16} = 2.9670$–7.9664 against the true $1/s = 2.3359$–6.3868, *tight to* 1.1323–1.2738, ceiling trend $x^{0.059 \pm 0.017}$ (at $K = 8$ already inside a factor 1.767). Directions checked before use: (M1) $1/s \le (S_L)_{11}$ with slack 8095–1.013×10^5 on 28/28 (Schur 1917 / Haynsworth 1968 — T157's entry is a ceiling on $1/s$, an inequality and not an identity), and no trial ever overshot s.

The entry is two-dimensional. The Green-column trial space is not a bound at all — it is the identity: $\mathrm{span}\{t_1, A^{-1} L_P t_1\}$ attains s exactly ($g/s = 1$ to 10^{-9} on 28/28), for a theorem reason — $L_P t_1 = \mu_1^P t_1$ (KMS 1953), so $A^{-1} L_P t_1$ *is* the Dirichlet maximiser. This settles that the entry is a *two-dimensional* quantity the moment one Green column is granted; that column costs an m-sized solve, and the fixed-size sine truncation pays exactly the factor 1.1323–1.2738 for not needing it. The object the chain actually wants: $L\, g_{16} = 0.2143$–0.2724 against the true $Ls = 0.2521$–0.3413, and with the certified pencil floor t in place of L, $t\, g_{16} = 0.0314$–0.0826 — the Dirichlet ladder lands in the band of T157's *measured* $p_1 = 0.2010$–0.3282, without an angle.

The growth pointer, refuted — an honest negative. The growth T157 pointed at is real and sub-quadratic: $f^{\mathrm{arch}}/(4\sin^2(\theta/2)) \sim \theta^{-q}$ with $q = 1.259$–1.438 ($q < 2$ says f^{arch} itself decays like $\theta^{0.56\text{–}0.74}$ — a partial cancellation of numerator and denominator, not a clean pole), and the binding vector of the domination pencil sits exactly where T157 said: 0.6818–0.9988 of its mass in the *lowest* dyadic band (centroid $\theta/\pi = 0.0191$–0.0845 — the localisation half of the pointer is confirmed). What is refuted is that the growth can pay for it: the atom negative mass grows like $\theta^{-1.546\ldots-1.744}$, *faster* than the certified arch floor's $\theta^{-1.203\ldots-1.458}$ — the low-θ limit belongs to the atom, and the growth is a liability, not a resource. In the lowest band the arch floor covers only 0.4161–0.4507 of the atom mass, band-local domination *fails in every band* on 21/21 windows, and the off-band coupling (Frobenius 1.612–27.97, the conservative direction) exceeds the available margin by 660–7.71×10^5: what carries the inequality is precisely the structure a dyadic argument throws away. T157's adaptive Lipschitz ceiling, re-run band by band, certifies 7 of 116 (band, window) pairs — exactly the large-θ bands where the growth is not the resource. The margin is 3.57×10^{-5}–2.86×10^{-3} with trend $x^{-1.649 \pm 0.643}$, shrinking: R1$''$ stays a per-window certificate, and the verdict gate reads it numerically.

The debt relabelled, the balance, the stress. The containment $\mathrm{span}\{t_1, A^{-1} L_P t_1\} \subset S_{16}$ is exact (5.84×10^{-16} on every window) — nothing in the T156 debt is about the spaces — but it cannot be cashed: Ritz defects are not monotone in the trial space (smallest principal angle to the eight bottom Ritz directions 65.04–73.04°). With the Dirichlet floor in place of the measured angle, $F(P, 1/(L\, g_{16})) = 0.7276$–$0.7857$ dominates the true defect $d_1 = 0.5392$–0.6241 with margin 1.1969–1.3568 on 28/28 ($x^{0.052 \pm 0.007}$) — both sides now fixed-size certified numbers, so the debt moves MEASURED → CERT-UNIF (not a theorem, and the probe does not pretend otherwise). End to end, priced like for like: the substitution is direction-clean and conservative (the substituted complement floor 4.3904–$6.5357\, \mu_1^P$ never exceeds the measured 5.7763–8.6051), the Temple recovery goes from 0.4951–1.0 to 0.2872–1.0 of $\lambda_1(A)$ — a cost of 1.000–1.724, above the 4×10^{-2} bar. The balance: **6 theorem** / 3 certified-uniform / 4 certified-per-window / **1 measured** (T157 stood at 5/2/3/3): p_1 is *retired* — the ladder replaces the angle outright — the model domination is relabelled, and the one remaining measured row ($\lambda_1(A)$ itself, needed only by the sharp L variant) vanishes entirely if $t\, g_{16}$ is used: the measured count falls from three to one. The obligatory stress breaks on *five* axes: p_1 collapses ($x^{-4.818 \pm 0.080}$), the true entry $1/s$ explodes to 1185–2.547×10^5, the fixed-size ceiling $1/g_{16}$ explodes *with* it ($x^{2.267 \pm 0.042}$ — the ceiling tracks the collapse instead of hiding it; the tightness stays at 1.001, so what breaks is the size of the thing bounded, not the sharpness), the floor $L\, g_{16}$ collapses ($x^{-0.275 \pm 0.047}$), and the tail bound reaches 1.56×10^3 — against flat on the real family. Controls: parity 4.8×10^{-14}; μ_1^P to 2.0×10^{-11} in the worthless configuration; new — for $A = L_P$ the ladder returns $s = 1$ to 1.9×10^{-13} *and* every partial sum $g_K = 1$ to 4.0×10^{-15}, exactly constant in K; the Dirichlet negative control hits its own $2/\sqrt{N}$ prediction to 7.4×10^{-5}.

After T158: the entry ceiling is a theorem plus one Cholesky — two rests remain

(O1) **an m-free upper bound on one quadratic form**: $x^\mathsf{T} B_{LL} x = 2.9670$–$7.9664$ for a nearly universal fixed sixteen-vector (pairwise alignment 0.8599–0.9976 across the windows, normalised to $x_1 = 1$) — nothing else is needed for R2″. The obstruction, named: the archimedean half carries $-9.323 \times 10^5 \ldots -2.394 \times 10^4$ and the atom half $+2.395 \times 10^4 \ldots +9.323 \times 10^5$, both of one sign across the surface, so the bound must be relatively accurate to 3.5×10^{-6}–2.3×10^{-4} — a requirement growing as $x^{2.135\pm0.021} \approx h^2$, which is the $1/\mu_1^P = N^2/(4\pi^2)$ normalisation of the pencil block: the explicit formula's own cancellation, localised in sixteen coordinates, not an artefact of the trial vector. *(O2)* **an m-free lower bound on $\lambda_{\min}(B_{HH})$ that is not band-diagonal**: band-diagonal minorants are refuted (the band-local inequality is false in every band), and pricing the off-band coupling by its norm is refuted too (660–7.7×10^5 times the available margin); what is left is the *sign structure* of the off-band arch entries. *(O3, optional)* a certified floor on $\lambda_1(A)/\mu_1^P$ better than t (the gap between $L\,g_{16}$ and $t\,g_{16}$ is a factor 3.25–8.47 — the only measured row left). What stands a priori on the measurement surface, precisely: for every m and every K, $1/s \le 1/g_K$ with g_K a sum of K strictly positive terms read off one Cholesky of the leading $K \times K$ block of B — the inequality, its direction, its monotonicity in K and the exactness of the two-dimensional span are a priori; the *size* of the terms is not. The five T158 candidates (the dual form plus the positive ladder; the two-dimensionality; the $1/g_{16}$ sharpness; the negative result on the growth pointer; the relabelling) were bundled with T159, and the theorem cores are now ledger-carried as v553. T159 (`exact_form_probe.py`) executed the cancellation algebraically — next subsection.

10.49 The cancellation executed algebraically: seven identities, the gauge identity, and the form is the weight (T159)

T159 (`exact_form_probe.py`, contract `EXACT.FORM`, 41/41, 2.9 s on 23 of 24 prime-power zones $h = 142 \ldots 1293$ (70–24574 atoms; the sign surface on 18 windows $h = 142 \ldots 963$), verdict FORM-RESISTS, the gates evaluated from the numbers: R2″ m-free FALSE — every pricing of the transformed sum grows, best rung $h^{+3.604}$, every closed vector grows, frozen $h^{+3.510}$; R1″ m-free FALSE — the sign structure sits entirely on the archimedean side — terms = 2) executes T158's rest O1 symbolically. The algebra is delivered; the bound is not: *the cancellation is localised, and it is the weight.*

The symbolic sixteen-form: seven machine-checked identities. All seven hold to 10^{-12} against the absolute scale $\sum_d |c_d\, w_d|$ — the honest bar, because relative to the cancelled total the same identities hold only to 1.0×10^{-12}–1.2×10^{-8}: the cancellation eats 4.0–8.1 decimal digits in double precision, and that measured loss *is* the quantitative reason no m-free bound can be read off numerically. (P1) $x^\mathsf{T} B_{LL} x = y^\mathsf{T} A y$ with $y = \sum_k (x_k/\sqrt{\mu_k^P})\, t_k$ — the pencil is gone (2.8×10^{-17}–1.4×10^{-15}). (P2) the *exact lag sum* $x^\mathsf{T} B_{LL} x = \sum_d c_d\, w_d$ with $w_d = T_d - H_d$ (T the autocorrelation, H the convolution of y): one explicit arithmetic sum with arithmetic (c) and geometry (w) separated. (P3) w_d in *closed form*: a 256-term Dirichlet-kernel expression (Dirichlet 1829, product-to-sum) reproduces the assembled weights to 1.42×10^{-15}–2.85×10^{-15} of $\sup|w|$ at every lag of every window — w_d is a fixed-size trigonometric function of (d, h), no m-dependence beyond $N = 2h + 1$. (P4) the *gauge identity*, the content of the module:

$$\sum_d w_d \;=\; 0 \qquad \text{exactly,}$$

because $A_{rs} = c_{|r-s|} - c_{M-1-r-s}$ annihilates constant lag vectors identically (0.0–9.4×10^{-14} of $\sup|w|$): *the form is blind to the lag mass of c and sees only its variation* — and the lag mass is exactly where the archimedean and atom halves are h^2-sized and cancel. (P5) the two closed scalars $w_0 = \sum_k x_k^2/\mu_k^P$ (the $N^2/4\pi^2$ factor itself, 2.655×10^5 at $h = 1119$, relative error $\le 4 \times 10^{-16}$) and $2w_0 - w_1 = \|x\|^2$ exactly — because L_P is the odd Toeplitz $(2, -1, 0, \ldots)$ and the t_k are its exact eigenvectors (KMS 1953): the huge and the $O(1)$ scale sit in the *same two weights*. (P6) the p-fold Abel identity exact at every rung $p = 1..5$ (2.6×10^{-20}–3.3×10^{-16}). (P7) the Toeplitz-minus-Hankel signature $A_{r,s} - A_{r+1,s+1} = f(r+s)$ (1.1×10^{-16}–1.5×10^{-16}) — the premise of P4. Reproduction: $x^\mathsf{T} B_{LL} x = 1.7648$–$4.9008$ flat ($h^{-0.130}$; same object as T158's 2.967–7.966 on a different surface), halves $-9.605 \times 10^5 \ldots - 9.828 \times 10^3$ against $+9.831 \times 10^3 \ldots + 9.605 \times 10^5$, cancellation depth 1.86×10^{-6}–3.56×10^{-4} falling $h^{-2.234}$; the sixteenth Cholesky rung stays sharp, $(1/g_{16})/(1/s) = 1.0694$–$1.2509$.

m-free not reached: four routes closed with exponents. The core negative first: **the h^2 does not telescope away** — |arch| and |atom| both grow $h^{+2.104}$, i.e. like $w_0 = 1/\mu_1^P$ itself. The h^2 is the common *weight* of both halves, not an archimedean artefact algebra could clear, and it cancels only in the sum — the gauge identity says why: each half alone carries the lag mass against an h^2-sized weight. The four routes: (R1) **T158's fixed sixteen-vector, refuted** — frozen at one window the form grows $h^{+3.510}$ (3.025–6.997×10^5), with the reason closed: $\mathrm{cond}(B_{LL}) = 1.223 \times 10^5$–$1.386 \times 10^8 \sim h^{+2.887}$, so a direction error ε costs $\varepsilon^2 \lambda_{\max} \sim \varepsilon^2 h^2$; a closed direction would have to be accurate to $\kappa^{-1/2} = 8.5 \times 10^{-5}$–$2.9 \times 10^{-3} \sim h^{-3/2}$, while the true optimiser drifts by $1 - \cos = 2.52 \times 10^{-2}$–$1.47 \times 10^{-1}$ between neighbouring zones — a refutation, not a difficulty. (R2) one sine refuted: $B_{11} = 1/g_1 = 1.9 \times 10^4$–$2.1 \times 10^7 \sim h^{+3.066}$, the price of collapsing sixteen rungs to one is 4224–1.3×10^7 — the sixteen-dimensional minimum is irreducible, the cancellation lies *across* the sixteen. (R3) the preregistered ansatz family (4 forms $\times$ 10 exponents, no fitting): the best member grows $h^{+3.001}$ — no closed power direction is flat. (R4) $\ell^1 \times \sup$ pricing of the Abel sum at every rung refuted: best rung $p = 1$ with bound 1.54×10^7–$3.95 \times 10^{10} \sim h^{+3.604}$, missing the target 1.765–4.901 by 5.63×10^6–2.21×10^{10}, higher rungs worse ($p = 5$: $h^{+7.718}$) — the transformation removes the lag mass (ℓ^1 masses fall) but the surviving weight tails grow exactly as fast ($\sup |W^1| = 3.45 \times 10^5$–$2.00 \times 10^8 \sim h^{+2.906}$, and $W_1^1 = -w_0$ identically, so $\sup |W^1| \geq w_0 \sim h^2$ is unavoidable: the transformation *moves* the cancellation from the lag mass into the weight tail, it does not remove it). Two m-free-shaped new pieces survive: $\|\Delta c^{\mathrm{arch}}\|_1 = 3.6487$–$5.3788$ **flat** ($h^{+0.165}$) — the T115 kernel has uniformly bounded variation ≤ 6.00 in the zone — and the atom budget is T150-Chebyshev-correct, $\|\Delta c^{\mathrm{atom}}\|_1 = 37.0$–$191.7 \sim h^{+0.727}$ inside $\sum \mu_j \leq 4B\sqrt{X}$ at the *atom cutoff* $X = e^{2\alpha}$ (a pedantic correction: not the window size $N = 2h+1$ — a different integer), occupancy 0.8767–0.9602.

The sign law: the archimedean block is a Z-matrix, exactly. (S1) holds raw: B_{HH}^{arch} **is a symmetric Z-matrix** — 100% of the off-band archimedean entries nonpositive, without similarity, on 18/18 windows; (S2) the checkerboard sign pattern is refuted (0.4890–0.4936 mass fraction = a coin flip). Collatz–Wielandt bites on the archimedean half: the *closed sign-based floor* $\min_k (B_{HH}^{\mathrm{arch}} \mathbf{1})_k = 0.8194$–$1.1647$, flat ($h^{+0.176}$), comfortably above the target $t = 0.25$ — row sums of a Z-matrix, an m-free-shaped object (Gantmacher–Krein 1950; the Perron root of the off-band part sits a factor 2.2740–2.7262 below its row-sum norm: the signs pay). And it is not enough: the atom block obeys *no* sign law (0.3067–0.4891 nonpositive = noise), its off-band mass exceeds the archimedean floor by 32.44–4555 growing $h^{+2.330}$, and on the full block the M-matrix criterion is vacuous: best power test vector $\min_k (B_{HH} v)_k / v_k = -9.324 \ldots - 0.6086$ against the exact $\lambda_{\min}(B_{HH}) = 0.0742$–$0.1974$. The verdict without hedging: the archimedean sign structure is real, exact and usable, and it does not suffice, because the object it would have to dominate has no sign structure.

Assembly, stress, and the numerical horizon. End to end the chain still certifies $\lambda_1(A) \geq t\, \mu_1^P$ per window with $t = 0.0705$–0.1583, recovery 0.0686–0.2164 of the exact $\lambda_1(A)$ (a deeper zone surface than T158's 0.287–1.0 — a surface difference, not a regression; and the decisive reading: M1/M2 did *not* move it — the new theorems are structural, not quantitative). The T145 no-go breaks 5/5 (no Z-law, signature defect 7.79×10^{-2}, M-matrix criterion vacuous despite positive definiteness, the localisation referee m/H_m diverges $h^{+0.821}$, diagonal constancy violated); three directions re-verified before use (Thomson maximum never overshot, Schur complement $\geq 1/s$, Ritz ceiling); Dirichlet and parity controls at 8×10^{-15} with the $2/\sqrt{N}$ bound *saturated* to 0.999708–0.999995. The balance moves 6/3/4/1 $\to$ **13 theorem** / 6 certified-uniform / 4 certified-per-window

/ **1 measured**: seven new theorems P1–P7 and three new certified-uniform rows (U1 the Z-law, U2 the archimedean floor ≥ 0.81, U3 the bounded variation ≤ 6), zero new per-window rows, zero new measured rows. And a **numerical horizon, declared as its own measurement**: one window dropped because $\mathrm{cond}(B_{LL}) > 10^{12}$, the first at $h = 1292$ — beyond it the sixteen-block is singular in double precision and the per-window optimiser does not numerically exist. That is the same obstruction the cancellation depth reports, seen from the other side. The theorem cores of T158 + T159 are ledger-carried as `v553` ([`verification/v553_exact_form_identities.py`]): the dual form with the positive ladder, the two-dimensional span, the gauge identity with the signature, the closed scalars with the lag sum, and the Z-law with the Collatz–Wielandt floor — per instance, with the pairing bound and m-freeness explicitly typed open.

After T159: one fully explicit pairing inequality remains

(O1) R2″ is now **one fully explicit inequality**: the form equals $\sum_{d\geq 1}(\Delta c)_d\, W^1_d$ with $\|\Delta c^{\mathrm{arch}}\|_1 \leq 6$ (U3, m-free-shaped), $\|\Delta c^{\mathrm{atom}}\|_1 = O(h^{0.73})$ inside the Chebyshev budget, and $\sup|W^1| = 53.88\text{–}504.72 \times w_0$. What is missing is a bound on the *one pairing* that uses the *correlation* between Δc and W^1 instead of their sizes: $\ell^1 \times \sup$ is off by $h^{3.7}$, and the cancellation to be recovered is exactly h^2. *(O2)* R1″ needs a third device for the atom off-band block that is neither a norm (refuted in T158 by 660–7.7×10^5) nor a sign (refuted here, purity 0.31–0.49). *(O3)* two cheap proofs would upgrade rows: U1 from the Hankel half of the closed T115 kernel, U3 as a total-variation bound on the same integral. Permanently off the list: the fixed sixteen-vector, the single sine, power ansätze, Abel-ℓ^1 at every rung, the checkerboard, the full-block M-matrix. T160 (`pairing_probe.py`, contract `PAIRING`) executed exactly that attack — next subsection.

10.50 The pairing attacked through correlation: the sampling identity — the last term is a prime exponential sum (T160)

T160 (`pairing_probe.py`, contract `PAIRING`, 46/46, 2.8 s on 19 of 20 prime-power zones $h = 199\ldots1256$ (187–19593 atoms per window; the R1″ sub-surface on 8 windows $h = 199\ldots534$; one window dropped at the declared T159 horizon $\mathrm{cond}(B_{LL}) > 10^{12}$, first at $h = 1292$ — not fitted through), verdict PAIRING-RESISTS, the gates evaluated from the numbers: arch half inside the bar TRUE (slack 1.95×10^{-8} of the $O(1)$ total); arch half non-growing at a fixed witness FALSE (0 of 120 fixed pairs — but floor-limited, see below); atom half inside the bar FALSE (best device 4.78×10^6); third R1″ device positive FALSE (floor -7.93) — terms = 2) attacks the one pairing $\sum_{d\geq 1}(\Delta c)_d\, W^1_d$ through its correlation structure. Two terms remain open — the same *count* as T159, and that is the honest headline — but both are now *located*, and one of them is no longer an algebra problem.

No oscillation: the sign-block count is closed, and the block device is refuted by its own numbers. W^1 has $J = 3$ sign blocks on every window, flat ($h^{+0.000}$), with a *closed* bound $J \leq 4(2\cdot 16)+2 = 130$ from the trigonometric zero count (w_d is a piecewise trigonometric polynomial of degree ≤ 32; Féjer 1915): W^1 is one smooth *three-lobe profile* with quasi-period 0.332–0.333 of the window, not an oscillation. The prefix ratio $\sup|\Psi|/\sup|W^1| = 83.5\text{–}569.8$ grows as $h^{+1.022}$ — the quantitative reason iterated Abel loses: each further summation by parts trades a factor h of tail size for a factor h of difference smallness, neutral in the exponent and strictly worse in the constant. Consequently the Leibniz block device the contract asked for is refuted *quantitatively*: block-Hölder overshoots as $h^{+4.115}$ and block-Abel as $h^{+4.131}$ — 5.2–16.5 times *worse* than the already-refuted $\ell^1 \times \sup$ pricing at $h^{+3.595}$, because with three blocks the block masses are of prefix size. The head peel is refuted too: no proper split clears the bar on any window ($K^* = M$ on 19/19), and the best improper one ($K/M = 0.97$) still overshoots the true pairing $Q = 1.7527\text{–}6.0895$ by 60.3–4.80×10^3, growing as $h^{+2.428}$ — the h^2 cancellation itself, essentially undiminished.

The moment laws, and the second numerical horizon. What carries on the smooth half are three *closed sign-definite moment laws*, new theorems, machine-checked: $m_0 = \sum_d w_d = 0$ (the T159 gauge identity), $m_1 = \sum_d d\, w_d = -[S_0^2 + 2\sum_j P_j^2] \leq 0$ (the $\max(r,s)$ identity: minus a sum of squares; 7.1×10^{-16}), and $m_2 = \sum_d d^2 w_d = -[2S_1 - (M-1)S_0]^2 \leq 0$ (a *perfect square*; 3.0×10^{-16}), with closure of m_p in the moments of y for every even p (verified at $p = 2, 4$): the pairing cannot see the lag mass, and it sees a linear or quadratic trend only with a sign known in advance. Against a *fixed* polynomial witness (fraction 0.2 of the lags, degree 48, the same pair on every window — a device, not a fit) the smooth archimedean half of the pairing is *reproduced* to 0–2.44×10^{-15} relative, worst slack 1.95×10^{-8} of the $O(1)$ total on all 19 windows. And a **second numerical horizon is declared** rather than hidden: the measured residual is 4.12–428 times the double-precision floor $\varepsilon \sup |c^{\mathrm{arch}}| \, \|w\|_1$, so the surface reports the arithmetic floor of the measurement and *not* the witness error — the arch half is honestly reported as *neither established nor refuted* on any double-precision surface; only a symbolic Chebyshev bound on the T115 kernel could decide it (rest item R-A below). Atom sparsity is refuted in one number: the support is *not* sparse — it fills 0.3226–0.6377 of the lag range, because the prime-power gaps $\log(1 + 1/n)$ fall below the grid spacing D as soon as $n > 1/D$; even the support-localised ℓ^1 (exact position *and* exact local value of every atom, absolute values taken) misses by 4.78×10^6–2.64×10^9 growing as $h^{+3.505}$, while the required relative precision on the atom half is the cancellation depth itself, $|\mathrm{total}|/|\mathrm{atom}| = 2.19 \times 10^{-6}$–$1.12 \times 10^{-4} \sim h^{-2}$.

The core of T160: the sampling identity. The atom half of the pairing is a *weighted sampling* of the closed weight vector at the prime-power positions — a theorem, machine-checked to 1.0×10^{-16}–2.9×10^{-15} of the absolute scale on every window:

$$\sum_d c_d^{\mathrm{atom}} \, w_d \;=\; -\frac{1}{2} \sum_{n \le X} \frac{2\Lambda(n)}{\sqrt{n}} \, \widehat{w}(\log n) \quad \text{(plus one reflected term below the grid),}$$

with $\widehat{w}$ the linear interpolant of the closed weight vector — and by T159's Dirichlet-kernel form, $\widehat{w}(u)$ is a *fixed* trigonometric polynomial with frequencies $\pi(k \pm l)/\alpha + O(1/h)$, $k, l \le 16$. So **the atom half of R2″ is, identically, a finite combination of sums $\sum_{n \le X} \Lambda(n)\, n^{-1/2} \cos(t \log n)$ at thirty-two explicit frequencies** t — needed to relative depth 2.19×10^{-6}–$1.12 \times 10^{-4} = h^{-2}$, and measured to cancel only to 0.0000–0.3659 of the trivial mass bound. The consequence, stated with the fence in place: *the h^2 cancellation is the intrinsic arithmetic hardness of the problem*, not an artefact of the T115 assembly — the geometric half is now evaluated down to the arithmetic floor, so nothing geometric is left to trade against the prime sum. $\sum_n \Lambda(n)\, n^{-1/2-it}$ is the *shape* of a partial sum of a Dirichlet series on the line $\mathrm{Re}\, s = \frac{1}{2}$; that is reported as a *diagnosis* of where the h^2 lives, never as an input: the sums are assembled from the finite von-Mangoldt table and nothing else, no zero of any L-function is read, generated or approximated, no equivalence is claimed, and Weil 1952 / Bombieri 2000 remains an address.

The third R1″ device: $\rho > 1$, flat — and its inequality form is refuted. On the sub-surface (8 windows, $\lambda_{\min}(B_{HH}) = 0.0748$–$0.2023$; the Z-matrix and its Collatz–Wielandt floor 0.8811–1.0988 reproduced exactly): the rank/support route is refuted — the effective negative rank of the atom block is 0.455–0.634 of the block dimension, so no fixed-size corner can contain it; the corner device is refuted — it prices the coupling by $\|M_{12}\|_2 = 1.554$–71.447 and is T158's refuted norm route in disguise. The surviving fact: $\rho = \min_{0 \ne v \in V_-} (v^{\mathsf{T}} B_{HH}^{\mathrm{arch}} v)/(-v^{\mathsf{T}} B_{HH}^{\mathrm{atom}} v) = \mathbf{1.0036}$–$\mathbf{1.0140} > 1$ on *every* window, flat ($h^{-0.011}$) — the archimedean block dominates the atom block on every destructive direction, an *angle* statement no norm can see, and the first R1″ fact that is neither a norm nor a sign nor a rank (MEASURED: a generalised eigenvalue of two size-h blocks). Its inequality form $B_{HH} \succeq B_{HH}^{\mathrm{arch}} - (1/\rho)\, P_- B_{HH}^{\mathrm{arch}} P_-$ is a theorem given ρ, with no arithmetic on the right — and as a device it is refuted: floor $-7.9261\ldots - 1.0919 < 0$ on 8/8, because the margin $(1 - 1/\rho)\lambda_{\min} = 3.76 \times 10^{-3}$–$1.24 \times 10^{-2}$ loses to the archimedean cross-coupling 2.808–18.15 between V_- and its complement by a factor 10^4. Any future device must use ρ *together with* the geometry of V_- inside B_{HH}^{arch} — pricing that geometry by a norm is exactly what fails.

Two cheap theorems, the reassembly, the stress. **U3 is upgraded cert-unif → theorem**: c^{arch} is nonpositive and monotone increasing on $d \ge 1$ (both hypotheses machine-checked; fraction 1.0000 of 25458 pairs), so the total variation equals the total rise and $\|\Delta c^{\mathrm{arch}}\|_1 \le |c_0^{\mathrm{arch}}| + 2 \sup_{d \ge 1} |c_d^{\mathrm{arch}}| < 6$ — measured 3.9101–5.4153 against the closed bound 3.9105–5.4153: *four digits*, the constant closed and window-independent. U1 is *partially* upgraded, and the honest statement is the interesting one: $K_{kl}(0) = 0$ and $\sum_d K_{kl}(d) = 0$ exactly, so $R_{kl}(0) = R_{kl}(1) = 0$ and the Abel step has no boundary term — the Z-law of B^{arch}_{HH} is *equivalent* to one *prime-free* trigonometric inequality $\sum_{d\geq 2}(\Delta c^{\mathrm{arch}})_d\, R_{kl}(d) \leq 0$ (the weight nonnegative by U3); it holds on every sampled pair, but the pointwise strengthening $R_{kl}(d) \leq 0$ is refuted (it holds at only 0.4848 of the lags): the monotone weight is essential, and the statement is *not* yet a theorem and not reported as one. End to end, the number is the verdict: the composite chain (arch half by the fixed witness, atom half by the support-localised ℓ^1) recovers 3.79×10^{-10}–2.10×10^{-7} of the exact per-window value — *if the atom half were evaluated as exactly as the arch half now is, the same chain would recover* 1.0000: the whole deficit of R2$''$ is carried by one term, of size 4.77×10^{6}–2.64×10^{9}. T159's certified per-window band 0.287–1.0 is unchanged — T160 moves the obstruction, it does not sharpen the number. The identities roll up worst-case at 2.95×10^{-12}; the T145 no-go breaks on **6/6** axes including the two new ones (A5 the gauge/moment structure — the no-go form does not annihilate constants, so the Abel step has no analogue on it; A6 the complementarity — ρ is undefined on it); Dirichlet and parity eigenpair controls at 8.2×10^{-15} with the $2/\sqrt{N}$ bound saturated. The balance moves 13/6/4/1 → **14 theorem** (6 new) / 4 certified-uniform / 3 certified-per-window / 3 measured, with 8 refuted device families; the candidates T160-P1…P7 (the moment laws; the sampling identity; the block count with its refutation; the U3 upgrade; the U1 partial upgrade; ρ; the second horizon) are all PENDING — no promotion here, and `v553` is untouched.

After T160: two classical rests, one arithmetic object, and the circularity question

(R-A) one classical estimate: a symbolic Chebyshev bound for $s \mapsto A(s, D)$ on $[0.2\cdot 2\alpha,\ 2\alpha]$ (the Bernstein ellipse of the T115 integral, whose only singularity is the $1/s$ head at the origin) — it decides the *arch* half of R2$''$ m-freely, which no double-precision surface can (T160-P7). *(R-B)* one trigonometric inequality, prime-free: $\sum_{d\geq 2}(\Delta c^{\mathrm{arch}})_d\, R_{kl}(d) \leq 0$ — it turns the Z-law (T159-U1) and with it the Collatz–Wielandt floor into a theorem. *(R-C)* the arithmetic term, and it is not an exercise: a bound on $\sum_{n\leq X} \Lambda(n)\, n^{-1/2} \cos(t \log n)$ at the thirty-two explicit frequencies $t = \pi(k \pm l)/\alpha$, to relative depth h^{-2} — by the sampling identity (T160-P2) this *is* the atom half of R2$''$, and nothing geometric is left to trade against it. *(R-D)* a fifth device for R1$''$: ρ together with the geometry of V_-, without pricing that geometry by a norm. *(R-E)* optional: closed odd-p moment functionals. T161 (`classical_closure_probe.py`, contract `CLASSICAL.CLOSURE`) executed exactly that programme — the two classical rests R-A/R-B plus the *circularity triage* for R-C — next subsection.

10.51 The circularity triage: the hardness sits in the split, not in the primes (T161)

T161 (`classical_closure_probe.py`, contract `CLASSICAL.CLOSURE`, 35/35, 1.2 s on 19 log-spaced prime-power zones of 75 admissible, $h = 50 \ldots 1445$ — a 28× lever arm — with $\alpha = 2.3979$–6.4938 and atom cut-off $X = e^{2\alpha} = n^2_{\mathrm{zone}}$ exactly $= 121$–4.37×10^5; $\psi(x)/x \leq 1.038830$ *verified* at every jump point up to $n = 400000$, the unconditional input the triage prices; verdict CLOSURE-RESISTS, triage verdict NEEDS-BEYOND-RH-STRENGTH) attacks R-A and R-B head-on and prices R-C against the classical ladder.

R-A: the analytic half closes m-free, and the head needs no peeling. Two theorems anchor it: the T115 arch kernel is *exactly* $A(s, D) = D\, \widehat{A}(s, D)$ with $\widehat{A}$ the triangle smear of $g(w) = e^{-w/2}/(1-e^{-2w})$ (4.2×10^{-16}–2.0×10^{-13} against the assembly on 19/19), and the analyticity width is set by the $1/s$ head alone — the poles of g sit exactly at $w = ik\pi$, and on $[0.4\alpha, 2\alpha]$ the first complex pole lands outside the Bernstein ellipse by margin 1.29–3.29 on every window. The Bernstein parameter is therefore *closed and scale-free*, $\rho^* = (3 + \sqrt{5})/2 = 2.618034$ (it depends only on the interval ratio 5; on the surface $\rho(D) = 2.5593$–2.6160), with the ellipse constant $M = 3.06$–5.93 uniform in h and the measured coefficients obeying the envelope at 0.055–$0.074 \leq 1$ — tight to 1.1 orders. The new structure T161 contributes: the smeared pole integrates in *closed form*, so for every lag $d \geq 1$

$$c^{\mathrm{arch}}_d \;=\; \Psi_d + D\, \widehat{G}(dD, D), \qquad \Psi_d \;=\; -\tfrac{1}{2}\big[(d+1)\log(d+1) - 2d \log d + (d-1)\log(d-1)\big],$$

with Ψ closed, D-free and m-free (the $\log D$ cancels identically; agreement 1.7×10^{-16}–4.8×10^{-16} at *every* lag of every window) and $\widehat{G}$ the smear of the regularised $g - 1/(2w)$ — no head peel, no fitted cut; on the full interval $[0, 2\alpha]$ the closed rate $\rho_{\text{reg}} = 2.05$–3.38 is *not* scale-free ($\to 1 + \sqrt{\pi/\alpha}$). The "fixed degree" hope is refuted with a number: both routes give a degree *schedule* $K(h) = O(\log h)$ ($K = 12$–19 peeled, 9–25 head-free; $dK/d\log h \approx 2.0$ and 4.5), because $\|w\|_1$ grows as $h^{+2.85}$ (fit) against $D \sim h^{-1}$. The residual of R-A, named exactly: the *log-moment* $\sum_d \Psi_d\, w_d = 2.15$–$3.14$ of the arch half $= -\frac{1}{2}\sum_d (d\log d)(\Delta^2 w)_d$ — prime-free, classical (a Clausen/Lerch-type object), and *not* in the polynomial moment ladder.

R-B: refuted in all four readings — and the fraction bound survives. The contract's inequality $\sum_{d\geq 2}(\Delta c^{\text{arch}})_d\, R_{kl}(d) \leq 0$ is tested in all four of its readings before any proof is attempted, and all four fail: per pair directly on 3168 of 4320 pairs (worst $+0.92$ of $\|\Delta c^{\text{arch}}\|_1$), per pair in the Abel-tail reading on 1229, the *aggregate* by *sign* — the arch half is negative on every window while its off-diagonal block is positive, exactly the harmful direction for the needed upper bound — and the Abel route's sufficient condition on 3801 of 4320 (worst $+763$). The reason is structural: R_{kl} is a difference of Dirichlet kernels and changes sign in d, and the coefficients $a_k a_l$ change sign too — a one-sign statement was never the right object. The by-products are positive: $(\Delta c^{\text{arch}})_d \geq 4.5 \times 10^{-8} > 0$ and decreasing, with a total-positivity reason from the Ψ split (Ψ is minus the triangle smear of the completely monotone $1/w$). What survives is weaker and usable: $|\text{off-diagonal}|/|Q^{\text{arch}}| = 0.1035$–$0.1713$, flat in h (exponent -0.116, fit) over a factor 24 — a certified *fraction* bound with the constant $\frac{1}{4}$, costing the chain a factor $1 + \frac{1}{4}$ in the constant and nothing in the structure; its m-free version is a 16×16 Gram positivity statement (Fejér 1915, Schur 1917), entirely prime-free.

The triage — and the answer to the circularity question. A new theorem first: the thirty-two frequencies satisfy $t\,(2\alpha) = 2\pi j$ *exactly* for $j = 1 \ldots 32$ — they are precisely the *Fourier harmonics of the log-window* $[0, 2\alpha]$, so the atom half of the pairing is the vector of the first 32 Fourier coefficients of the measure $\sum_n (2\Lambda(n)/\sqrt{n})\,\delta_{\log n}$. The measured 0.00–0.37 cancellation of T160 is then *identified*: at the harmonics $\theta = t\log X = 2\pi j$ collapses the partial-summation main term to the closed, parameter-free prediction $(\sqrt{X} - 1)/(\frac{1}{4} + t^2)$, which matches $S(t)$ to 0.014–0.232 of the trivial mass, improving as $X^{-0.39}$ (fit) — the $1/(\frac{1}{4} + t^2)$ is the Mellin factor of $x^{-1/2}$, *not* an arithmetic saving; the genuinely arithmetic part is the residual against $d\psi(x) - dx$. The δ arithmetic, compactly (the demand is $h^{-2} = X^{-\delta}$ with $\delta = \log h/\alpha$ exactly, since $\log X = 2\alpha$):

strength	**δ it buys**
trivial / Chebyshev 1852	0
zero-free region (de la Vallée Poussin)	$c/\sqrt{\log X} \to 0$ (no power saving)
van der Corput 1921 / Vinogradov 1937	inapplicable ($t = 0.5$–42 is $O(1)$; $j \leq 32$ turns)
harmonic-frequency geometry ($\sin\theta = 0$)	0.29–0.92 (measured suppression 7.0–19.8)
RH strength	$\frac{1}{2} - o(1)$
needed (arch/atom split)	**1.1482–1.8809**
needed (arch+smooth / fluctuation re-split)	**0.9839–1.3767**

The sharpest form of the same fact: the required *absolute* precision is 0.012–0.31 of the *last term* of the sum — below the boundary term $|\psi(X) - X|\, X^{-1/2}$ that every partial-summation bound carries, so no strengthening of the $\psi(x) - x$ input (zero-free region, RH, or beyond) lowers the floor of the method to it. And the re-split proves the demand is *split-dependent*: subtracting the smooth prime term $-\int e^{u/2}\widehat{w}(u)\,du$ cancels the arch half to 0.50–0.76 (the explicit-formula pairing, confirmed), and the demand moves from $\delta = 1.148$–1.881 to $\delta_{\text{eff}} = 0.984$–$1.377$ — it *moves* with the split, but stays above $\frac{1}{2}$ on 18/18.

The honest reading, stated prominently and soberly: the chain is not circular. It does not secretly need RH — it needs *more* than RH-strength input would supply, and the demand is not even reachable by the partial-summation framework at any input strength. The measured cancellation at the thirty-two frequencies is fully the prime-number-theorem main term, with no arithmetic surplus; the frequency geometry is genuinely special (worth $\delta = 0.29$–0.92, already inside the measured numbers) and does not close the gap. So the h^2 is located in the *split*, not in the primes: T161 refutes two splits with numbers, and finding a third — one with $\delta_{\text{eff}} < \frac{1}{2}$

— is the whole remaining R2″ task. The controls are re-verified rather than inherited (Dirichlet kernel identity 1.3×10^{-15} including the degenerate branch, KMS eigenpairs 7.1×10^{-15}, Chebyshev reconstruction 6.1×10^{-15}, odd_toeplitz$(c^L) = L_P$ exact); the balance moves by +6 theorem / + 6 certified-uniform / + 1 certified-per-window / + 6 measured / + 3 refuted to **20**/10/4/9/3. The closed cores of T160+T161 — the sampling identity, the moment laws with the TV theorem, the harmonic-frequency theorem with the PNT prediction, the head split with ρ^*, and the fraction bound with the sign refutation as its wired negative control — are load-bearing as `v554` [`verification/v554_sampling_harmonics_identities.py`] (Section 4.20); the δ triage itself is a comparison of strengths, a documentation item, and is promoted nowhere.

After T161: one moment, one matrix inequality, one split, one device

(R-A′) one closed prime-free moment: the log-moment $\sum_d \Psi_d\, w_d = -\frac{1}{2}\sum_d (d\log d)(\Delta^2 w)_d$, 2.15–3.14 of the arch half and outside the polynomial moment ladder (Clausen/Lerch address) — everything else in R-A is done, and a *fixed* witness degree is provably not available, only the schedule $K(h) = O(\log h)$. *(R-B′)* one matrix inequality, prime-free: the m-free version of the fraction bound $|\text{off-diagonal}| \le \frac{1}{4}|Q^{\text{arch}}|$ — a positivity/norm statement for the 16×16 Gram form $(\Delta c^{\text{arch}}, R^1)_{kl}$ (Fejér 1915, Schur 1917), certified on every window today. *(R-C′)* one *split*: not an exponential-sum bound — the demand sits below the boundary term of every partial summation — but a split of the pairing with $\delta_{\text{eff}} < \frac{1}{2}$; the arch/atom split gives 1.148–1.881 and the arch+smooth/fluctuation split 0.984–1.377, both above $\frac{1}{2}$, and finding a third is the whole remaining R2″ problem. *(R-D)* a fifth device for R1″: $\rho = 1.0036$–$1.0140 > 1$, flat, stays the surviving structure fact. T162 (`third_split_probe.py`, contract `THIRD.SPLIT`) executed exactly that search — next subsection.

10.52 The third split: the Mellin ladder saturates, and the Fejér split goes sub-RH at a price (T162)

T162 (`third_split_probe.py`, contract `THIRD.SPLIT`, 30/30, 5.2 s on the T161 surface rebuilt *independently* from the declared recipe: 19 log-spaced prime-power zones of 75 admissible, $h = 50\ldots1445$, of which 18 windows sit inside the declared condition horizon $\text{cond}(B_{LL}) \le 10^{12}$, $h = 50\ldots1218$; the pairing identity re-checked to 9.98×10^{-18}–3.01×10^{-14} of the large scale and the gauge identity to 2.5×10^{-16}; $\psi(x)/x \le 1.038830$ verified at every jump point up to $n = 400000$, i.e. $\kappa = 0.038821$ — the *one* unconditional arithmetic input every number below is priced against; verdict DELTA-REDUCED) searches for the third split, and the search returns both the split and its limit.

The currency, declared before any candidate. Two numbers are kept strictly apart from here on, because T161 used only the first: $\delta_{\text{val}} = \log(|R|_{\text{meas}}/|Q|)/\log X$ is the measured residual *value* — a floor — and $\delta_{\text{bnd}} = \log(E_{\text{triv}}\, C_w/|Q|)/\log X$ is what a *proof* must beat, with E_{triv} the Chebyshev-strength (unconditional) size of the arithmetic factor. RH strength is $\frac{1}{2}$ in both currencies, as a comparison of strengths and never as a claim; always $\delta_{\text{bnd}} \ge \delta_{\text{val}}$. The taxonomy, in both currencies (smooth part closed on every row):

split	δ_{val} **(floor)**	δ_{bnd} **(demand)**
C0 arch / atom	1.1482–1.8809	— (no factorisation)
C1 arch+smooth / fluctuation	0.9839–1.3767	— (no cell-level input)
C2 = C1 + Mellin ladder at K^*	0.6247–0.9307	— (same)
C5 = C2 + one Abel step	0.6247–0.9307	1.3965–1.7867
C3 spectral (ground pair, ≤ 16 modes)	—	0.4602–1.3145 (price $\sim h^{+2.93}$)
C4 Fejér taper (`fejer_2`)	—	**0.1332–0.4169** (price $\sim h^{+2.86}$)
frontier, price ≤ 2		1.319–1.650
frontier, price $\le 10^2$		0.412–1.126
frontier, price $\le 10^4$		−0.079–0.681

The Mellin ladder is the third genuine gain, and it is an asymptotic series. Past the PNT main term the smooth part of $d\psi$ is the derivative of the *elementary* closed function $-\frac{1}{2}\log(1 - x^{-2})$, so the k-th rung pays $+\sum_d w_d\, C_d(-(2k+\frac{1}{2}))$ in closed cell moments (Mellin 1896) — no zero data anywhere. The residual is minimised at $K^*=2$ on all 18 windows (relative drop 0.9015–0.9256 — a genuine factor), and past K^* it *rises* by 15.5–25.1 up to $K=20$: the predicted log-divergence of the formal sum $\sum_k x^{-2k-1} = 1/(x(x^2-1))$, not a numerical artefact. What it buys: $\delta_{\text{val}} = 0.6247$–$0.9307$ against C1's 0.9839–1.3767, a forced (not fitted) gain of 0.359–0.446 — and it *cannot* be iterated to the missing $X^{0.12}$–$X^{0.43}$, because the series diverges. C0 and C1 reproduce T161's 1.148–1.881 and 0.984–1.377 to 0.01 at both ends on the independently rebuilt surface.

One Abel step, and R-C′ stops being an arithmetic problem. Level 0 admits no unconditional input at all (no cell-level Chebyshev bound — T161's granularity finding restated); one Abel step, licensed by the gauge identity, gives $|V_d| \le 2\kappa\sqrt{X}$ by two partial summations, and the ladder identity $\sum_d v_d w_d = \sum_d V_d^{(k)}(\Delta^k w)_d$ is machine-checked at every level $k = 1\ldots4$ (3.9×10^{-15}–3.1×10^{-14}, boundary free). The optimal level is *exactly one* on 18/18, for a closed reason: each further step multiplies the envelope by $\sim 2/D$ and divides $\|\Delta^k w\|_1$ by $\sim\omega_{16}$, and the product is $32\pi/\alpha = 16.7$–$41.9 > 1$ — partial summation is a single step, and it has been taken. At that level the demand factorises *exactly* as

$$\delta_{\text{bnd}} \;=\; \tfrac{1}{2} + \frac{\log(2\kappa\,\|\Delta w\|_1/|Q|)}{\log X},$$

so every arithmetic input sits in the single constant κ and the remaining question is the closed, *prime-free* ratio $\|\Delta w\|_1/|Q| = 2.25\times10^3$–$1.15\times10^6$, growing as $h^{+1.769}$ (fit); an RH-strength input misses the corresponding test statistic by 4.8–8.2 orders of magnitude.

The Fejér split, and the price of going below one half. The Thomson dual makes every trial vector x with $x_1 = 1$, $Q(x) > 0$ admissible at the price $P(x) = Q(x)/Q(x^*) \ge 1$ in the $1/s$ ceiling — and the objective must be priced, or it is gamed: an *unpriced* search does find $\delta_{\text{bnd}} < 0$ here, by making $Q(x)$ huge and the bound vacuous. Priced (778 trial vectors per window, the fast path re-assembled against a literal double loop at 0.0), the result is an exchange rate: a sub-$\frac{1}{2}$ demand *exists* on every window, at minimum price $P_{\min} = 7.2\times10^1$–8.2×10^4 growing as $h^{+2.0}$ (fit), and every unit of δ costs 1.17–3.31 *decades* of price. The structured families beat the sweep: the spectral truncation C3 reaches 0.4602–1.3145 (below $\frac{1}{2}$ on 5/18), and the **Fejér taper C4** — x_k weighted by $1-k/n_b$ (Fejér 1915), which by self-adjointness of the smoothing *is* the contract's "pairing against Fejér-averaged Λ-mass" — reaches $\delta_{\text{bnd}} = \mathbf{0.1332}$–$\mathbf{0.4169}$, *below* $\frac{1}{2}$ *on all* 18 *windows*, lowering the demand by 1.15–1.51 in δ against the untapered optimiser. The catch, stated without hedging: its price grows as $h^{+2.86}$ (fit), so the sub-RH demand is bought with an asymptotically diverging loss in the $1/s$ bound, and a per-window certificate at growing price is not a uniform statement — R2″ is *relocated* into the price, not closed.

R-A′ closes, R-B′ is refuted. The log-moment $L = \sum_d \Psi_d w_d$ — the one non-polynomial moment of the arch half — agrees on three independent routes: the direct sum; the double-Abel form *with* its Wronskian boundary term, $L = -\frac{1}{2}\sum_d (d\log d)(\Delta^2 w)_d - \frac{1}{2}M\log M\, w_{M-1}$ exactly (2.1×10^{-15}–4.2×10^{-14}; the boundary term is 3.2×10^{-6}–3.5×10^{-2} of L and not optional — T161's identity is completed here, not quoted); and the Lerch/Frullani integral

$$L \;=\; -\tfrac{1}{2}\int_0^\infty \frac{(1-e^{-y})^2}{y^2}\,G(y)\,dy, \qquad G(y) = \sum_{d\ge1} w_d\, e^{-(d-1)y} \text{ closed},$$

agreeing with the direct sum to 0.0–6.0×10^{-16} — machine precision — and converging at $y = 0$ for exactly one reason, the gauge identity; the $d = 1$ term peels in closed form via $\int_0^\infty ((1-e^{-y})/y)^2 dy = 2\log 2$, and the same constant reproduces $\Psi_1 = -\log 2$ to 0.0, an independent confirmation of the whole derivation (Frullani 1828 / Lerch 1887 / Clausen 1832). R-A′ moves from open to certified-per-window; it is m-indexed, and no m-freeness is claimed. R-B′ is *refuted*, and the refutation is a result: the 16×16 Gram form *is* the arch half ($a^\top G a = Q^{\text{arch}}$ to 0.0–1.9×10^{-15}, a theorem) and it is *indefinite on every window*, $\lambda_{\min}/|\lambda_{\max}| = -13.34\ldots-1.59$, zero of 18 windows semidefinite in either sign; the mechanism is measured — the diagonal is strictly positive but small, the unweighted Gershgorin ratio is 7.97–580.76, so Gershgorin 1931 does not apply and no semidefiniteness certificate exists. What survives is the a-weighted quarter bar: off-diagonal mass against the arch half $= 0.1435$–0.2902, flat (exponent -0.102, fit), holding on 14 of 18 windows and exceeded by 16.1% on four — reported honestly as 0.29 and not as $\frac{1}{4}$: a bound on a *contribution*, not a spectral property.

The assembly, the stress, and the anti-fitting refusal. The full chain $Q = (Q^{\text{arch}} + Q_K^{\text{smooth}}) + R$ closes end to end at 6.1×10^{-17}–1.8×10^{-15} of the large scale $\max(|\text{arch}|, |\text{atom}|)$ — one unbroken chain of identities followed by exactly one inequality, which holds with 2.0–3.2 orders of slack; the gap $\delta_{\text{bnd}} - \delta_{\text{val}} = 0.77$–$0.86$ *is* the price of factorising, and it is bigger than the 0.12–0.43 separating the residual value from RH strength: the factorisation, not the arithmetic, now dominates the loss. The must-break controls break — uniform surrogate prime powers (same mass multiset) drive the total to 5.2×10^2–4.6×10^7 of $|Q|$, and violating the gauge breaks the Abel identity by *precisely* $c_0 \sum_d w_d$ (reproduced to 1.9×10^{-14}, a positive control on the mechanism) — while a control that did *not* break is declared as information: $4\times$ under-resolution leaves the total $O(1)$ (0.41–1.83), so the cancellation does not rest on resolving individual prime powers. And the anti-fitting refusal, the most uncomfortable number of the probe, reported in full: the exponent $\beta = \frac{1}{2}$ in the smooth prime term is *forced* by the $n^{-1/2}$ weighting; the empirical minimiser over $[0.2, 0.8]$ is $\beta^* = 0.600$–$0.800 \neq \frac{1}{2}$, where the residual would look up to a hundred times better — and it is *refused*: the drift is a property of the residual being large (50–76% of the object), not evidence about the exponent. The verdict, in the two currencies: the δ trajectory $1.88 \to 1.38 \to 0.93$ is real and monotone and does *not* reach $\frac{1}{2}$ (the best provable demand after the one Abel step is $X^{0.90}$–$X^{1.29}$ beyond RH strength), and the exhaustion *saturates rather than converges* — three independent ladders all turn around at their first or second step. The balance of T162: 10 theorem / 2 certified-uniform / 3 certified-per-window / 9 measured, zero fit rows (the only fitted numbers are h-exponents, labelled in place); the candidates T162-P1...P6 (the closed log-moment; the Abel form with its boundary term; the quarter bar; the Gram indefiniteness — a negative result that closes a route; the Abel-level theorem; the exchange rate) are all PENDING — no promotion here, and v554 is untouched.

After T162: one front, one substitute, one device

(R-C″) the Pareto front, now prime-free: find an admissible trial vector x with $\delta_{\text{bnd}}(x) < \frac{1}{2}$ at a price $Q(x)/Q(x^*)$ the T159/T160 chain can pay — such x exist at price $\geq 7.2 \times 10^1$ growing as $h^{+2.0}$, and both ways out are closed questions: fix the affordable price from the chain and read the frontier, or enlarge the search space beyond the 16 low modes, where the total variation may drop at constant price. *(R-B″)* a substitute for the refuted Gram positivity: the chain wanted a *sign* and has a contribution bound (the a-weighted quarter bar, certified-uniform via flatness). *(R-D)* a fifth device for R1″: $\rho = 1.0036$–$1.0140 > 1$, flat, untouched by T162. T163 (`pareto_front_probe.py`, contract `PARETO.FRONT`) executed exactly that survey — next subsection.

10.53 The Pareto front: the front resists as a theorem, and the h^2 is the parity spectral gap (T163)

T163 (`pareto_front_probe.py`, contract `PARETO.FRONT`, 32/32, 1.9 s on a surface rebuilt with declared caps: 28 log-spaced prime-power zones of 70 admissible, $h = 142\ldots1445$ — a $10\times$ lever arm, with $H_{\min} = 128$ declared as twice the largest mode block $K = 64$ — of which 27 windows sit inside the condition horizon $\operatorname{cond}(B_{LL}) \leq 10^{12}$; the closed weight vector re-derived as a *correlation form*, $w_0 = A_0$, $w_d = 2A_d - H_{M-1-d}$ with A the autocorrelation and H the self-convolution of $v = T_K^{\mathsf{T}} a$, to 1.6–2.0×10^{-15} — $O(h^2)$ and block-size-free, which is what makes the sweeps affordable; pairing identity to 5.9×10^{-17}–9.5×10^{-14} of the large scale, cancellation headroom ≥ 10 decades over double precision, declared; $\psi(x)/x \leq 1.038830$ verified at every jump point up to $n = 400000$, $\kappa = 0.038821$; verdict FRONT-RESISTS) fixes the affordable price *from the chain* and reads the front — and the answer arrives as a theorem rather than as a search result. The theorem-shaped cores of T162 and T163 are promoted as v555 [`verification/v555_pareto_tv_identities.py`] (Section 4.21).

The price axis is chain-derived, not invented. $Q(x^*) = 1/g_{16}$ certifies that the T160 optimiser *is* the top rung of the T158 Cholesky ladder (2.3×10^{-13}–2.1×10^{-8}, below the conditioning bar the solve carries); the rung prices $P_{\text{rung}}(K) = g_{16}/g_K \geq 1$ are prices the chain has *already paid*, since $1/g_K$ is an upper bound on $1/s$ it already accepts; and the affordable price is preregistered as $P_{\text{aff}} = P_{\text{rung}}(1) = g_{16}B_{11} = 3.93 \times 10^3$–$1.01 \times 10^7$, growing as $h^{+3.049}$ (fit) — the T157 route-(0) certificate $1/s \leq B_{11}$, the loosest rung the chain has ever used. The Fejér knob $t_k(\sigma) = \max(0, 1 - k/\sigma)$ (Fejér 1915; 50 geometric grid points on $[1, 4096]$ plus ∞) has *both endpoints in the chain*: $\sigma = 1$ is exactly the $K = 1$ rung (0–3.9×10^{-15}) and $\sigma = \infty$ the untapered optimiser, i.e. the $K = 16$ rung. Price and demand are *measured* monotone in opposite directions along the knob at all 27×50 grid points (every point admissible), so the knob curve *is* the Pareto front: there is no interior optimum to find, and the only question is where the chain can afford to sit. Read at the declared prices, the flat caps 1.25/2.00/10.00 give $\delta_{\text{bnd}} = 1.310$–$1.756$ / 1.263–1.686 / 1.102–1.375, and the flat rung tiers the same — 0/27 **windows below the yardstick $\frac{1}{2}$ at any flat price**; only the full P_{aff} reaches $\delta_{\text{bnd}} = -0.004$–$0.019$ on 27/27. The crossing price is $P_{\text{cross}} = 8.05 \times 10^2$–$1.01 \times 10^5 \sim h^{+1.906}$ (fit), and $P_{\text{cross}}/P_{\text{aff}} = 0.0066$–$0.2047$: the crossing sits 5–152× *strictly inside* the certificate the chain already accepts, with the margin *widening* in h — the one genuinely new positive of the part, stated without inflation, because a per-window bound at a price growing like h^2 is not a uniform statement.

The price anatomy is an identity, not a fit. Substituting $P(x) = g_{16}|Q(x)|$ into the definition of the demand gives, for every admissible trial vector and with no asymptotics,

$$\delta_{\text{bnd}}(x) \;=\; \tfrac{1}{2} + \frac{\log\!\big(2\kappa\, g_{16}\, \mathrm{TV}(x)/P(x)\big)}{\log X}, \qquad \delta_{\text{bnd}}(x) < \tfrac{1}{2} \iff P(x) > 2\kappa\, g_{16}\, \mathrm{TV}(x),$$

machine-checked at all 27×50 grid points (0.0–7.0×10^{-14}): **price and demand are two coordinates on one object**, the trial vector enters only through the single ratio $\mathrm{TV}(x)/|Q(x)|$, and the "exchange rate" T162 measured is a change of variables, not a second lever. The crossing exponent *decomposes additively* — TV at the crossing grows as $h^{+2.013}$, g_{16} as $h^{-0.061}$, sum $+1.952$ against the directly fitted $+1.906$ — so the h^2 sits in the total variation and nowhere else. The taper's structural gain in TV is real and *bounded*: a factor 2.75–4.64, h-flat, worth at most 0.276 of δ and decaying like $1/\log X$; where the price moves, $\tau_{\text{TV}} = -0.0191$ (median) says every remaining unit of δ is paid one-for-one out of the price. And the second half of the anatomy identifies the object: the cancellation quality $\mathrm{CANC}(x^*) = |Q|/\max(|Q^{\text{arch}}|, |Q^{\text{atom}}|) = 2.2 \times 10^{-6}$–$3.6 \times 10^{-4}$ ($h^{-2.039}$) jumps to 0.449–0.934 at $\sigma = 2$ — *the taper destroys the arch/atom cancellation*, and the price is the cancellation depth read backwards.

The search space, and then the theorem that makes the sweeps unnecessary. Enlarging the block, $K = 16 \to 32 \to 64$, lowers TV only by 0.868–0.958 at a price *dividend* (0.81–0.94) and makes the demand *worse* (up to $+0.0135$); the TV exponent is unmoved ($+1.832/+1.820/+1.810$); the best closed candidate — the Fejér-weighted 64-mode sum — gives a TV factor 0.843–0.931 and no crossing; and the pooled flat-price best over everything built is $\delta_{\text{bnd}} = 1.331$–$1.788$. Then the sweeps can be put aside, because the question they were probing has an exact answer:

$$\textbf{(T1)}\ w_0(x) = \|a(x)\|^2 \quad \textbf{(T2)}\ \mathrm{TV}(x) \geq |w_0(x)| \quad \textbf{(T3)}\ x_1 = 1 \Rightarrow \|a\|^2 \geq 1/\mu_1^P \implies \mathrm{TV}(x) \geq \frac{1}{4\sin^2(\pi/N)} \sim h^2$$

for *every* admissible trial vector in the parity sector — (T1) because the parity rows are orthonormal (0–5×10^{-16}), (T2) by telescoping with $w_M = 0$, (T3) because the Thomson normalisation is not free. Verified on all 1674 trial vectors built anywhere in the probe (50 taper widths × 4 weightings × 3 block sizes), with slack $\mu_1^P\mathrm{TV} = 5.00$–$23.20$ — the floor is tight to a factor of order ten — and the floor itself 2.06×10^3–$1.77 \times 10^5 \sim h^{+1.997}$ (fit; exactly h^2 in closed form since $N = 2h + 1$). **(T4)** Combined with the crossing criterion: every x with $\delta_{\text{bnd}} < \frac{1}{2}$ must pay $P > 2\kappa g_{16}/\mu_1^P = 5.46 \times 10^1$–$5.87 \times 10^3 \sim h^{+1.937}$, already 5.5–590× above the largest declared flat cap — *a flat-price sub-$\frac{1}{2}$ demand is provably impossible on this surface*, for every admissible trial vector and not merely for the families swept. R-C‴ ("bounded total variation at bounded price") is thereby **closed negatively**, by an inequality.

Never in the primes — the sober reading. The relocation T162 suspected is exact, and its h^2 is now identified: the $h^{2.86}$ of T162's Fejér price, the $h^{1.91}$ of the crossing price and T161's granularity are *all* the reciprocal smallest parity eigenvalue — the spectral gap of the parity Laplacian (Kac–Murdock–Szegő 1953) meeting the entry normalisation $x_1 = 1$. Sub-$\frac{1}{2}$ demand at flat price is impossible in the parity sector as long as $1/s$ must stay $O(1)$, which is the whole point of the T152–T160 chain; the statement involves no prime, no taper and no truncation — and the mode sweep confirms it from the other side, since $16 \to 64$ modes buys a price dividend, a worse demand, and leaves the TV exponent at $h^{+1.81}$, a bounded factor above the theoretical floor.

R-B″, R-D, the stress and the anti-fitting control. The chain never needed the sign: in the upper direction $1/s \le Q(x)$ is Legendre duality plus an *evaluated* positive number, with headroom $\ge 10^{14.9}$ over the rounding floor on every window; T162's indefiniteness is reproduced ($a^{\mathsf{T}}Ga = Q^{\text{arch}}$ to 9.3×10^{-13}, $\lambda_{\min}/|\lambda_{\max}| = -14.6 \ldots -2.2$); what the sign would have bought is *uniformity* — the certified cancellation quality at the operating point drifts ($h^{-0.172}$) — so R-B″ reduces to R-B‴: make the positivity margin h-uniform (T162's a-weighted quarter bar is the right shape of object). R-D is a type mismatch: the taper acts on the trial vector while the block floor $1/g_K$ is a property of the *operator*; smoothing the operator instead (a Fejér 3-tap average of the lag coefficients) degrades the floor by 2.3×10^3–$2.7 \times 10^5 \sim h^{+2.036}$ — the same h^2 again, for the same reason — and bounds a different number: a measured negative, not a device. The must-break controls break: the uniform surrogate drives the total to 7.7×10^2–4.7×10^7 of $|Q|$, the gauge violation to 3.4×10^{-2} of $\|w\|_1$, and the T145 resolution no-go breaks *monotonically* over $\nu = 4 \to 2 \to 1 \to 0.5 \to 0.25$ with damage $1.0 \to 9.1 \to 25.4 \to 129 \to 434$ (maxima; the $\nu = 4$ column an exact null control) — monotone in the collision rate, which is what makes it the T145 mechanism rather than a generic loss of accuracy. And the anti-fitting control is enforced rather than promised: a single globally fixed knob setting $\sigma = 3$, calibrated on the smallest window alone, delivers $\delta_{\text{bnd}} = 0.176$–$0.380$ out of sample on all 26 remaining windows at 0.14–0.23 of P_{aff} — no per-window tuning anywhere. The balance of T163: 11 theorem / 1 certified-uniform / 4 certified-per-window / 10 measured, zero fit rows (the only fitted numbers are h-exponents, labelled in place).

After T163: one successor with two prime-free arms

(R-C‴) **closed, negatively, on this construction**: bounded total variation at bounded price is impossible in the parity sector — $\mathrm{TV}(x) \ge 1/\mu_1^P \sim h^2$ for every x with $x_1 = 1$, an inequality over the whole admissible set. *(R-E)* the successor the theorem points at, both arms prime-free: *arm A* — show that the downstream T159/T160 chain tolerates a *growing* $1/s$ ceiling (a question about the chain, not about the pairing); *arm B* — represent the entry functional in a sector whose lowest eigenvalue does *not* vanish like h^{-2}. *(R-B‴)* make the positivity margin h-uniform (the certified CANC at the operating point drifts, exponent -0.172; the a-weighted quarter bar is the right object). *(R-D)* a fifth device for R1″: $\rho = 1.0036$–$1.0140 > 1$, flat, untouched by the Fejér perspective for the type reason above. T164 (`sector_change_probe.py`, contract `SECTOR.CHANGE`) is running — the sector change of exactly that successor question.

11 Status and outlook

11.1 What is certified, what is measured, what is hypothesis input

The three-way split, stated once

Certified (exact identities or closed-form bounds, machine-checked): the twenty-one ledger modules of Section 4; and, in the sandbox arc, the bare bound b_0 in closed form, the parity superselection $JQJ = Q$, $JB = -BR$, the pole splitting $ab^{\mathsf{T}} + ba^{\mathsf{T}} = ss^{\mathsf{T}} - tt^{\mathsf{T}}$, the channel inertia, the avoidance law $Q_{\text{full}} \succeq 0 \Rightarrow \|B^{\mathsf{T}}v_i\|^2 \le w_i\Lambda_-$, the growth inequality, the accumulation bounds, the Sherman–Morrison/Woodbury reduction of (R) to $r \le 1$, the three T108 identities (ε as a $Q|_{\text{odd}}$ energy, the rank-one overlap $hh^{\mathsf{T}}/\varepsilon$, and ε as the square of the last Cholesky pivot), the two T109 certificates (the graded matrix cap $\omega_{\text{cert}} = 0.2561$–$0.7569$, no hypothesis input, and the residual certificate for the boundary value, sharpness 1.0000–2.4937, 16/16 in budget), the three T110 pieces of the propagation circle (the exact splitting of the growth step, embedding error $\leq 2.6 \times 10^{-14}$, with the new atom's restriction to the old window the exact zero matrix; the graded Loewner minorant step, valid exactly under the induction hypothesis, holding 15/15 with retention 1.0000; and the base-case Cholesky certificate at $n = 2$, $\lambda_{\min} \geq 6.93 \times 10^{-2}$ at three resolutions), and their T111 deep-ladder extension (117 handover certificates at retention 1.000000, largest single loss 5.96×10^{-8}; atom entry the exact zero matrix on 117/117; and the arithmetic ladder-wall statement at frozen cell width, gap $> 2D$ needed against the twin-prime gap 0.003831 at $521 \to 523$), the T114 margin-free step certificates (Albert/pseudo-inverse Schur, 27/27 ladder steps to $n = 1331$ including all seven torn zones, with the exact 4×4 Schur complement at 42–67% of the block scale — no margin division anywhere), and the T115 compression certificates (the exact identity $X_{\text{mixed}} = Q_{\text{old,mixed}}$; Albert on 66/66 (zone, q) combinations; the one-sided second-order compression error; the deep margin-free step at $n = 155\,921$ and the ten-step chains — certificates for a coarser Galerkin space, declared as such), and the T116 boundary-recursion identities (the partial-minimisation identity $t^{\mathsf{T}} T^{-1} t = r^{\mathsf{T}} \Sigma^{-1} r + \sigma$ on six zones; the Haynsworth double-Albert equivalence 6/6 with sharp negative controls; the exact Riccati chaining, bit-exact on real windows; and the exact carriage of the global rank-one pole in the bordered state — the deep march itself certifies the band model's cost geometry, declared as such), and the T117 epsilon-theorem package (the four Galerkin identities — the cell functional of $2\sinh(x/2)$, the exact nesting $T_c = P^{\mathsf{T}} T_f P$, the two-level Pythagoras, the dual-norm residual form, all to 10^{-10} or better; the two-level lower-bound chain, every link an identity or a completed Cholesky, positive on all 19 nested pairs; and the exact jump formulas — the Levinson bordering drop r_0^2/s_0 and the rank-$(k_0{+}1)$ Woodbury corner update, both to 10^{-10}), and the T118 symbol package (the closed-form two-grid symbols — σ_S the weighted harmonic and σ_z the arithmetic mean of the aliasing pair; the saturation identity $\varepsilon_c - \varepsilon_f = y^{\mathsf{T}} S y$ to 4.9×10^{-11}; the CBS identity $\lambda_{\min}(L_z^{-1} S L_z^{-\mathsf{T}}) = 1 - \gamma^2$ to 2.1×10^{-15}; the isometry ordering onto $T_h(\sigma_z)$; $\inf \sigma_z > 0$ with explicit second-order margins on 7/14 windows; and the pole-mass identity in the oscillation space, $\leq 2.1 \times 10^{-5}$ via the DTFT factor $|\sin(\theta/2)|$), and the T119 arithmetic package (the closed-form comb symbol — one cosine per prime-power atom, linearly interpolated over the two nearest lags — with the parity rule for σ_z; the atom-count bound $|\sigma_z^{\text{comb}}| \leq \Xi(\alpha)$, uniform in D; the archimedean growth identity $\inf \sigma_z^{\text{arch}} = \log(1/D) + B$ with universal $B = -1.0474$; the resulting D_0 theorem; the large-sieve section certificate on one window (floor $+0.396$ where $\inf \sigma_z = -0.349$); the Levinson corner identity and innovation mass budget; the identity $\kappa_{\text{end}} = 1/(1 + R)$ to 1.1×10^{-16}; and the coarse-graining fence $\lambda_{\min}(T_h(\sigma * K)) \geq \lambda_{\min}(T_h(\sigma))$ by unitary averaging), and the T120 Harnack package (the exact increment system $Kv = s - u_0 G\mathbf{1}$ to 2.4×10^{-12}; the exact symbol identity $\sigma_c \sigma_z - \tau^2 = f_1 f_2$ to 3.9×10^{-16}; the pairing certificate (C1) — a proved implication whose single hypothesis is one sign of the corner increments, giving $|R - 1| \leq 0.04745$ and $\kappa_{\text{end}} \geq 0.480827$ for $n_C \geq 8$; the frame identity $\alpha_o = u_k/2 + O(D)$ over all 732 handovers; and the universal-B evaluation of the D_0 criterion on all 1492 zones, $\alpha^* = 0.693$), and the T121 wide-window package (the shifted-Cholesky section floors on the dense frame windows; the Christoffel/Fejér per-dip budget, positive on 8/16 frame windows up to $\alpha = 3.446$ at sharpness 0.20–0.76; and the exact deficit identity $y^{\mathsf{T}} S y / \text{SUPPLY} = r_{\text{ray}} r_{\text{cbs}} / (1 - \gamma^2)$ to 2.8×10^{-14}), and the T122 net-balance package (the closed form of the compressed Hankel term, $(Z^{\mathsf{T}} H Z)_{jk} = b_{j+k}$, to 7.0×10^{-16} on 36/36 rows; the two-grid isometry identity $A_z = W^{\mathsf{T}} \text{Toeplitz}_M(c) W$ with $W^{\mathsf{T}} W = I$ exact; the certified band bound (5R) and its uniform y-free variant, non-vacuous on 36/36 rows to $\alpha = 6.31$; the structural Rayleigh bound, positive and sound on 36/36 rows with residual slack 1.00–1.03; and the exact residual decomposition of the fitted balance, closing to 2.9×10^{-16}), and the T123 structural-CBS package (the two-isometry completion $V = BP$ with $V^{\mathsf{T}} V = I$ and $V^{\mathsf{T}} W = 0$ exact, the three-compression identities to 3.4×10^{-15}; the closed forms of the coupling — half a first difference of the lag sequence to 1.8×10^{-15}, the aliasing-difference symbol to 5.7×10^{-16}; the uniform oscillation positivity (5R*), $\lambda_{\min}(E_z) = +0.303\ldots + 5.017 > 0$ on every row at slack 1.0007–1.007; the exact dyadic depth-layer split of $y^{\mathsf{T}} G y$ to 1.1×10^{-15}; and the two-term residual split of the certified balance, closing to 2.8×10^{-17}), and the T125 assembly package (the integer window nesting $M_n(k) = M_o(k + 1)$ on all 29 consecutive pairs of the composing run with the incoming atom block bit-exactly zero and the composition residual 2.9×10^{-14}; the four assembly identities on 88/88 rungs of 52/52 zones; (8R) certified on 88/88 rungs at $\text{cb}/\delta = 0.9071$–$0.9971$; the ε lower bound on 52/52 zones at 3.2×10^{1}–8.7×10^{10} over the declared floating-point bound; the base-case equivalence $\operatorname{sign} \lambda_{\min}(Q^{(L)}) = \operatorname{sign} \varepsilon$ on 52/52; and the margin-free handover on 62/62 steps with the exact Schur complement 3.0×10^{10}–4.6×10^{13} above its floor while the norm surrogate is negative throughout — 430 completed Cholesky certificates in total), and the T126 seam package (the exact DP decision of the partition question on the real gap table — $\rho_{\text{crit}} = 2.0258$, no partition at 1.83, refinement-only infeasible at any ratio ≤ 64; the Haynsworth reproduction of $\lambda_{\min}(S)$ to 2.8×10^{-14} with the transport bracket containing the fine value on 30/30 real pairs; the twelve certified free-resolution seams, eleven in one re-gridding and the one at $n = 31$ through a three-stage ladder; the end-to-end seam at $\rho = 2.251$; and the exact fractional-Dirichlet identity of the window form).

Measured (holds on the resolved windows; no uniform constant): ρ_-; the ratio $r = 0.0053$–0.1814 and $\hat{r} < 1$ at all 92 resolution points; $\omega = 0.2366$–0.7531; the strict odd-channel margin $\lambda_{\min}(Q|_{\mathrm{odd}}) = 1.4 \times 10^{-5}$–$5.1 \times 10^{-2}$; the margin map $m_k \geq$ need109 on 16/16 zones (ratio 2.09–179.6), extended by T111 to 199 zones with the crossing *measured* between $n = 461$ and $n = 463$ ($n^* \approx 462$, an upper bound; the T110 extrapolation $n \approx 170$ was a fit artefact); the reserve trend $f_{\mathrm{crit}} \sim n^{-0.39}$ (fit); the requirement wall $n = 727$; the closure map 44/64; the handover margins; the two race slopes; the T114 ratio closure $r = 2 \times 10^{-4}$–0.103 on 27/27 steps and the D-powers of ε and κ (fits); the T115 transport split (the bracket's lower end positive on 7/11 regrid pairs, threshold $\rho^* = 1.83$ against a median ratio 2.001) and the fall laws $\lambda_{\min}(S) \sim \rho^{-1.69}$, $\varepsilon \sim \rho^{-1.71}$ (fits); the T116 law $\varepsilon \approx 8.34\, D^{1.790} \alpha^{-6.04}$ (jackknife ≤ 0.041), the factor-120 jumps at prime-power entries, the comb truncation budget growing like $n^{2.17}$, and the criticality $\hat{t}^{\mathsf{T}} T^{-1} \hat{t} \to 0.999975745$ (fits/measurements); the T117 rates $\theta'(\mathrm{sum}) = 1.74$ against $\theta = 1.76$ (loss 0.04 powers of D) with bound/$\varepsilon \in [0.111, 0.185]$, the growth of $\lambda_{\min}(S)$ under refinement, the entry direction (all 23 prime-power entries *raise* ε — correcting the T116 factor-120 reading, a sweep artefact), and the saturation ratio in $[0.675, 0.744]$; the T118 measurements: the corner-exponent drift $+0.238$ per halving (3–7.4σ), the concentration growth $\sim h^{0.99}$, the CBS floor $1 - \gamma^2 \geq 0.181$, the computed saturation band $\beta \in [0.252, 0.336]$ on 20 triples, the log-versus-power growth decision on the 128× FFT lever (fits with jackknife bands), the reduction sharpness 32–76% on 14/14 windows, and the zero crossings of $\inf \sigma_z$ on 3/5 zones; the T119 measurements: the mass distribution of $\|y\|^2$ (corner 0.18–0.42, middle 0.19–0.38, inner half 0.36–0.48), the exponents $\varepsilon \sim D^{1.761}$, $\|y\|^2 \sim D^{1.961}$, $\lambda_{\min}(S)\|y\|^2 \sim D^{1.757}$ (fits), the flat band $\kappa_{\mathrm{end}} = 0.4906$–$0.5093$ with drift ≤ 0.0019 per level, the ratio $R = 0.9634$–1.0349, and the wide-zone crossing depths $M \sim 10^{13}$–10^{187} (extrapolations of the certified growth law); the T120 measurements: the corner sign structure (the T^{-1} corner block positive share 1.0000 against a TP2 minor share of only 0.51–0.88 and a K^{-1} positive share of 0.44–0.51 — the two refutations), the sign purity 1.0000 on all 3915 rows with $R \in [0.9098, 1.1011]$, the per-cell ratio $|a_j/w_j|$ up to 4.8×10^3 (the per-cell inequality is false), the (C1) decay $n_C^{-0.58\pm0.05}$ (fit), the comb-supremum incoherence 0.038–0.603 of Ξ (a grid supremum), the large-sieve floor usable on 2/6 zones, and the γ drift $1 - \gamma^2 = 0.079$–0.283 with fitted slope -0.297 ± 0.013 in α; the T121 measurements: section positivity on all 16 frame windows to $\alpha = 6.277$ (the ten large rows Lanczos measurements, upper bounds by construction), the net balance SUPPLY/$\varepsilon_c \sim D^{-0.177\pm0.096} \alpha^{-1.565\pm0.135}$ with per-resolution refits $-1.88 \ldots -1.62$ (fits), the link-(5) ratio 0.357–0.916 on 42/42 rows (the Hankel refutation), the Rayleigh and CBS slacks 1.29–13.19 and 3.42–16.67, the one-sign run 24/24 at $n_C = h/16$ against 21/24 at $h/8$ (run-share slope -0.0258 ± 0.0042, a fit), the Fisher–Hartwig drift $-0.008 \ldots + 2.934$ against the allowed -0.120 (inconclusive, declared as the result), the no-cancellation ratio 4.6×10^{-8}–7.1×10^{-5}, and the pairing quotient $\delta = 0.0126$–0.1331 over 360 rows ($\kappa_{\mathrm{end}} \geq 0.4688$, $\delta \sim n_C^{+0.29}$); the T122 measurements: the Hankel/Toeplitz norm ratio 0.123–0.781 (the Weyl floor positive on only 12/36 rows), the band-bound recovery 0.429–0.995, the high-band mass of Wy (99.3–100.0% against 0.1–1.7% for y itself), the actual CBS coupling $\kappa_y = 0.0067$–0.5311 against $\gamma^2 = 0.7173$–0.9647, the certified balance $S_2/\varepsilon_c \sim D^{-0.003\pm0.075} \alpha^{-0.729\pm0.090}$ with the measured-CBS balance $\alpha^{-0.113\pm0.063}$ against the ceiling $\alpha^{-0.116\pm0.062}$ (fits), and the band margin 0.024–$0.363 \sim D^{-0.042} \alpha^{-0.525}$ (a fit); the T123 measurements: the band mass of Wy (99.64–100.00% at $d = |\theta - \pi| < \pi/2$ against 8.9×10^{-6}–1.5×10^{-3} in the $\theta = 0$ cell) and of the worst-case y^* (81.22–100.00% middle-band), the coupling range $\kappa_y = 0.0067$–0.4622 against $\gamma^2 = 0.7248$–0.9467 (a factor 2–109), the coarse conditioning $\mathrm{cond}(A_c) = 2.1 \times 10^4$–$3.1 \times 10^7$ with $\lambda_{\min}(A_c) = 1.7 \times 10^{-5}$–$2.9 \times 10^{-4}$ (envelope-margin ratio 8.4×10^2–4.0×10^5, rising like $\alpha^{+4.202}$, a fit), the two-factor overshoot 1.20–3.51 with the certified margin positive on 5/24 rows, the certified balance $S_6/\varepsilon_c \sim D^{-0.069\pm0.099} \alpha^{-0.544\pm0.113}$ against the ceiling $\alpha^{-0.044\pm0.076}$ (fits; gap $\alpha^{+0.500}$), and the lid share $\varepsilon_f/\varepsilon_c = 0.177$–$0.546 \sim \alpha^{+0.030}$ (a fit); the T125 measurements: the retention against the measured ε_0 (0.0013–0.9694) and against the L-level ceiling (0.9140–0.9929), the pairing band $C_1 = 0.01063$–0.20927 over the ladder (worst case a single shallowest zone; $C_1 \leq 0.06721$ on the remaining 51), $R = 0.8243$–1.0377 and $\kappa_{\mathrm{end}} = 0.4907$–$0.5481$, the head-room minimum 31.6 at $n = 243$, the rung-level balance $\mathrm{cb}/\delta \sim D^{-0.010} \alpha^{+0.001}$ and the run-level $\sum \mathrm{cb}/\sum \delta \sim \alpha^{-0.012}$ (fits), and the (S4) cancellation ratio $|r_{\mathrm{corr}}| = 0.3888$–$14.8018$ with the norm-only variant returning exactly 0; the T126 measurements: the transport reach (positive to $\rho = 2.251$, first failure at 2.061, an overlap band rather than a clean split), the fixed-ratio zone sweep (12/22 zones at 2.0256), the free-resolution overhead $D_{\mathrm{free}}/\mathrm{cap} = 0.021$–$1.000$ (median 0.201) with re-gridding at 764/9 700 boundaries, the seam retention (median 0.578), the pencil alignment 0.866–0.987 (median 0.975), the Galerkin consistency orders $D^{1.798\pm0.057}$ and $D^{1.802\pm0.065}$

with Richardson limit 2.1256×10^{-4} (fits with jackknife bands), and the diverging continuity constants of the refuted naive continuum route; every scaling exponent quoted in this document.

Hypothesis input / classical (used, named, not proved here): Kneser, Witt/Eichler, Shimura, Waldspurger/Baruch–Mao, Cohen, Weil–Guinand, Bombieri, Yoshida, Connes–Consani, Douglas, Slepian–Landau–Pollak, Gronwall/Robin (unconditional form only), Grenander–Szegő, Sherman–Morrison/Woodbury, Christoffel–Darboux, Szegő–Levinson, the Weyl bound, Combes–Thomas/Jaffard, Prager–Synge, first-order Kato perturbation, Albert (the pseudo-inverse Schur criterion), Haynsworth (the partial-minimisation form of the Schur complement), Céa/Strang's first lemma, Bramble–Hilbert, Brandt/Yserentant two-scale spaces, Szegő–Kolmogorov, Aubin–Nitsche duality, Payne–Weinberger, Levinson/Durbin, Kac–Murdock–Szegő/Widom/Fisher–Hartwig, Fejér–Riesz, Hartman/Peller (Hankel smoothing), Gershgorin, Parseval, the local Fourier / two-grid symbol (Brandt, Hackbusch, Fiorentino–Serra), Yserentant's strengthened Cauchy–Schwarz constant, Axelsson/Axelsson–Vassilevski (the band-split strengthened-CBS mechanism), the Bank–Smith/Dörfler–Nochetto saturation assumption (with Bornemann–Erdmann–Kornhuber as the honest caveat that saturation can fail), the Montgomery–Vaughan/Selberg large sieve (upper direction only), Chebyshev/Mertens (the atom count), Trench/Böttcher–Silbermann (explicit Toeplitz inverses), Delsarte–Genin (the split Levinson recursion), Wilkinson/Higham backward error.

11.2 The asymptotic mountain

Three separate reasons why the cascade of Figure 8 is not a route to a theorem as it stands, and why the document does not present it as one. Since T110 the mountain has a *precise* formulation — the three gaps of Section 7: no reserve, no scalar step law (structurally excluded by the boundary layer), and no uniformity in k. Since T111 it also has a *measured* one — the three walls: the margin wall at $n^* \approx 462$ (measured, an upper bound; the earlier $n \approx 170$ was a fit artefact), the arithmetic ladder wall at the twin-prime pair $521 \to 523$, and the requirement wall at $n = 727$. The mechanism itself never failed on the deep ladder — all 117 handovers certify at retention 1.000000, and the reserve even opens with depth ($f_{\text{crit}} \sim n^{-0.39}$, fit) — so the actionable content of the measurement is the reinterpretation: the operating variable is the window depth, not the zone index, and a proof along this route must beat the exponent gap $-0.974 = -0.681 - 0.293$ and couple the depth δ_0 to the local prime gap. T112 (`adaptive_scaling_probe.py`) then performed that coupling: the ladder wall and the depth dependence of the requirement wall fall *structurally* in the gap-coupled frame, but the margin wall is frame-invariant at exponent -0.974, with the prime-gap dependence disclosed (Bertrand–Chebyshev suffices upward; Zhang/Maynard force θ-uniformity downward). T113 (`limit_operator_probe.py`) then settled what that invariant number *is*: the wall carries the same exponent in every admissible currency — it is substance — but it measures the *discretisation*, not the spectrum: at a fixed window both leading eigenvalues carry the same cell-width power under refinement (the continuum window form has no gap), the positive floor is a cancellation of relative size 10^{-7}–10^{-4}, and the T109 requirement chain divides by an artifact margin. T114 (`margin_free_step_probe.py`) then discharged the leading part of that balance: the margin-free step certificate exists (Albert 1969, the exact Schur complement), runs eleven steps past the old wall to $n = 1331$, certifies all seven torn zones, and shows the wall was an $O(1)$ numerator divided by an artifact floor — chains now stop only at the $h \le 1500$ window cost, never at a step, and the (R) demand itself is grid-bound and deleted ([P5]). T115 (`schur_transport_probe.py`) then split the two-object core: the transport of the exact Schur complement across a real regrid certifies only mild refinement ($\rho \lesssim 1.8$) — and for a principled reason, since on nested ladders, where the transport error is exactly zero, $\lambda_{\min}(S)$ itself falls like $\rho^{-1.7}$ — while the two-scale compression breaks the compute cap: a certified margin-free step at $n = 155\,921$ (117× deeper), ten-step chains, the stopper always cost and never a step, with the contiguous reach moving from $n \le 125$ to $n \le 5437$ (the $n \log n$ barrier divided, not broken). T116 (`boundary_formulation_probe.py`) then ran the boundary reformulation itself: the induction step *is* a boundary process — the global pole rides exactly in a $12 \times 12 + 12 + 1$ state and one unbroken Riccati march ran $169\,236$ prepends to 1.35 million cells (903× the old cap) at flat cost, declared as a cost-geometry demonstration and not a Weil certificate — but the state cannot be bounded, because the prime comb itself refuses compression (every incoming cell reflects off every prime power in the window; the truncation budget grows like $n^{2.17}$). T117 (`epsilon_theorem_probe.py`) then made that inequality theorem-shaped: the Galerkin reading is four exact identities, the lower-bound direction is a certified two-level chain that loses no power of D ($\theta' = 1.74$ against $\theta = 1.76$), the jumps have exact closed forms — and, correcting T116, the prime-power entries *raise* ε, so the factor-120 falls were a sweep artefact and the jump side-condition disappears. Since T117 the mountain is three named analytic lemmas about one symbol, each with a classical address — the Kac–Murdock–Szegő/Widom corner asymptotics of T^{-1}, a one-level lower bound for $\lambda_{\min}(S)$ at the Nyquist frequency, and the Bank–Smith saturation constant, two of the three constants rather than rates — plus the comb as the honest support obstruction. T118 (`symbol_lemmas_probe.py`) then measured the three against their addresses, and two of the three stand: saturation is an *identity* here; the Schur floor is rescued by the strengthened Cauchy–Schwarz shift onto the oscillation Gram, certified on 7/14 windows with the remainder shown to be under-resolved rather than obstructed on the 15 680-point FFT lever; and the corner lemma corrects itself — a log singularity is the boundary of all powers, so the chain is repaired at its last step and the rate now lives on the oscillation mass $\|y\|^2$. Since T118 the mountain is *one* genuinely new analytic statement — (H2), a $D^{1.75}$ lower bound for the oscillation mass — plus the standard CBS uniformity (H1), two arithmetic reductions for the comb dips, and the [**SUPPORT**] obstruction. T119 (`oscillation_mass_probe.py`) then closed the arithmetic half as a theorem — $\inf \sigma_z > 0$ for all $D < D_0(\alpha) = \exp(-(\Xi(\alpha)+B))$, with Ξ the prime-power atom count and $B = -1.0474$ universal — proved the energy route to (H2) empty (the hypothesis is genuinely new content), and reduced the oscillation-mass half to one discrete Harnack inequality through the exact identity $\kappa_{\text{end}} = 1/(1 + R)$. T120 (`harnack_probe.py`) then *explained* that inequality — the two increment families are one sequence offset by a single fine cell, so given one sign of the corner increments the summed bound $|R - 1| \leq 0.047$ is a proved two-line implication, while the per-cell version is provably false and the two obvious structural routes to the sign (total positivity, the discrete maximum principle) are refuted by measurement — and at the same time sharpened the wide-window defects: the frame consumes $\alpha = u_k/2$ by construction, so the unconditional D_0 route covers not one handover of the real ladder ($\alpha^* = 0.693$, below the first admissible zone), and $1 - \gamma^2$ decays with α. Since T120 the mountain is four named defects — one sign of the corner increments (a corner-localised Fisher–Hartwig estimate), a uniform bound on the pairing ratio (a discrete gradient estimate), D_0 for the whole ladder, and the α-dependence of γ — plus the [**SUPPORT**] obstruction; the chain contains no zeta input anywhere. T121 (`wide_window_probe.py`) then ran the σ_z route against the real ladder and restructured two of the four: the route survives as a *section* statement ($\lambda_{\min}(T_h(\sigma_z)) > 0$ on 16/16 frame windows to $\alpha = 6.277$, half of them where the symbol infimum is negative; the certified Christoffel budget reaches $\alpha = 3.446$) and dies as a symbol statement, and the net α-balance of the whole chain is one poly-log ($\alpha^{-1.57}$, uniform in D), split exactly into the Rayleigh and CBS steps with nothing in γ — while one link recorded as classical was refuted (the odd sector's Hankel term). Since T121 the mountain is the same four defects changed in kind — the corner sign (now a head-versus-tail inequality at $n_C = h/16$), the pairing bound, the section lemma with the Hankel term, and one poly-log of α-slack with a structural address — plus the [**SUPPORT**] obstruction. T122 (`net_balance_repair_probe.py`) then repaired the two worst-case steps that poly-log named: the Hankel term is the reflection half of an exact isometry ($A_z = W^{\mathsf{T}}\mathrm{Toeplitz}_M(c)W$, $W^{\mathsf{T}}W = I$), so link (5) is retired by an identity — and the per-dip Nikolskii budget with it — while the certified band floor reaches $\alpha = 6.31$ where the Christoffel budget tore at 3.45; the structural Rayleigh bound is sharp (residual slack 1.00–1.03, flat in α), the certified deficit halves to $\alpha^{-0.73}$ (exactly uniform in D), and with the measured CBS coupling the balance closes to within the chain's own ceiling ($\alpha^{-0.113}$ against $\alpha^{-0.116}$). Since T122 the mountain is the two unchanged sign/pairing defects plus two scalar estimates of the same band-profile kind — the band margin (F3) and a structural CBS bound (F4) — plus the [**SUPPORT**] obstruction. T123 (`structural_cbs_probe.py`) then ran the same band machinery at both scalars, and they *split*: the margin closes almost for free (the reduction to the band profile of Wy costs 0.7%, and the oscillation block gains uniform certified positivity, (5R*)), but the coupling resists structurally — every certified route to κ_y needs the coarse form from below, whose near-null direction is smooth and lives on a cancellation no pointwise symbol minorant can see — and the remaining $\alpha^{+0.500}$ gap to the ceiling is identified with the ε_f lid itself, the same object 4–8 recursion levels deep. Since T123 the mountain is the unchanged Harnack pair (F1)/(F2), one band-profile lemma (F3), and coarse-level positivity (F4) — no longer an estimate to sharpen but the recursion base case — plus the [**SUPPORT**] obstruction. T124 (`multilevel_telescope_probe.py`) then built that recursion, and the telescope carries: the ladder is one window form on nested spaces ($P^\mathsf{T} A^{(\ell+1)} P = A^{(\ell)}$ exact — the two-level system is literally one rung), the rung contribution is a residual in the inverse norm — a maximum, not a minimum — so it is certified from *above*, where the envelope works ((8R), valid on 400/400 rungs at a drift seven times weaker than T123's), and (F4) collapses from a quantitative estimate to a sign the coarse-to-fine induction already carries, with base case pure semidefiniteness — $\lambda_{\min}(A_c)$, $\mathrm{cond}(A_c)$ and γ^2 leave the chain entirely. T125 (`grand_assembly_probe.py`, the series finale) then mounted all of it on one ladder and changed the *shape* of the mountain rather than its height: on 52 of 52 zones — 30 of them a literally composing run of consecutive atoms on one common resolution — the load-bearing spine [A][B][C][E] is at CHOLESKY-CERT, so the Harnack pair (F1)/(F2) *leaves the spine* and survives only in the second, independent route (iv). Since T125 the mountain is therefore *one* declared accounting convention ((C-F5)) on the spine, the Harnack pair as an alternative route rather than a bottleneck, and **(F6) uniformity in the zone index** — no uniform-in-k handover Schur complement, no uniform lower bound on cb_ℓ — plus the [**SUPPORT**] obstruction. That uniformity gap, not any missing estimate, is the honest distance from the chain to any infinite statement (Section 9), and it is where the series closes. T126 (`uniformity_seams_probe.py`, the first part of the second phase, Section 10) then opened the attack on exactly that gap and finished the seam architecture: the segment-partition question dissolves into a monotone free-resolution construction (re-gridding only at record-small gaps, all twelve affordable real seams certified, a full seam end to end at the ratio the arithmetic demands), the continuum step closes through an exact fractional-Dirichlet identity onto Lipschitz test functions in the reached window, and the audit splits (F6) into three named lemmas of which exactly *two* are genuinely new — the U3 direction lemma and the U5 zone-uniform seam floor. T127 (`two_inequalities_probe.py`) then dissected both, and both changed: the U5 separating variable is pure interval geometry (the border protrusion $s_{l,r}$, not the atom and not the gap), U5 as posed is *refuted* and replaced by a band covering 99.26% of the real seams with the 8 exceptions enumerated (Mersenne/Fermat pairs and one twin), and U3 collapses to a coarse floor through the harmonic-mean identity, with exactly one direction type to exclude. T128 (`teml_probe.py`) then worked the four-point list cheapest first, and three of the four stand at their preregistered bars: (L) is derived and closed (the three affordable seams certified, the five open ones each lacking exactly one Cholesky beyond the cap), (T) becomes an exact identity plus majorants valid on every transport, (E) certifies through the Q-inner-product chain on all nine in-band seams, and (M)'s boundary-layer exclusion is now a proof with an 11.6× floor margin — while the fourth point, the protrusion concentration κ, missed its own bar by 3.6%, systematically in the ratio. T129 (`kappa_deep_seams_probe.py`) then tested the κ law that miss preregistered, and broke it productively: the fitted law falls once on 331 fresh transports with the bar frozen, but the structure underneath it is solved with identities — the flat border vector sits at $\kappa = 1$ exactly, a linear density profile at $\kappa = 2$ exactly, everything above is curvature, and the curvature chain is a per-transport theorem with an explicit constant holding on all 436 transports — while the two affordable deep seams ($n = 127$, 256) carry complete certificates on the graded two-scale space, honestly downgraded for a measured 8% false-positive rate of the compression, declared before the results. T130 (`curvature_bridge_probe.py`) then attacked the two named pieces and exactly one of the two stands: the graded-to-uniform *bridge* stands as an identity — the matrix-form Céa/Strang defect reproduces the directly computed uniform floor on 84 pairs with zero overshoot, explains the 8% false positives completely (they sat in a bracket the bridge makes unnecessary), and carries both deep seams to positive fine floors at up to 3.8× the factorisation cap — while the curvature bound honestly broke its frozen shape band on 13 of 545 disjoint test transports and is reduced to a zone-uniform bound on *one* exponent. T131 (`self_supply_probe.py`) then built the self-supply loop and left it one number short of closed: the $\varepsilon \to \lambda_{\min}$ secular sandwich is a *theorem* — sharp to a factor ≈ 1.3, with the sign half an equivalence — that replaces the Lanczos estimate on all 84 bridge pairs with zero brackets lost (and exposes that the old Ritz value overestimated the floor by up to 7.9×), sign constancy became a theorem via Perron–Frobenius on S^{-1}, the one-hump property honestly broke at depth, and M25 is reduced to positivity of the pole-free section with nine decades of slack. Since T131 the mountain is the map V6: two irreducibles — the word "for all" (M19) and the RH address itself (M21) — plus **M25 reduced-to-pole-free**, M22 collapsed onto one measured exponent, the vacuous M17 assembly with its named whitening fix, the conditional comb bound (M18), the per-side law (M24), the (C-F5) convention, the all-α step that no computation can close, and the **[SUPPORT]** obstruction. T134 (`pole_free_floor_probe.py`, PARTIAL) then closed the *existence* half of that pole-free floor — every Cholesky pivot positive on 79/79 windows, per-window certificates throughout — while every cheap lower-bound route fails by *sign*, not size: the surviving opening is an M-matrix question, and the whitening honestly corrected T131's diagnosis (the comb dominates the pole in the norm; what is missing is a localisation statement). T136 (`m_matrix_pair_probe.py`, ONE-CARRIES) then took that question and closed one of its three items outright — Varga's regular splitting makes $\rho(J) = \tau/(1+\tau)$ an *identity* and Collatz–Wielandt bounds it a priori to 1.00–1.03 of the measured gap on 900/900 blocks — while the exact bookkeeping $\lambda_{\min} \sim D^{-0.56} \cdot D^{2.72} \cdot D^{0.12}$ shows the whole degradation sitting in the margin and M17 closes *negatively* (the bad subspace is delocalised). T137 (`long_lag_probe.py`, BOTH-RESIST) made the perturbation explicit (comb-generated anti-diagonal stripes at the prime-power atoms, each a perfect matching, amplitude certified) and *killed a whole family by certificate*: $\rho(|E|) \geq 1.32$ from below on 35/35 windows, so no absolute-value bound can ever supply a floor — the residue is one named object, a sign-preserving bound. T138 (`sign_compensation_probe.py`, PAIR-EXACT) kept the signs and found the mechanism — the coupling sign follows the *interval geometry* of the two edges, and the m-paired Neumann certificate removes the arithmetic wall on all 77 dead blocks (pool 563 → 875/900) — while T139 (`green_decay_probe.py`, DENSE-RESISTS) refuted the classical decay lemma *at its hypothesis*, arithmetically, derived the sign law from one exact telescoping identity and killed the never-truncating layer series from below: the measured core shrinks to *one signed inequality at stripe distance* $b \leq 16$. T140 (`signed_band_probe.py`, FINITE-CORE) then gave that inequality an exact finite core per zone — $\rho(W) = \lambda_{\max}(K^{1/2} H K^{1/2})$, a closed-geometry covering kernel times a mass-plus-Dirichlet form, with the checkerboard split replacing the $O(nb)$ Weyl steps by three D-independent ones and *all* the D-dependence located in the geometry; what remains is a zone-uniform discrete Hardy inequality. T141 (`discrete_hardy_probe.py`, HARDY-RESISTS) attacked exactly that inequality: four exact identities put it in classical two-weight shape, but the certified constant is *not* zone-uniform ($D^{-0.366\pm0.036}$ against a bar of 0.25) while the object it bounds *is* ($D^{-0.229\pm0.007}$) — the growth sits in the diagonal profile; the additive shape is dead as a *shape* by its own exact Weyl floor and the joint shape fails at the normalisation alone ($\Omega = 20.7$–2724), so the residue is one closed conductance profile. The identity and structure blocks underneath the map are no longer sandbox: they are promoted as v542, v543, v544 and v545 (Sections 4.8–4.11). And the *reverse-flow* parts have landed — T132 (`bd_seam_probe.py`) made the Beurling–Deny triad an operator discriminator for the seam DtN, SPECTRUM-ONLY with the gap in the killing measure, T133 (`cert_floor_probe.py`) audited the suite's own PSD rows and produced the exact positive-mixture Gram certificate that hardened v379 in this same round, and T135 (`comb_compressibility_probe.py`, BOUNDED-STATE) showed the seam DtN admits a bounded faithful state ($m_{\text{cert}} = 12$, from the pre-declared set) where the Weil window provably has none — with the driver named as weight summability and `QEC.SEAM.01` explicitly *not* advanced.

1. **Sixteen zones are sixteen zones.** Everything below the top line of the cascade is established on the resolved zones $n = 2 \ldots 29$ (T111's deep ladder extends the measurements to $n = 521$, T114's margin-free ladder to $n = 1331$, T115's compressed run to $n = 155\,921$, T116's boundary march to 1.35×10^6 cells with atom reach $n = 173$ — finitely many zones either way). Not one constant in the chain is certified uniformly in k. The T101 requirement list (A)–(D) is

unchanged: the hardness would sit in (A), a sharp, non-perturbative arithmetic lower bound at a Weil-positivity edge.

2. **Finite resolution is a wall, and a measured one.** Numerics as a witness is exhausted beyond $\alpha \approx 0.55$ ($O(M^{-1.8})$, T96); the finite-block route was judged the structurally wrong instrument, not an under-resourced one (T92); $\lambda_{\min}$ collapses geometrically by a factor 11.4 per mode, so double precision ends at $N \approx 10$ while $N = 24$ would need 10^{-27} entries.

3. **The induction base is not free either.** The recursion terminates arithmetically (240/240 in ≤ 4 steps to the classical zone, T99; 960/960 with longest chain 13 steps, T101), but the zone peaks need percent-level bounds, not 0.1% (T99: extrapolated peak margins $+2.7 \times 10^{-2}$, $+1.4 \times 10^{-2}$, $+1.7 \times 10^{-2}$, i.e. 2.2–5.5% of $\mu_k/2$, with zone 5 saturating exactly).

11.3 Extraction candidates: the community-checkable path

The useful question at this point is not "how close is the program to RH" — it is not close, and no such claim is made — but "which pieces of it are unconditional theorems that someone else can check". Three candidates are named in the diary and are repeated here as the outlook; the third one is the finale's own recommendation and now leads:

- **The assembled chain, as a conditional lemma paper in numerical linear algebra** (the T125 recommendation). Stated without one word of this project's vocabulary: *a certified multilevel lower bound for the Galerkin prediction error of a log-singular Toeplitz form with a prime-power comb*, with the (8R) direction reversal as the theorem and the margin-free bordering induction as its corollary — the comb is then merely the example that makes the symbol log-singular, and no arithmetic conjecture is mentioned anywhere. The reassessment scored it against the matching-lemma package on the five criteria of the July extraction analysis and found it *stronger* on three (prior-art independence — the ingredients are classical but the assembly is a corollary of none of them; self-containedness — decisive, since the statement needs no project vocabulary at all; referee surface — three Schur-complement facts, Parseval, Galerkin orthogonality and one floating-point convention) and *comparable* on the other two (machine-checkability: 430 Cholesky certificates and 440 identities against `v541`'s 33/33 with a 10^6 enumeration; honest boundary). Two conditions come with the recommendation: carry (C-F5) in the *abstract* rather than in a remark, together with the measured failure of the solution-free rung as a separate proposition — that negative result is publishable content and it is what keeps the gap from looking overlooked — and do *not* extract (F1)/(F2) with it, since they are window certificates outside the spine and their classical address belongs in a separate note whose whole content is a discrete Harnack inequality for the increments of the inverse.
- **Zone extension beyond $\log 2$.** Unconditional Weil positivity is classically known exactly up to support width $\log 2$ (Yoshida 1992; Bombieri 2000). T91's target inequality — for $a \in (\log 2, a^*]$ and $\|f\| = 1$, $(P_{\text{pole}} + A_{\text{arch}})(f) \geq \sqrt{2} \log 2 \cdot h_f(\log 2)$ — is typed as *provable-shaped* and would, if proved, be a standalone classical result: a zone extension past Bombieri's boundary. Its typing is explicit and unflattering: RH $\Rightarrow$ (T), but (T) $\not\Rightarrow$ RH. T92 showed the finite-block route to it is the wrong instrument; that narrows, rather than closes, the target.
- **The matching lemma.** Already proved exact-integer on $[4, 10^6]$ with an explicit margin inside `v541`, and reduced by T77 to a restricted σ_{-1} divisor bound in the Gronwall regime — with the note that Robin's *unconditional* 1983 bound suffices at the measured headroom, so the RH-equivalent sharpening is not needed. Closing the tail (the correlated-cancellation lemma on credit-rich non-coherent atoms) would be a self-contained classical statement.

Final fence

The prime line of TFPT has produced seven machine-checked modules about lattice-native Hecke structure, a finite relative-trace identity, and an exactly closing transport ledger. It has *not* produced evidence for the Riemann Hypothesis, and the one place where RH appears — I5 — appears as an equivalence typing that names the residual difficulty rather than reducing it. The corresponding open gates in the ledger (`ZETA.CENSUS.TO.GL1`, `ZETA.HP.CARRIER`) are untouched by everything in Sections 5–7, and no marker moves anywhere on account of this document.

A Probe index, diary parts T11–T163

Every row is one exploratory probe in `experiments/tfpt-discovery/`; the file column omits the common suffix `_probe.py`. "Checks" is the probe's own assertion count, all green. Verdict strings are the diary's preregistered verdict enums. Rows marked "→ vN" were later consolidated into the corresponding load-bearing module; the promotion, not the probe, is the load-bearing artefact. The index covers the 153 parts T11–T163 that the current diary file records (154 runs — T104 was a double run with two independent arms — and 4075 green checks); the ten earlier probes T1–T10 predate that file and are not indexed here. Series totals as recorded by the diary at T125, the final part of the series: 125 **probes,** 3139/3139 **sandbox checks**, plus v535–v555 promoted; with T126–T163, the first thirty-eight parts of the second phase, the running totals are 163 **probes and** 4267/4267 **checks**. T164 (`sector_change_probe.py`) is running and not yet indexed.

Part	Probe	Contract / finding	Verdict	Checks
T11	`e8_glue_chi4_signed_theta`	the χ_4-signed glue theta welds parts 6 and 7	NO PROMOTION	19/19
T12	`e8_glue_lseries_selfdual`	L-series of the signed glue count; Apéry/$\zeta(3)$ bridge; Poisson self-dual width 2π	NO PROMOTION	16/16
T13	`e8_glue_character_atlas`	glue-character atlas of E_8	NO PROMOTION	11/11
T14	`seam_selfdual_width`	SEAM.SELFDUAL.WIDTH; unimodularity gate	DEFLATED	13/13
T15	`zeta_arith_completion`	ZETA.ARITH.COMPLETION; shell Hamiltonian	PARTIAL	12/12
T16	`zeta_local_euler`	ZETA.LOCAL.EULER; one weight-4 Hecke polynomial	PARTIAL	16/16
T17	`zeta_funceq_duality`	ZETA.FUNCEQ.DUALITY; Poisson involution	PARTIAL	9/9
T18	`zeta_weil_recovery_trace`	ZETA.TRACE.COMPLETENESS; Guinand–Weil balance	BENCHMARK-STANDING	17/17
T19	`prime_thinning_seam_truncation`	PRIMON.SEAM.TRUNCATION; thinning vs. the finite tower	MIXED	19/19
T20	`seam_archimedean_kernel`	archimedean kernel from the seam density of states	KILLED-AS-NAIVE	25/25
T21	`prime_prediction_mechanics`	four prediction channels quantified	MAPPED	19/19
T22	`seam_halfcut_dilation_arch`	non-DOS successor of the T20 kill	PARTIAL (K1, K2 pass; K3 fails)	25/25
T23	`seam_boost_arch_dictionary`	arch dictionary via the seam boost	DEAD	23/23
T24	`seam_truncated_trace_rvm`	truncated trace / Riemann–von Mangoldt route	DEAD	25/25
T25	`seam_tate_phase`	scattering phase as the last observable	DEAD (arch route closed)	29/29
T26	`shell_multiplicity_one_quotient`	shell quotient towards multiplicity one	PARTIAL	18/18
T27	`e8_pneighbor_hecke`	Kneser p-neighbours carry the Eisenstein half	PARTIAL → v535	27/27
T28	`e8_kneser_transport_holonomy`	cusp coefficient from mod-3 lattice geometry	TRANSPORT-VISIBLE	22/22
T29	`primon_transport_category`	primon monoid relations	RELATIONS-VERIFIED	16/16
T30	`e8_transport_cusp_uniformity`	uniformity of the T28 reading at $p = 5, 7$	BROKEN (reading); transport holds	32/32

Part	Probe	Contract / finding	Verdict	Checks
T31	`e8_neighbor_operator_decomposition`	cusp extraction is a rule $\nu_p = a\,\mathrm{Id} + b\,T_p$	REPAIRED-BY-DICTIONARY → v535	49/49
T32	`census_newform_recovery`	census redundancy is 2-adic oldform structure	REDUNDANCY-IS-OLDFORM → v535	24/24
T33	`e8_nu_rule_many_primes`	hardening of the ν-rule to $p \leq 100$; Eichler finding	HARDENED → v536	26/26
T34	`compiler_functor_theta`	project ZETA.COMPILER.FUNCTOR founded	FOUNDED-WITH-ANCHOR	13/13
T35	`rankin_selberg_functor`	Rankin–Selberg translates the abelian shadow	RS-TRANSLATES-EISENSTEIN	13/13
T36	`e8_lambda_charsum_evaluator`	cusp remainder isolated geometrically	TWO-SIDED → v536	29/29
T37	`e8_salie_signed_cusp`	signed cusp channel, scalable	SCALABLE-AND-SIGNED	37/37
T38	`cuspidal_bridge`	Shimura preimage of f_8 inside the compiler monoid	BRIDGE-FOUND → v537	13/13
T39	`zeta_center_atlas`	the abelian drop	ABELIAN-DROP-CLOSED	13/13
T40	`weil_recovery_compiler_sector`	Weil-recovery sector terrain	TERRAIN-MAPPED	24/24
T41	`functor_wiring_halfintegral`	the half-integral bridge is Hecke-equivariant	FUNCTOR-WIRED → v537	13/13
T42	`e8_local_densities`	local densities closed	CLOSED → v536	25/25
T43	`kohnen_plus_waldspurger`	Kohnen plus space	PARTIAL	12/12
T44	`kohnen_operator_projection`	central-value route via Kohnen plus	OUTSIDE-KOHNEN-SCOPE → v537	17/17
T45	`baruch_mao_metaplectic`	central-value family as $b(d)^2$; R constant	CENTRAL-VALUES-WIRED → v537	13/13
T46	`stage4_prolate_prototype`	stage-4 prolate prototype	IMPL-OPEN	16/16
T47	`stage4_prolate_uv_dirac`	pointwise UV/Dirac prototype	UV-NO-SIGNAL	17/17
T48	`lambda_scale_uniqueness`	does the compiler fix the Λ scale?	SCALE-TORSOR (claim killed)	17/17
T49	`rankin_positivity_miniature`	the Deligne/Rankin mechanism runs in-suite	MINIATURE-RUNS	28/28
T50	`half_integral_selfconvolution`	the central-value family aggregates exactly	FAMILY-AGGREGATED	17/17
T51	`waldspurger_family_kernel`	first non-collapsing infinite carrier candidate	INFINITE-CARRIER-CANDIDATE	22/22
T52	`hecke_orbit_transfer`	path space of Hecke words; fibre determinants	TRANSFER-REALIZED	16/16
T53	`relative_trace_compiler`	v535/v536/v537 are three projections of one RTF identity	ONE-FORMULA → v538	16/16
T54	`waldspurger_canonical_measure`	the "canonical discriminant measure" claim	MARGINAL-WEIGHT (killed)	26/26
T55	`rtf_period_pairing`	canonical distributional limit of the finite RTF	DISTRIBUTIONAL-RTF-TYPED → v539	36/36
T56	`rtf_pairing_sharpening`	all T55 identifications to sub-percent	SHARPENED	27/27
T57	`infinite_rtf_packing`	both infinity directions in one double series $Z(s,w)$	PACKED	24/24
T58	`zeta_rtf_xi_residue`	GL(1) shadows of $Z(s,w)$	GL1-SHADOW-CLASSICAL (route A closed)	24/24
T59	`weil_gns_identification`	residual factor in closed form	TRANSFORM-REQUIRED	19/19
T60	`plancherel_descent`	Sato–Tate moment contraction	PARTIAL	21/21
T61	`st_isotype_gl1_core`	the trivial Sato–Tate isotype is a GL(1) core	GL1-CORE-EXACT → v539	24/24
T62	`core_isolation`	d-twist mixing vanishes fibrewise	CORE-ISOLATED → v539	24/24

Part	Probe	Contract / finding	Verdict	Checks
T63	`weil_core_completion`	Plancherel correction irreducibly non-automorphic	EXTRA-TERMS → v539	33/33
T64	`rtf_stabilization`	$\det_2$ absorbs obstruction 2; the minus stays	HALF-STABLE	34/34
T65	`pi_digits_prime_distribution`	prime distribution in the digits of π	PI-NULL	16/16
T66	`pi_prime_four_level`	the π/prime question on four levels	FOUR-LEVEL-NULL	23/23
T67	`amplitude_dirac_sqrt`	Dirac square root of the amplitude plane	DIRAC-SQRT-EXACT → v540	27/27
T68	`geometric_polarization`	geometric polarisation $b = N_+ - N_-$	RESHUFFLE-ONLY → v540	34/34
T69	`mixed_channel_full_weight`	even-k deletion is a determinant invariant	DELETION-UNIVERSAL → v540	33/33
T70	`linear_measure_weil`	positive linear Θ carrier; pure plus combination	LINEAR-PLUS-COMBINATION → v540	29/29
T71	`transport_terrain`	FE of the linear family exactly anchored	TERRAIN-MAPPED → v540	40/40
T72	`cone_enlargement`	the cone library saturates at 5/24	COMPLEMENTARY-PAIR → v540	34/34
T73	`continuous_cone`	per-direction hybrid cones in the continuum	HYBRID-GAINS	28/28
T74	`door_a_spectral_polarization`	the spectral side is provably sign-blind	VACUUM-STRUCTURE	32/32
T75	`door_b_lambda_transport`	the gap functional λ^* has its own structure	LAMBDA-STRUCTURED	28/28
T76	`hybrid_universality`	the recipe certifies 91/91 adversarial Weil directions	RECIPE-UNIVERSAL-ON-BATTERY	22/22
T77	`matching_lemma_structure`	the matching lemma reduces to a restricted divisor bound	LEMMA-CLASSICAL-SHAPED	21/21
T78	`lemma_window_proof`	matching lemma exact-integer on $[4, 10^6]$	WINDOW-PROVED → v541	25/25
T79	`transport_ledger`	the transport ledger closes; I5 named and typed	LEDGER-CLOSES → v541	22/22
T80	`tail_correlation_lemma`	character-exact signed envelope; tail to $N^* \approx 10^{23}$	RESERVE-PARTIAL → v541	29/29
T81	`recipe_coherent_avoidance`	reachability theorem: coherent targets need coherent m	AVOIDANCE-FAILS → v541	31/31
T82	`heat_arch`	Δ_{arch} is the internal Γ difference	ARCH-INTERNAL → v541	22/22
T83	`fe_symmetrization`	symmetrisation absorbed; the wall is FE-transversal	INVARIANT-NULL	27/27
T84	`gaussian_lift`	Grössencharacter phases give $L(1, \lambda_k)$ convergence	LIFT-WORKS-UNANCHORED	29/29
T85	`lambda_equivariant_design`	the λ design closes the coherent class	LEMMA-CLOSES-LAMBDA → v541	24/24
T86	`correlated_cancellation`	the Q-pairing involution closes the non-coherent tail	LEMMA-FULLY-CLOSED	29/29
T87	`i5_connes_dictionary`	I5 one-family form vs. the semi-local Connes structure	DICTIONARY-EXACT-CORE	22/22
T88	`i5_tightness`	eight near-vertical tight curves; Γ-side spacing	TIGHT-SET-PARAMETRIZED	27/27
T89	`sonin_crossing`	window compression is Bombieri's classical object	CROSSING-MAPPED	19/19
T90	`residual_dissection`	the residual vectors are explicit Gaussian modes	CORE-DISSECTED	17/17
T91	`thin_band_analytic`	the band is the one-atom zone; atom rescue	BAND-PARTIAL (3/4 closed)	19/19
T92	`zone_extension_proof`	the finite-block route is the wrong instrument	T-SKELETON	36/36
T93	`band_reconciliation`	third independent implementation of the T89/T91 band map	MIXED (map recalibrated)	41/41
T94	`prime_blind_demo`	753/753 primes of an unseen window from lattice counting	BLIND-100	16/16
T95	`continuum_extension`	C1 chain proved; band top behaves as a positivity edge	T-CONTINUUM-NUMERIC	28/28
T96	`relay_test`	the T95 edge was a map artefact; the relay is real	EDGE-ARTIFACT + RELAY-CONFIRMED	21/21
T97	`relay_induction`	the induction step has proof shape; one cross-block lemma	ALIGNMENT-ONLY	105/105

Part	Probe	Contract / finding	Verdict	Checks
T98	`cross_lemma`	the $\sqrt{\lambda_0}$ law is Douglas range inclusion (circular)	LAW-CONFIRMED-MECHANISM-OPEN	44/44
T99	`dk_recursion`	exact mirror-parity selection rule; recursive inequality	DECAY-LAW-FOUND	23/23
T100	`bulk_remainder`	closure 11/24 → 24/24 samples; the drift was a grid artefact	REMAINDER-CLOSES-ZONES	27/27
T101	`k_asymptotics`	16 zones resolved; primitives flat, instrument loses the race	CROWDING-TRENDS	31/31
T102	`arithmetic_bound`	the onset is manufactured by the Schur dressing; C/g was a proxy	MECHANISM-IDENTIFIED	42/42
T103	`instrument`	slope halves; closure map 7/64 → 44/64	INSTRUMENT-IMPROVED	29/29
T104	`schur_profile_bound`	arm B: naive margin route dead, threshold split closes 16/16	CHAIN-PARTIAL	21/21
T104	`schur_profile_chain`	arm A: independent twin of the same contract	CHAIN-PARTIAL	47/47
T105	`bare_wing_bound`	bare bound certified in closed form; avoidance is a theorem	ONE-OF-TWO	28/28
T106	`friedrichs_angle`	parity split of the Weil pole; hardness in the odd channel	DENSITY-MAPPED	32/32
T107	`odd_channel_closure`	Sherman–Morrison/Woodbury reduction to $r = \kappa/\varepsilon$	SCALAR-TRACTABLE	30/30
T108	`ratio_certificate`	ε is an exact $Q\|_{\mathrm{odd}}$ energy; (R) on two scalars; first non-vacuous margin chain	EPSILON-IDENTITY	44/44
T109	`boundary_decay`	cancellation, not decay; both scalars certified (graded cap + residual certificate); one strict-margin input left	BOUNDARY-CERTIFIED	29/29
T110	`margin_propagation`	the induction circle closes on the measured zones (certified base case + graded Loewner minorant; atom entry free); three gaps named	MARGIN-PROPAGATES	28/28
T111	`deep_zone_stress`	the crossing measured, not extrapolated ($n^* \approx 462$, upper bound); three walls (margin / twin-prime ladder / requirement); mechanism never fails, 117/117	CROSSING-CONFIRMED	23/23
T112	`adaptive_scaling`	gap-coupled frame: ladder wall and ω depth-dependence fall structurally; margin wall frame-invariant (-0.974); trade to [P1]/[P2], prime-gap inputs disclosed	SCALING-PARTIAL	20/20
T113	`limit_operator`	the wall is currency-invariant (-1.168 in all five currencies) but measures the discretisation: no continuum gap, floor a 10^{-7}–10^{-4} cancellation, need109 an artifact margin; new balance [P1]–[P4]	SUBSTANCE-CONFIRMED	27/27
T114	`margin_free_step`	the margin-free step: Albert/pseudo-inverse Schur certifies 27/27 steps (11 past the old wall, to $n = 1331$) and all 7 torn zones; the wall was an $O(1)$ numerator over an artifact floor; (R) grid-bound, deleted ([P5]); new two-object core	WALL-DISSOLVES	22/22
T115	`schur_transport`	transport certifies only mild refinement (split at $\rho^* \approx 1.8$; on nested ladders $\lambda_{\min}(S)$ itself falls like $\rho^{-1.7}$), but the two-scale compression breaks the cap: margin-free step at $n = 155\,921$, ten-step chains, stopper always cost; remaining list three points, one inequality (Szegő–Levinson)	TRANSPORT-BLOCKED	26/26
T116	`boundary_formulation`	the induction as a boundary process: the (Σ, r, σ) state carries the global pole exactly and marches $169\,236$ prepends to 1.35×10^6 cells at flat cost (cost-geometry demo, not a Weil certificate); the prime comb refuses compression (budget $\sim n^{2.17}$); $\varepsilon \approx 8.34\, D^{1.790} \alpha^{-6.04}$ with factor-120 jumps — a Galerkin error under Aubin–Nitsche duality	RICCATI-PARTIAL	33/33

Part	Probe	Contract / finding	Verdict	Checks
T117	`epsilon_theorem`	the one inequality theorem-shaped: four exact Galerkin identities ($\varepsilon > 0 \iff \tilde{t} \notin T(V_c)$); a certified two-level lower bound on 19/19 pairs at $\theta' = 1.74$ against $\theta = 1.76$ (no power of D lost); exact jump formulas (Levinson bordering, rank-(k_0+1) Woodbury) — entries *raise* ε, the factor-120 falls were a sweep artefact; three named analytic lemmas remain, two of them constants	THEOREM-SHAPED	23/23
T118	`symbol_lemmas`	the three lemmas against their classical addresses: saturation is an *identity* here ($\varepsilon_c - \varepsilon_f = y^\mathsf{T} S y$, $\beta \in [0.252, 0.336]$); the S-symbol is a weighted *harmonic* aliasing mean, vacuous on the comb — rescued by the CBS shift onto the oscillation Gram, whose *arithmetic* mean suppresses the comb quadratically ($\inf \sigma_z > 0$ on 7/14 windows; on the 15 680-point FFT lever the floor rises logarithmically and crosses zero on 3/5 zones — under-resolved, not obstructed); the corner exponent drifts (log is the boundary of all powers) and the chain is repaired on $\|y\|^2$; one analytic statement remains — (H2), $\|y\|^2 \geq c\, D^{1.75}$	TWO-OF-THREE	36/36
T119	`oscillation_mass`	the arithmetic half closes as a theorem: the comb symbol in closed form (one cosine per prime-power atom; parity rule under σ_z), the atom-count bound $\lvert\sigma_z^{\mathrm{comb}}\rvert \leq \Xi(\alpha)$ uniform in D, the archimedean growth *identity* $\inf \sigma_z^{\mathrm{arch}} = \log(1/D) + B$ with universal $B = -1.0474$ — hence $\inf \sigma_z > 0$ for all $D < D_0(\alpha) = e^{-(\Xi(\alpha)+B)}$, deep-verified ($M$ up to 65 536, matrix-free); the large sieve certifies one window's section *without* pointwise positivity (+0.396 where $\inf \sigma_z = -0.349$) and dies structurally for $\alpha \geq 2.5$ (a moment problem); the energy route to (H2) is proven empty (genuinely new content); and $\kappa_{\mathrm{end}} = 1/(1+R)$ *exactly* (1.1×10^{-16}) — one discrete Harnack inequality remains	ARITHMETIC-DONE	27/27
T120	`harnack`	the Harnack core is proof-shaped: a_j and w_j are the odd/even half of *one* increment sequence, offset by a single fine cell, so one sign of the corner increments gives $\lvert R-1\rvert \leq \sum \lvert v_{2j+1} - v_{2j}\rvert / \sum \lvert v_{2j}\rvert \leq 0.04745$ *unconditionally* (C1; the per-cell inequality is provably *false*, $\lvert a_j/w_j\rvert$ up to 4.8×10^3); $R \in [0.9098, 1.1011]$ on 3915 rows, $\kappa_{\mathrm{end}} \geq 0.480827$ for $n_C \geq 8$; two structural routes refuted (T^{-1} corner positive but not TP2; the discrete maximum principle false — u oscillates in the bulk); the α-audit makes (D2) total ($\alpha_o = u_k/2$ by construction, D_0 holds on 3/1492 zones, $\alpha^* = 0.693$ below the first admissible handover); γ gains an exact symbol identity but $1-\gamma^2$ falls with α (slope -0.297 ± 0.013); theorem V4: 18 links, defect list $3 \to 4$	HARNACK-EXPLAINED	21/21

Part	Probe	Contract / finding	Verdict	Checks
T121	`wide_window`	the σ_z route survives the real ladder as a *section* statement and dies as a symbol statement: $\lambda_{\min}(T_h(\sigma_z)) > 0$ on 16/16 frame windows ($n = 7 \ldots 283\,303$, α up to 6.277, h up to 107 790; 8 with $\inf \sigma_z < 0$, the section at 0.64–3.23 $\times$ $\lvert \inf \sigma_z \rvert$ positive), the Christoffel/Fejér per-dip budget certified on 8/16 up to $\alpha = 3.446$ (each dip 0.156–0.720 of one Fejér width; the dip count grows like $e^{2.41\alpha}$, a fit); the net balance SUPPLY$/\varepsilon_c \sim D^{-0.18}\alpha^{-1.57}$ — one poly-log, uniform in D, split exactly (2.8×10^{-14}) into the Rayleigh ($\alpha^{-1.02}$) and CBS ($\alpha^{-0.51}$) steps, nothing in γ; link (5) refuted on 42/42 (ratio 0.357–0.916 — the odd sector's Hankel term); one-sign 24/24 at $n_C = h/16$ (only 21/24 at $h/8$; certificates restated, free since $\delta \sim n_C^{+0.29}$); the FH comparison inconclusive at reachable h (drift $-0.008 \ldots +2.934$ against an allowed -0.120); $\delta = 0.0126$–0.1331 over 360 rows, $\kappa_{\text{end}} \geq 0.4688$; theorem V5: 19 links, the defect count stays 4 but every entry changes kind	WIDE-RESTRUCTURED	21/21
T122	`net_balance_repair`	the Hankel term is the reflection half of an isometry, not an error term: $A_z = W^{\mathsf{T}} \text{Toeplitz}_M(c) W$ with $W = BZ$, $W^{\mathsf{T}} W = I$ exact (the compressed Hankel is half a second difference of the lag sequence, closed form to 7.0×10^{-16}; norm ratio 0.123–0.781, $O(1)$ — Z suppresses both halves equally; the Weyl floor positive on only 12/36); the certified band bound $\lambda_{\min}(A_z) \geq \text{thr} - \lambda_{\max}(G)$ non-vacuous on 36/36 rows to $\alpha = 6.31$ (the T121 Christoffel budget tore at 3.45; recovery 0.429–0.995; caveat: the optimum sits at $\text{supp}(g) \approx 1$); the structural Rayleigh bound is *sharp* ($q_{\text{str}} = 1.00$–1.03, drift $\alpha^{-0.002}$), removing the whole $r_{\text{ray}} \sim \alpha^{+0.83}$ slack; the certified balance halves the deficit, $S_2/\varepsilon_c \sim D^{-0.003}\alpha^{-0.729}$, exactly D-uniform, and with the measured CBS coupling ($\kappa_y = 0.0067$–0.5311 against γ^2) it reads $\alpha^{-0.113}$ against the chain's own ceiling $\alpha^{-0.116}$ (residual decomposition closes to 2.9×10^{-16}); theorem V6: the count stays 4, link (5) retired by an identity, (F3) the band margin (0.024–$0.363 \sim D^{-0.042}\alpha^{-0.525}$) and (F4) a structural CBS estimate absorb (E3*)/(E4*)	NET-IMPROVED	20/20

Part	Probe	Contract / finding	Verdict	Checks
T123	`structural_cbs`	the coupling resists structurally: the coarse block is the other half of the T122 construction ($V = BP$ an isometry, $V^\mathsf{T} V = I$, $V^\mathsf{T} W = 0$ exact — A_c, B_x, A_z three compressions of one window Toeplitz form, identities to 3.4×10^{-15}); the coupling is half a *first* difference of the lag sequence, in symbol form $-\frac{i}{2}\sin(\varphi/2)[\sigma(\varphi/2) - \sigma(\varphi/2 + \pi)]$ — the aliasing difference, vanishing where the coarse space lives, which explains the factor 2–109 between $\kappa_y = 0.0067$–0.4622 and $\gamma^2 = 0.7248$–0.9467; both certified routes to κ_y need the coarse form from below (band-split CS is correct 24/24 and non-vacuous 0/24, costing $\mathrm{cond}(A_c)$ up to 3.1×10^7; the Schur route yields (5R*) $\lambda_{\min}(E_z) = +0.303 \ldots + 5.017 > 0$ on every row at slack 1.0007–1.007 but needs $E_c \succ 0$, true on 0/24) — the near-null direction of the window form is smooth, and its eigenvalue is a cancellation no pointwise symbol minorant sees (envelope margin 8.4×10^2–$4.0 \times 10^5 \times \lambda_{\min}(A_c)$, rising $\alpha^{+4.202}$); the band margin (F3) reduces to the band profile of Wy at a cost of 0.7% (dyadic split exact to 1.1×10^{-15}, overshoot 1.20–3.51); the certified balance $S_6/\varepsilon_c \sim D^{-0.069}\alpha^{-0.544}$ against the ceiling $\alpha^{-0.044}$ — gap $\alpha^{+0.500}$, all of it the coupling (two-term residual split closes to 2.8×10^{-17}) — and the gap is identified as the ε_f lid itself, 4–8 recursion levels deep; theorem V7: the count stays 4, restructured — (F1)/(F2) the Harnack pair, (F3) one band-profile lemma, (F4) coarse positivity = the recursion base case	CBS-RESISTS	20/20
T124	`multilevel_telescope`	the telescope carries: the ladder is one window form on nested spaces ($P^\mathsf{T} A^{(\ell+1)} P = A^{(\ell)}$ exact to 2.4×10^{-14} — the T122/T123 two-level system is one rung), every rung is the saturation identity and the Galerkin energy of the level correction ($\delta_\ell = \|u_{\ell+1} - Pu_\ell\|_A^2$ to 3.7×10^{-8}; telescope $\sum \delta_\ell = \varepsilon_0 - \varepsilon_L$ to 9.2×10^{-10}; top rung 0.27–0.88, geometric decay, $L = 5$ ceiling 0.9911–0.9987); the rung is a *maximum* (dual residual form $\delta_\ell = \hat{s}^\mathsf{T} S^{-1}\hat{s}$; $r_\ell = W\hat{s}$ by Galerkin), so it wants S from *above*: $S \preceq A_z$ needs only $A_c \succeq 0$ (Haynsworth), $A_z \preceq U = Z^\mathsf{T} T_M(\mathrm{up})Z$ is Parseval against the certified majorant — (8R) $\delta_\ell \geq \hat{s}^\mathsf{T} U^{-1}\hat{s}$ valid on 400/400 rungs (80 chains, 20 zones), $\mathrm{cb}/\delta = 0.4739$–$0.9938$ at drift $\alpha^{-0.080}$, 7× weaker than T123's $\alpha^{-0.544}$ ($\lambda_{\min}(E_c) < 0$ on 399/400 is irrelevant — (8R) never forms E_c; the Levinson bordering route falls: refinement is not a bordering, $\mathrm{tail}/\delta = 13$–$863$); the near-null direction stays smooth and self-similar (prolonged Rayleigh quotient exactly $\lambda_{\min}(A^{(\ell)})$, overlap 0.944–1.000), and (8R) needs only the *sign* of A_c — the previous rung's conclusion: the non-circular induction runs coarse → fine, base case $\varepsilon_L \geq 0$; balance $\sum \mathrm{cb}/\varepsilon_0 = 0.6404$–$0.9729$ against the ceiling, two-level $\mathrm{cb}/\varepsilon_c \sim D^{-0.090}\alpha^{-0.100}$ against T123's $\alpha^{-0.544}$ — $+0.444$ of the $\alpha^{+0.500}$ gap; theorem V8: (F1)/(F2) the Harnack pair, (F3) closed, (F4) *collapsed* to a sign, (F5) new, a bookkeeping convention — the solution-free rung provably fails	TELESCOPE-CARRIES	28/28

Part	Probe	Contract / finding	Verdict	Checks
T125	`grand_assembly`	the series finale — the chain composes: all five stages ([A] base Cholesky, [B] telescope with (8R) rungs, [C] ε lower bound, [D] κ_{end}/parity, [E] margin-free handover) complete on 52/52 ladder zones plus 10 handover-only zones to $n = 1331$, and ladder 1 is a *literally composing* run — 30 consecutive atoms $n = 2 \ldots 79$ on ONE common resolution $D = 3.472446 \times 10^{-3}$, $M_n(k) = M_o(k+1)$ as an integer identity on 29/29 pairs, the incoming atom block bit-exactly zero, so $X = Q_{\text{old}}$ and the next zone's leading block is that X to 2.9×10^{-14} (T114's [P2] seam constructively *circumvented*: 0/84 gap ratios dyadic, so one frame is chosen for the whole run, at the price of its length); the load-bearing finding: the weakest stage of the whole mounting is [D] at WINDOW-CERT, but the SPINE [A][B][C][E] is CHOLESKY-CERT on 52/52 — *the Harnack pair (F1)/(F2) no longer carries*, it lives only in the independent second route (iv); (8R) on 88/88 rungs at cb/δ = 0.9071–0.9971 (primal form above δ on 88/88 — an upper bound, never a supply), base-case equivalence $\operatorname{sign} \lambda_{\min}(Q^{(L)}) = \operatorname{sign} \varepsilon$ on 52/52 at head-room 4.4×10^5–5.6×10^{10}, ε lower bound on 52/52 at 0.9140–0.9929 of the L-level ceiling and 3.2×10^1–8.7×10^{10} over the declared fp bound, Albert Schur 3.0×10^{10}–4.6×10^{13} above its floor on 62/62 steps while the norm surrogate is negative throughout; 430 completed Choleskys, 440 identities; theorem V-final printed with line-by-line attribution (25 lines: 10 identity, 9 Cholesky, 3 window — all in (iv) —, 1 hypothesis = (C-F5)) and a five-point "what is not claimed"; series balance: kill list 24 routes over 18 parts (4 of them one cause), cascade 7 stations, certification degree 90.3% chain / **96.2**% spine identity-or-Cholesky, RH distance in six checkable points; new defect (F6) uniformity; extraction reassessment: stronger on 3/5 criteria than the matching-lemma candidate	ASSEMBLY-GREEN	34/34

Part	Probe	Contract / finding	Verdict	Checks
T126	`uniformity_seams`	the first part of phase 2 ("the full proof") — the seam architecture is finished: the partition question is decided *exactly* by DP on the real gap table (9 700 zones, $n \le 10^5$) and refuted at T115's $\rho^* = 1.83$ (dies at $n = 127$, the twin-after-a-large-gap pattern; true requirement $\rho_{\mathrm{crit}} = 2.0258$, essentially the single quotient $g(125)/g(127)$; refinement-only infeasible at any ratio ≤ 64; coarsening fails at the destination grid, and ladders make it worse); the decisive new measurement: at fixed ratio 2.0256 the certificate survives on only 12/22 zones — the seam dies at the *zone*, not the ratio, so no threshold closes the programme; what carries is the *free-resolution route*: the running minimum $D_k = \min(\mathrm{cap}_k, D_{k-1})$ is monotone by construction, re-gridding at only 764/9 700 boundaries (7.88%, record-small gaps), all 12 affordable real seams certified (11 in one re-gridding, $n = 31$ via a three-stage ladder, median retention 0.578), a full seam end to end at $\rho = 2.251 \ge 2.0256$; the uniformity audit (35 rungs, 14 zones) gives every spine constant a status — U1 self-certifying, U2 one Bernstein bound, U3 measured-flat but *directional* (alignment 0.975), U4 a compute wall, U5 *falls* (the load-bearing gap), U6 resolved; the continuum step closes via the exact fractional-Dirichlet identity (86.1% negative off-diagonals; consistency $D^{1.798}/D^{1.802}$, Richardson limit $2.1256 \times 10^{-4} > 0$); the full-proof map has 16 points, 5 done/classical, 9 conditional, exactly 2 genuinely new — the (U3) direction lemma and the (U5) zone-uniform seam floor at ratio ≤ 2.026	SEAMS-CERTIFIED	31/31

Part	Probe	Contract / finding	Verdict	Checks
T127	`two_inequalities`	both remaining inequalities are proof-shaped, and neither is the statement T126 named: the fixed-ratio sweep reproduces (12/22, T126's list — the *certificate* dies, not the positivity: $\lambda_{\min}(S_f) > 0$ on all 22); of 29 candidate separating variables 5 separate perfectly (22/22), all the same *geometric* quantity — the border protrusion $s_{l,r}$, i.e. what the fine border block protrudes past the coarse block's inner edge (corr $+0.997$ with $\tau_{\rm dn}$; every arithmetic candidate fails, best 0.727); U5 as posed *refuted*, replaced by **U5-band**: zone-uniform for every $\rho \leq 1.49531$, at least one zone dies for $\rho \geq 1.49766$ (bisected to 2.3×10^{-3}, binding $n = 19$), covering 99.26% of real seams — the remaining 8 are an enumerated list of next-atom-distance ≤ 2 pairs (Mersenne $3, 7, 31, 127, 8191$; Fermat-like $4, 16, 256, 65536, 131071$; twin $101, 103$; Mihăilescu 2004 cited, not used); the band buys $\tau_{\rm dn} \approx 1$, not small η ($\eta = \eta_{\rm cons} + \eta_{\rm proj}$ exact; max $\lvert\eta\rvert/\lambda = 1.146$ at the tightest ratio; floor $c \geq 0.0090$ over 307 in-band transports); the sharpest finding: the near-identity census (43 deep zones, $\rho = 1.00140$) fails *once* ($n = 479$, $\lvert\eta\rvert/\lambda = 1.472$) — the band is a *proxy* for the border geometry, repaired by a three-stage ladder (retention 0.5027), and the risk indicator $\tau_{\rm dn}/\tau_{\rm flat}$ (Spearman $+0.749$, the failure its argmax) sees it before any factorisation; U3 defuses via the harmonic-mean identity ${\rm cb}/\delta = 1/\sum w_i/\mu_i$ ($\hat{s}$-mass on $\{\mu \geq \frac{1}{2}\} \geq 0.9665$ over 42 rungs, so a *crude* floor $\mu_{\min} \geq 1.62 \times 10^{-2}$ gives ${\rm cb}/\delta \geq \frac{1}{4}$); "$\hat{s}$ avoids the bad direction" refuted — flatness is generic, exactly one direction type fails (a boundary layer at the window edge, concentration $3.1\times$ vs $0.03\times$ for $\hat{s}$); map V2: 16 points, 10 carried, 2 conditional-classical, the 2 genuinely new reformulated as (T)/(E)/(M) plus the enumeration (L) — three inequalities of the same craft as the proven links, residual risk in the word *uniform*	BOTH-SHAPED	28/28

Part	**Probe**	**Contract / finding**	**Verdict**	**Checks**
T128	`teml`	the four-point list of map V2, worked cheapest first, and three of the four stand at their preregistered bars: **(L)** is *derived* and closed — the out-of-band class $\rho > 1.49531$ has exactly 8 seams up to $n = 300\,000$, all at next-atom distance 1 with identical border geometry ($\mathrm{gc}_c = 2$, $\mathrm{gc}_f = 4$); $n = 7, 16$ certified in one step (retention 0.179/0.156), $n = 31$ through a two-rung ladder (0.091), and 127/256/8191/65536/131071 BUDGET-OPEN at $h_f = 2\,476 \ldots 6.2 \times 10^6$ (1.7×–4119× over the cap; each lacks exactly *one* Cholesky, nothing conceptual; Mihăilescu 2004 cited as the address, not used); **(T)** from predictor to bound — the exact identity $\tau_{\mathrm{dn}} = \|y\|^2 - m_{\mathrm{prot}} + m_{\mathrm{fill}} - V_{\mathrm{bord}}$ (9.0×10^{-13} against an independent full projection, 126 transports), $\tau_{\mathrm{dn}} \geq 1 - m_{\mathrm{prot}} - V_{\mathrm{bord}}$ with explicit majorants valid 126/126 (exact bound positive 126/126, fully explicit 125/126; slack median 0.275, the oscillation majorant the loose part at 7×) — *but* the protrusion concentration $\kappa = m_{\mathrm{prot}}\,\mathrm{gc}_f/(t_l + t_r) = 0.086$–$2.071$ misses the preregistered bar $\kappa \leq 2$ on 1/120 by 3.6%, *systematically* in ρ (group maxima 1.889/1.965/2.071 at $\rho = 1.001/1.250/1.495$); the bar is not moved, the next hypothesis is $\kappa \leq 2 + C(\rho - 1)$; and κ splits by side (1.61–2.07 right against 0.09–0.25 left) — (T) is an *eigenvector* statement; **(E)+(T)** jointly: the ℓ_2 route certifies 0/12 (median 55× overshoot), the Q-inner-product chain $\sqrt{\lambda_f} \geq \sqrt{\lambda_c \tau^* - \lvert\eta_{\mathrm{cons}}\rvert} - \sqrt{e_d}$ ($e_d/\lambda_c = 0.041$–0.154) certifies 9/9 in-band seams at floor ≥ 0.0850, 11/12 overall; the one tear ($n = 31$, already on the (L) list) is η-limited ($\lvert\eta_{\mathrm{cons}}\rvert/\lambda_c = 0.392 > \tau^* = 0.363$ — no sharper (T) would have helped, nor a smaller ratio), repaired by the two-rung ladder, every rung closing the chain by itself; **(M)** earned: the boundary-layer exclusion is a *proof* from the closed form (adjacent-component ratio in $[e^{-D/2}, 1]$, outer-k-cell share $\leq kD(1 + \cosh\alpha)/\sinh\alpha$, separation $\geq 84\times$), $\mu_{\min}(S, A_z) = 1 - \gamma^2$ exact (5.9×10^{-15}), $U \preceq (1 + \varepsilon_{\mathrm{env}})A_z$ with $\varepsilon_{\mathrm{env}} \leq 0.038$ Cholesky-certified — constructed floor 0.188 against the 1.62×10^{-2} demanded, an 11.6× margin on all 31 rungs; the comb residual stays measured but is measurably no layer (half its mass needs 12–39% of the half-window); map V3: 21 points — 10 done, 3 certified on real objects, 3 conditional-classical, 5 genuinely open (M9 the κ bar, M17 the pencil mass, M18 the comb, M19 “for all”, M21 the RH address)	THREE-OF-FOUR	27/27

Part	Probe	Contract / finding	Verdict	Checks
T129	`kappa_deep_seams`	the most productive break of the phase: the fitted law falls, the theorem underneath stands — $C^* = 0.3506$ frozen from T128's three calibration groups (a fit, labelled a fit, not refitted; calibration itself 0/105, not the test); on 331 *new* transports (14 ratios 1.05–4.00, none in the calibration set; $h_f = 29$–1450) the law $\kappa \le 2 + C^*(\rho - 1)$ is violated **once** ($n = 7$ at $\rho = 3.00$, slack -0.0334; the bare unmoved bar 2 would fall 78 times), the per-side law on κ_r 1/293, the side asymmetry reproduces ($\kappa_r = 1.70$–2.91 vs $\kappa_l = 0.05$–0.21); the bar is not moved; **the structural ground is solved with identities**: $\sum(1 - g_i) = t_l + t_r$ (7.1×10^{-13}) makes the flat border vector $\kappa = 1$ *exactly* (T128's bar 2 was twice the flat value), the profile identity $p_{N-1} = 2 - p_0 + 2E$ (6.7×10^{-16}) makes a linear density profile $\kappa = 2$ *exactly* (everything above is curvature; $2E > p_0 \iff p_{N-1} > 2$, 114/436 with no exception), and the chain $\kappa \le 2 - p_0 + \frac{N-2}{N} \sum \lvert \Delta p - \overline{\Delta p} \rvert + (p_{\max} - p_{N-1})$ is a *theorem* with an explicit constant (Hölder/telescope/Abel), valid on all 436 (slack median 0.344; last term vanishes 392/436); depth block: 14 graded transports at $h_f = 1656$–$17\,669$ (12× the cap, matrix-free, $m/h = 0.064$–0.512), 0 violations — the residual risk is a ratio effect, not a resolution effect; **the deep seams**: machinery verified (two-scale assembly $= J^{\mathsf{T}} Q J$ to 2.7×10^{-16}, $X = Q_{\text{old}}$ to 2.4×10^{-17}), $n = 127$, 256 compressed to $m/h = 0.507$, 0.255 carry *complete* graded seam certificates (X PD, Schur floors above Wilkinson, bracket positive, retentions 0.307/0.417), the floor survives 6 further coarsening rungs — honestly downgraded to GRADED-CERTIFIED: Rayleigh–Ritz points the wrong way (1.53–5.18×, no exception) and a measured 8% false-positive rate (4/48 pairs graded-positive where uniform is not), declared before the seams; $(L) =$ [3 uniform, 2 graded, 3 budget-open]; **M18**: the Cauchy–Schwarz row bound is vacuous (0/40, reported as the failure it is), the S-procedure operator bound works (40/40 below 1, 0.170–0.774, 1.3–182× above the truth, median 7.8; CERTIFIED-CONDITIONAL on $\theta_1 \sim D^{1.219}$, a fit; saturates at $(kD)^{0.263}$ instead of decaying); map V4: 9 entries — M9 reduced, L2 graded, M18 bounded, M17 untouched, **M19 "for all" and M21 the RH address irreducible**, M22/M23/M24 new and named; promotion: *ready in principle* as a narrow module on the five identity/theorem pieces	KAPPA-WILD	28/28

Part	Probe	Contract / finding	Verdict	Checks
T130	`curvature_bridge`	the two named pieces attacked head on, and exactly one stands — the bridge, not the bound: **the Céa/Strang defect identity** $S_{\text{graded}} - S_{\text{uniform}} = R^{\mathsf{T}} X^{-1} R$ verified to 1.3×10^{-13}, matrix-free (FFT matvec + BPX-preconditioned CG, one graded Cholesky under the cap); the *scalar* form is vacuous (loses 3–28×), the *matrix* form reproduces the directly computed uniform Schur floor on **84 pairs (21 zones) to** 2.5×10^{-11} **relative with zero overshoot** — T129's 8% false positives explained completely (16/84 at bracket level, **0/84** at floor level: they sat in the transport bracket, which the bridge makes unnecessary); **both deep seams carry**: $n = 127 \to \lambda_f = 0.2255$, $n = 256 \to \lambda_f = 0.2614$ ($h_f = 2476/5694$, $1.7\times/3.8\times$ the cap), BRIDGED-CERTIFIED-CONDITIONAL on *one* measured number per grid (the Lanczos floor for $\lambda_{\min}(X_{\text{fine}})$; the Grenander–Szegő route for it *dead*, symbol indefinite, floor $-76 \ldots -27$; conclusion unmoved at $100\times/10^4\times$ smaller floor); **the curvature front**: the TV rewrite $\sum \lvert\Delta p - \overline{\Delta p}\rvert = \mathrm{TV}(P) \le (2n_{\text{run}} - 2)\cdot \text{sag}$ is *exact* (one hump on 83/765, sign purity 765/765), the power profile wins 635/635 ($a = 0.703$–0.999, median 0.856), the T108 Levinson corner identity independently reproduced (24 zones to $M = 1236$, 4.2×10^{-9}) but the pole solve only a proxy (cosine 0.992–0.996, curvature ratio 1.5–5.3) — and the frozen shape band ($a \in [0.734, 0.897]$, $S^* = 1.1926$) **breaks on 13/545** disjoint test transports (bar not moved; per-transport exponent 520/545) ⇒ M22 reduced to a zone-uniform bound on *one* exponent; theorem battery: 0 violations of the exact chain on 765, the fully explicit chain closes 635/635 but 0 under the bare bar 2 ($\kappa > 2$ measured on 130/765); map V5 (12 items): I1–I4 + **I5 new** certified, I6 bridge + I7 seams certified-conditional, M23 conditionally closed, **M25 new** (a certified positive floor for $\lambda_{\min}(X_{\text{fine}})$), promotion check list (1)–(6); M19/M21 untouched	ONE-OF-TWO	30/30

Part	Probe	Contract / finding	Verdict	Checks
T131	`self_supply`	the self-supply loop is built, one number short of closed: **the $\varepsilon \to \lambda_{\min}$ secular sandwich is a theorem** — $\varepsilon/(\|u\|^2 + \varepsilon/\mu_1) \le \lambda_{\min}(Q) \le \varepsilon/\|u\|^2$ with the sign half an *equivalence* (Albert 1969) — *sharp* (relative width 1.2663–1.3894, true value at 1.06–1.31× the lower bound, 0 violations on 10 dense windows), ε itself certified at the deep grid (Céa/Strang + CG, retention 1.0000); **correction to T130**: the Lanczos Ritz value overestimates $\lambda_{\min}(X)$ by up to 7.9× (an upper bound by construction, never a floor); the chain floor replaces it on **84 bridge pairs with 0 brackets lost** (retention 0.766–1.059 against the certified Cholesky floor, median 0.875), both deep seams CHAIN-POSITIVE ($n = 127$: $\varepsilon \ge 2.63 \times 10^{-7} \to$ floor 1.08×10^{-6}; $n = 256$: $3.83 \times 10^{-8} \to 2.52 \times 10^{-7}$), μ_1 tolerating **nine decades** of underestimate ⇒ **M25 reduced-to-pole-free**; the mirror rung dead ($T(\ell)$ indefinite 4/4 — the Szegő indefiniteness; the telescope runs coarse-from-fine only); **M22**: sign constancy a **theorem** via Perron–Frobenius on S^{-1} (entrywise positive 575/575; Metzler fails 11/575), the one hump **broken at depth** (530/575 with more than two monotone runs; the certified $(2n_{\text{run}} - 2) \cdot$ sag majorant replaces it on 575/575), exponent and S^* stay measurements ($a = 0.727$–0.999, 55/492 outside the old band; the deeper sweep lifts the required S^* to 1.8472, the frozen 1.1926 exceeded 16 times); **M17** assembled and *vacuous* (majorant 27/27, non-vacuous 0/27; dominant loss the U-metric mismatch 2.18; named fix: M18 whitened); map V6 (13 items), promotion list now **9 items** (new: the sandwich, the Perron sign theorem, the run-count chain); shortest remaining list four points	SUPPLY-PARTIAL	25/25

Part	Probe	Contract / finding	Verdict	Checks
T132	`bd_seam`	the first *reverse-flow* part: the phase's toolkit turned on the theory side, and the Beurling–Deny triad becomes an **operator discriminator** where the spectrum is blind — the BD identity exact to 2.4×10^{-16} at every N, the triad reconstructing the *operator* (1.4×10^{-14} at $d = 256$), diffusion part identically 0 (finite atomic space; the continuum DtN is the pure-jump $\frac{1}{2}$-Laplacian, Caffarelli–Silvestre 2007, fitted decay 1.980), and the **killing measure = the μ_4 mark-sum profile pointwise** (1.3×10^{-14}): the marks enter the operator *only* through the killing term; the **Markov remainder is exactly zero** with the closed form $J_{rs} = 1/(d \sin^2(\pi(r-s)/d))$ for odd $r-s$ and 0 for even — which explains T126's 86.1% count fraction; discriminator verdict SPECTRUM-ONLY: spectra agree to 7.5×10^{-13} but the triad gap sits in the **killing measure at** 0.1746 **at every** N (spread 1.9×10^{-14}) while the jump gap shrinks ($7.7 \times 10^{-3} \to 3.5 \times 10^{-4}$), so "same spectrum + μ_4-equivariance" is provably too weak; controls break the triad (0.520–0.982 against mark-local 7.5×10^{-16}) and the grading too; covariance limit named: random orthogonal (90.4) and generic μ_4-equivariant conjugation (86.5) destroy the triad, the mark-preserving lattice symmetries (D_4: clock rotation, sheet reflection) leave it **exactly** invariant (1.6×10^{-14}) $\Rightarrow$ `QGEO.MARKS.01` and `QGEO.KERNEL.01` are **coupled**; ceiling stated: this is the *model* DtN, `QGEO.KERNEL.01` **not** closed	SPECTRUM-ONLY	21/21
T133	`cert_floor`	the audit that changed a load-bearing module: shifted-Cholesky certificates (Wilkinson 1968 / Higham 2002 / Rump 2006 bars) applied to the suite's own RP/PSD rows, machinery itself validated on **840 controls** (0 false certifications, 0 false negative-direction fires, sharpness median 1.000) — the 10×10 Hankel v379 builds is **as a matrix of doubles certifiably NOT PSD** (certified negative direction -7.75×10^{-18} at 60 digits; also at $n = 14, 20$) and the module's $\lambda_{\min} > -10^{-10}$ passes it; the cause is priced, not guessed (cond $\approx 1.8 \times 10^{18}$ over the certifiability ceiling 1.0×10^{13}; entry bar 7.6×10^{-12} dominates every fp floor; 16/17 window variants MEASUREMENT-ONLY), the blind band exhibited (a certified negative direction passes the test), and the *mathematical* matrix is fine (certified floor 2.7×10^{-25} at 40 digits, while rounding to doubles moves it 3.5×10^{-17} — eight orders more; precision ladder 25/40/70/70 digits at $n = 6 \ldots 20$); **the exact repair needs no precision**: $M = V^{\mathsf{T}} \mathrm{diag}(1/1600) V$ is a positive-mixture Gram, so $M \succeq 0$ in exact arithmetic with no bar — the route v171 already takes; other rows clean (v171 exactly rank-3 Vandermonde Gram, $\lambda_{\min} = 0$ exactly; v527/v519 CERT-UP, $N = 8, 16$ even in binary64); recommendation table explicit: **no marker moves**, method and wording only — executed in the same round as the v379 hardening and the `SEAM.S3.RP.01` rewording	MIXED	23/23

Part	Probe	Contract / finding	Verdict	Checks
T134	`pole_free_floor`	the three items of T131's rest list, and nothing else — **the floor is never an existence problem**: every Cholesky pivot of the *pole-free* form positive on 79/79 windows (T119's negative pivot belonged to the form *with* the pole), per-window Cholesky certificates R4 79/79 / R2b 68/68 — but **every cheap route fails by sign, not size** (Dirichlet+Gershgorin, global, Jacobi dominance, Ostrowski all 0/68; path comparison infeasible, congestion up to 4.7×10^5): the nine decades of slack are worthless to them; anatomy: the good half $\mathrm{diag}(w) + L_N$ definite 68/68 ($\lambda_{\min} = 0.13$–3.06, 8–85,000× above target), the *entire* loss the positive-off-diagonal Laplacian L_P at the long lags ($\lambda_{\max}(L_P)/\mu_1$ up to 4.7×10^5, $k_{\text{eff}} = 24$–397 — not low-rank, T131's rank-one trick does not generalise); R5 measured: compression and Schur both *above* $\lambda_{\min}$ on 56/56 — no coarse recursion can supply a floor; degradation cert $\sim D^{2.26}$ (fit); **the inverse-positivity gets its class**: the lumped Stieltjes comparison $S_B = S + L_\Delta$ is Stieltjes and definite by construction, $\rho(\text{Jacobi}) = 0.709$–$0.990 < 1$ on 900/900 $\Rightarrow S_B^{-1}$ a convergent nonnegative series (discrete Green function; naive split only 659/900; killing reading refuted, row sums negative 900/900; Neumann majorant 558/558, coverage 26.9%; $S^{-1} > 0$ certified a posteriori 900/900 at residual $\leq 2.6 \times 10^{-11}$; RP-on-floor-eigenspace weakening refuted 535/900); **whitening corrects T131**: the 2.18 metric mismatch was not the dominant loss — the comb dominates the pole by 4.7–81×; first finite bound 2.81–5.04 (sharp 2.52–4.26) vs bar 0.50, $\sigma_b = 0.77$–1.00 exhausts the budget alone $\Rightarrow$ the rest is a localisation statement; promotion list $9 \to 15$, rest list five points	PARTIAL	21/21

Part	Probe	Contract / finding	Verdict	Checks
T135	`comb_compressibility`	the third *reverse-flow* trace: the T116 Riccati machinery ported *verbatim* to the seam DtN Λ_Σ (bordered identity 1.5×10^{-16}, banded m-march = dense state to 8.2×10^{-16}; pole candidate the rank-8 Calderón polarisation on the 16 residue classes, documented as a *model*) — census not a kill: 100.00% of the off-band mass on the mod-4 stripes (complement machine zero), stripe count exactly $\lfloor W/4 \rfloor = \Theta(W)$ (exponent 1.0000 ± 0.0000) against the true prime comb's $\pi(W)$-like 0.726 ± 0.018; **the core number** $m_{\text{cert}} = 12$: error/margin 3.3×10^{-16} exact ($h = 64$–1500) and 2.1×10^{-17} certified ($h = 1000$–100,000, *falling* in h) against T116's 1.3×10^2–1.1×10^6 *above* the margin; $m = 4 \to 48$ divides the error by 4.7×10^8 (T116: no decay); $12 \in \{8, 12, 16, 24, 32\}$ pre-declared $\Rightarrow$ MATCHED, not derived (bar-fragile: $8 \le m_{\text{cert}} \le 16$ defensible); controls: mod-3 also closes ($m = 8$) $\Rightarrow$ the driver is **weight summability**, not comb periodicity; the *true* prime comb *fails* under identical machinery (error/margin 0.16–0.27 rising, 12× state buys 1.69×) — non-vacuity established; sober: `QEC.SEAM.01` *not* advanced, m_{cert} partly circular (16 channels by hand), model DtN not the raw seam; new: finite-dimensionality visible in a *recursion*, not only a spectrum, with the Weil window as the proved contrast	BOUNDED-STATE	13/13
T136	`m_matrix_pair`	the M-matrix question T134 left open, and **one of three goals closes outright**: the classical row-sum criterion is *vacuous* at the border (900/900 negative row sums — the same sign failure), but **Varga's regular splitting makes** $\rho(J) = \tau/(1+\tau)$ **an identity** for the lumped Stieltjes comparison and the Collatz–Wielandt bound at the M-matrix anchor vector reproduces the measured $\rho = 0.709$–0.990 with a gap shortfall of only 1.00–1.03, *flat* in D and in the zone on 900/900 (a priori $\rho \le 0.7160$–0.9903, bar 10) $\Rightarrow$ rest item *(3)* **closed**, two solves instead of a spectrum; the pair itself resists with an *exact* anatomy: anchor half certifiable losing ≤ 1.51, but margin $1 - \rho(E)$ only 2.5×10^{-6}–2.5×10^{-2} wide against cheap $\rho(E)$ bounds at 1.82–3.11, the shifted ladder empty *by* monotonicity of M-matrix inverses (Berman–Plemmons), and the three-way split $\lambda_{\min}(A) \sim D^{2.27} = D^{-0.56} \times D^{2.72} \times D^{0.12}$ (anchor × gap × slack; fitted exponents) putting the *whole* degradation in the margin against an anchor that improves with depth; **M17 closes negatively**: the bad pencil subspace is *delocalised* (participation 0.54–0.98, 91% of cells for 90% of the mass, flatness 0.98), its dimension tracking the atom count ($n_{\text{bad}}/n_{\text{at}} = 1.64$) not the grid, so $\sigma_b(k) \ge 0.646$ and no cut brings M18 under 0.50 (best 2.76–7.00) — what M18 needs is a localisation *statement*; promotion list 15 → 22 (→ v543)	ONE-CARRIES	30/30

Part	Probe	Contract / finding	Verdict	Checks
T137	`long_lag`	the long lags named, and **a whole bounding family killed by certificate**: the archimedean half carries *no* positive off-diagonal (0 of 40 windows), so L_Δ is entirely **comb-generated** — supported on the anti-diagonal stripes $r + s = M - 1 - \mathrm{round}(u_j/D)$ at the prime-power atoms (1.00–1.00 of the positive entries within one grid cell of an atom, 4–192 stripes on 3–99 atoms, 1.33–1.97 per atom), each stripe a *perfect matching* (orthogonal edges, 2×2 blocks), with the certifiable amplitude law $\Delta \le c_{\mathrm{comb}}(\mathrm{anti})$ (the single-atom $\mu_j/2$ only *typical*, worst ratio 2.268); **the $\lvert E\rvert$-envelope family is dead by a lower bound**: Collatz–Wielandt from below gives $\rho(\lvert E\rvert) \ge 1.3218$–$2.2963$, above 1 on 35/35, so no Gershgorin bound, no row-sum norm and no rescaling can *ever* produce a floor; the surviving edge/Gram family (second differences of the Green function) reaches 1.1530–2.3502 against a true ρ of 0.9752–1.0000 and clears the target on 0 of 35 (overshoot 24–7.1×10^5 gaps), the coarsening ladder crossing 1 only when the entire Gram is one block $\Rightarrow$ *the inter-stripe coupling is the object*, decay algebraic $\mathrm{dist}^{-0.63}$ (fit; Demko–Moss–Smith *not* applicable, A_B dense); **rest item (2) delivered**: the Neumann series bounds the Green function from below with *no* error term (recovering 0.011–0.406 of the minimum entry) and the anchor-weighted ceiling from above (overshoot 2.49–12.24), yet the $S^{-1} > 0$ certificate still stops at 563/900 (716/900 k-exact, 66/900 structural) because $\rho(\lvert F\rvert) \ge 1$ on 77 blocks — *the same wall*: the absolute value destroys the cancellation, and the residue is a sign-preserving bound; promotion list $22 \to 30$ ($\to$ v544)	BOTH-RESIST	22/22

Part	Probe	Contract / finding	Verdict	Checks
T138	`sign_compensation`	the cancellation **kept instead of majorised**, and the mechanism found: the sign of an inter-stripe coupling follows the **interval geometry** of its two edges — nested positive on 0.76–0.94 of the area, crossing 0.59–0.63, disjoint only 0.06–0.36 (mostly negative) — *not* the lag distance (flip rate near 0.5), with global cancellation $\lvert\Sigma\rvert/\Sigma\lvert\cdot\rvert = 0.29$–$0.70$ (the one-number reason for every absolute-value overshoot); honest negatives: Karlin TP_2 *fails* (0.54–0.65 of the minors), Green rank-one dominated but *not* semiseparable; the best **certified** signed bound (band-exact Cholesky + block row-sum tail) reaches 0.9758–1.1020 against target 0.998556–0.999979 — 1/16 windows, overshoot 0.02–565 gaps against T137's 27–8×10^5, three orders better and no floor; the clamp *localises*: the band crosses the target at $b = 2$–8 with the certified tail still 0.46–0.70 (never small together), the far tail is **net negative** (truncation raises ρ on 59/142 rungs), Leibniz pairing buys nothing, two-body exact but thin ($\leq 0.6\%$; three-body up to 2.5× — a *many-body* effect); **the m-paired Neumann certificate** ($(I-F)^{-1} = (\sum_{j<m} F^j)(I-F^m)^{-1}$, only the outer factor majorised, condition $\rho(\lvert F^m\rvert) < 1$) removes the arithmetic wall on *all* 77 dead blocks, certifies 52, pool 563 → 875/900 ($\max\lvert F^m\rvert/\lvert F\rvert^m$ median 0.295); M17 whitened: $\mathrm{St} = I - W_S$, $W_S \succeq 0$, direct Kantorovich harmonic-mean route with no bad subspace, margin relaxed 0.5 → 0.8284 (854/900, HARMONIC-PARTIAL); promotion list 30 → 37; the margin question returns one level down ($\rho(W_S)$)	PAIR-EXACT	26/26

Part	Probe	Contract / finding	Verdict	Checks
T139	`green_decay`	the decay lemma **refuted at its hypothesis, arithmetically**: the certified kernel envelope decays $d^{-0.15}$–$d^{-0.04}$ where Jaffard 1990 needs $p > 1$ — the archimedean lags sit *on* the borderline ($\sim 1/d$), the prime-power atoms lay $O(1)$ spikes across the window (no envelope sees the comb's thinness); constant chain constructive: Demko–Moss–Smith *verified where applicable* (75/75 banded rungs), operator gate 0.97–1.83 (1/75), algebra gate 3.4–1.6×10^3 (0/75); exponent *not* inherited (kernel $d^{-1.5\ldots-1.1}$ vs inverse $d^{-0.52\ldots-0.37}$); the replacement is an **identity**: $b_e^\mathsf{T} G b_{e'} = \sum_{\text{box}} H$ exactly (4×10^{-15}), $H < 0$ on 0.71–0.87 off-diagonal, $H > 0$ on 100% of the diagonal $\Rightarrow$ **T138's whole sign law derived** from one kernel plus box geometry (Gantmacher–Krein one-pair quantitatively dead); the never-truncating layer series **killed from below**: 3.388–6.171 vs target 0.999182–0.999972 (0/15, overshoot 681–46900$\times$ = the effective layer count), zero trace forbids dropping any layer, Gram $\preceq$ Band$_b$ *never* (85/85), Weyl floor $\max_b \text{Rayleigh}(\text{Band}_b) = 1.0303$–$1.1116 >$ target on 15/15 — no decay lemma can rescue it; clean-up: m-ladder to $m = 48$ certifies 619/620 border blocks (the one open: need 2.15, tightness at index distance 15), $\rho(W_S) = 0.667$–0.939 inherits the structure verbatim, three-body cumulant negative on 0.93–1.00 of 4200 triples (certified *benign*); promotion list 37 $\to$ 47; **the core shrinks to one signed inequality at stripe distance** $b \leq 16$	DENSE-RESISTS	30/30

Part	Probe	Contract / finding	Verdict	Checks
T140	`signed_band`	the one signed inequality acquires an **exact finite core per zone**: the H-telescope lifted from entries to the *form*, Gram $= CHC^\mathsf{T}$ exactly (3×10^{-15}, $C = \mathrm{diag}(\sqrt{\Delta})M$ with the edge-interval incidence M), so rank(Gram) $\leq h - 1$ (44–202 vs 128–1499 edges) and $\rho(W) = \lambda_{\max}(K^{1/2}HK^{1/2})$ exactly (6×10^{-14}), $K = M^\mathsf{T}\Delta M$ the covering kernel in closed geometric form $K_{rs} = W([r \wedge s, r \vee s])$ (nonnegative, monotone by inclusion; Green-like, *not* one-pair); energy reordering $H = \mathrm{diag}(s) + L_N$ exact (5×10^{-16}) — the sign law as mass + long-range Dirichlet form (weights positive on 0.71–0.87); the certified chain **tears at the drop of the positive H part** (6.4–239× alone; the Hardy step adds 6.4–9.0×, input $\max_k Q_k = 18.5$–339; 0/16 on all four routes); **the checkerboard split solves R2**: $\mathrm{Band}_b = D + A_{\mathrm{even}} + A_{\mathrm{odd}}$ entrywise exact ⇒ three D-independent Weyl steps instead of $O(nb)$; blocks zone-uniform ($D^{0.13}$, $D^{0.02}$) vs $\lambda_{\max}(K) \sim D^{-2.99}$ — **all D-dependence in the geometry**; band from below > target on 16/16 (rank inflation 1.2–10.1× as mechanism), index-grid far part Loewner-*positive* on 256/256 (no truncation licence in either grid); Floquet not applicable (spread 0.15–0.69), Bendixson certified at 1.07–1.08 × $\rho(W)$; border pool 420, m-ladder to 24 certifies 402, the 18 open far-carried (0.635–0.912); core inherited one level down ($\rho(W_S)$ to 4×10^{-15}, = 0.541–0.925); map V12 (18 rows), 17 new promotion items (stock 47 → 64); **the rest is one zone-uniform discrete Hardy inequality**	FINITE-CORE	31/31

Part	Probe	Contract / finding	Verdict	Checks
T141	`discrete_hardy`	the Hardy ingredient **resists, and the resistance gets an address**: four exact identities put R1$''$ in classical two-weight shape — the K^+ Rayleigh form $\rho(W) = \sup\{u^\top Hu/u^\top K^+u\}$ (3×10^{-11}), the signed crossing kernel $L_N = D^\top BD$, $B_{kl} = \sum_{r \le k \wedge l,\, k \vee l < s} N_{rs}$ (7×10^{-16}), $Q = B_+\mathbf{1}$ (3×10^{-15}; T140's Cauchy–Schwarz step *is* the Schur/Gershgorin diagonal step) and $DKD^\top = L_\Delta$ on the interior nodes (2×10^{-13}; the Hardy denominator is the original Laplacian, its diagonal the *endpoint edge mass*); **the uniformity separates**: the certified constant $A_M = 2.20$–31.23 runs as $D^{-0.366\pm0.036}$ (spread 14.21× over a D range of 11.21×) — *not* zone-uniform at bar 0.25 — while the object it bounds, the exact Dirichlet share, is uniform in the strict sense ($D^{-0.229\pm0.007}$, band 0.235): the growth sits in the diagonal profile, and the closed Muckenhoupt product ($D^{-2.687}$) and the Gershgorin variant ($D^{-0.993}$) overshoot the signed constant by 25–15903× and 1.2–14.3×; **the additive shape is dead as a shape** — its own exact two-term Weyl floor is already 1.694–3.855× the target (0/60; additive Hardy 2.775–32.152×, the Hardy step costing only 1.62–16.86× over the exact share, so the shortfall is the *split*); **the joint shape fails at the normalisation alone** — $\Lambda(H, Y_{\mathrm{tri}}) = 0.500$–$0.942$ is fine, $\Omega = 20.71$–2723.99 (ideal 1, $D^{-1.867}$) is not, with an entrywise off-tridiagonal defect of K^+ of only 0.027–0.285: Gantmacher–Krein returns, the *profile* is the whole remaining freedom; R3$'$ shaped as a first-moment statement (cancellation gain of the signed kernel 31.95–15561.41× against the T140 drop cost); border pool 256, ladder to $m = 24$ certifies 245, 11 open and far-carried (0.835–0.902); map V13, 13 statements ripe (stock 62 → 75); **the rest list collapses to R1b: one closed conductance profile with $Y \preceq K^+$ and $\Omega \approx 1$** (→ v545)	HARDY-RESISTS	22/22

Part	Probe	Contract / finding	Verdict	Checks
T142	`conductance_profile`	the optimal Hardy weight **found exactly, not guessed**: the capacity decomposition $K^{-1} = D^{\mathsf{T}} J^{-1} D + xx^{\mathsf{T}}/\mathrm{cap}$ ($J = DKD^{\mathsf{T}} = L_{\Delta}$ the endpoint Laplacian, $x = K^{-1}\mathbf{1}$ the equilibrium charge, $\mathrm{cap} = \mathbf{1}^{\mathsf{T}} K^{-1}\mathbf{1}$) verified to 5.5×10^{-12} on 26/26 windows; the Dirichlet half is an *orthogonal projection*, so $\Omega = 1$ **exactly** (T141 guessed 20.7–2724, $D^{-1.87}$) — the weight is geometry (Miclo 1999, Maz'ya capacity), and the exact capacity Rayleigh form follows with no inequality taken; the variational class optimum is $\Lambda\Omega = 1.3160$–1.4235 at $D^{-0.029\pm0.014}$ — **the failure is a constant factor, not a power** — with the stable shape $c_k \sim 1/J_{kk}$ (correlation 0.988–0.995) and essentially no mass; the certified chain reads $\Lambda\Omega = 2.2671$–2.4536 vs target $\sim$1 (0/26, flat at $D^{0.006\pm0.004}$; gain over guessing 31.9–137×); **the rank ladder closes the comparison path**: the diagonal minorant + r dominant residual modes stalls at 1.39–1.41 for all $r \leq 128$ and crosses 1 only at $r^*/m = 0.995$–0.998 (the tautology $Y = K^{-1}$; anchored to ρ at 1.7×10^{-11}), and since $\rho(W) = 1 - \Theta(D^3)$ with equality only at $Y \propto K^{-1}$, **no comparison argument can deliver D-uniformity**; R3′ corrected — only 0.03% of the first moment of $(-H)_+$ inside distance 8 (support radius 204–873), the long-range Loewner step certified, its constant $\sim D^{-1.73}$ not uniform ⇒ R3″; border pool 72, 68 certified, 4 open (required far shrink 0.656–1.052); map V14, 9 statements ripe (stock 75 → 84); **rest list: R1c the sharp capacity–Rayleigh route, R1d the LP/SDP dual, R3″, R4**	PROFILE-RESISTS	24/24

Part	Probe	Contract / finding	Verdict	Checks
T143	`sharp_capacity`	the sharp route **carries**: the exact capacity–Rayleigh form $1-\rho(W)=\inf_v[d^\top(J^{-1}-B)d+(x^\top v)^2/\mathrm{cap}-\sum_k s_k v_k^2]/[d^\top J^{-1}d+(x^\top v)^2/\mathrm{cap}]$, $d=Dv$, is an *identity* on 26/26 windows (assembly $10^{-11}/5\times10^{-12}$, eigenvalue 3×10^{-6} relative under the declared 1/gap conditioning), with a structurally new bookkeeping — the minimiser *orthogonal* to the equilibrium charge (the capacity rank one carries nothing), the mass share *negative* (-0.0698 to -0.0029), the gap a cancellation between Green share (exactly 1) and crossing share (1.0029–1.0698), the naive split certified dead at 6.19–66.63 $\times\,\rho$; **Maz'ya's criterion lands inside its window**: $\Phi_{\mathrm{sup}}\lambda=0.5438$–$0.6457$ on 26/26 (window $[1/4,1]$) for a *non-Markovian* form (72–84% positive off-diagonals), the bound $\lambda\geq 1/(4\Phi_{\mathrm{sup}})$ captures 0.387–0.460 of the true gap with a **zone-uniform loss factor** $D^{-0.048\pm0.010}$ (bar 0.25); the supremum lives on closed families (0.9606–1.0000 of the best, local search $\leq 1.041\times$; in node coordinates *intervals* dominate by 8.3–129.5 — Muckenhoupt's one-dimensional structure) and is *certified by enumerating all* 255–16383 *subsets* on 6 border blocks ($\Phi_{\mathrm{sup}}\lambda=0.6087$–$0.9016$, inside the window); Miclo's constructive chain loses 46–2561 ($D^{-2.13}$) — the conclusion holds, the classical proof mechanism does not; R1d dead as a direction sample (the LP dual saturates at the trivial floor, $\rho(W)$ to a factor 1.000000; needs the full SDP constraint), border pool 36, 32 certified, 4 open, the capacity argmax entirely the far tail; map V15, 10 statements ripe (stock 84 $\to$ 94); **rest list: S1 one named capacity inequality $\mathrm{cap}_E(A)\geq\lvert A\rvert\lambda_0/c_0$ with absolute c_0, S2 the Muckenhoupt/interval route to it, S3 a closed capacity lower bound, R1d$'$, R4**	SHARP-CARRIES	24/24

Part	Probe	Contract / finding	Verdict	Checks
T144	`capacity_inequality`	the interval route **carries, and the chain has exactly one unproven input**: with the *Jacobi whitening* as the single comparison licence ($\kappa_{\mathrm{up}} = 1.1195$–$1.3162$ certified per window; Gershgorin would allow ≤ 1.983; a denominator comparison, untouched by T142's obstruction), two interval-reduction statements separate — the *pointwise hull comparison is false* (up to 425, drifting $D^{-1.137}$, scattered level sets as offenders $\Rightarrow$ S2 must be a testing condition) while the *best-interval comparison holds*, $c_{\mathrm{glob}} = 1.0000$–$1.6876$ zone-uniform at $D^{-0.068\pm0.024}$, measured on 2.24 million sets incl. *all* 16383 subsets of 136 enumerated blocks (capacity monotonicity holds on all of them, though the form is not Dirichlet); **the two-weight sup is exact, not sampled**: 11 390 676 intervals exhausted via a Cholesky prefix-sum identity (verified to 10^{-11}), giving the core number $B_{\mathrm{res}}\hat{\lambda} = 0.6694$–$0.7813$ inside the Maz'ya window $[1/4, 1]$ on the whole surface, flat at $D^{0.013\pm0.005}$ and coordinate-robust (T143: 0.5438–0.6457), with the free half $\Phi_{\mathrm{sup}}\hat{\lambda} \leq 1$ passing on all 2.24M sets; **the chain** $\lambda \geq 1/(c_0\kappa_{\mathrm{up}}c_{\mathrm{glob}}B_{\mathrm{res}})$ certifies 0.1002–0.2653 of the exact gap per window, the residual D-dependence in the Jacobi transfer ($D^{-0.178}$), not the capacity calculus; dead: the naive Muckenhoupt trial-energy transcription (89–6995, $D^{-1.885}$); honest counter-side exact: $\Phi_{\mathrm{int}}\hat{\lambda} = 0.0928$–$0.3578$ drifts $D^{0.654\pm0.016}$ — intervals saturate S1 only from above; **the Markov perturbation route is certified dead** ($\lambda_{\min}(E-P) < 0$ everywhere, $t^* \sim D^{3.96}$; 3–12% positive off-diagonal entries carrying 36.7–64.5% of the mass); **S3 stronger than asked**: the Cauchy–Schwarz floor makes Φ_{sup} a max-density-subgraph value (Charikar 2000; Goldberg 1984), $\Psi_{\mathrm{all}}\hat{\lambda} = 0.7399$–$0.8515$ at $D^{0.015\pm0.005}$ over *all* 2^m sets — the flattest number of the probe, argmax on intervals (B_{res} only 1.078–1.129 below); R4: 22 blocks, 19 certified, 3 open, $c_{\mathrm{glob}} = 1.0000$ exact on 5 enumerated; map V16, 9 statements ripe (stock 93); **rest list: S1′ — S1 under a Green mean-density bound, a Muckenhoupt-type instead of a Markov-type hypothesis — S2′ as testing condition, R4, R5; the only unproven input is the absolute constant** c_0	INTERVAL-CARRIES	31/31

Part	Probe	Contract / finding	Verdict	Checks
T145	`mazya_proof`	the proof attempt for S1′, **one step missing, and it has a name**: the classical Maz'ya proof transcribes step by step onto the non-Markovian form — M1 level arithmetic (theorem, factor 1.649–2.670 against the ceiling 3), M2 the Green floor (theorem, Cauchy–Schwarz), M3 truncation admissibility (theorem, $\times 4$, slack ≤ 1.036 on 168 levels) — and **the Markov property sits in exactly one line**, M4 truncation subadditivity, which is *void* here: 3.2–14.9% positive off-diagonals carrying 18.3–58.7% of the mass, the conductance share of the minimiser's energy negative on 61/64 windows, while the same code on the Stieltjes surrogate $E - P$ reproduces the classical step; **but M4 splits** — the mass half is a theorem ($\sum_k f_k^2 \leq \psi^2$ pointwise; ratio 0.2639–0.2995) and dominates, so $\sigma_{\mathrm{tot}} = 0.2145$–$0.4425 < 1$ everywhere; two routes make c_0 **explicit**: energy $12\sigma_{\mathrm{tot}} = 2.5734$–$5.3099$, sign-free $G_{\mathrm{dy}} = 2.1559$–6.3937, better route $c_0 = 2.2480$–4.2265 at $D^{0.028\pm0.017}$ (zone-uniform) against Maz'ya's Dirichlet value 4, and **S1′ is certified per window**: cert_lam_max(R) $\leq c_0\Psi$ on 64/64 (sign price $\lvert R\rvert$ vs R^+: 1.12–1.41, witness built in); the **no-go** $R = aa^\mathsf{T} + \varepsilon I$, $a_i = i^{-1/2}$ (entrywise nonnegative, proved density ceiling $4+\varepsilon$, $\lambda_{\max} \sim \log m$) shows $\Psi_{\mathrm{all}}\hat{\lambda}$ decaying $0.702 \to 0.488$ and G_{dy} growing $7.6 \to 36.8$ — a level-profile hypothesis is *necessary*; S2′ retired as a requirement ($c_{\mathrm{test}} = 1.0782$–$1.1327$ against Charikar's cited 2), R5 two-sided but width $D^{0.543}$ (downgraded; used side flat $D^{0.065}$), R4 all 24 blocks closed; map V17, 11 statements ripe (stock 104); **rest: L1 the level lemma — an a-priori bound for σ_{tot} or G_{dy}, either alone closes S1′ — L2 the profile route to it, R4 bookkeeping**; identity spine of T142–T145 promoted as v546	ONE-STEP-MISSING	33/33

Part	Probe	Contract / finding	Verdict	Checks
T146	`level_lemma`	the proof attempt for L1, **and the level lemma stands**: closed on the measurement surface as a chain of theorems and certified window inequalities, no step reading the minimiser — four hand-written closed profiles (sine, ramp, pole vector, equilibrium potential) lose by **5–9 orders of magnitude** (every guess sees $O(1)$, the gap is $O(D^3)$); what survives is the **Green column** itself ($\lambda_{\mathrm{up}}/\hat{\lambda} = 1.0497$–$1.3764$), and **Davis–Kahan is instrumented and discarded** (informative on 0/64: the spectrum bottom is a near-degenerate block, $\lambda_2/\hat{\lambda} = 1.2542$–$3.4471$, separation in the denominator) because $\psi = \lambda R\psi$ is an *identity* — no transfer step exists in the argument; the lever: $\Vert\psi\Vert_\infty \le \lambda_{\mathrm{up}} \max_j \Vert Re_j\Vert$ within 1.055–1.426 of the truth, the trivial ceiling would give $\Gamma \le \sqrt{m} = 5.1$–$34.2$, measured $\Gamma = 1.8356$–2.2797 (columns 2.7–16.0 below trivial — that factor *is* the delocalisation, read off the form); the localisation contradiction stays qualitative (leak tolerance $\le 2.38 \times 10^{-3}$), Perron only on the lumped partner; **the cake base is free** (the chain uses only $\vert\psi\vert \le \psi_t$, Maz'ya's dyadic 8 falls to $2b^2 \approx 2$, gain 3.76, price zero via the $O(m)$ union identity) and $c_0^{\mathrm{ap}} = 2b^2\Gamma \min(1, \Gamma_1) + \varepsilon = \mathbf{3.9042}$–$\mathbf{4.8488}$ ($b = 1.031$, 64/64) against T145's measured 2.156–6.394 — at the size of Maz'ya's Dirichlet value 4 — chain loss 0.0422–0.1586; stress tests discriminate (no-go Γ 3.676 $\to$ 13.800, $m^{0.420}$, unbounded; theorem half survives, slack 1.087–1.242; Dirichlet control flat 1.7183–1.7315); border pool 24/24 closed, a-priori chain positive on 24 ($c_0^{\mathrm{ap}} = 2.855$–$3.176$); map V18, stock 108; **rest: all-D asymptotics of the Green columns ($\max_j \Vert Re_j\Vert\lambda_{\min}$), σ_{tot} stays measured (not a hole), the actual 3 T134 blocks**; a-priori core promoted as v547	LEVEL-CARRIES	30/30

Part	Probe	Contract / finding	Verdict	Checks
T147	`green_asymptotic`	the asymptotics itself, **and the last open object becomes an exact identity**: $\Gamma = \sqrt{Q^\star}\, S_w$ (verified to 2.3×10^{-16} on all 85 windows $m = 26$–1491 and both stress forms) — D-uniformity *equivalent* to the boundedness of two named scalars, $S_w = \lambda_{\rm up}\|R\|_F$ purely spectral (effective bottom multiplicity $n_{\rm eff} = 1.251$–2.831, certified ≤ 4.6438 at a certified cut by LDL^T inertia) and $Q^\star = m \max_j (R^2)_{jj} / \operatorname{tr}(R^2)$ purely geometric, scale-free (certified ≤ 2.8634 against the localised extreme $m = 1491$); **the classical decay route is computationally dead** (Demko–Moss–Smith $q = 0.994815$–0.999999, envelope $q^{m/2} = 0.9346$–0.9995 above the 0.05 bar everywhere; Jaffard exponent of E itself 0.06–0.82) and the columns are *delocalised* (half-mass radius 0.048–0.105 of the window) — delocalisation *is* the bound: 0.9864–0.9996 of $\|Re_j\|^2$ from the bottom block alone, $Q_B/\lvert B\rvert = 1.375$–$1.839$ against the exact floor (naive $Q_B \sim m$ wrong by 5–258), T146's unexplained factor 2.7–16.0 identified as $\sqrt{m/Q_B}$ up to $\lambda_{\rm up}/\lambda_{\rm lo}$; honest negative: the crude free-tail split grows $m^{0.538\pm0.020}$, reported, not used; **the mechanism is named**: Toeplitz-minus-Hankel section, bottom eigenvectors Fourier modes by Szegő/Widom (Kac–Murdock–Szegő 1953, Widom 1958, Basor–Ehrhardt 2009), sharp prediction $Q_B \le 2\lvert B\rvert$ **holds**, chain $Q_B \le \sum_B m\|\psi_k\|_\infty^2 \le 2\sum_B \|a_k\|_1^2$ pure arithmetic (sparsity ≤ 8.773 per mode, 0.999+ of the energy in the lowest $4\lvert B\rvert$ modes); four D-strata with certified maxima, six trend exponents flat at bar 0.25, all fits; congruence structure discarded (scatter 0.465); $\sigma_{\rm tot}$ does not fall but is located (block leakage 0.9627 vs required 2.7×10^{-7}); **the 3 open R4 border blocks close** (8/8 rebuilt border windows, $c_0^{\rm ap} = 2.9135$–3.0383, same mechanism); stress separates: no-go diverges and hits $Q_B = m/H_m$ to 1.0000, Dirichlet control flat (1.4142–1.4376 vs sine value 2); honest limit: translation covariance broken by the whitening (Toeplitz defect 0.21–0.54), the spectral factor needs its own a-priori input; map V19, stock 114; **rest: a Szegő theorem for the reweighted section (lifts the chain to all D), an a-priori bound for S_w, the $\sigma_{\rm tot}$ classification**	ASYMPTOTIC-SHAPED	39/39

Part	Probe	Contract / finding	Verdict	Checks
T148	`szego_bottom`	the lifting statement dissected, **and the second factor closes on the surface**: the passage from the pure to the reweighted section classified in seven typed steps — the order-2 model ladder and the pure section's Fourier sparsity THEOREMS (KMS 1953, Widom 1958, Basor–Ehrhardt 2009; model $\nu = 4.236$–4.257 flat), the ℓ^1 ceiling $\|a\|_1 \le 2\sqrt{\nu}+1$ CERTIFIED ($\nu = (m+1)^{3/2}\|L_0\psi\|_1/(2\sqrt{2})$, every m-power cancelled; excess -2.6×10^{-6}), the chain $Q^\star \le 2\langle\|a\|_1^2\rangle_w \le 2\langle\mathrm{ceil}^2\rangle_w$ CERTIFIED 48/48, the commutator/Davis–Kahan route DEAD (5.4×10^2–5.3×10^5, 2.7–5.7 orders vacuous), the live route the LIOUVILLE transform $m\|\psi\|_\infty^2 \le 2\kappa_\Lambda \mathrm{ceil}(\hat\varphi)^2$ with $\kappa_\Lambda = 1.733$–3.933 certified (gain 50.6), and $\nu \le F(\text{form})$ THE OPEN INPUT (a-priori ceiling 81.4–469.0, grows $m^{0.615}$; ν_L needs $\lesssim 34$, measures 282, $x^{0.166}$); **the hypothesis isolated**: weight-class experiment at identical $\kappa_\Lambda = 4$ — two BV classes flat ($x^{0.000}/x^{-0.007}$), the rough class $x^{1.994}$ — $\mathrm{TV}(\log\Lambda)$ is the named scalar, measured 3.63–11.93 on the real surface, $x^{0.444}$ NOT flat; S_w **closes**: LDL^T layer cake $\Rightarrow S_w \le 1.9587$ certified (true 1.282–1.683), $x^{0.007\pm0.007}$ flat, per D-stratum; **honest negative**: the Toeplitz symbol has $f(0) < 0$ on 48/48 $(-87.3\ldots-22.5)$ — positive definiteness is a finite-section effect of minus-Hankel + lumping, the KMS order-2 hypothesis survives only as a measurement ($\alpha = 1.643$–1.989 in $[1.5, 2.5]$); no-go violates exactly the smoothness hypothesis (ν diverges, $n_{\text{eff}} = 1.0000$ innocent), Dirichlet exact ($\nu \to \pi$, $m\|\psi\|_\infty^2 \to 2$); **the big number**: the certified chain delivers 8.36–15.86% of the true gap (median 11.74%, no window loses more than 12×), a-priori-shaped factors still a lower bound at factor 10.5; balance 6 certified / 2 measured / 2 fits, fits in no inequality; σ_{tot} RETIRED AS A ROUTE; map V20, stock 114; core promoted as v548; **rest: $\nu_L \le F(\text{form})$ m-free (= bounded $\mathrm{TV}(\log\Lambda)$ or weaker, target $\lesssim 34$ vs 282); cosmetic: an order-2 theorem from the minus-Hankel structure**	ONE-INPUT-MISSING	31/31

Part	Probe	Contract / finding	Verdict	Checks
T149	`weight_smoothing`	the one open input attacked through the gauge freedom, **and the hypothesis is refuted rather than weakened**: the whitening diagonal is a FREE GAUGE (identity plus one Rayleigh step, 9 preregistered candidates, every one a valid lower bound, family maximum valid); the preregistered winner `const` (geometric mean of the Jacobi diagonal) delivers the three ordered numbers — sandwich $\sigma \leq 5.5789$ CERTIFIED ($x^{0.154}$, bar 8.0), TV($\log \tilde{\Lambda}$) **exactly** 0.0000 (T148's 11.93 at $x^{0.444}$ ELIMINATED, not reduced), $\nu_{\tilde{L}} = 222.70$ in target on 21/44 windows only ($\rho = 1.347 \to 0.485$) — at price $\tilde{\kappa}_{\mathrm{up}} \leq 2.3146$ vs 1.2647; **the two-regime decomposition**: flutter-killers cheap (sandwich 1.129–1.572) but move $\nu_{\tilde{L}}$ by $\leq 0.9\%$, only macro-removers improve it (factor 1.27) — TV($\log \Lambda$) was the MISLOCATED hypothesis, the roughness sits in the FORM; chain: $\Gamma = \sqrt{Q^\star} S_w$ gauge-invariant at 4.4×10^{-16} (89 assemblies), family maximum 9.52–18.45% of the true gap certified (median 12.68%), a priori 0.800–1.566% (up to 1.284× better), Jacobi still wins certified 44/44; $Q^\star$ ceiling 697.9 → 149.1, exponent 0.551 → 0.330, NOT flat (driver $\nu_{\tilde{L}} = 265.2$, $x^{0.772}$, above the bottom block); stress: no-go already smooth, ν diverges $x^{1.420}$ under EVERY gauge; Dirichlet exact ($\nu \to \pi$, 5.9×10^{-14}) and CORRECTS the question to the ladder $\nu_k \leq Ck^2$ with m-free C ($C = 18.66$–44.61, $x^{0.272}$); TV anatomy: 85%+ flutter, amplitude 0.064–0.182 FLAT, edges innocent; 7 promotion candidates, unpromoted; **rest: m-free C in $\nu_k \leq Ck^2$ (load-bearing); an a-priori flutter-amplitude bound (closes it outright); cosmetic order-2 theorem**	PARTIAL-SMOOTHING	30/30

Part	Probe	Contract / finding	Verdict	Checks
T150	`mode_ladder`	the ladder attacked through the flutter amplitude, **and the mechanism has a name — parity**: the form is EXACTLY the compression of the full symmetric Toeplitz section onto its antisymmetric parity sector ($U_-^{\mathsf{T}} T_M U_- = T - H$, cross block $\le 4.0 \times 10^{-16}$ on real windows; Basor–Ehrhardt 2009 the address) and **the entire negative inertia of the sign-changing symbol sits in the EVEN sector** (odd 0 / even 1 on 72/72, LDL$^{\mathsf{T}}$) — $f(0) < 0$ EXPLAINED, not worked around; the diagonal is an EXPLICIT zone function ($\Lambda_r = c_0 - c_{M-1-2r} + \sum \max(c_{\lvert r-s\rvert} - c_{M-1-r-s}, 0)$, residual 0.0 on 72/72) with the exact split $c = c^{\text{arch}} + c^{\text{atom}}$, $c^{\text{atom}} \le 0$; the flutter amplitude a CERTIFIED form functional, fit-free ($\text{amp}_{\text{loc}} = 0.0606$–$0.1993$, $x^{-0.165\pm0.015}$, largest log step 0.1927; chain $\le \text{half} \cdot d_1^{\lambda} / \min \Lambda = 1.345$–$3.954$; atom budget $4B\sqrt{N}$ CLOSED); honest negative: the archimedean diagonal is NOT the macro profile (level shift 0.734–2.647, $x^{0.244}$); **the perturbation step precisely dead, twice**: multiplicative (Loewner) transfers with $\exp(\text{amp}) \le 14.11$ — the gauge step CLOSES — additive (Weyl / Bauer–Fike / Davis–Kahan) misses by 2.4–5.8 orders ($\lVert A^{\text{atom}} \rVert = 4.3$–$12.3\%$ of the section) AND the archimedean-only section is itself INDEFINITE (2–7 negatives) — **the atoms are co-responsible for the positivity**; third valid ceiling from the parity sines ($\nu_k = \pi k^2$ on the control to 0.0002, gain 1.000–1.130); BUT $C = 14.24$–$43.39 = 13.81\pi$, $x^{0.258\pm0.009}$ — the one gate falls; max at $k = 1$ (67/72), the gap $\pi \to 13.81\pi$ exactly the ARITHMETIC LEAKAGE of the bottom mode (ℓ^1 1.41–1.79, $m\lVert\psi\rVert_\infty^2 = 2.45$–$3.41$ vs 2); chain 0.0680–0.1845 certified (T149's best reproduced exactly), ladder-shaped $Q^\star$ ceiling valid by monotonicity (price $\le 2.555\times$ for ONE constant); 5/10 factors flat; stress LOCALISED: the no-go violates the ladder alone ($x^{1.840}$, multiplier smooth), controls $C = \pi$ (0.0004/0.0002); 9 candidates (stock $121 + 9 - 6 = 124$); core of T149+T150 promoted as v549; **rest: $\nu_k \le Ck^2$ with m-free C for the ODD parity sector at a sign-changing symbol — THE one inequality; a closed bound on $\min \Lambda^{\text{arch}}$; Basor–Ehrhardt at** $f(0) < 0$	ONE-TERM-MISSING	36/36

Part	Probe	Contract / finding	Verdict	Checks
T151	`odd_ladder`	the one inequality attacked head-on, **and it closes by REROUTING OFF THE LADDER**: the positivity of the odd sector is a GRID fact — the parity sines live on $\theta_k = 2\pi k/N$ with $\mu_k^P = 2 - 2\cos\theta_k$ exact, and the symbol's negative window ($\theta_c = 7.06 \times 10^{-4}$–$1.32 \times 10^{-2}$) lies UNDER the odd spacing θ_1 (ratio 0.328–0.407, $x^{-0.057}$) — stepped over on 72/72; BUT the symbol language fails ($f''(0) = 3.8 \times 10^5$–6.8×10^8, f moves 0.47–0.85 across ONE mode spacing, Fejér averaging instead of sampling, deviation 1.39–1.83 at $k = 1$, margin vacuous 72/72) — **T150's rest 3 answered NEGATIVELY: the local model is a MATRIX statement** — $\lambda_k(A) \le S\mu_k^P$ for $k \le 8$ via LDL$^\mathsf{T}$ (Sylvester), $S = 1.1019$–2.3870, $f(0) < 0$ ABSORBED (Rayleigh floor $L_P \ge \mu_1^P I$); leakage = LOW-k ROTATION ($\sin\angle(\psi, t_1) = 0.816$–$0.894$, shares 0.355/0.687/0.005 at $k = 1/2$–$4/{>}64$), no arithmetic sideband; **the Sobolev step IS the multiplicative mode comparison**: virtual node $\Rightarrow \|\psi\|_\infty^2 \le 2\|\psi\|_2\|\nabla\psi\|_2$, certified pencil floor $\kappa = 0.2273$–0.4502 $\Rightarrow b_k \le 2m\sqrt{K_{\mathrm{bot}}\mu_k^P/\kappa} \le C_S k$ — **LINEAR,** $C_S = 11.5137$–19.5731 **(median** 13.82**), trend** $x^{0.020\pm0.007}$ **NON-GROWING** vs 13.81π at $x^{0.258}$; factor 7.5–34.9 better at the bottom; rotation route priced and discarded ($4\sqrt{m}$, overshoots 4.2×10^3–1.0×10^6); global pencil useless ($x^{2.695}$) — ALL carried by $R = K_{\mathrm{bot}}/\kappa = 3.3634$–$9.7108$ ($x^{0.037\pm0.015}$), the ONE fit; $\min \Lambda^{\mathrm{arch}} = c_0^{\mathrm{arch}} - c_{M-1}^{\mathrm{arch}} = 2.5928$–$4.8981$ CLOSED AND ATTAINED (rest 2 done); $Q^\star_{\mathrm{sob}} = 12.39$–$23.94$ beats the ℓ^1 ceiling 2.69–10.13×, end to end $\to 2.01 \times 10^{-2}$–3.52×10^{-2} (gain up to 3.18×), dominant loss now $\Psi = 4.7 \times 10^3$–6.1×10^6; no-go breaks IN THE PENCIL only ($x^{1.986}$), controls exact (pencil $(1,1)$ to 3.7×10^{-12}, Sobolev price EXACTLY π); 10 candidates (stock $124 + 10 - 5 = 129$); core promoted as v550; **rest: the m-freeness of $R = K_{\mathrm{bot}}/\kappa$ — ONE scalar; a closed κ floor from the lag vector;** Ψ	ODD-CARRIES	27/27

Part	Probe	Contract / finding	Verdict	Checks
T152	`pencil_ratio`	the one scalar attacked head-on, **and both hoped-for arch gifts are REFUTED while the floor closes structurally**: A^{arch} is NOT positive in the odd sector on any window ($\lambda_{\min} = -2.8132$–-1.8396, certified by completed Cholesky; in pencil normalisation $\kappa_{\mathrm{ar}} = -5.28 \times 10^5$–$-4.00 \times 10^3 = O(-m^2)$, the lowest parity sine having L_P-energy $O(m^{-2})$ against an $O(1)$-negative form) and the atom part is negative too (-1408–-25.6): **odd-sector positivity is a CANCELLATION between geometry and arithmetic, not a property of either half** (at t_1: arch -5.26×10^5–-3.98×10^3 plus atom 8.19×10^3–2.88×10^6 — the atoms OVERSHOOT); additive Weyl misses κ by 1.5×10^4–1.8×10^6, the ℓ^1 budget by a further 2.3×10^5–9.9×10^7; BUT the arch diagonal is negative on only $k \leq 4$–7 modes (m-independent), above them quotient $\geq 0.989 \Rightarrow$ **the Schur two-block criterion (Schur 1917, both Cholesky floors subtracted) with a FIXED 16-mode low block certifies $\kappa \geq t = 0.2250$–0.2500 on every window, $m^{0.006\pm0.036}$, quartile medians identical — the floor is no longer a fit**, consuming ONE unproven inequality $B_{HH} \succeq tI$ (entrywise/Gershgorin provably insufficient: -8.99×10^5–-58 even after the cut; median row coupling 3.9–19.6 vs diagonal 0.36–0.89); the arch part does not carry the CEILING either ($K^{\mathrm{arch}}_{\mathrm{bot}} = 24.6$–$1326$, $m^{1.519}$, one to two orders above the true $K_{\mathrm{bot}} = 1.102$–2.118 — the atoms CREATE the alignment); new number $R \leq K_{\mathrm{bot}}/t = 4.408$–$8.471$**, $m^{0.099\pm0.167}$, certified on BOTH ends**, quartile walk 1.269 over a 4.54× m-range; two structural m-free routes lose 4–6 orders (Courant–Fischer: atom overshoot $O(m^2)$; two-sided bulk: KMS ratio a theorem, 285.95–288.97 $\leq$ 289, but $T_{HH} = 26.8$–3116 grows $m^{2.044}$); no-go breaks exactly at the unclosed step ($m^{1.994}$), control $(1,1,1)$ to 3.5×10^{-11}; $\Psi \leq 8.70 \times 10^3$–$6.24 \times 10^5$ IS $1/\lambda_{\min}(E)$ to within 1.12–1.19, grows $m^{2.18}$ because $\mu_1^P = O(m^{-2})$, and 89–92% of trace(E^{-1}) sits on the eight lowest modes — the ladder-sum rebuild is worth 3.46–5.29× end to end; honest dual reading printed (band estimator alone: RATIO-RESISTS); T151's $x^{0.037}$ was fitted against the zone variable, refit in m: 0.099; 10 map entries, NO promotion (the candidates consume the unproven block positivity); **rest: $B_{HH} \succeq tI$ as a block statement; an m-free K_{bot} ceiling; the Ψ rebuild, subset sup → ladder sum**	ONE-TERM-MISSING	37/37

Part	Probe	Contract / finding	Verdict	Checks
T153	`psi_ladder`	the mapped rebuild carried out, **and refuted by exactly its own factor — Ψ is PINNED**: for the subset maximising the first mode every term of $\sum_k \lambda_k^{-1}\langle v_k, \mathbf{1}_S\rangle^2$ is nonnegative, so $a_1/\lambda_1 \le \Psi \le 1/\lambda_1$ with $a_1 = 0.7695$–0.8306 — exactly the $8/\pi^2 = 0.8106$ of the first parity sine — Ψ determined to within $1/a_1 = 1.204$–1.300: **T152's** 3.46–5.29× **reserve never existed**, and the head/tail replacement lies 2.31–6.12× BELOW the hard lower bound (the maximising subset is aligned with v_1 — the head IS the value of Ψ); three rigorous rebuilt forms, all upper bounds on the same Ψ: mode-resolved ladder sum 1.195–1.761×, arithmetic-free operator form $(\mathrm{const}/t)\Psi_P$ and theorem-grade collapse $\Psi \le \mathrm{const}/(t\mu_1^P)$ both 3.73–9.70×; net gain 0.50–1.11× (median 0.62), end to end 1.01×10^{-2}–3.92×10^{-2} (the retired $c_0^{\mathrm{ap}} = 4.157$–$4.870$ and the collapse cost cancel); collapse cost named: $\lambda_1(A)/(t\mu_1^P) = 3.25$–$8.40$ — the pencil minimum not at the bottom mode; Ψ **CLOSED as a term** (form (c) retires Charikar's licence and every per-window diagonalisation, needing nothing beyond t); **SIGN REVERSAL on the bulk**: the arch part is POSITIVE on the bulk parity block ($\lambda_{\min}(B_{HH}^{\mathrm{arch}}) = 1.0362$–$1.4143 > t = 0.25$, peeling ≤ 8 modes, m-free) while the ATOMS are negative there (-484.6–-14.95); four block candidates dead (entrywise Gershgorin -1929–-21.35; block Gershgorin -333.4–-7.68, long-range coupling; mode-index T+H residual 0.70–0.95; recursive Schur scale-invariant, $t = 0.25$ at levels 0/1/2), the live Kato-type route positive per window ($1 + \lambda_{\min}(C) = 8.3 \times 10^{-4}$–$2.9 \times 10^{-2}$) but shrinking $x^{-1.778}$; t certified 0.2400–0.2500 on 18 windows, flat; ceiling: no fixed family flat (sines $x^{2.47}$, Hann $x^{2.46}$, Fejér degenerate; the overshoot is the HANKEL REFLECTION half, +9.45 vs Toeplitz −1.99), but **two inverse-iteration steps give FLAT** $K^F = 1.432$–2.369 **on a fixed-size-**8 **certificate** vs true $K_{\mathrm{bot}} = 1.102$–1.896; 9 map entries (v580–v588), NO promotion; **rest: TWO terms — the block inequality (sharp form: ONE number, the Kato distance against the m-free arch reserve** 1.04–1.41**) and the Green/alignment estimate of** $(A^{-1}L_P)^2 t_k$**,** $k \le 8$ **— both losing to the SAME missing object**	REBUILD-RESISTS	33/33

Part	Probe	Contract / finding	Verdict	Checks
T154	`green_align`	the Green/alignment estimate attacked head-on, **and the CEILING CLOSES, exactly and at fixed size** — with a direction lesson: the ceiling needs NO residual argument (Ritz values are UPPER bounds of the same-index eigenvalues, Courant–Fischer 1920/Cauchy 1829; Temple 1928/Kato 1949 is a FLOOR device) — the certificate $\mathrm{span}\{t_1..t_8\} + A^{-1}L_P\,\mathrm{span}\{t_1..t_8\}$ (16 columns, FIXED size, 8 Choleskys $\leq 8 \times 8$) carries $K^F = 1.1019$–1.9964 flat ($x^{0.094\pm0.058}$) **agreeing with the inertia-certified K_{bot} to** 5.17×10^{-7} **on** 12/12 — the 8 size-m LDL^T counts RETIRED from the ceiling step (sines alone: 11.2–2.1×10^4, $x^{2.47}$); FLOOR at fixed size REFUTED ($\|R\|^2/(\beta\lambda_1) = 6.3 \times 10^3$–$5.6 \times 10^9$, $x^{4.50}$; scan closes 1/12 at $k_c/m = 0.64$; scalar Temple vacuous 12/12); **the obstruction NAMED: seven bottom directions agree to** 0.15–1.35° **(median), ONE sits at** 82.93–89.79° **— that single misalignment IS the collapse price**, recovered IN FULL per window by one Cholesky (4.408–7.985); the missing fixed-size ingredient reduces to ONE arithmetic-free number (the L_P floor on the Ritz complement, flat 5.93–8.25 μ_1^P, worth 91–100%); Kato anatomy: the loss is NOT the floor ($\lambda_{\min}(B_{HH}) = 0.266$–$0.415 > t$) but misalignment (1.97–121 at 24.9–89.8°; minimiser 96.5–100% on the 8 modes above the cut), Hankel culprit refuted TWICE (both halves hurt at the minimiser; without the reflection $\lambda_1 = -7.49 \ldots -2.64$), the only working peel needs 0.40–0.69 of the window; end to end, two numbers never conflated: m-free-in-shape UNCHANGED 1.01×10^{-2}–3.92×10^{-2}, **per-window certified** 4.45×10^{-2}–3.13×10^{-1} **— inside the target band**; no-go: floor survives ($t = 0.05$), ceiling explodes $x^{1.99}$, certificate WITH it ($x^{2.29}$, overshoot 1.12–2.19) — attainment is a property of the REAL section; parity control 3.5×10^{-6}, direction violated 0/26; **rest: TWO uniformity terms — R1′ the m-free block floor (direction-aware) and R2′ the m-free bottom-mode floor on the Ritz complement (arithmetic-free)**; certificate core promoted as v551	ALIGN-RESISTS	29/29

Part	Probe	Contract / finding	Verdict	Checks
T155	`bottom_floor`	the two uniformity floors attacked head-on, **and the complement floor becomes an EXACT fixed-size certificate**: $\min_{v\perp W} v^{\mathsf{T}} L_P v \geq \mu^P_{K+1} - \lambda_{\max}(M^{1/2}(I - GG^{\mathsf{T}})M^{1/2})$, $M = \mathrm{diag}(\mu^P_{K+1} - \mu^P_k)$, $G = T_K Q_W$ — an identity plus one Rayleigh bound, a THEOREM for every m and every subspace, direction checked (the certified $\lambda_{\max}$ is SUBTRACTED and the certificate never exceeds the exact value) — reproducing the m-sized number to 0.999999783–0.999999958 on 16/16 windows, $K = 12$ sufficient everywhere (ladder $8 \ldots 64$ preregistered); controls hit from both sides ($\mathrm{span}\{t_1..t_8\} \to \mu^P_9$ at 1.05×10^{-12}; the WORTHLESS configuration $\mathrm{span}\{t_2..t_9\} \to \mu^P_1$ at 8.89×10^{-11}, reported rather than flattered); **the defect is localised at $k = 1$, the bottom mode itself**: free scale 80.74–81.00 against measured 5.82–7.86 (factor 10.31–13.92), largest uncovered fraction 0.539–0.611 ($x^{-0.063\pm0.016}$), and the 82–90° direction is DEMYSTIFIED — it lives on modes 9–12 (L_P Rayleigh 81.2–103.4 μ^P_1, $\leq 1.9 \times 10^{-2}$ of its mass on modes 1–8): exactly why $K = 12$; **the collapse price** 3.250–8.471 **is recovered** 78.8–100.0% **AT FIXED SIZE** (in full on 12/16, $\geq 90\%$ on 14/16; 32 Green columns + an 8×8 Ritz block + one 12×12 certificate, no size-m factorisation anywhere); two repairs refuted with mechanism (the pencil-ceiling chain dies on $\kappa = \lambda_{\max}(B) = 3.1 \times 10^3$–$2.7 \times 10^6$, $x^{2.82}$, attained where L_P is small; wider $r = 12/16/24$ raises the floor ×1.19–3.71 but the residual ×12.4–8514, Temple 0/32); block side: $\lambda_{\min}(B_{HH}) = 0.2430$–$0.4249$ certified positive, T154's anatomy belongs to a DIFFERENT vector (the B_{HH} minimiser puts only 1.5×10^{-5}–0.69 on the eight-mode window), the direction-aware 2×2 split dies on the coupling ($\lVert B_{wd} \rVert = 5.81$–$389.41$, ×16.8–908 too large, $x^{1.94}$), the local atom norm exceeds the local arch reserve by 1.50–5.12, the deeper cut does not terminate; **the arch reserve IS the symbol infimum** (1.000022–1.000521 on 14/14; band 1.0004–1.4049, factor 4.0–5.6 over t — theorem candidate, Szegő 1915/Widom 1958) while **the FULL symbol infimum is $-714.2 \ldots -7.6$, negative on every window against a positive section floor: NO symbol argument can produce R1′** — the mechanism must be finite-section Fejér cancellation (min diagonal 0.3840–0.7444); end to end 3.28×10^{-2}–2.83×10^{-1} CERTIFIED AT FIXED SIZE, the declared 4×10^{-2} bar missed by 1.22 and reported (like-for-like: the share 78.8–100%); no-go breaks on TWO axes (ceiling $x^{2.010}$; the new instrument reads defect 0.999–1.000 at exactly 90.00°, recovery 0/8); parity control 4.85×10^{-14}, Dirichlet hits its own $2/\sqrt{N}$ prediction to 4.1×10^{-7}; **rest: R2″ an m-free upper bound on ONE 12×8 object; R1″ the m-free atom part on the bulk, with every symbol argument excluded**	FLOORS-RESIST	27/27

Part	Probe	Contract / finding	Verdict	Checks
T156	`twelve_by_eight`	the two rest objects attacked head-on, **and both become SINGLE SCALARS**: on $\mathrm{span}\{t_1, A^{-1}L_P t_1\}$ the coupling is the identity $t_1^\intercal A y_1 = t_1^\intercal L_P t_1 = \mu_1^P$ (no A in it), so the t_1-loss of the bottom Ritz mixing is an EXACT closed function $F(P, r)$ of the Kantorovich product $P = (t_1^\intercal A t_1)(t_1^\intercal A^{-1} t_1) = 5.6 \times 10^2$–$1.0 \times 10^6$ and the inverse moment ratio $r = \|A^{-1}t_1\|^2/(t_1^\intercal A^{-1} t_1)^2 = 2.7158$–$3.4089$, FLAT ($x^{-0.047\pm0.020}$; reduction verified to 2.2×10^{-16}, against the true 2×2 Ritz pair to 4.4×10^{-16}); the box route is VACUOUS despite the new m-free bound $P - 1 \geq L^2 s^2 (r - 1)$ ($\sup F = 1.0000$ where ≤ 0.632 is needed, attained at $P \to 1$); the best route is the two-line theorem $r \leq 1/(Ls) \leq 1/p_1$, TIGHT to 1.03–1.16, so **R2″ is ONE m-free lower bound on** $p_1 = \cos^2 \angle(t_1, e_1(A)) = 0.2010$–$0.3282$ (55.05–63.36°, flat $x^{0.054\pm0.029}$; loss at the ceiling 0.6718–0.7990 against truth 0.5392–0.6316); interlacing refuted in BOTH forms (rank hole at $K = 12$: eigenvalue 1 exactly; empty at $K = 8$: floor 1.02–2.58 μ_1^P against 5.78–7.83); the model dominates the truth 16/16 (ratio 0.833–0.917) — MEASURED, and the stress kills it; block side: **J2a in T155's ≥ 1 form REFUTED** — $\inf f^{\mathrm{arch}}/(4\sin^2(\theta/2)) = 0.8226$–$1.3973$, BELOW 1 on 3/12 (T155's 1.00002–1.00052 was the upper surface); survives as the weaker $\inf \geq t$ (factor 3.29–5.59, non-decreasing, grid-stable 1.4×10^{-5}, minimum at $\theta/\pi = 0.990$–1.000); Fejér damping real but ONE order (5–31 where the split needs 3–1981 growing); additive split DIVERGENT ($\|B_{HH}^{\mathrm{atom}}\| = 2.2$–$2673.4$, $h^{2.31}$, closes 0/12); **the mechanism is ALIGNMENT**: on the minimiser arch 1.4685–91.7130 + atom $-91.2694\ldots - 1.1151$ $= \lambda_{\min}(B_{HH}) = 0.2661$–$0.4436$, the minimiser seeing 6.2×10^{-3}–5.0×10^{-1} of the atom operator ($x^{-1.044}$) at 52.7–90.0° from the atom-extremal vector; end to end UNCHANGED 3.28×10^{-2}–2.83×10^{-1}, recovery median 99.8% ($\geq 75\%$ on 12/16, like-for-like); balance 9 THEOREM (3 new), 6 CERTIFIED, 3 MEASURED (zero-MEASURED target missed); no-go breaks on THREE axes (K^F $x^{2.047}$; p_1 COLLAPSES to 7.4×10^{-15}–6.6×10^{-10}, $x^{-4.818}$; r-ceiling tightness 12.5–24.6 vs 1.03–1.16) and kills the domination step (16/16 real, 0/8 no-go); controls: parity 4.8×10^{-14}, μ_9^P at 2.3×10^{-13}, μ_1^P at 2.0×10^{-11} in the worthless configuration, Dirichlet $2/\sqrt{N}$ to 7.4×10^{-5}; **rest: R2″ one m-free lower bound on the angle p_1; R1″ one m-free alignment statement (the arch half ready: a one-variable grid infimum)**	TERMS-RESIST	37/37

Part	Probe	Contract / finding	Verdict	Checks
T157	`angle_floor`	the two angles attacked head-on, **and neither falls, but both change shape**: the best p_1 route is the RESOLVENT, $p_1 \geq \hat{g}_1^2(1-\text{tail}) = 0.1968$–$0.3228$ (97.9% of the measured 0.2010–0.3282), splitting into a NEW THEOREM times ONE measured number — **the sine-block confinement** $\lVert\gamma_H\rVert^2 \leq \lambda_1/(t\mu_{17}^P) \leq (S/t)/\rho_{17} = 0.0165$–$0.0293$ (from the certified floor t and ladder S alone: $e_1(A)$ is 97.1–98.4% inside the first 16 parity sines; with κ instead of S the bound is VACUOUS 43.5–38632) times $\hat{g}_1^2 = 0.2010$–0.3282 (flat, ONE entry of the bottom vector of a 16×16 Schur null problem); three routes refuted with numbers: the classical Rayleigh angle bound EMPTY already measured (16/16 negative, $\lambda_2/\lambda_1 = 1.73$–$5.87$ nearly degenerate), its block version empty at every J although $p_{\text{blk}}(2) = 0.9369$–$0.9993$ (t_1 lives in the bottom PAIR — the T156 question answered), the angle-free Cauchy–Schwarz route loses exactly the Kantorovich $P = 5.6 \times 10^2$–1.0×10^6; **the structural gem:** $1/s = (S_L)_{11} = 2.3359$–$6.2049$ **FLAT** — R2$''$ is a bound on ONE diagonal entry of the 16×16 Schur complement the chain already forms, and the inversion-free CS ceiling is off by 5.4×10^2–5.2×10^5 (the cancellation is nearly complete); **the Lipschitz deckel is EXECUTED**: adaptive bisection with the local constant certifies $\inf f^{\text{arch}}/(4\sin^2(\theta/2)) \geq t$ on 12/12 (0.2500–0.5353, 255–3139 evaluations, cost $x^{0.850}$) — the arch half is CERT-UNIF for the first time; mechanism localised: atom extremal $\theta/\pi = 0.0158$–0.3762 against arch 0.9901–0.9995 — OPPOSITE ends, so a proof must use the θ-GROWTH of the arch ratio, not its infimum at π; the domination $B_{HH}^{\text{arch}} - tI \succeq (-B_{HH}^{\text{atom}})_+$ holds 12/12 (quotient 1.0003–1.0907, margin 7.3×10^{-4} at $h = 970$, trend shrinking — per window, NOT uniform); substitution costs 1.00–1.62; balance 9 THEOREM, 2 CERT-UNIF, 2 CERT-WINDOW, 3 MEASURED (target 1 missed — but all three now fixed-size in the statement); no-go breaks on FIVE axes (p_1 $x^{-4.818}$; S $x^{2.289}$; confinement to vacuum; resolvent identically ZERO; $(S_L)_{11}$ $x^{2.268}$), floor survives; parity 4.8×10^{-14}, Dirichlet $2/\sqrt{N}$ to 7.4×10^{-5}; **rest: an m-free UPPER bound on $(S_L)_{11}$ (cancellation-seeing); an m-free non-shrinking R1$''$ domination (pointer: θ-growth); the T156 model-domination debt**; the four instrument candidates of T155/T157 promoted as v552	ANGLES-RESIST	32/32

Part	Probe	Contract / finding	Verdict	Checks
T158	`schur_entry`	the cancellation-seeing bound attacked head-on, **and it EXISTS, as a THEOREM**: the Thomson dual form $s=(B^{-1})_{11}=\max_x(2x_1-x^{\mathsf T}Bx)$ (Maz'ya 1985) makes every trial vector a LOWER bound on s, hence an UPPER bound on $1/s$ — exactly the variational structure Cauchy–Schwarz lacks (CS evaluates a MAXIMUM at one direction and missed by 3.13×10^3–5.18×10^5; reproduced, direction intact 28/28); the Cholesky ladder $g_K=\sum_{j\le K}y_j^2$, $y=L_K^{-1}e_1$: terms STRICTLY POSITIVE, partial sums monotone 28/28, $1/g_1=\hat a=$ T157's route; at $K=16$ **certified** $1/g_{16}=2.9670$–7.9664 **against true** $1/s=2.3359$–6.3868**, TIGHT to** 1.1323–1.2738**, flat** **(**$x^{0.059\pm0.017}$**)**; the Green route is the IDENTITY — $\mathrm{span}\{t_1,A^{-1}L_Pt_1\}$ attains s to 10^{-9} because $L_Pt_1=\mu_1^Pt_1$ (KMS 1953): **the entry is TWO-dimensional** once one Green column is granted, the sine truncation pays exactly 1.13–1.27; floor $L\,g_{16}=0.2143$–0.2724 lands in the band of T157's measured p_1 WITHOUT an angle; **the T157 growth pointer REFUTED**: the growth is real and sub-quadratic ($q=1.259$–1.438) and the binding vector does sit in the lowest dyadic band (0.6818–0.9988 of its mass — localisation confirmed), BUT the atom negative mass grows FASTER ($\theta^{-1.546\ldots-1.744}$ vs arch $\theta^{-1.203\ldots-1.458}$), band-local domination fails in EVERY band 21/21 (lowest-band cover 0.4161–0.4507), the off-band coupling exceeds the margin by 660–7.71×10^5 — what carries the inequality is the structure a dyadic argument discards; margin 3.57×10^{-5}–2.86×10^{-3}, SHRINKING ($x^{-1.649\pm0.643}$); the T156 debt RELABELLED MEASURED → CERT-UNIF ($F(P,1/(L\,g_{16}))=0.7276$–$0.7857$ vs $d_1=0.5392$–0.6241, margin 1.1969–1.3568 on 28/28, both sides fixed-size); end to end survives the substitution (0.2872–1.0 of $\lambda_1(A)$, cost 1.000–1.724); balance 6 **THEOREM / 3 CERT-UNIF / 4 CERT-WINDOW / 1 MEASURED** (was 5/2/3/3 — the measured count falls three → one, p_1 RETIRED); no-go breaks FIVE-FOLD (p_1 $x^{-4.818}$; $1/g_{16}$ $x^{2.267}$ tracks the collapse; $L\,g_{16}$ $x^{-0.275}$; tightness stays 1.001 — the size breaks, not the sharpness); parity 4.8×10^{-14}, ladder exactly constant in K on $A=L_P$ (4.0×10^{-15}), Dirichlet $2/\sqrt{N}$ to 7.4×10^{-5}; **rest: an m-free upper bound on ONE quadratic form** $x^{\mathsf T}B_{LL}x=2.9670$–$7.9664$ **(halves cancel to depth h^2 — the explicit formula's own cancellation); the SIGN STRUCTURE of the off-band arch entries**; five candidates (P1–P5) bundled with T159, theorem cores promoted	ENTRY-RESISTS → v553	36/36

Part	Probe	Contract / finding	Verdict	Checks
T159	`exact_form`	the cancellation executed ALGEBRAICALLY — **seven machine-checked identities** (to 10^{-12} against the absolute scale $\sum \lvert c_d w_d\rvert$; relative to the cancelled total only 10^{-12}–1.2×10^{-8}: the cancellation eats 4.0–8.1 decimal digits, reported as a measurement): (P1) $x^\mathsf{T} B_{LL} x = y^\mathsf{T} A y$, pencil gone; (P2) the EXACT LAG SUM $= \sum_d c_d w_d$, $w_d = T_d - H_d$, arithmetic and geometry separated; (P3) w_d in CLOSED form (256-term Dirichlet kernel, 1.4×10^{-15}); (P4) **the GAUGE IDENTITY $\sum_d w_d = 0$ EXACTLY** — Toeplitz-minus-Hankel annihilates constant lag vectors: the form is BLIND to the lag mass, exactly where the h^2 halves live; (P5) $w_0 = \sum x_k^2/\mu_k$ and $2w_0 - w_1 = \lVert x\rVert^2$; (P6) p-fold Abel exact to $p = 5$; (P7) the TmH signature $A_{r,s} - A_{r+1,s+1} = f(r+s)$; m**-free NOT reached, four routes closed with exponents**: the h^2 does NOT telescope (both halves grow $h^{+2.104}$ = the common WEIGHT); T158's fixed sixteen-vector REFUTED ($h^{+3.510}$, $\mathrm{cond}(B_{LL}) \sim h^{+2.887}$, needed accuracy $h^{-3/2}$ vs optimiser drift 2.5×10^{-2}–1.5×10^{-1}); one sine $h^{+3.066}$; ansatz family $h^{+3.001}$; Abel-$\ell^1\times$sup best rung $h^{+3.604}$ (misses by 5.6×10^{6}–2.2×10^{10}, $W_1^1 = -w_0$ makes $\sup\lvert W^1\rvert \geq h^2$ unavoidable); two m-free-shaped pieces NEW: $\lVert\Delta c^{\mathrm{arch}}\rVert_1 = 3.65$–$5.38$ flat (BV ≤ 6) and the T150 budget at the atom cutoff $X = e^{2\alpha}$; **(S1) EXACT: B_{HH}^{arch} is a symmetric Z-MATRIX** (100% off-band nonpositive, raw, 18/18), checkerboard refuted (0.489–0.494 = coin flip); Collatz–Wielandt BITES on the arch half: **closed sign-based floor $\min_k(B_{HH}^{\mathrm{arch}}\mathbf{1})_k = 0.819$–$1.165$, flat, above** $t = 0.25$ — but the atom block has NO sign law (0.31–0.49) and the full-block M-matrix criterion is vacuous ($-9.32\ldots-0.61$ vs exact 0.074–0.197); end to end $t = 0.0705$–0.1583 unmoved (M1/M2 structural, not quantitative); no-go breaks 5/5; balance 6/3/4/1 → 13 **THEOREM** / 6 **CERT-UNIF** / 4 **CERT-WINDOW** / 1 **MEASURED** (seven new theorems, three new CERT-UNIF); NUMERICAL HORIZON declared: $\mathrm{cond}(B_{LL}) > 10^{12}$ from $h = 1292$, one window dropped, the same obstruction as its own measurement; **rest: ONE fully explicit pairing inequality $\sum(\Delta c)_d W_d^1$** needing CORRELATION instead of sizes ($\ell^1\times$sup off by $h^{3.7}$, the cancellation exactly h^2); a third device for the atom off-band block; theorem cores promoted	FORM-RESISTS → v553	41/41

Part	Probe	Contract / finding	Verdict	Checks
T160	`pairing`	the one pairing $\sum(\Delta c)_d W^1_d$ attacked through its CORRELATION structure, **and the watershed of the phase arrives — the SAMPLING IDENTITY: the last term IS a prime exponential sum** (19 of 20 zones $h = 199$–1256, 1 dropped at the declared horizon); W^1 **does NOT oscillate**: $J = 3$ sign blocks, flat, CLOSED bound $J \le 130$ (trigonometric zero count) — one smooth three-lobe profile; **block/Leibniz device REFUTED**: block-Hölder $h^{+4.115}$, block-Abel $h^{+4.131}$ — 5.2–16.5× WORSE than $\ell^1 \times \sup$ at $h^{+3.595}$; head peel refuted ($K^* = M$ on 19/19; best overshoot 60.3–$4.80 \times 10^3 \sim h^{+2.428}$ = the h^2 itself); **NEW THEOREMS — the sign-definite MOMENT LAWS** $m_1 = -[S_0^2 + 2\sum_j P_j^2] \le 0$, $m_2 = -[2S_1 - (M-1)S_0]^2 \le 0$, m_p closed for every even p; the fixed witness (frac 0.2, degree 48) reproduces the arch half to worst slack 1.95×10^{-8} of the $O(1)$ total — **FLOOR-LIMITED at** 4.12–428× **the double-precision floor: SECOND HORIZON declared**, the arch half neither established nor refuted on any double-precision surface; sparsity refuted (support fills 0.32–0.64; support-localised ℓ^1 misses $h^{+3.505}$); **the SAMPLING IDENTITY** (1.0×10^{-16}–2.9×10^{-15}): $\sum_d c_d^{\text{atom}} w_d = -\frac{1}{2}\sum_{n \le X}(2\Lambda(n)/\sqrt{n})\,\widehat{w}(\log n)$ $\Rightarrow$ **the atom half IS a finite combination of** $\sum_{n\le X}\Lambda(n)n^{-1/2}\cos(t\log n)$ **at** 32 **EXPLICIT frequencies** $t = \pi(k \pm l)/\alpha$, needed to depth $h^{-2} = 2.19 \times 10^{-6}$–$1.12 \times 10^{-4}$, measured to cancel only to 0.00–0.37 of the trivial bound — **the h^2 cancellation is the INTRINSIC ARITHMETIC HARDNESS**, not an assembly artefact (finite von-Mangoldt table only; no zero read, no equivalence claimed); R1″: rank/support and corner (T2) refuted (the norm route in disguise, $\lVert M_{12}\rVert_2 \le 71.4$); **the surviving fact** $\rho = 1.0036$–$1.0140 > 1$ **on every window, FLAT** — arch dominates atom on every destructive direction; its inequality form refuted (floor $-7.93\ldots-1.09$; margin vs cross-coupling = factor 10^4); **U3 UPGRADED to THEOREM**: TV < 6, measured 3.9101–5.4153 vs closed 3.9105–5.4153 (FOUR digits); U1 PARTIAL: Z-law ⇔ ONE prime-free trigonometric inequality (pointwise form refuted — the weight is essential); end to end 3.79×10^{-10}–2.10×10^{-7}, with an exact atom half the SAME chain gives 1.0000 (deficit in ONE term); no-go breaks 6/6; balance 13/6/4/1 → 14 **THEOREM (6 new) / 4 CERT-UNIF / 3 CERT-WINDOW / 3 MEASURED / 8 refuted families**; T160-P1…P7 PENDING, no promotion; **rest: R-A the symbolic Chebyshev bound; R-B the prime-free inequality; R-C the exponential-sum bound at the 32 frequencies (not an exercise); R-D a fifth R1″ device**	PAIRING-RESISTS	46/46

Part	Probe	Contract / finding	Verdict	Checks
T161	`classical_closure`	the two classical rests plus the circularity triage, **and the triage returns its most consequential answer: THE CHAIN IS NOT CIRCULAR** (19 log-spaced zones $h = 50$–1445, a 28× lever arm; $X = e^{2\alpha} = n_{\mathrm{zone}}^2$ exactly; $\psi(x)/x \le 1.038830$ verified at every jump to 400000); **R-A closes m-free**: $A = D\widehat{A}$ exactly (4.2×10^{-16}–2.0×10^{-13}), only the $1/s$ head binds ($i\pi$ margin 1.29–3.29) ⇒ **scale-free Bernstein rate $\rho^* = (3+\sqrt{5})/2 = 2.618034$** (interval ratio 5 only), $M = 3.06$–5.93 uniform, envelope 0.055–$0.074 \le 1$; **the head needs NO peeling**: $c_d^{\mathrm{arch}} = \Psi_d + D\widehat{G}(dD, D)$ with $\Psi_d = -\frac{1}{2}[(d{+}1)\log(d{+}1) - 2d\log d + (d{-}1)\log(d{-}1)]$ CLOSED, D-free, m-free (1.7–4.8×10^{-16} at every lag); full-interval rate $\rho_{\mathrm{reg}} = 2.05$–$3.38$ NOT scale-free; "fixed degree" REFUTED: schedule $K(h) = O(\log h)$ both routes ($K = 12$–19 / 9–25); residual the prime-free LOG-MOMENT $\sum \Psi_d w_d = 2.15$–3.14 of the arch half, outside the polynomial ladder; **R-B REFUTED in all four readings** (per pair 3168/4320, worst +0.92; Abel tail 1229; aggregate by SIGN — arch negative, off-diagonal positive = the harmful direction; Abel condition false 3801/4320) — **survives: the FRACTION bound** $\lvert\mathrm{off}\rvert/\lvert Q^{\mathrm{arch}}\rvert = 0.1035$–$0.1713 \le \frac{1}{4}$**, flat** ($h^{-0.116}$), m-free version a 16×16 Gram question (Fejér 1915, Schur 1917); **NEW THEOREM: $t(2\alpha) = 2\pi j$ exactly — the 32 frequencies ARE the Fourier harmonics of the log-window**; the measured cancellation is FULLY the PNT main term $((\sqrt{X}-1)/(\frac{1}{4}+t^2)$ to 0.014–0.232 of mass, improving $X^{-0.39}$ — a Mellin factor, no arithmetic saving); **δ arithmetic: needed $\delta = 1.1482$–1.8809 against RH strength $\frac{1}{2}$** — gap $X^{0.648}$–$X^{1.381}$; absolute demand 0.012–0.31 of the LAST TERM, below the boundary term of every partial summation ⇒ no $\psi(x) - x$ input reaches it; frequency geometry worth $\delta = 0.29$–0.92 (suppression 7.0–19.8, matched control), already inside the numbers; **re-split moves the demand** (smooth prime term cancels the arch half to 0.50–0.76; $\delta_{\mathrm{eff}} = 0.984$–$1.377$, still $> \frac{1}{2}$ on 18/18) ⇒ **the h^2 sits in the SPLIT, not in the primes**; balance 20/10/4/9/3; closed cores of T160+T161 promoted; **rest: R-A′ the log-moment; R-B′ the Gram fraction bound; R-C′ ONE split with $\delta_{\mathrm{eff}} < \frac{1}{2}$; R-D a fifth device**	CLOSURE-RESISTS → v554	35/35

Part	Probe	Contract / finding	Verdict	Checks
T162	`third_split`	the search for the third split, **and it EXISTS while the exhaustion SATURATES** (the T161 surface rebuilt independently: 19 zones $h = 50$–1445, 18 windows inside the cond horizon $h = 50$–1218; two currencies declared before any candidate: δ_{val} the measured residual value, δ_{bnd} what a proof must beat, RH strength $\frac{1}{2}$ in both); C0/C1 REPRODUCE T161 to 0.01 (1.1482–1.8809 / 0.9839–1.3767); **C2 the MELLIN LADDER — the third genuine gain, an ASYMPTOTIC series**: closed cell moments $C_d(-(2k+\frac{1}{2}))$, optimum $K^* = 2$ on 18/18, gain 0.359–0.446 ($\delta_{\mathrm{val}} = 0.6247$–$0.9307$), past K^* the residual RISES $\times 15.5$–25.1 (the predicted log-divergence) — not iterable; **C5 the Abel ladder: optimal level EXACTLY ONE** ($32\pi/\alpha = 16.7$–$41.9 > 1$, closed), $\delta_{\mathrm{bnd}} = 1.3965$–$1.7867$, and after the step the demand is EXACTLY $\frac{1}{2} + \log(2\kappa\lVert\Delta w\rVert_1/\lvert Q\rvert)/\log X$ — **R-C′ is PRIME-FREE**, all arithmetic in $\kappa = 0.038821$; **C4 the FEJÉR split — the actual find:** $\delta_{\mathrm{bnd}} = 0.1332$–$0.4169$ **BELOW** $\frac{1}{2}$ **on** 18/18 (taper = smoothing of the Λ-mass by self-adjointness; lowering 1.15–1.51) — **at a price growing $h^{+2.86}$: RELOCATED, not closed**; frontier $P_{\min} = 7.2 \times 10^1$–$8.2 \times 10^4 \sim h^{+2.0}$, 1.17–3.31 decades per δ unit; **R-A′ CLOSED**: the log-moment on three routes to machine precision (0.0–6.0×10^{-16}; Lerch/Frullani integral, $d = 1$ peel $= 2\log 2$ reproduces $\Psi_1 = -\log 2$; Wronskian boundary term completes T161); **R-B′ REFUTED**: the 16×16 Gram form INDEFINITE on 18/18 ($\lambda_{\min}/\lvert\lambda_{\max}\rvert = -13.34\ldots-1.59$; Gershgorin inapplicable at 7.97–580.76) — survives: the a-weighted quarter bar 0.1435–0.2902 flat ($h^{-0.102}$), 14/18; end to end 6.1×10^{-17}–1.8×10^{-15} of the large scale; the factorising gap 0.77–0.86 > 0.12–0.43 to RH strength; surrogate breaks $\times 5.2 \times 10^2$–4.6×10^7, gauge break EXACTLY $c_0 \sum w_d$, under-resolution does NOT break (declared as INFO); **anti-fitting:** $\beta^* = 0.600$–$0.800 \neq \frac{1}{2}$ **would look** 100× **better and is REFUSED** ($\frac{1}{2}$ forced by the $n^{-1/2}$ weighting); balance 10 THEOREM / 2 CERT-UNIF / 3 CERT-WINDOW / 9 MEASURED, zero FIT rows; T162-P1…P6 PENDING, no promotion; **rest: R-C″ the Pareto front; R-B″ a substitute for the Gram positivity; R-D a fifth device**	DELTA-REDUCED	30/30

Part	Probe	Contract / finding	Verdict	Checks
T163	`pareto_front`	the survey of the Pareto front, **and the front resists AS A THEOREM, not as a failed search** (28 zones $h = 142$–1445 with declared caps, 27 windows inside the cond horizon; the closed weight vector re-derived as a CORRELATION form, $O(h^2)$ and block-size-free, 1.6–2.0×10^{-15}; $\psi(x)/x \le 1.038830$ verified at every jump to 400000, $\kappa = 0.038821$); **the price axis is CHAIN-DERIVED**: $Q(x^*) = 1/g_{16}$ (the T160 optimiser IS the T158 top rung), $P_{\text{aff}} = P_{\text{rung}}(1) = g_{16}B_{11} = 3.9 \times 10^3$–$1.0 \times 10^7 \sim h^{+3.049}$; the Fejér knob has BOTH endpoints in the chain ($\sigma = 1$ exactly the K=1 rung, $\sigma = \infty$ the K=16 rung) and is measured MONOTONE in both coordinates at all 27×50 grid points $\Rightarrow$ **the knob curve IS the front**; flat caps 1.25/2/10 and every flat rung tier: 0/27 **below** $\frac{1}{2}$; only P_{aff} reaches 27/27 (-0.004–0.019); crossing $P_{\text{cross}} = 8.1 \times 10^2$–$1.0 \times 10^5 \sim h^{+1.906}$ at 0.0066–0.2047 of P_{aff} — 5–152× strictly INSIDE the accepted certificate, margin widening; **the price anatomy is an IDENTITY**: $\delta_{\text{bnd}} = \frac{1}{2} + \log(2\kappa g_{16}\text{TV}/P)/\log X$ exact at all 27×50 points ($\le 7.0 \times 10^{-14}$) $\Rightarrow$ $\delta_{\text{bnd}} < \frac{1}{2} \iff P > 2\kappa g_{16}\text{TV}$, crossing exponent decomposes additively (TV +2.013, g_{16} -0.061); taper gain in TV bounded and h-flat (2.75–4.64, worth ≤ 0.276 of δ, decaying $1/\log X$; τ_{TV} median -0.0191); the taper DESTROYS the cancellation (CANC 2.2×10^{-6}–$3.6 \times 10^{-4} \to 0.449$–$0.934$) — the price is the cancellation depth read backwards; mode sweep $K = 16 \to 64$: price dividend, WORSE demand (+0.0135), TV exponent unmoved ($+1.83/+1.82/+1.81$); **NEW THEOREM (T1–T4):** $w_0 = \lVert a \rVert^2$**,** TV $\ge \lvert w_0 \rvert$ **(telescoping),** $x_1 = 1 \Rightarrow \text{TV}(x) \ge 1/(4\sin^2(\pi/N)) \sim h^2$ **for EVERY admissible trial vector** — verified on all 1674 built, slack 5.0–23.2; hence $\delta_{\text{bnd}} < \frac{1}{2}$ forces $P > 2\kappa g_{16}/\mu_1^P = 5.5 \times 10^1$–$5.9 \times 10^3$, 5.5–590× above the largest flat cap — **R-C‴ CLOSED NEGATIVELY by an inequality; the h^2 IDENTIFIED: T162's $h^{2.86}$, the crossing's $h^{1.91}$ and T161's granularity are ALL the reciprocal smallest parity eigenvalue — the spectral gap of the parity Laplacian (KMS 1953) meeting $x_1 = 1$: NEVER IN THE PRIMES**; R-B″: the chain never needed the sign (duality + evaluated number, headroom $\ge 10^{14.9}$) — what remains is UNIFORMITY (R-B‴); R-D type mismatch (operator smoothing degrades $\times 2.3 \times 10^3$–$2.7 \times 10^5 \sim h^{+2.04}$ — the same h^2); no-go staircase MONOTONE $1.0 \to 9.1 \to 25.4 \to 129 \to 434$; out-of-sample $\sigma = 3$: $\delta_{\text{bnd}} = 0.176$–$0.380$ on 26/26 at 0.14–0.23 of P_{aff}; balance 11 THEOREM / 1 CERT-UNIF / 4 CERT-WINDOW / 10 MEASURED, zero FIT rows; **rest: R-E the successor (both arms prime-free); R-B‴; R-D**	FRONT-RESISTS → v555	32/32

B Symbols used in Sections 5–7

All of these are sandbox objects; they are listed so the figures and formulas above can be read without the diary.

Symbol	Meaning
Q_{Weil}, Q_{cert}, Δ_2, Δ_{arch}	the four terms of the transport ledger; $Q_{\mathrm{Weil}} = Q_{\mathrm{cert}} + \Delta_{\mathrm{arch}} + \Delta_2$ closes at 4.4×10^{-16} (v541)
A_{fam}, A_{shift}	the two archimedean halves of the heat family; $\Delta_{\mathrm{arch}} = A_{\mathrm{fam}} - A_{\mathrm{shift}}$ (T82)
I5	the one remaining coupled inequality, $Q_{\mathrm{cert}} + \Delta_2 + A_{\mathrm{fam}} - A_{\mathrm{shift}} \geq 0$ (T79)
μ_k, w_k, g_k	atom strength, handover window width, atom spacing at zone k
$D_k(\alpha)$	the Schur-complement defect of the induction step; target $D_k \leq \mu_k/2$ (T98)
$\sigma_k(\delta)$	the E_- Schur profile of the genuine Weil form just above atom entry (T102)
E_-, E_0, E_+	the three atom eigenspaces; the k-th atom acts as $\mathrm{diag}(-\frac{1}{2}, 0, +\frac{1}{2})$
bare_k, b_0	$\lambda_{\min}(Q_{\mathrm{full}}\|_{E_-})$ and its certified excess over $\mu_k/2$ (T105)
L, Λ_-	the soft dressing scalar and its free cap (5.63–7.52, divergent) (T105)
J	the mirror-parity involution; $JQJ = Q$, $JB = -BR$ (T105)
$\rho_\pm$	the Friedrichs-angle cosines of the even / odd channel (T106)
(R)	$Q_{\mathrm{full}}(\alpha_k)\|_{J=-1} \succeq (\mu_k/2)P_-\|_{J=-1}$, the remaining Loewner statement (T106)
s^*, τ, κ, ε	Sherman–Morrison scalar, its Woodbury split $s^* = \tau + \kappa$, and $\varepsilon = 1 - \tau$ (T107)
r	the residual ratio κ/ε; (R) $\iff$ $r \leq 1$; measured 0.0053–0.1814 (T107)
x, h, x_0	the pole response $x = T_{\mathrm{odd}}^{-1}\tilde{t}$, its demand compression $h = V^{\top}x$, and the single edge entry that carries $\\|h\\|$ on 8 zones (T108)
ω, ω_{cert}	the wing scalar $(\mu/2)\lambda_{\max}(\Gamma)$ with $\Gamma = V^{\top}T_{\mathrm{odd}}^{-1}V$, and its graded-matrix-cap certificate (T108/T109)
m_k, need109	the strict odd-channel margin $\lambda_{\min}(Q\|_{\mathrm{odd}})$ at zone k, and the explicit demand $(\mu/2)H_{\mathrm{cert}}^2/((1-\omega_{\mathrm{cert}})\kappa)$ of the T109 chain; $m_k \geq$ need109 on 16/16 measured zones (T110)
(Σ, r, σ); θ, φ	the boundary state of the Riccati recursion (partial minimisation of the pole-free form over the interior; carries the pole exactly), and the two exponents of the measured law $\varepsilon \approx 8.34\, D^{\theta}\alpha^{\varphi}$, $\theta = 1.790$, $\varphi = -6.04$ (T116)
δ_ℓ, $\hat{s}_\ell$, U	the rung contribution of the level telescope ($\varepsilon_\ell = \varepsilon_{\ell+1} + \delta_\ell$), its oscillation-space residual data ($r_\ell = W\hat{s}_\ell$), and the certified Parseval majorant $U = Z^{\top}T_M(\mathrm{up})Z$ of the per-rung bound (8R) (T124)